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| United States Patent | 12409197 |
| Kind Code | B2 |
| Date of Patent | September 09, 2025 |
| Inventor(s) | Henn; Matthew R. et al. |

## Synergistic bacterial compositions and methods of production and use thereof


### Abstract

Provided are therapeutic compositions containing microbial populations for prevention, treatment and reduction of symptoms associated with a dysbiosis of a mammalian subject such as a human.



**Inventors:** **Henn; Matthew R. (Somerville, MA), Litcofsky; Kevin Daniel (Boston, MA), D'Onofrio; Anthony Mario (Northborough, MA), Ohsumi; Toshiro K. (Cambridge, MA), Mckenzie; Mary-Jane Lombardo (Arlington, MA), Von Maltzahn; Geoffrey (Boston, MA), Cook; David N. (Brooklyn, NY), Berry; David Arthur (Brookline, MA), Afeyan; Noubar B. (Lexington, MA), Aunins; John Grant (Doylestown, PA)**

**Applicant:** **Seres Therapeutics, Inc.** (Cambridge, MA)



**Family ID:** **52278746**

**Assignee:** **Seres Therapeutics, Inc. (Cambridge, MA)**

**Appl. No.:** **18/424354**

**Filed:** **January 26, 2024**

### Prior Publication Data

| Document Identifier | Publication Date |
|---|---|
| US 20240325464 A1 | Oct. 03, 2024 |

### Related U.S. Application Data

division parent-doc US 17588122 20220128 US 11918612 child-doc US 18424354
division parent-doc US 16230807 20181221 US 11266699 20220308 child-doc US 17588122
division parent-doc US 15039007 US 10258655 20190416 WO PCT/US2014/067491 20141125 child-doc US 16230807
us-provisional-application US 61908702 20131125
us-provisional-application US 61908698 20131125
us-provisional-application US 62004187 20140528


### Publication Classification

**Int. Cl.:** **A61K39/02** (20060101); **A23L33/135** (20160101); **A61K35/74** (20150101); **A61K35/742** (20150101); **A61K35/744** (20150101); **A61K45/06** (20060101); **C12N1/20** (20060101); A61K35/00 (20060101)

**U.S. Cl.:**

CPC **A61K35/742** (20130101); **A23L33/135** (20160801); **A61K35/74** (20130101); **A61K45/06** (20130101); **C12N1/20** (20130101); A61K2035/11 (20130101); A61K2035/115 (20130101); Y02A50/30 (20180101)

### Field of Classification Search

**USPC:** None

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*Primary Examiner:* Graser; Jennifer E

*Attorney, Agent or Firm:* STERNE, KESSLER, GOLDSTEIN & FOX P.L.L.C.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application of U.S. patent application Ser. No. 17/588,122, filed Jan. 28, 2022 (now U.S. Pat. No. 11,918,612, issued on Mar. 5, 2024), which is a divisional application of U.S. patent application Ser. No. 16/230,807, filed Dec. 21, 2018, 2016 (now U.S. Pat. No. 11,266,699, issued on Mar. 8, 2022), which is a divisional application of U.S. patent application Ser. No. 15/039,007, filed May 24, 2016 (now U.S. Pat. No. 10,258,655, issued on Apr. 16, 2019), which is the National Stage of International Application No. PCT/US2014/067491, filed Nov. 25, 2014, which claims the benefit of U.S. Provisional Application No. 61/908,698, filed Nov. 25, 2013; U.S. Provisional Patent Application No. 61/908,702, filed Nov. 25, 2013; and U.S. Provisional Patent Application No. 62/004,187, filed May 28, 2014, the entire disclosures of which are hereby incorporated by reference in their entirety for all purposes.

REFERENCE TO A SEQUENCE LISTING SUBMITTED ELECTRONICALLY

(1) The content of the electronically submitted sequence listing in (Name: 4268_0390007_SequenceListing_ST26.xml; Size: 5,546,501 bytes; and Date of Creation: May 23, 2024) filed with the application is herein incorporated by reference in its entirety.

BACKGROUND

(2) Mammals are colonized by microbes in the gastrointestinal (GI) tract, on the skin, and in other epithelial and tissue niches such as the oral cavity, eye surface and vagina. The gastrointestinal tract harbors an abundant and diverse microbial community. It is a complex system, providing an environment or niche for a community of many different species or organisms, including diverse strains of bacteria. Hundreds of different species may form a commensal community in the GI tract in a healthy person, and this complement of organisms evolves from the time of birth to ultimately form a functionally mature microbial population by about 3 years of age. Interactions between microbial strains in these populations and between microbes and the host, e.g., the host immune system, shape the community structure, with availability of and competition for resources affecting the distribution of microbes. Such resources may be food, location and the availability of space to grow or a physical structure to which the microbe may attach. For example, host diet is involved in shaping the GI tract flora.

(3) A healthy microbiota provides the host with multiple benefits, including colonization resistance to a broad spectrum of pathogens, essential nutrient biosynthesis and absorption, and immune stimulation that maintains a healthy gut epithelium and an appropriately controlled systemic immunity. In settings of 'dysbiosis' or disrupted symbiosis, microbiota functions can be lost or deranged, resulting in increased susceptibility to pathogens, altered metabolic profiles, or induction of proinflammatory signals that can result in local or systemic inflammation or autoimmunity. Thus, the intestinal microbiota plays a significant role in the pathogenesis of many diseases and disorders, including a variety of pathogenic infections of the gut. For example, subjects become more susceptible to pathogenic infections when the normal intestinal microbiota has been disturbed due to use of broad-spectrum antibiotics. Some of these diseases and disorders are chronic conditions that significantly decrease a subject's quality of life and ultimately some can be fatal.

(4) Fecal transplantation has been shown to sometimes be an effective treatment for subjects suffering from severe or refractory GI infections and other disorders by repopulating the gut with a diverse array of microbes that control key pathogens by creating an ecological environment inimical to their proliferation and survival. Such approaches have demonstrated potential to decrease host susceptibility to infection. Fecal transplantation, however, is generally used only for recurrent cases because it has the potential to transmit infectious or allergenic agents between hosts, involves the transmission of potentially hundreds of unknown strains from donor to subject, and is difficult to perform on a mass scale. Additionally, fecal transplantation is inherently nonstandardized and different desired and/or undesired material may be transmitted in any given donation. Thus, there is a need for defined compositions that can be used to decrease susceptibility to infection and/or that facilitate restoration of a healthy gut microbiota.

(5) In addition, practitioners have a need for safe and reproducible treatments for disorders currently treated on an experimental basis using fecal transplantation.

SUMMARY OF THE INVENTION

(6) To meet the need for safe, reproducible treatments for disorders that can be modulated by the induction of a healthy GI microbiome and to treat diseases associated with the GI microbiome, Applicants have designed bacterial compositions of isolated bacterial strains with a plurality of functional properties, in particular that are useful for treating dysbiosis (e.g., restoring a GI microbiome to a state of health), and for treating disorders associated with infection or imbalance of microbial species found in the gut that are based on Applicants discoveries related to those bacterial strains and analysis and insights into properties related to those strains and combinations of those strains, leading to the inventions disclosed herein.

(7) In a first aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium capable of forming a spore, a second type of isolated bacterium capable of forming a spore and optionally a third type of isolated bacterium capable of forming a spore, wherein the first type, the second type and the optional third type are not identical, and wherein at least two of the first type, the second type and the optional third type are capable of synergistically decreasing and/or inhibiting the growth and/or colonization of at least one type of pathogenic bacteria. In some embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria capable of forming spores. In other embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria not containing at least one sporulation-associated gene. In further embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria in spore form. In further embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria in vegetative form. In further embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria in spore form, and wherein the bacterial composition further comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria in vegetative form. In further embodiments, the bacterial composition comprises at least about 5 types of isolated bacteria and at least about 20% of the isolated bacteria are capable of forming spores or are in spore form. In further embodiments, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 2 of the isolated bacteria are capable of forming spores or are in spore form. In further embodiments, the first type, second type and optional third type are present in the composition in approximately equal concentrations. In further embodiments, the first type and the third type are present in the composition in approximately equal concentrations. In further embodiments, the second type and the third type are present in the composition in approximately equal concentrations. In further embodiments, the first type is present in the composition in at least about 150% the concentration of the second type and/or the third type. In further embodiments, the first type, second type and optional third type are individually present in the composition in at least about 150% the concentration of the third type. In further embodiments, the composition consists essentially of between two and about twenty types of isolated bacteria, wherein at least two types of the isolated bacteria are independently capable of spore formation. In further embodiments, at least two types of the isolated bacteria are in spore form. In further embodiments, the first, second and third types are independently selected from Table 1. In further embodiments, the first, second and third types comprise an operational taxonomic unit (OTU) distinction. In further embodiments, the OTU distinction comprises 16S rRDNA sequence similarity below about 95% identity. In further embodiments, the first, second and third types independently comprise bacteria that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1.

(8) In another aspect, provided are compositions comprising an effective amount of a bacterial composition comprising a first type of isolated bacterium; a second type of isolated bacterium; and a third type of isolated bacterium, wherein at least one of the first, second and third types are capable of forming a spore, wherein the first, second and third types are not identical, and wherein a combination of at least two of the first, second and third types are inhibitory to at least one type of pathogenic bacteria. In some embodiments, a combination of the first, second and third types is capable of being inhibitory to the pathogenic bacterium. In other embodiments, a combination of the first, second and third types is capable of being cytotoxic or cytostatic to the pathogenic bacterium. In further embodiments, a combination of the first, second and third types is capable of being cytotoxic or cytostatic to the pathogenic bacterium. In further embodiments, a combination of the first, second and third types is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least equal to the concentration of the combination of

the first, second and third types. In further embodiments, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Pseudomonas, Neisseria, Mycoplasma, Mycobacterium, Listeria, Leptospira, Legionella, Helicobacter, Haemophilus, Francisella, Escherichia, Enterococcus, Klebsiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Brucella, Borrelia,* and *Bordetella*. In further embodiments, the first, second and third types synergistically interact. In further embodiments, at least one of the first, second and third types are capable of independently forming a spore. In further embodiments, at least two of the first, second and third types are capable of independently forming a spore. In further embodiments, the first, second and third types are capable of independently forming a spore.

(9) In further embodiments, wherein the first, second and third types are capable of functionally populating the gastrointestinal tract of a human subject to whom the composition is administered. In further embodiments, the functional populating of the gastrointestinal tract comprises preventing a dysbiosis of the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises treating a dysbiosis of the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises reducing the severity of a dysbiosis of the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises reducing one or more symptoms of a dysbiosis of the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises preventing colonization of the gastrointestinal tract by a pathogenic bacterium. In further embodiments, the functional populating of the gastrointestinal tract comprises reducing colonization of the gastrointestinal tract by a pathogenic bacterium. In further embodiments, the functional populating of the gastrointestinal tract comprises reducing the number of one or more types of pathogenic bacteria in the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises increasing the number of one or more non-pathogenic bacteria in the gastrointestinal tract. Also provided are single dose units comprising the bacterial compositions provided herein, for example, dose units comprising at least 1×10.sup.7, 1×10.sup.8, 1×10.sup.9, 1×10.sup.10, 1×10.sup.11, or 1×10.sup.12 colony forming units (CFUs) of viable bacteria. Also provided are pharmaceutical formulations comprising an effective amount of the compositions provided herein, and further comprising an effective amount of an anti-bacterial agent, a pharmaceutical formulation comprising an effective amount of the bacterial composition, and further comprising an effective amount of an anti-fungal agent, a pharmaceutical formulation comprising an effective amount of the bacterial composition, and further comprising an effective amount of an anti-viral agent, and a pharmaceutical formulation comprising an effective amount of the bacterial composition, and further comprising an effective amount of an anti-parasitic agent.

(10) In another aspect, provided are methods comprising administering to a human subject in need thereof an effective amount of the bacterial compositions, and further comprising administering to the human subject an effective amount of an anti-biotic agent. In some embodiments, the bacterial composition and the anti-biotic agent are administered simultaneously. In other embodiments, the bacterial composition is administered prior to administration of the anti-biotic agent. In further embodiments, provided are methods in which the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is not detectably increased or is detectably decreased over a period of time. In other embodiments, the human subject is diagnosed as having a dysbiosis of the gastrointestinal tract. In other embodiments, the human subject is diagnosed as infected with a pathogenic bacterium selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Pseudomonas, Neisseria, Mycoplasma, Mycobacterium, Listeria, Leptospira, Legionella, Helicobacter, Haemophilus, Francisella, Escherichia, Enterococcus, Klebsiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Brucella, Borrelia,* and *Bordetella*. In other embodiments, the anti-bacterial agent is administered to the human subject prior to administration of the bacteria composition. In other embodiments, the number of pathogenic bacteria present in or excreted from the gastrointestinal tract of the human subject is detectably reduced within two weeks of administration of the bacterial composition.

(11) In another aspect, provided are methods of functionally populating the gastrointestinal tract of a human subject, comprising administering to the subject an effective amount of the bacterial composition of the present invention, under conditions such that the first, second and third types functionally populate the gastrointestinal tract of the human subject. In some embodiments, the bacterial composition is orally administered, rectally administered, or the combination of orally and rectally administered. In other embodiments, the bacterial composition is topically or nasally administered or inhaled.

(12) Also provided are methods of preparing a comestible product, comprising combining with a comestible carrier the bacterial compositions of the present invention, wherein the comestible product is substantially free of non-comestible materials.

(13) In one aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium capable of forming a spore and a second type of isolated bacterium capable of forming a spore, wherein the first type, second type and optional third type are not identical, and wherein at least one of the first type, second type and optional third type are capable of decreasing and/or inhibiting the growth and/or colonization of at least one type of pathogenic bacteria. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9, or 10 types of isolated bacteria and at least 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprise at least about 5 types of isolated bacteria and at least 2 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises i) at least about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 or more types of isolated bacteria capable of forming spores, ii) at least about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 or more types of isolated bacteria not known to be capable of forming spores, or iii) any combination of i) and ii). In an embodiment, the first type, the second type and the optional third type are present in the composition in approximately equal concentrations or activity levels. In an embodiment, the first type, the second type and the optional third type are present in the composition in not substantially equal concentrations. In an embodiment, the first type is present in the composition in at least about 150% the concentration of the second type, or wherein the second type is present in the composition in at least about 150% the concentration of the first type. In an embodiment, the composition consists essentially of: i) between two and about twenty types of isolated bacteria, wherein at least two types of isolated bacteria are independently capable of spore formation; ii) between two and about twenty types of isolated bacteria, wherein at least two types of isolated bacteria not known to be capable of spore formation, or iii) any combination of i) and ii). In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium are selected from Table 1. In an embodiment, the first type of isolated bacterium, the second type of isolated bacterium and the optional third type of isolated bacterium comprise an operational taxonomic unit (OTU) distinction. In an embodiment, the OTU distinction comprises 16S rDNA sequence similarity below about 95% identity. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium independently comprise bacteria that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, a combination of the first type, second type and optional third type are: i) cytotoxic, ii) cytostatic, iii) capable of decreasing the growth of the pathogenic bacterium, iv) capable of inhibiting the growth of the pathogenic bacterium, v) capable of decreasing the colonization of the pathogenic bacterium, vi) capable of inhibiting the colonization of the pathogenic bacterium, or vii) any combination of i)-vi). In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacteria present at a concentration at least equal to the concentration of the combination of the first type, the second type and the optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about twice the concentration of the combination of the first type, the second type and the optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the

pathogenic bacterial present at a concentration at least about ten times the concentration of the combination of the first type, the second type and the optional third type. In an embodiment, the combination is capable of proliferating in the presence of the pathogenic bacteria. In an embodiment, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the first type, the second type and the optional third type synergistically interact. In an embodiment, the first type, the second type and the optional third type synergistically interact to inhibit the pathogenic bacterium. In an embodiment, the composition comprises a combination of bacteria described in any row of Table 4a or Table 4b, or a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or a combination of bacteria described in any row of Table 4a that has a 75.sup.th percentile designation.

(14) In another aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium, a second type of isolated bacterium and an optional third type of isolated bacterium, wherein only one of the first type, the second type and the optional third type is capable of forming a spore, and wherein at least one of the first type, the second type and the optional third type is capable of decreasing the growth and/or colonization of at least one type of pathogenic bacteria.

(15) In another aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium, a second type of isolated bacterium and an optional third type of isolated bacterium, wherein the first type, the second type and the optional third type are not spores or known to be capable of forming a spore, and wherein at least one of the first type, the second type and the optional third type are capable of decreasing the growth and/or colonization of at least one type of pathogenic bacteria.

(16) In an embodiment, at least one of the first type, second type and optional third type are capable of reducing the growth rate of at least one type of pathogenic bacteria. In an embodiment, at least one of the first type, second type and optional third type are cytotoxic to at least one type of pathogenic bacteria. In an embodiment, at least one of the first type, second type and optional third type are cytostatic to at least one type of pathogenic bacteria. In an embodiment, the first type, second type and optional third type are selected from Table 1. In an embodiment, the first type, second type and optional third type comprise different species. In an embodiment, the first type, second type and optional third type comprise different genera. In an embodiment, the first type, second type and optional third type comprise different families. In an embodiment, the first type, second type and optional third type comprise different orders. In an embodiment, the first type, second type and optional third type comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75.sup.th percentile designation.

(17) In another aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium and a second type of isolated bacterium, wherein: i) the first type, second type and optional third type are independently capable of forming a spore; ii) only one of the first type, second type and optional third type is capable of forming a spore or iii) neither the first type nor the second type is capable of forming a spore, wherein the first type, second type and optional third type are not identical, wherein the first type, second type and optional third type are capable of functionally populating the gastrointestinal tract of a human subject to whom the composition is administered. In an embodiment, the first type, second type and optional third type comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75.sup.th percentile designation. In an embodiment, the functional populating of the gastrointestinal tract comprises preventing a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises treating a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing the severity of a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing one or more symptoms of a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises preventing growth and/or colonization of the gastrointestinal tract by a pathogenic bacterium. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing growth and/or colonization of the gastrointestinal tract by a pathogenic bacterium. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing the number of one or more types of pathogenic bacteria in the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises increasing the number of one or more non-pathogenic bacteria in the gastrointestinal tract. In an embodiment, the bacterial composition comprises 0, 1, 2, 3 or greater than 3 types of isolated bacteria capable of forming spores. In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria capable of forming spores. In an embodiment, the bacterial composition comprises at least about 7 types of isolated bacteria capable of forming spores. In an embodiment, the first type, second type and optional third type are present in the composition in not substantially equal concentrations. In an embodiment, the first type, second type and optional third type are present in the composition in approximately equal concentrations. In an embodiment, the first type is present in the composition in at least about 150% the concentration of the second type. In an embodiment, the second type is present in the composition in at least about 150% the concentration of the first type. In an embodiment, the composition consists essentially of between two and about ten types of isolated bacteria, wherein at least one type of isolated bacteria are independently capable of spore formation. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium are selected from Table 1. In an embodiment, the first type of isolated bacterium, the second type of isolated bacterium and the optional third type of isolated bacterium comprise an operational taxonomic unit (OTU) distinction. In an embodiment, the OTU distinction comprises 16S rDNA sequence similarity below about 95% identity. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium independently comprise bacteria that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, a combination of the first type, second type and optional third type are cytotoxic or cytostatic to the pathogenic bacterium. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacteria present at a concentration at least equal to the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about twice the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacteria present at a concentration at least about ten times the concentration of the combination of the first type, second type and optional third type. In an embodiment, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the first type, second type and optional third type synergistically interact to be cytotoxic to the pathogenic bacterium. In an embodiment, wherein the first type, second type and optional third type synergistically interact to be cytostatic to the pathogenic bacterium.

(18) In another aspect, provided are single dose units comprising the compositions of the present invention. In an embodiment, the dose unit

comprises at least 1×10.sup.4, 1×10.sup.5, 1×10.sup.6, 1×10.sup.7, 1×10.sup.8, 1×10.sup.9, 1×10.sup.10, 1×10.sup.11 or greater than 1×10.sup.11 colony forming units (CFUs) of either spores or vegetative bacterial cells. In an embodiment, the dose unit comprises a pharmaceutically acceptable excipient, an enteric coating or a combination thereof. In an embodiment, the dose unit further comprises a drug selected from corticosteroids, mesalazine, mesalamine, sulfasalazine, sulfasalazine derivatives, immunosuppressive drugs, cyclosporin A, mercaptopurine, azathiopurine, prednisone, methotrexate, antihistamines, glucocorticoids, epinephrine, theophylline, cromolyn sodium, anti-leukotrienes, anti-cholinergic drugs for rhinitis, anti-cholinergic decongestants, mast-cell stabilizers, monoclonal anti-IgE antibodies, vaccines, and combinations thereof, wherein the drug is present in an amount effective to modulate the amount and/or activity of at least one pathogen. In an embodiment, the dose unit is formulated for oral administration, rectal administration, or the combination of oral and rectal administration, or is formulated for topical, nasal or inhalation administration. In an embodiment, the dose unit comprises a of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75.sup.th percentile designation.
(19) In another aspect, provided are kits comprising in one or more containers: a first purified population of a first type of bacterial spores substantially free of viable vegetal bacterial cells; a second purified population of a second type of bacterial spores substantially free of viable vegetal bacterial cells; and optionally a third purified population of a third type of bacterial spores substantially free of viable vegetal bacterial cells, wherein the first type, second type and optional third type of bacterial spores are not identical, and wherein the first type, second type and optional third type of bacterial spores, when co-localized in a target region of a gastrointestinal tract of a human subject in need thereof, are capable of functionally populating the gastrointestinal tract. In an embodiment, the first purified population and the second purified population are present in a single container. In an embodiment, the first purified population, the second purified population and the optional third purified population present in two or optionally three containers. In an embodiment, the first purified population and the second purified population are lyophilized or substantially dehydrated. In an embodiment, the kit further comprises in one or more containers an effective amount of an anti-bacterial agent, an effective amount of an anti-viral agent, an effective amount of an anti-fungal agent, an effective amount of an anti-parasitic agent, or a combination thereof in one or more containers. In an embodiment, the kit further comprises a pharmaceutically acceptable excipient or diluent. In an embodiment, the first purified population, the second purified population and the optional third purified population comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75.sup.th percentile designation.
(20) Also provided are pharmaceutical formulations comprising an effective amount of the compositions of the invention, and further comprising an effective amount of an anti-bacterial agent, an effective amount of an anti-fungal agent, an effective amount of an anti-viral agent, an effective amount of an anti-parasitic agent.
(21) Also provided are comestible products comprising a first purified population of a first type of bacterial spores, a second purified population of a second type of bacterial spores and optionally a third purified population of a third type of bacterial spores, wherein the first type, second type and optional third type of bacterial spores are not identical, wherein the comestible product is substantially free of viable vegetal bacterial cells, and wherein the first type, second type and optional third type of bacterial spores, when administered to a human subject in need thereof, are capable of functionally populating the gastrointestinal tract of the human subject. In an embodiment, the comestible product comprises a food or food additive, a beverage or beverage additive, or a medical food. In an embodiment, the comestible product comprises at least 1×10.sup.4, 1×10.sup.5, 1×10.sup.6, 1×10.sup.7, 1×10.sup.8, 1×10.sup.9, 1×10.sup.10, 1×10.sup.11 or greater than 1×10.sup.11 colony forming units (CFUs) of viable spores. In an embodiment, the comestible product comprises a first type of bacterial spores and a second type of bacterial spores selected from Table 1, or where the first type of bacterial spores and the second type of bacterial spores independently comprise bacterial spores that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, the first purified population, the second purified population and the optional third purified population comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any row of Table 4a that has a 75.sup.th percentile designation.
(22) Also provided are methods comprising administering to a human subject in need thereof an effective amount of a bacterial composition comprising at least a first type of isolated bacterium, a second type of isolated bacterium and optionally a third type of isolated bacterium, wherein: the first type, second type and optional third type are independently capable of forming a spore; only one of the first type, second type and optional third type is capable of forming a spore; or none of the first type, the second typeand optional third type is capable of forming a spore, wherein the first type, second type and optional third type are not identical, and wherein at least one of the first type, second type and optional third type exert an inhibitory-effect on a pathogenic bacterium present in the gastrointestinal tract of the human subject, such that the number of pathogenic bacteria present in the gastrointestinal tract is not detectably increased or is detectably decreased over a period of time. In an embodiment, the composition comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75.sup.th percentile designation. In an embodiment, the human subject is diagnosed as having a dysbiosis of the gastrointestinal tract. In an embodiment, the human subject is diagnosed as infected with a pathogenic bacterium selected from the group consisting of *Yersinia*, *Vibrio*, *Treponema*, *Streptococcus*, *Staphylococcus*, *Shigella*, *Salmonella*, *Rickettsia*, *Orientia*, *Pseudomonas*, *Providencia*, *Proteus*, *Propionibacterium*, *Neisseria*, *Mycoplasma*, *Mycobacterium*, *Morganella*, *Listeria*, *Leptospira*, *Legionella*, *Klebsiella*, *Helicobacter*, *Haemophilus*, *Fusobacterium*, *Francisella*, *Escherichia*, *Ehrlichia*, *Enterococcus*, *Coxiella*, *Corynebacterium*, *Clostridium*, *Chlamydia*, *Chlamydophila*, *Campylobacter*, *Burkholderia*, *Brucella*, *Borrelia*, *Bordetella*, *Bifidobacterium*, *Bacillus*, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the bacterial composition is administered simultaneously with i) an antibiotic, ii) a prebiotic, or iii) a combination of i) and ii). In an embodiment, the bacterial composition is administered prior to administration of i) an antibiotic, ii) a prebiotic, or iii) a combination of i) and ii). In an embodiment, the bacterial composition is administered subsequent to administration of i) an antibiotic, ii) a prebiotic, or iii) a combination of i) and ii). In an embodiment, the number of pathogenic bacterium present in or excreted from the gastrointestinal tract of the human subject is detectably reduced within one month, within two weeks, or within one week of administration of the bacterial composition. In an embodiment, the number of pathogenic bacterium present in or excreted from the gastrointestinal tract of the human subject is detectably reduced within three days, two days or one day of administration of the bacterial composition. In an embodiment, the human subject is detectably free of the pathogenic bacterium within one month, two weeks, one week, three days or one day of administration of the bacterial composition. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9, or 10 types of isolated bacteria. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9, or 10 types of isolated bacteria and at least 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 2 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises: i) at least about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 or more types of isolated bacteria capable of forming spores, ii) at least about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 or more types of isolated bacteria not known to be capable of forming spores, or iii) any combination of i) and ii). In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 1 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial

composition comprises at least about 5 types of isolated bacteria and at least 1 of the isolated bacteria is not capable of forming spores. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria, wherein i) at least 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria are capable of forming spores, ii) at least 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria are not capable of forming spores, or iii) any combination of i) and ii). In an embodiment, the first type, second type and optional third type are present in the composition in approximately equal concentrations. In an embodiment, the first type, second type and optional third type are present in the composition in not substantially equal concentrations. In an embodiment, the first type is present in the composition in at least about 150% the concentration of the second type, or wherein the second type is present in the composition in at least about 150% the concentration of the first type. In an embodiment, the composition consists essentially of between two and about ten types of isolated bacteria, wherein at least two types of isolated bacteria are independently capable of spore formation. In an embodiment, the composition consists essentially of between two and about ten types of isolated bacteria, wherein at least two types of isolated bacteria are not capable of spore formation. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium are selected from Table 1. In an embodiment, the first type of isolated bacterium, the second type of isolated bacterium and the optional third type of isolated bacterium comprise an operational taxonomic unit (OTU) distinction. In an embodiment, the OTU distinction comprises 16S rDNA sequence similarity below about 95% identity. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium independently comprise bacteria that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, a combination of the first type, second type and optional third type are cytotoxic or cytostatic to the pathogenic bacterium. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least equal to the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about twice the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about ten times the concentration of the combination of the first type, second type and optional third type. In an embodiment, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the first type, second type and optional third type synergistically interact to be cytotoxic to the pathogenic bacterium. In an embodiment, the first type, second type and optional third type synergistically interact to be cytostatic to the pathogenic bacterium.

(23) Also provided are methods of functionally populating the gastrointestinal tract of a human subject, comprising administering to the subject an effective amount of a bacterial composition comprising at least a first type of isolated bacterium, a second type of isolated bacterium, and optionally a third type of isolated bacterium wherein i) the first type, second type and optional third type are independently capable of forming a spore; ii) only one of the first type, second type and optional third type is capable of forming a spore or iii) none of the first type, the second type and the optional third type is capable of forming a spore, wherein the first type, second type and optional third type are not identical, under conditions such that the first type, second type and optional third type functionally populate the gastrointestinal tract of the human subject. In an embodiment, the composition comprises a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any row of Table 4a that has a 75.sup.th percentile designation. In an embodiment, the bacterial composition is orally administered, rectally administered, or the combination of orally and rectally administered. In an embodiment, the bacterial composition is topically or nasally administered or inhaled. In an embodiment, the first type of isolated bacteria and the second type of isolated bacteria are selected from Table 1. In an embodiment, the bacterial composition consists essentially of spores, wherein the spores comprise spores of the first type of isolated bacteria, spores of the second type of isolated bacteria and spores of the optional third type of isolated bacteria. In an embodiment, the first type of isolated bacteria and the second type of isolated bacteria independently comprise bacterial spores that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, the functional populating of the gastrointestinal tract comprises preventing a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises treating a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing the severity of a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing one or more symptoms of a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises preventing colonization of the gastrointestinal tract by a pathogenic bacterium. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing colonization of the gastrointestinal tract and/or growth by a pathogenic bacterium. In an embodiment, wherein the functional populating of the gastrointestinal tract comprises reducing the number of one or more types of pathogenic bacteria in the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises increasing the number of one or more non-pathogenic bacteria in the gastrointestinal tract. In an embodiment, the bacterial composition comprises at least about 3, 5, 7 or 9 types of isolated bacteria capable of forming spores. In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 20% of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 2 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria, wherein i) at least 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria are capable of forming spores, ii) at least 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria are not capable of forming spores, or iii) any combination of i) and ii). In an embodiment, the first type, second type and optional third type are present in the composition in approximately equal concentrations. In an embodiment, the first type, second type and optional third type are present in the composition in not substantially equal concentrations. In an embodiment, the first type is present in the composition in at least about 150% the concentration of the second type, or wherein the second type is present in the composition in at least about 150% the concentration of the first type. In an embodiment, the composition consists essentially of between two and about ten types of isolated bacteria, wherein i) at least one type of isolated bacteria is capable of spore formation, ii) at least one type of isolated bacteria is not capable of spore formation, or iii) a combination of i) and ii). In an embodiment, a combination of the first type, second type and optional third type are inhibitory to the pathogenic bacterium. In an embodiment, the combination reduces the growth rate of the pathogenic bacterium. In an embodiment, the combination is cytostatic or cytotoxic to the pathogenic bacterium. In an embodiment, the combination is capable of inhibiting growth of the pathogenic bacterial present at a concentration at least equal to the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting growth of the pathogenic bacterial present at a concentration at least about twice the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about ten times the concentration of the combination of the first type, second type and optional third type. In an embodiment, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium,*

*Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the first type, second type and optional third type synergistically interact to reduce or inhibit the growth of the pathogenic bacterium. In an embodiment, the first type, second type and optional third type synergistically interact to reduce or inhibit the colonization of the pathogenic bacterium. In an embodiment, the method comprises administering to the human subject a single dose unit comprising at least 1×10.sup.4, 1×10.sup.5, 1×10.sup.6, 1×10.sup.7, 1×10.sup.8, 1×10.sup.9, 1×10.sup.10, 1×10.sup.11 or greater than 1×10.sup.11 colony forming units (CFUs) of viable bacteria. In an embodiment, the dose unit comprises a bacterial population substantially in the form of spores. In an embodiment, the dose unit comprises a pharmaceutically acceptable excipient and/or an enteric coating. In an embodiment, the unit dose is formulated for oral administration, rectal administration, or the combination of oral and rectal administration. In an embodiment, the unit dose is formulated for topical or nasal administration or for inhalation.
(24) In another aspect, provided are methods of reducing the number of pathogenic bacteria present in the gastrointestinal tract of a human subject, comprising administering to the subject an effective amount of a pharmaceutical formulation comprising an effective amount of the composition of the present disclosure, and further comprising an effective amount of an anti-microbial agent, under conditions such that the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about one month of administration of the pharmaceutical formulation. In an embodiment, the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about two weeks of administration of the pharmaceutical formulation. In an embodiment, the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about one week of administration of the pharmaceutical formulation. In an embodiment, the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about three days of administration of the pharmaceutical formulation. In an embodiment, the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about one day of administration of the pharmaceutical formulation. In an embodiment, the anti-microbial agent comprises anti-bacterial agent. In an embodiment, the anti-microbial agent comprises anti-fungal agent. In an embodiment, the anti-microbial agent comprises anti-viral agent. In an embodiment, the anti-microbial agent comprises anti-parasitic agent.
(25) In another aspect, provided are methods of preparing a comestible product, comprising combining with a comestible carrier a first purified population comprising at least a first type of isolated bacterium, a second purified population comprising at least a second type of isolated bacterium and optionally a third purified population comprising at least a third type of isolated bacterium, wherein: i) the first type, second type and optional third type are independently capable of forming a spore; ii) only one of the first type, second type and optional third type is capable of forming a spore or iii) none of the first type, the second type and the optional third type is capable of forming a spore, wherein the first type, second type and optional third type of bacteria are not identical, wherein the comestible product is substantially free of non-comestible materials. In an embodiment, at least one of the first purified population, the second purified population and the optional third purified population consist essentially of viable spores. In an embodiment, the first purified population, the second purified population and the optional third purified population consist essentially of viable spores. In an embodiment, the comestible product is substantially free of viable vegetal bacterial cells. In an embodiment, the viable spores, when the comestible product is consumed by a human subject in need thereof, are capable of functionally populating the gastrointestinal tract of the human subject. In an embodiment, the comestible product comprises a food or food additive. In an embodiment, the comestible product comprises a beverage or beverage additive. In an embodiment, the comestible product comprises a medical food. In an embodiment, the comestible product comprises at least 1×10.sup.4, 1×10.sup.5, 1×10.sup.6, 1×10.sup.7, 1×10.sup.8, 1×10.sup.9, 1×10.sup.10, 1×10.sup.11 or greater than 1×10.sup.11 colony forming units (CFUs) of viable spores. In an embodiment, the first purified population, the second purified population and the optional third purified population comprise a combination of bacteria described in any row of Table 4a or Table 4b, or any row of Table 4a that has a ++++ designation or a +++ designation, or any row of Table 4a that has a 75.sup.th percentile designation. In an embodiment, spores are of a bacterium selected from Table 1. In an embodiment, the first purified population and the second purified population independently comprise bacterial spores that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1.
(26) Also provided are methods of reducing the abundance of a pathogen in the gastrointestinal tract of a subject comprising administering a composition of in a therapeutically effective amount and allowing the bacterial composition to compete with the pathogen in the gastrointestinal tract of a subject.
(27) Further provided are methods of treating diarrhea comprising administering a bacterial composition in a therapeutically effective amount and allowing the bacterial composition to reduce the diarrheal effect of a pathogen in the gastrointestinal tract of a subject. In an embodiment, the pathogen is *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the pathogen is *Clostridium difficile, Salmonella* spp., pathogenic *Escherichia coli*, or vancomycin-resistant *Enterococcus* spp. In an embodiment, the pathogen is *Clostridium difficile*. In an embodiment, the composition is administered orally. In an embodiment the composition comprises a combination of bacteria described in any row of Table 4a, or Table 4b, or any row of Table 4a that has a ++++ designation or a +++ designation, or any row of Table 4a that has a 75.sup.th percentile designation.
(28) In some aspects, the invention relates to a composition comprising a network ecology selected from Table 10. In some embodiments, the network ecology comprises network clades provided in Table 10. In other embodiments, the network ecology comprises network OTUs provided in Table 10. In some cases the composition comprises *Blautia producta, Clostridium disporicum, Clostridium innocuum, Clostridium mayombei, Clostridium orbiscindens, Clostridium symbiosum*, and *Lachnospiraceae bacterium* 5_1_57FAA. In some embodiments, the composition the composition is effective for treating at least one sign or symptom of a dysbiosis, for example, the is effective for reducing at least one sign or symptom of infection or dysbiosis associated with *C. difficile, Klebsiella pneumonii, Morganella morganii*, or vancomycin-resistant Enterococci (VRE).
(29) In another aspect, the invention relates to a composition comprising a bacterial heterotrimer selected from a heterotrimer identified in Table 4a, Table 4b, or Table 12, such that the heterotrimer can e.g., inhibit growth of a pathobiont in a CivSim assay.
(30) In some aspects, the invention relates to a composition comprising a bacterial heterotrimer selected from a heterotrimer identified in Table 14, Table 15, Table 16, Table 17, Table 17, Table 18, Table 19, Table 20, or Table 21, such that the organisms of the heterotrimer can augment and/or engraft in a human gastrointestinal tract. In some embodiments, the engraftment and/or augmentation can occur after administration of the composition to a human having a dysbiosis. In some embodiments, the dysbiosis is associated with the presence of *C. difficile* in the gastrointestinal tract of the human.
(31) Additional objects and advantages will be set forth in part in the description which follows, and in part will be obvious from the description, or may be learned by practice of the embodiments. The objects and advantages will be realized and attained by means of the elements and

combinations particularly pointed out in the appended claims.
(32) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the claims.
(33) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments and together with the description, serve to further explain the embodiments.
BRIEF DESCRIPTION OF TABLES
(34) Table 1 is a list of Operational Taxonomic Units (OTU) with taxonomic assignments made to genus, species, and phylogenetic clade. Clade membership of bacterial OTUs is based on 16S sequence data. Clades are defined based on the topology of a phylogenetic tree that is constructed from full-length 16S sequences using maximum likelihood methods familiar to individuals with ordinary skill in the art of phylogenetics. Clades are constructed to ensure that all OTUs in a given clade are: (i) within a specified number of bootstrap supported nodes from one another, and (ii) within 5% genetic similarity. OTUs that are within the same clade can be distinguished as genetically and phylogenetically distinct from OTUs in a different clade based on 16S-V4 sequence data, while OTUs falling within the same clade are closely related. OTUs falling within the same clade are evolutionarily closely related and may or may not be distinguishable from one another using 16S-V4 sequence data. Members of the same clade, due to their evolutionary relatedness, play similar functional roles in a microbial ecology such as that found in the human gut. Compositions substituting one species with another from the same clade are likely to have conserved ecological function and therefore are useful in the present invention. All OTUs are denoted as to their putative capacity to form spores and whether they are a pathogen or pathobiont (see Definitions for description of “Pathobiont”). NIAID (National Institute of Allergy and Infectious Disease) Priority Pathogens are denoted as ‘Category-A’, ‘Category-B’, or ‘Category-C’, and opportunistic pathogens are denoted as ‘OP’. OTUs that are not pathogenic or for which their ability to exist as a pathogen is unknown are denoted as ‘N’. The ‘SEQ ID Number’ denotes the identifier of the OTU in the Sequence Listing File and ‘Public DB Accession’ denotes the identifier of the OTU in a public sequence repository.
(35) Table 2 provides phylogenetic clades and their members determined using 16S full-length and V4 sequencing.
(36) Table 3 is a list of human diseases, disorders and conditions for which the provided bacterial compositions are useful.
(37) Table 4a. Provides representative combinations of the present invention tested in vitro.
(38) Table 4b. Provides representative combinations of the present invention tested in vitro
(39) Table 5 provides data from testing of representative ternary OTU combinations of the present invention in a CivSim assay and in vivo.
(40) Table 6 provides data on the ability of a 15 member bacterial composition to inhibit VRE in vitro.
(41) Table 7 provides data on the ability of a 15 member bacterial composition to inhibit *K. pneumoniae* in vitro.
(42) Table 8 provides data on the ability of a 15 member bacterial composition to inhibit *M. morganii* in vitro.
(43) Table 9 provides data demonstrating the efficacy of combinations of the present invention against *C. difficile* infection in a preventive murine model.
(44) Table 10. Provides exemplary combinations of the present invention that were tested against *C. difficile* infection in a preventive murine model.
(45) Table 11. Provides bacterial OTUs associated with a bacterial composition used to treat patients with *C. difficile* associated diarrheal disease, and to OTUs comprising the OTUs undergo engraftment and ecological augmentation to establish a more diverse microbial ecology in patients post-treatment. OTUs that comprise an augmented ecology are not present in the patient prior to treatment and/or exist at extremely low frequencies such that they do not comprise a significant fraction of the total microbial carriage and are not detectable by genomic and/or microbiological assay methods. OTUs that are members of the engrafting and augmented ecologies were identified by characterizing the OTUs that increase in their relative abundance post treatment and that respectively are: (i) present in the ethanol-treated spore preparation and absent in the patient pretreatment, or (ii) absent in the ethanol-treated spore preparation, but increase in their relative abundance through time post treatment with the preparation due to the formation of favorable growth conditions by the treatment. Notably, augmenting OTUs can grow from low frequency reservoirs in the subject, or be introduced from exogenous sources such as diet. OTUs that comprise a “core” composition in the treatment bacterial composition are denoted.
(46) Table 12 provides bacterial compositions that exhibited inhibition against *C. difficile* as measured by a mean log inhibition greater than the 99% confidence interval (C.I.) of the null hypothesis (see Example 6, ++++) and that are identified in at least one spore ecology treatment or in a human subject microbiome after treatment with a composition.
(47) Table 13 provides exemplary of 4-mer to 10-mer bacterial compositions that were comprised in a bacterial therapy administered to subjects with *C. difficile*-associated diarrheal disease.
(48) Table 14 provides exemplary ternary OTUs that either engrafted or augmented in at least one patient (of 29 that responded to treatment) after treatment with a spore ecology composition. Each ternary combination was either in all doses or the organisms of the ternary combination were present together in all subjects at some post-treatment time.
(49) Table 15 provides exemplary OTUs that engrafted in at least one subject. The ternary combinations were found in 95% of the doses of administered spore ecology compositions.
(50) Table 16 provides exemplary OTUs that augmented in at least one patient post treatment with a spore ecology composition. The ternary combinations were found together in at least 75% of the subjects at some post-treatment timepoint.
(51) Table 17 provides exemplary OTU combinations that were present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTU *Clostridiales* sp. SM4/1 as either augmenting or engrafting in the subjects given doses containing the ternary composition.
(52) Table 18 provides exemplary ternary OTU combinations that were present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTU *Clostridiales* sp. SSC/2 as either augmenting or engrafting in the subjects given a composition containing the ternary combination.
(53) Table 19 provides exemplary ternary combinations of OTUs that were present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTU *Clostridium* sp. NML 04A032 as either augmenting or engrafting in the subjects given a composition containing the ternary combination.
(54) Table 20 provides exemplary ternary combinations of OTUs that were present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTUs *Clostridium* sp. NML 04A032, *Ruminococcus lactaris*, and *Ruminococcus torques* as either augmenting or engrafting in the subjects given a composition containing the ternary combination.
(55) Table 21 provides exemplary ternary combinations of OTUs that are present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTUs *Eubacterium* rectale, *Faecalibacterium prausnitzii*, *Oscillibacter* sp. G2, *Ruminococcus lactaris*, and *Ruminococcus torques* as either augmenting or engrafting in the subjects given a composition containing the ternary combination.
(56) Table 22 provides alternate names of organisms found in OTUs of the embodiments of the present invention.

## Description

BRIEF DESCRIPTION OF FIGURES

(1) FIG. **1** shows an exemplary phylogenetic tree and the relationship of OTUs and Clades. A, B, C, D, and E represent OTUs, also known as leaves in the tree. Clade 1 comprises OTUs A and B, Clade 2 comprises OTUs C, D and E, and Clade 3 is a subset of Clade 2 comprising OTUs D and E. Nodes in a tree that define clades in the tree can be either statistically supported or not statistically supported. OTUs within a clade are more similar to each other than to OTUs in another clade; the robustness the clade assignment is denoted by the degree of statistical support for a node upstream of the OTUs in the clade.

(2) FIG. **2** provides a schematic of 16S rDNA gene and denotes the coordinates of hypervariable regions 1-9 (V1-V9), according to an embodiment of the invention. Coordinates of V1-V9 are 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294, and 1435-1465 respectively, based on numbering using *E. coli* system of nomenclature defined by Brosius et al., Complete nucleotide sequence of a 16S ribosomal RNA gene (16S rRNA) from *Escherichia coli*, PNAS 75 (10): 4801-4805 (1978).

(3) FIG. **3** highlights in bold the nucleotide sequences for each hypervariable region in the exemplary reference *E. coli* 16S sequence (SEQ ID NO: 2047) described by Brosius et al., supra.

(4) FIG. **4** provides representative combinations of the present invention tested in vitro and their respective inhibition of pathogen growth.

(5) FIG. **5** shows an in vivo hamster *Clostridium difficile* relapse prevention model to validate efficacy of network ecology bacterial composition, according to an embodiment of the invention.

(6) FIG. **6** shows the increase in the total microbial diversity (measured using the Chao-1 diversity index) in the gut of human subjects with recurrent *Clostridium difficile* associated disease pretreatment and post-treatment with a microbial spore ecology.

(7) FIG. **7** shows the compositional change in the microbiome (measured using the Bray-Curtis PCoA metric) in the gut of human subjects with recurrent *Clostridium difficile* associated disease pretreatment and post-treatment with a microbial spore ecology.

DEFINITIONS

(8) As used herein, the term “antioxidant” refers to, without limitation, any one or more of various substances such as beta-carotene (a vitamin A precursor), vitamin C, vitamin E, and selenium that inhibit oxidation or reactions promoted by Reactive Oxygen Species (“ROS”) and other radical and non-radical species. Additionally, antioxidants are molecules capable of slowing or preventing the oxidation of other molecules. Non-limiting examples of antioxidants include astaxanthin, carotenoids, coenzyme Q10 (“CoQ10”), flavonoids, glutathione, Goji (wolfberry), hesperidin, lactowolfberry, lignan, lutein, lycopene, polyphenols, selenium, vitamin A, vitamin C, vitamin E, zeaxanthin, or combinations thereof.

(9) “Backbone Network Ecology” or simply “Backbone Network” or “Backbone” are compositions of microbes that form a foundational composition that can be built upon or subtracted from to optimize a Network Ecology or Functional Network Ecology to have specific biological characteristics or to comprise desired functional properties, respectively. Microbiome therapeutics can be comprised of these “Backbone Networks Ecologies” in their entirety, or the “Backbone Networks” can be modified by the addition or subtraction of “R-Groups” to give the network ecology desired characteristics and properties. “R-Groups” as used herein, can be defined in multiple terms including, but not limited to: individual OTUs, individual or multiple OTUs derived from a specific phylogenetic clade or a desired phenotype such as the ability to form spores, or functional bacterial compositions that comprise. “Backbone Networks” can comprise a computationally derived Network Ecology in its entirety or can be subsets of the computed network that represent key nodes in the network that contributed to efficacy such as but not limited to a composition of Keystone OTUs. The number of organisms in the human gastrointestinal tract, as well as the diversity between healthy individuals, is indicative of the functional redundancy of a healthy gut microbiome ecology. See The Human Microbiome Consortia. 2012. Structure, function and diversity of the healthy human microbiome. Nature 486:207-214. This redundancy makes it highly likely that non-obvious subsets of OTUs or functional pathways (i.e., “Backbone Networks”) are critical to maintaining states of health and or catalyzing a shift from a dysbiotic state to one of health. One way of exploiting this redundancy is through the substitution of OTUs that share a given clade (see below) or of adding members of a clade not found in the Backbone Network.

(10) “Bacterial Composition” refers to a consortium of microbes comprising two or more OTUs. Backbone Network Ecologies, Functional Network Ecologies, Network Classes, and Core Ecologies are all types of bacterial compositions. A “Bacterial Composition” can also refer to a composition of enzymes that are derived from a microbe or multiple microbes. As used herein, Bacterial Composition includes a therapeutic microbial composition, a prophylactic microbial composition, a Spore Population, a Purified Spore Population, or ethanol treated spore population.

(11) “Clade” refers to the OTUs or members of a phylogenetic tree that are downstream of a statistically valid node in a phylogenetic tree (FIG. **1**). The clade comprises a set of terminal leaves in the phylogenetic tree (i.e., tips of the tree) that are a distinct monophyletic evolutionary unit and that share some extent of sequence similarity. Clades are hierarchical. In one embodiment, the node in a phylogenetic tree that is selected to define a clade is dependent on the level of resolution suitable for the underlying data used to compute the tree topology. Exemplary clades are delineated in Table 1 and Table 2. As used herein, clade membership of bacterial OTUs is based on 16S sequence data. Clades are defined based on the topology of a phylogenetic tree that is constructed from full-length 16S sequences using maximum likelihood methods familiar to individuals with ordinary skill in the art of phylogenetics. Clades are constructed to ensure that all OTUs in a given clade are (i) within a specified number of bootstrap supported nodes from one another, and (ii) within 5% genetic identity. OTUs that are within the same clade can be distinguished as genetically and phylogenetically distinct from OTUs in a different clade based on 16S-V4 sequence data, while OTUs falling within the same clade are closely related. OTUs falling within the same clade are evolutionarily closely related and may or may not be distinguishable from one another using 16S-V4 sequence data. Members of the same clade, due to their evolutionary relatedness, play or are predicted to play similar functional roles in a microbial ecology such as that found in the human gut. In some embodiments, one OTU from a clade can be substituted in a composition by a different OTU from the same clade.

(12) The “Colonization” of a host organism includes the non-transitory residence of a bacterium or other microscopic organism. As used herein, “reducing colonization” of a host subject's gastrointestinal tract (or any other microbiotal niche) by a pathogenic or non-pathogenic bacterium includes a reduction in the residence time of the bacterium the gastrointestinal tract as well as a reduction in the number (or concentration) of the bacterium in the gastrointestinal tract or adhered to the luminal surface of the gastrointestinal tract. The reduction in colonization can be permanent or occur during a transient period of time. Reductions of adherent pathogens can be demonstrated directly, e.g., by determining pathogenic burden in a biopsy sample, or reductions may be measured indirectly, e.g., by measuring the pathogenic burden in the stool of a mammalian host.

(13) A “Combination” of two or more bacteria includes the physical co-existence of the two bacteria, either in the same material or product or in physically connected products, as well as the temporal co-administration or co-localization of the two bacteria.

(14) The term “consisting essentially of” as used herein conforms to the definition as provided in the Manual of Patent Examination and Procedure (MPEP; March 2014). The basic and novel characteristics of inventions claimed herein include the ability to catalyze changes in a microbiome ecology of a mammalian subject, e.g., a human, from dysbiotic to a more normative state, and to promote engraftment and

augmentation of microbiome component as set out in the specification, e.g., see Tables 14-21. A more normative state can include, in a non-limiting example, a decrease in a sign or symptom of a disease or disorder associated with a dysbiosis.
(15) "Cytotoxic" activity of bacterium includes the ability to kill a bacterial cell, such as a pathogenic bacterial cell. A "cytostatic" activity or bacterium includes the ability to inhibit, partially or fully, growth, metabolism, and/or proliferation of a bacterial cell, such as a pathogenic bacterial cell. Cytotoxic activity may also apply to other cell types such as but not limited to Eukaryotic cells.
(16) "Dimer" refers to a combination of bacteria that is comprised of two OTUs. The descriptions "homodimer" and "heterodimer" refer to combinations where the two OTUs are the same or different, respectively.
(17) "Dysbiosis" refers to a state of the microbiota or microbiome of the gut or other body area, including mucosal or skin surfaces in which the normal diversity and/or function of the ecological network is disrupted. Any disruption from a preferred (e.g., ideal) state of the microbiota can be considered a dysbiosis, even if such dysbiosis does not result in a detectable decrease in health. This state of dysbiosis may be unhealthy, it may be unhealthy under only certain conditions, or it may prevent a subject from becoming healthier. Dysbiosis may be due to a decrease in diversity, the overgrowth of one or more pathogens or pathobionts, symbiotic organisms able to cause disease only when certain genetic and/or environmental conditions are present in a subject, or the shift to an ecological network that no longer provides a beneficial function to the host and therefore no longer promotes health.
(18) "Ecological Niche" or simply "Niche" refers to the ecological space in which an organism or group of organisms occupies. Niche describes how an organism or population or organisms responds to the distribution of resources, physical parameters (e.g., host tissue space) and competitors (e.g., by growing when resources are abundant, and when predators, parasites and pathogens are scarce) and how it in turn alters those same factors (e.g., limiting access to resources by other organisms, acting as a food source for predators and a consumer of prey).
(19) "Germinant" is a material or composition or physical-chemical process capable of inducing vegetative growth of a bacterium that is in a dormant spore form, or group of bacteria in the spore form, either directly or indirectly in a host organism and/or in vitro.
(20) "Inhibition" of a pathogen or non-pathogen encompasses the inhibition of any desired function or activity of the bacterial compositions of the present invention. Demonstrations of inhibition, such as decrease in the growth of a pathogenic bacterium or reduction in the level of colonization of a pathogenic bacterium are provided herein and otherwise recognized by one of ordinary skill in the art. Inhibition of a pathogenic or non-pathogenic bacterium's "growth" may include inhibiting the increase in size of the pathogenic or non-pathogenic bacterium and/or inhibiting the proliferation (or multiplication) of the pathogenic or non-pathogenic bacterium. Inhibition of colonization of a pathogenic or non-pathogenic bacterium may be demonstrated by measuring the amount or burden of a pathogen before and after a treatment. An "inhibition" or the act of "inhibiting" includes the total cessation and partial reduction of one or more activities of a pathogen, such as growth, proliferation, colonization, and function. Inhibition of function includes, for example, the inhibition of expression of pathogenic gene products such as a toxin or invasive pilus induced by the bacterial composition.
(21) "Isolated" encompasses a bacterium or other entity or substance that has been (1) separated from at least some of the components with which it was associated when initially produced (whether in nature or in an experimental setting), and/or (2) produced, prepared, purified, and/or manufactured by the hand of man. Isolated bacteria include those bacteria that are cultured, even if such cultures are not monocultures. Isolated bacteria may be separated from at least about 10%, about 20%, about 30%, about 40%, about 50%, about 60%, about 70%, about 80%, about 90%, or more of the other components with which they were initially associated. In some embodiments, isolated bacteria are more than about 80%, about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more than about 99% pure. In some embodiments, isolated bacteria are separated from 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of the other components with which they were initially associated. In some embodiments, isolated bacteria are more than 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more than 99% pure. As used herein, a substance is "pure" if it is substantially free of other components. The terms "purify," "purifying" and "purified" refer to a bacterium or other material that has been separated from at least some of the components with which it was associated either when initially produced or generated (e.g., whether in nature or in an experimental setting), or during any time after its initial production. A bacterium or a bacterial population may be considered purified if it is isolated at or after production, such as from a material or environment containing the bacterium or bacterial population, or by passage through culture, and a purified bacterium or bacterial population may contain other materials up to about 10%, about 20%, about 30%, about 40%, about 50%, about 60%, about 70%, about 80%, about 90%, or above about 90% and still be considered "isolated." In other embodiments, a purified bacterium or bacterial population may contain other materials up to 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 90% and still be considered "isolated." In some embodiments, purified bacteria and bacterial populations are more than about 80%, about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more than about 99% pure. In some embodiments, purified bacteria and bacterial populations are more than 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more than 99% pure. In the instance of bacterial compositions provided herein, the one or more bacterial types present in the composition can be independently purified from one or more other bacteria produced and/or present in the material or environment containing the bacterial type. Bacterial compositions and the bacterial components thereof are generally purified from residual habitat products.
(22) "Keystone OTU" or "Keystone Function" refers to one or more OTUs or Functional Pathways (e.g., KEGG or COG pathways) that are common to many network ecologies or functional network ecologies and are members of networks that occur in many subjects (i.e., are pervasive). Due to the ubiquitous nature of Keystone OTUs and their associated Functions Pathways, they are central to the function of network ecologies in healthy subjects and are often missing or at reduced levels in subjects with disease. Keystone OTUs and their associated functions may exist in low, moderate, or high abundance in subjects. "Non-Keystone OTU" or "non-Keystone Function" refers to an OTU or Function that is observed in a Network Ecology or a Functional Network Ecology and is not a keystone OTU or Function.
(23) "Microbiota" refers to the community of microorganisms that occur (sustainably or transiently) in and on an animal subject, typically a mammal such as a human, including eukaryotes, archaea, bacteria, and viruses (including bacterial viruses, i.e., phage).
(24) "Microbiome" refers to the genetic content of the communities of microbes that live in and on the human body, both sustainably and transiently, including eukaryotes, archaea, bacteria, and viruses (including bacterial viruses (i.e., phage)), wherein "genetic content" includes genomic DNA, RNA such as ribosomal RNA, the epigenome, plasmids, and all other types of genetic information.
(25) "Microbial Carriage" or simply "Carriage" refers to the population of microbes inhabiting a niche within or on humans. Carriage is often defined in terms of relative abundance. For example, OTU1 comprises 60% of the total microbial carriage, meaning that OTU1 has a relative abundance of 60% compared to the other OTUs in the sample from which the measurement was made. Carriage is most often based on genomic sequencing data where the relative abundance or carriage of a single OTU or group of OTUs is defined by the number of sequencing reads that are assigned to that OTU/s relative to the total number of sequencing reads for the sample. Alternatively, Carriage may be measured using microbiological assays.
(26) "Microbial Augmentation" or simply "augmentation" refers to the establishment or significant increase of a population of microbes that are (i) absent or undetectable (as determined by the use of standard genomic and microbiological techniques) from the administered therapeutic microbial composition, (ii) absent, undetectable, or present at low frequencies in the host niche (for example: gastrointestinal tract, skin, anterior-nares, or vagina) before the delivery of the microbial composition, and (iii) are found after the administration of the microbial composition or significantly increased, for example, 2-fold, 5-fold, 1×10.sup.2, 1×10.sup.3, 1×10.sup.4, 1×10.sup.5, 1×10.sup.6, 1×10.sup.7, or greater than

1×10.sup.8, in cases where they were present at low frequencies. The microbes that comprise an augmented ecology can be derived from exogenous sources such as food and the environment, or grow out from micro-niches within the host where they reside at low frequency. The administration of a bacterial microbial composition induces an environmental shift in the target niche that promotes favorable conditions for the growth of these commensal microbes. In the absence of treatment with a bacterial composition, the host can be constantly exposed to these microbes; however, sustained growth and the positive health effects associated with the stable population of increased levels of the microbes comprising the augmented ecology are not observed.

(27) "Microbial Engraftment" or simply "engraftment" refers to the establishment of OTUs present in the bacterial composition in a target niche that are absent in the treated host prior to treatment. The microbes that comprise the engrafted ecology are found in the therapeutic microbial composition and establish as constituents of the host microbial ecology upon treatment. Engrafted OTUs can establish for a transient period of time, or demonstrate long-term stability in the microbial ecology that populates the host post treatment with a bacterial composition. The engrafted ecology can induce an environmental shift in the target niche that promotes favorable conditions for the growth of commensal microbes capable of catalyzing a shift from a dysbiotic ecology to one representative of a health state.

(28) As used herein, the term "Minerals" is understood to include boron, calcium, chromium, copper, iodine, iron, magnesium, manganese, molybdenum, nickel, phosphorus, potassium, selenium, silicon, tin, vanadium, zinc, or combinations thereof.

(29) "Network Ecology" refers to a consortium of clades or OTUs that co-occur in some number of subjects. As used herein, a "network" is defined mathematically by a graph delineating how specific nodes (i.e., clades or OTUs) and edges (connections between specific clades or OTUs) relate to one another to define the structural ecology of a consortium of clades or OTUs. Any given Network Ecology will possess inherent phylogenetic diversity and functional properties. A Network Ecology can also be defined in terms of its functional capabilities where for example the nodes would be comprised of elements such as, but not limited to, enzymes, clusters of orthologous groups (COGS; http://www.ncbi.nlm.nih.gov/books/NBK21090/), or KEGG Orthology Pathways (www.genome.jp/kegg/); these networks are referred to as a "Functional Network Ecology". Functional Network Ecologies can be reduced to practice by defining the group of OTUs that together comprise the functions defined by the Functional Network Ecology.

(30) "Network Class" and "Network Class Ecology" refer to a group of network ecologies that in general are computationally determined to comprise ecologies with similar phylogenetic and/or functional characteristics. A Network Class therefore contains important biological features, defined either phylogenetically or functionally, of a group (i.e., a cluster) of related network ecologies. One representation of a Network Class Ecology is a designed consortium of microbes, typically non-pathogenic bacteria, that represents core features of a set of phylogenetically or functionally related network ecologies seen in many different subjects. In some occurrences, a Network Class, while designed as described herein, exists as a Network Ecology observed in one or more subjects. Network Class ecologies are useful for reversing or reducing a dysbiosis in subjects where the underlying, related Network Ecology has been disrupted.

(31) To be free of "non-comestible products" means that a bacterial composition or other material provided herein does not have a substantial amount of a non-comestible product, e.g., a product or material that is inedible, harmful or otherwise undesired in a product suitable for administration, e.g., oral administration, to a human subject. Non-comestible products are often found in preparations of bacteria from the prior art.

(32) "Operational taxonomic units" and "OTU" (or plural, "OTUs") refer to a terminal leaf in a phylogenetic tree and is defined by a nucleic acid sequence, e.g., the entire genome, or a specific genetic sequence, and all sequences that share sequence identity to this nucleic acid sequence at the level of species. In some embodiments the specific genetic sequence may be the 16S sequence or a portion of the 16S sequence. In other embodiments, the entire genomes of two entities are sequenced and compared. In another embodiment, select regions such as multilocus sequence tags (MLST), specific genes, or sets of genes may be genetically compared. In 16S embodiments, OTUs that share ≥97% average nucleotide identity across the entire 16S or some variable region of the 16S are considered the same OTU. See e.g., Claesson et al., 2010. Comparison of two next-generation sequencing technologies for resolving highly complex microbiota composition using tandem variable 16S rRNA gene regions. Nucleic Acids Res 38: e200. Konstantinidis et al., 2006. The bacterial species definition in the genomic era. Philos Trans R Soc Lond B Biol Sci 361:1929-1940. In embodiments involving the complete genome, MLSTs, specific genes, other than 16S, or sets of genes OTUs that share ≥95% average nucleotide identity are considered the same OTU. See e.g., Achtman and Wagner. 2008. Microbial diversity and the genetic nature of microbial species. Nat. Rev. Microbiol. 6:431-440; Konstantinidis et al., 2006, supra. The bacterial species definition in the genomic era. Philos Trans R Soc Lond B Biol Sci 361:1929-1940. OTUs can be defined by comparing sequences between organisms. Generally, sequences with less than 95% sequence identity are not considered to form part of the same OTU. OTUs may also be characterized by any combination of nucleotide markers or genes, in particular highly conserved genes (e.g., "house-keeping" genes), or a combination thereof. Such characterization employs, e.g., WGS data or a whole genome sequence. As used herein, a "type" of bacterium refers to an OTU that can be at the level of a strain, species, clade, or family.

(33) Table 1 below shows a List of Operational Taxonomic Units (OTU) with taxonomic assignments made to genus, species, and phylogenetic clade. Clade membership of bacterial OTUs is based on 16S sequence data. Clades are defined based on the topology of a phylogenetic tree that is constructed from full-length 16S sequences using maximum likelihood methods familiar to individuals with ordinary skill in the art of phylogenetics. Clades are constructed to ensure that all OTUs in a given clade are: (i) within a specified number of bootstrap supported nodes from one another, and (ii) within 5% genetic similarity. OTUs that are within the same clade can be distinguished as genetically and phylogenetically distinct from OTUs in a different clade based on 16S-V4 sequence data, while OTUs falling within the same clade are closely related. OTUs falling within the same clade are evolutionarily closely related and may or may not be distinguishable from one another using 16S-V4 sequence data. Members of the same clade, due to their evolutionary relatedness, play similar functional roles in a microbial ecology such as that found in the human gut. Compositions substituting one species with another from the same clade are likely to have conserved ecological function and therefore are useful in the present invention. All OTUs are denoted as to their putative capacity to form spores and whether they are a Pathogen or Pathobiont (see Definitions for description of "pathobiont"). NIAID Priority Pathogens are denoted as 'Category-A', 'Category-B', or 'Category-C', and Opportunistic Pathogens are denoted as 'OP'. OTUs that are not pathogenic or for which their ability to exist as a pathogen is unknown are denoted as 'N'. The 'SEQ ID Number' denotes the identifier of the OTU in the Sequence Listing File and 'Public DB Accession' denotes the identifier of the OTU in a public sequence repository.

(34) "Pathobionts" or "opportunistic pathogens" refers to specific bacterial species found in healthy hosts that may trigger immune-mediated pathology and/or disease in response to certain genetic or environmental factors (Chow et al., 2011. Curr Op Immunol. Pathobionts of the intestinal microbiota and inflammatory disease. 23:473-80). A pathobiont is an opportunistic microbe that is mechanistically distinct from an acquired infectious organism. The term "pathogen" as used herein includes both acquired infectious organisms and pathobionts.

(35) "Pathogen," "pathobiont" and "pathogenic" in reference to a bacterium or any other organism or entity that includes any such organism or entity that is capable of causing or affecting a disease, disorder or condition of a host organism containing the organism or entity, including but not limited to pre-diabetes, type 1 diabetes or type 2 diabetes.

(36) "Phenotype" refers to a set of observable characteristics of an individual entity. As example an individual subject may have a phenotype of "health" or "disease". Phenotypes describe the state of an entity and all entities within a phenotype share the same set of characteristics that describe the phenotype. The phenotype of an individual results in part, or in whole, from the interaction of the entity's genome and/or

microbiome with the environment, especially including diet.

(37) “Phylogenetic Diversity” is a biological characteristic that refers to the biodiversity present in a given Network Ecology or Network Class Ecology based on the OTUs that comprise the network. Phylogenetic diversity is a relative term, meaning that a Network Ecology or Network Class that is comparatively more phylogenetically diverse than another network contains a greater number of unique species, genera, and taxonomic families. Uniqueness of a species, genera, or taxonomic family is generally defined using a phylogenetic tree that represents the genetic diversity all species, genera, or taxonomic families relative to one another. In another embodiment phylogenetic diversity may be measured using the total branch length or average branch length of a phylogenetic tree. Phylogenetic Diversity may be optimized in a bacterial composition by including a wide range of biodiversity.

(38) “Phylogenetic tree” refers to a graphical representation of the evolutionary relationships of one genetic sequence to another that is generated using a defined set of phylogenetic reconstruction algorithms (e.g., parsimony, maximum likelihood, or Bayesian). Nodes in the tree represent distinct ancestral sequences and the confidence of any node is provided by a bootstrap or Bayesian posterior probability, which measures branch uncertainty.

(39) “Prediabetes” refers a condition in which blood glucose levels are higher than normal, but not high enough to be classified as diabetes. Individuals with pre-diabetes are at increased risk of developing type 2 diabetes within a decade. According to CDC, prediabetes can be diagnosed by fasting glucose levels between 100-125 mg/dL, 2 hour post-glucose load plasma glucose in oral glucose tolerance test (OGTT) between 140 and 199 mg/dL, or hemoglobin A1c test between 5.7%-6.4%.

(40) “rDNA,” “rRNA,” “16S-rDNA,” “16S-IRNA,” “16S,” “16S sequencing,” “16S-NGS,” “18S,” “18S-rRNA,” “18S-rDNA,” “18S sequencing,” and “18S-NGS” refer to the nucleic acids that encode for the RNA subunits of the ribosome. rDNA refers to the gene that encodes the rRNA that comprises the RNA subunits. There are two RNA subunits in the ribosome termed the small subunit (SSU) and large subunit (LSU); the RNA genetic sequences (rRNA) of these subunits is related to the gene that encodes them (rDNA) by the genetic code. rDNA genes and their complementary RNA sequences are widely used for determination of the evolutionary relationships amount organisms as they are variable, yet sufficiently conserved to allow cross organism molecular comparisons. Typically 16S rDNA sequence (approximately 1542 nucleotides in length) of the 30S SSU is used for molecular-based taxonomic assignments of prokaryotes and the 18S rDNA sequence (approximately 1869 nucleotides in length) of 40S SSU is used for eukaryotes. 16S sequences are used for phylogenetic reconstruction as they are in general highly conserved, but contain specific hypervariable regions that harbor sufficient nucleotide diversity to differentiate genera and species of most bacteria.

(41) “Residual habitat products” refers to material derived from the habitat for microbiota within or on a human or animal. For example, microbiota live in stool in the gastrointestinal tract, on the skin itself, in saliva, mucus of the respiratory tract, or secretions of the genitourinary tract (i.e., biological matter associated with the microbial community). Substantially free of residual habitat products means that the bacterial composition no longer contains the biological matter associated with the microbial environment on or in the human or animal subject and is 100% free, 99% free, 98% free, 97% free, 96% free, or 95% free of any contaminating biological matter associated with the microbial community. Residual habitat products can include abiotic materials (including undigested food) or it can include unwanted microorganisms. Substantially free of residual habitat products may also mean that the bacterial composition contains no detectable cells from a human or animal and that only microbial cells are detectable. In one embodiment, substantially free of residual habitat products may also mean that the bacterial composition contains no detectable viral (including bacterial viruses, i.e., phage), fungal, mycoplasmal contaminants. In another embodiment, it means that fewer than $1\times10-2\%$, $1\times10-3\%$, $1\times10-4\%$, $1\times10-5\%$, $1\times10-6\%$, $1\times10-7\%$, $1\times10-8$ of the viable cells in the bacterial composition are human or animal, as compared to microbial cells. There are multiple ways to accomplish this degree of purity, none of which are limiting. Thus, contamination may be reduced by isolating desired constituents through multiple steps of streaking to single colonies on solid media until replicate (such as, but not limited to, two) streaks from serial single colonies have shown only a single colony morphology. Alternatively, reduction of contamination can be accomplished by multiple rounds of serial dilutions to single desired cells (e.g., a dilution of 10-8 or 10-9), such as through multiple 10-fold serial dilutions. This can further be confirmed by showing that multiple isolated colonies have similar cell shapes and Gram staining behavior. Other methods for confirming adequate purity include genetic analysis (e.g., PCR, DNA sequencing), serology and antigen analysis, enzymatic and metabolic analysis, and methods using instrumentation such as flow cytometry with reagents that distinguish desired constituents from contaminants.

(42) “Synergy” refers to an effect produced by a combination, e.g., of two microbes (for example, microbes or two different species or two different clades) that is greater than the expected additive effectives of the combination components. In certain embodiments, “synergy” between two or more microbes can result in the inhibition of a pathogens ability to grow. For example, ternary combinations synergistically inhibit *C. difficile* if their mean log inhibition is greater than the sum of the log inhibition of homotrimers of each constituent bacterium divided by three to account for the three-fold higher dose of each strain or for binary combinations, the log inhibition of a homodimer of each constituent bacterium divided by two. In another example, synergy can be calculated by defining the OTU compositions that demonstrate greater inhibition than that represented by the sum of the log inhibition of each bacterium tested separately. In other embodiments, synergy can be defined as a property of compositions that exhibit inhibition greater than the maximum log inhibition among those of each constituent bacterium's homodimer or homotrimer measured independently. As used herein, “synergy” or “synergistic interactions” refers to the interaction or cooperation of two or more microbes to produce a combined effect greater than the sum of their separate effects.

(43) “Spore” or a population of “spores” includes bacteria (or other single-celled organisms) that are generally viable, more resistant to environmental influences such as heat and bacteriocidal agents than vegetative forms of the same bacteria, and typically capable of germination and out-growth. Spores are characterized by the absence of active metabolism until they respond to specific environmental signals, causing them to germinate. “Spore-formers” or bacteria “capable of forming spores” are those bacteria containing the genes and other necessary abilities to produce spores under suitable environmental conditions.

(44) “Spore population” refers to a plurality of spores present in a composition. Synonymous terms used herein include spore composition, spore preparation, ethanol-treated spore fraction and spore ecology. A spore population may be purified from a fecal donation, e.g., via ethanol or heat treatment, or a density gradient separation or any combination of methods described herein to increase the purity, potency and/or concentration of spores in a sample. Alternatively, a spore population may be derived through culture methods starting from isolated spore former species or spore former OTUs or from a mixture of such species, either in vegetative or spore form.

(45) In one embodiment, the spore preparation comprises spore forming species wherein residual non-spore forming species have been inactivated by chemical or physical treatments including ethanol, detergent, heat, sonication, and the like; or wherein the non-spore forming species have been removed from the spore preparation by various separations steps including density gradients, centrifugation, filtration and/or chromatography; or wherein inactivation and separation methods are combined to make the spore preparation. In yet another embodiment, the spore preparation comprises spore forming species that are enriched over viable non-spore formers or vegetative forms of spore formers. In this embodiment, spores are enriched by 2-fold, 5-fold, 10-fold, 50-fold, 100-fold, 1000-fold, 10,000-fold or greater than 10,000-fold compared to all vegetative forms of bacteria. In yet another embodiment, the spores in the spore preparation undergo partial germination during processing and formulation such that the final composition comprises spores and vegetative bacteria derived from spore forming species.

(46) “Sporulation induction agent” is a material or physical-chemical process that is capable of inducing sporulation in a bacterium, either

directly or indirectly, in a host organism and/or in vitro.
(47) To “increase production of bacterial spores” includes an activity or a sporulation induction agent. “Production” includes conversion of vegetative bacterial cells into spores and augmentation of the rate of such conversion, as well as decreasing the germination of bacteria in spore form, decreasing the rate of spore decay in vivo, or ex vivo, or to increasing the total output of spores (e.g., via an increase in volumetric output of fecal material).
(48) “Subject” refers to any animal subject including humans, laboratory animals (e.g., non-human primates, rats, mice), livestock (e.g., cows, sheep, goats, pigs, turkeys, and chickens), and household pets (e.g., dogs, cats, and rodents). The subject may be suffering from a dysbiosis, that contributes to or causes a condition classified as diabetes or pre-diabetes, including but not limited to mechanisms such as metabolic endotoxemia, altered metabolism of primary bile acids, immune system activation, or an imbalance or reduced production of short chain fatty acids including butyrate, propionate, acetate, and branched chain fatty acids.
(49) “Trimer” refers to a combination of bacteria that is comprised of three OTUs. The descriptions “homotrimer” and “heterotrimer” refer to combinations where all three OTUs are the same or different, respectively. A “semi-heterotrimer” refers to combinations where two constituents are the same with a third that is different
(50) As used herein the term “vitamin” is understood to include any of various fat-soluble or water-soluble organic substances (non-limiting examples include vitamin A, vitamin B1 (thiamine), vitamin B2 (riboflavin), vitamin B3 (niacin or niacinamide), vitamin B5 (pantothenic acid), vitamin B6 (pyridoxine, pyridoxal, or pyridoxamine, or pyridoxine hydrochloride), vitamin B7 (biotin), vitamin B9 (folic acid), and Vitamin B12 (various cobalamins; commonly cyanocobalamin in vitamin supplements), vitamin C, vitamin D, vitamin E, vitamin K, K1 and K2 (i.e., MK-4, MK-7), folic acid and biotin) essential in minute amounts for normal growth and activity of the body and obtained naturally from plant and animal foods or synthetically made, pro-vitamins, derivatives, analogs.
(51) “V1-V9 regions” or “16S V1-V9 regions” refers to the first through ninth hypervariable regions of the 16S rDNA gene that are used for genetic typing of bacterial samples. These regions in bacteria are defined by nucleotides 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294 and 1435-1465 respectively using numbering based on the *E. coli* system of nomenclature (Brosius et al., 1978. Complete nucleotide sequence of a 16S ribosomal RNA gene from *Escherichia coli*, PNAS USA 75 (10): 4801-4805). In some embodiments, at least one of the V1, V2, V3, V4, V5, V6, V7, V8, and V9 regions are used to characterize an OTU. In one embodiment, the V1, V2, and V3 regions are used to characterize an OTU. In another embodiment, the V3, V4, and V5 regions are used to characterize an OTU. In another embodiment, the V4 region is used to characterize an OTU. A person of ordinary skill in the art can identify the specific hypervariable regions of a candidate 16S rDNA by comparing the candidate sequence in question to a reference sequence and identifying the hypervariable regions based on similarity to the reference hypervariable regions, or alternatively, one can employ Whole Genome Shotgun (WGS) sequence characterization of microbes or a microbial community.
DETAILED DESCRIPTION
(52) Applicants have discovered combinations of bacteria that, when present, are associated with improvement in the microbiome of a subject, e.g., a subject having a dysbiosis such as a dysbiosis associated with *C. difficile*. In addition, combinations of microorganisms have been identified that are associated with the engraftment and/or augmentation of organisms associated with a healthy microbiome. Applicants have also identified combinations of microorganisms that can be useful for treating antibiotic-resistant or other pathogenic bacterial conditions. In some embodiments the microbial content of such a composition comprises the organisms, in other embodiments, the microbial content of the composition consists essentially of the organisms, and in other embodiments, the microbial content of the composition consists of the organisms. In all cases, the composition may include non-microbial components. In some cases, the composition comprises at least two organisms (e.g., three organisms) or more, as are described herein.
(53) Emergence of Antibiotic Resistance in Bacteria
(54) Antibiotic resistance is an emerging public health issue (Carlet et al., 2011. Society's failure to protect a precious resource: antibiotics. Lancet 378:369-371). Numerous genera of bacteria harbor species that are developing resistance to antibiotics. These include but are not limited to vancomycin resistant *Enterococcus* (VRE) and carbapenem resistant *Klebsiella* (CRKp). *Klebsiella pneumoniae* and *Escherichia coli* strains are becoming resistant to carbapenems and require the use of old antibiotics characterized by high toxicity, such as colistin (Cantón et al. 2012. Rapid evolution and spread of carbapenemases among Enterobacteriaceae in Europe. Clin Microbiol Infect 18:413-431). Additional multiply drug resistant strains of multiple species, including *Pseudomonas aeruginosa*, *Enterobacter* spp., and *Acinetobacter* spp. are observed clinically including isolates that are highly resistant to ceftazidime, carbapenems, and quinolones (European Centre for Disease Prevention and Control: EARSS net database. http://ecdc.europa.eu.). The Centers for Disease Control and Prevention in 2013 released a Threat Report (http://www.cdc.gov/drugresistance/threat-report-2013/) citing numerous bacterial infection threats that included *Clostridium difficile*, carbapenem-resistant Enterobacteriaceae (CRE), multidrug-resistant *Acinetobacter*, drug-resistant *Campylobacter*, extended spectrum ß-lactamase producing Enterobacteriaceae (ESBLs), vancomycin-resistant *Enterococcus* (VRE), multidrug-resistant *Pseudomonas aeruginosa*, drug-resistant non-typhoidal *Salmonella*, drug-resistant *Salmonella typhi*, drug-resistant *Shigella*, methicillin-resistant *Staphylococcus aureus* (MRSA), drug-resistant *Streptococcus pneumoniae*, vancomycin-resistant *Staphylococcus aureus* (VRSA), erythromycin-resistant Group A *Streptococcus*, and clindamycin-resistant Group B *Streptococcus*. The increasing failure of antibiotics due the rise of resistant microbes demands new therapeutics to treat bacterial infections. Administration of a microbiome therapeutic bacterial composition offers potential for such therapies.
(55) Applicants have discovered that subjects suffering from recurrent *C. difficile* associated diarrhea (CDAD) often harbor antibiotic resistant Gram-negative bacteria, in particular Enterobacteriaceae and that treatment with a bacterial composition effectively treats CDAD and reduces the antibiotic resistant Gram-negative bacteria. The gastrointestinal tract is implicated as a reservoir for many of these organisms including VRE, MRSA, *Pseudomonas aeruginosa, Acinetobacter* and the yeast *Candida* (Donskey, Clinical Infectious Diseases 2004 39:214, The Role of the Intestinal Tract as a Reservoir and Source for Transmission of Nosocomial Pathogens), and thus as a source of nosocomial infections. Antibiotic treatment and other decontamination procedures are among the tools in use to reduce colonization of these organisms in susceptible subjects including those who are immunosuppressed. Bacterial-based therapeutics would provide a new tool for decolonization, with a key benefit of not promoting antibiotic resistance as antibiotic therapies do.
(56) Bacterial Compositions
(57) Provided are bacteria and combinations of bacteria of the human gut microbiota with the capacity to meaningfully provide functions of a healthy microbiota or catalyze an augmentation to the resident microbiome when administered to mammalian hosts. In particular, provided are synergistic combinations that treat, prevent, delay or reduce the symptoms of diseases, disorders and conditions associated with a dysbiosis. Representative diseases, disorders and conditions potentially associated with a dysbiosis, which are suitable for treatment with the compositions and methods as described herein, are provided in Table 3. Without being limited to a specific mechanism, it is thought that such compositions inhibit the growth, proliferation, and/or colonization of one or a plurality of pathogenic bacteria in the dysbiotic microbiotal niche, so that a healthy, diverse and protective microbiota colonizes and populates the intestinal lumen to establish or reestablish ecological control over pathogens or potential pathogens (e.g., some bacteria are pathogenic bacteria only when present in a dysbiotic environment). Inhibition of pathogens includes those pathogens such as *C. difficile, Salmonella* spp., enteropathogenic *E. coli*, multi-drug resistant bacteria such as

*Klebsiella*, and *E. coli*, carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE).
(58) The bacterial compositions provided herein are produced and the efficacy thereof in inhibiting pathogenic bacteria is demonstrated as provided in further detail herein.
(59) In particular, in order to characterize those antagonistic relationships between gut commensals that are relevant to the dynamics of the mammalian gut habitat, provided is an in vitro microplate-based screening system that demonstrates the efficacy of those bacterial compositions, including the ability to inhibit (or antagonize) the growth of a bacterial pathogen or pathobiont, typically a gastrointestinal microorganism. These methods provide novel combinations of gut microbiota species and OTUs that are able to restore or enhance ecological control over important gut pathogens or pathobionts in vivo.
(60) Bacterial compositions may comprise two types of bacteria (termed "binary combinations" or "binary pairs") or greater than two types of bacteria. Bacterial compositions that comprise three types of bacteria are termed "ternary combinations". For instance, a bacterial composition may comprise at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, or at least 21, 22, 23, 24, 25, 26, 27, 28, 29 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or at least 40, at least 50 or greater than 50 types of bacteria, as defined by species or operational taxonomic unit (OTU), or otherwise as provided herein. In one embodiment, the composition comprises at least two types of bacteria chosen from Table 1.
(61) In another embodiment, the number of types of bacteria present in a bacterial composition is at or below a known value. For example, in such embodiments the bacterial composition comprises 50 or fewer types of bacteria, such as 49, 48, 47, 46, 45, 44, 43, 42, 41, 40, 39, 38, 37, 36, 35, 34, 33, 32, 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, or 10 or fewer, or 9 or fewer types of bacteria, 8 or fewer types of bacteria, 7 or fewer types of bacteria, 6 or fewer types of bacteria, 5 or fewer types of bacteria, 4 or fewer types of bacteria, or 3 or fewer types of bacteria. In another embodiment, a bacterial composition comprises from 2 to no more than 40, from 2 to no more than 30, from 2 to no more than 20, from 2 to no more than 15, from 2 to no more than 10, or from 2 to no more than 5 types of bacteria.
(62) In some embodiments, bacterial compositions are provided with the ability to exclude pathogenic bacteria. Exemplary bacterial compositions are demonstrated to reduce the growth rate of one pathogen, *C. difficile*, as provided in the Examples, wherein the ability of the bacterial compositions is demonstrated by assessing the antagonism activity of a combination of OTUs or strains towards a given pathogen using in vitro assays.
(63) In some embodiments, bacterial compositions with the capacity to durably exclude *C. difficile*, are developed using a methodology for estimating an Ecological Control Factor (ECF) for constituents within the human microbiota. The ECF is determined by assessing the antagonistic activity of a given commensal strain or combination of strains towards a given pathogen using an in vitro assay, resulting in observed levels of ecological control at various concentrations of the added commensal strains. The ECF for a commensal strain or combination of strains is somewhat analogous to the longstanding minimal inhibitory concentration (MIC) assessment that is employed in the assessment of antibiotics. The ECF allows for the assessment and ranking of relative potencies of commensal strains and combinations of strains for their ability to antagonize gastrointestinal pathogens. The ECF of a commensal strain or combination of strains may be calculated by assessing the concentration of that composition that is able to mediate a given percentage of inhibition (e.g., at least 10%, 20%, 50%, 70%, 75%, 80%, 85%, 90%, 95%, or 100%) of a target pathogen in the in vitro assay. Provided herein are combinations of strains or OTUs within the human microbiota that are able to significantly reduce the rate of gastrointestinal pathogen replication within the in vitro assay. These compositions are capable of providing a safe and effective means by which to affect the growth, replication, and disease severity of such bacterial pathogens.
(64) Bacterial compositions may be prepared comprising at least two types of isolated bacteria, wherein a first type and a second type are independently chosen from the species or OTUs listed in Table 1. Certain embodiments of bacterial compositions with at least two types of isolated bacteria containing binary pairs are reflected in Table 4a. Additionally, a bacterial composition may be prepared comprising at least two types of isolated bacteria, wherein a first OTU and a second OTU are independently characterized by, i.e., at least 95%, 96%, 97%, 98%, 99% or including 100% sequence identity to, sequences listed in Table 1. Generally, the first bacteria and the second bacteria are not the same OTU. The sequences provided in the sequencing listing file for OTUs in Table 1 are full 16S sequences. Therefore, in one embodiment, the first and/or second OTUs may be characterized by the full 16S sequences of OTUs listed in Table 1. In another embodiment, the first and/or second OTUs may be characterized by one or more of the variable regions of the 16S sequence (V1-V9). These regions in bacteria are defined by nucleotides 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294 and 1435-1465 respectively using numbering based on the *E. coli* system of nomenclature. (See, e.g., Brosius et al., Complete nucleotide sequence of a 16S ribosomal RNA gene from *Escherichia coli*, PNAS 75 (10): 4801-4805 (1978)). In some embodiments, at least one of the V1, V2, V3, V4, V5, V6, V7, V8, and V9 regions are used to characterize an OTU. In one embodiment, the V1, V2, and V3 regions are used to characterize an OTU. In another embodiment, the V3, V4, and V5 regions are used to characterize an OTU. In another embodiment, the V4 region is used to characterize an OTU.
(65) Bacterial Compositions Described by Species
(66) In some embodiments, compositions are defined by species included in the composition. Methods of identifying species are known in the art.
(67) Bacterial Compositions Described by Operational Taxonomic Units (OTUs)
(68) OTUs may be defined either by full 16S sequencing of the rDNA gene, by sequencing of a specific hypervariable region of this gene (i.e., V1, V2, V3, V4, V5, V6, V7, V8, or V9), or by sequencing of any combination of hypervariable regions from this gene (e.g., V1-3 or V3-5). The bacterial 16S rDNA is approximately 1500 nucleotides in length and is used in reconstructing the evolutionary relationships and sequence similarity of one bacterial isolate to another using phylogenetic approaches. 16S sequences are used for phylogenetic reconstruction as they are in general highly conserved, but contain specific hypervariable regions that harbor sufficient nucleotide diversity to differentiate genera and species of most microbes. Using well known techniques, in order to determine the full 16S sequence or the sequence of any hypervariable region of the 16S sequence, genomic DNA is extracted from a bacterial sample, the 16S rDNA (full region or specific hypervariable regions) amplified using polymerase chain reaction (PCR), the PCR products cleaned, and nucleotide sequences delineated to determine the genetic composition of 16S gene or subdomain of the gene. If full 16S sequencing is performed, the sequencing method used may be, but is not limited to, Sanger sequencing. If one or more hypervariable regions are used, such as the V4 region, the sequencing may be, but is not limited to being, performed using the Sanger method or using a next-generation sequencing method, such as an Illumina (sequencing by synthesis) method using barcoded primers allowing for multiplex reactions.
(69) Bacterial Compositions Exclusive of Certain Bacterial Species or Strains
(70) In one embodiment, the bacterial composition does not comprise at least one of *Enterococcus faecalis* (previously known as *Streptococcus faecalis*), *Clostridium innocuum*, *Clostridium ramosum*, *Bacteroides ovatus*, *Bacteroides vulgatus*, *Bacteroides thetaoiotaomicron*, *Escherichia coli* (1109 and 1108-1), *Clostridium bifermentans*, and *Blautia producta* (previously known as *Peptostreptococcus productus*).
(71) In another embodiment, the bacterial composition does not comprise at least one of *Acidaminococcus intestinalis*, *Bacteroides ovatus*, two species of *Bifidobacterium adolescentis*, two species of *Bifidobacterium longum*, *Collinsella aerofaciens*, two species of *Dorea longicatena*, *Escherichia coli*, *Eubacterium eligens*, *Eubacterium limosum*, four species of *Eubacterium rectale*, *Eubacterium ventriosumi*, *Faecalibacterium prausnitzii*, *Lactobacillus casei*, *Lactobacillus paracasei*, *Paracateroides distasonis*, *Raoultella* sp., one species of *Roseburia* (chosen from

*Roseburia faecalis* or *Roseburia faecis*), *Roseburia intestinalis,* two species of *Ruminococcus torques,* and *Streptococcus mitis.*
(72) In another embodiment, the bacterial composition does not comprise at least one of Barnesiella intestinihominis; *Lactobacillus reuteri*; a species characterized as one of *Enterococcus hirae*, *Enterococus faecium*, or *Enterococcus durans*; a species characterized as one of *Angerostipes caccae* or *Clostridium indolis*; a species characterized as one of *Staphylococcus warneri* or *Staphylococcus pasteuri*; and *Adlercreutzia equolifaciens.*
(73) In another embodiment, the bacterial composition does not comprise at least one of *Clostridium absonum, Clostridium argentinense, Clostridium baratii, Clostridium bifermentans, Clostridium botulinum, Clostridium butyricum, Clostridium cadaveris, Clostridium camis, Clostridium celatum, Clostridium chauvoei, Clostridium clostridioforme, Clostridium cochlearium, Clostridium difficile, Clostridium fallax, Clostridium felsineum, Clostridium ghonii, Clostridium glycolicum, Clostridium haemolyticum, Clostridium hastiforme, Clostridium histolyticum, Clostridium indolis, Clostridium innocuum, Clostridium irregulare, Clostridium limosum, Clostridium malenominatum, Clostridium novyi, Clostridium oroticum, Clostridium paraputrificum, Clostridium perfringens, Clostridium piliforme, Clostridium putrefaciens, Clostridium putrificum, Clostridium ramosum, Clostridium sardiniense, Clostridium sartagoforme, Clostridium scindens, Clostridium septicum, Clostridium sordellii, Clostridium sphenoides, Clostridium spiroforme, Clostridium sporogenes, Clostridium subterminale, Clostridium symbiosum, Clostridium tertium, Clostridium tetani, Clostridium welchii*, and *Clostridium villosum.*
(74) In another embodiment, the bacterial composition does not comprise at least one of *Clostridium innocuum, Clostridum bifermentans, Clostridium butyricum, Bacteroides fragilis, Bacteroides thetaiotaomicron, Bacteroides uniformis*, three strains of *Escherichia coli*, and *Lactobacillus* sp.
(75) In another embodiment, the bacterial composition does not comprise at least one of *Clostridium bifermentans, Clostridium innocuum, Clostridium butyricum*, three strains of *Escherichia coli*, three strains of *Bacteroides*, and *Blautia producta* (previously known as *Peptostreptococcus productus*).
(76) In another embodiment, the bacterial composition does not comprise at least one of *Bacteroides* sp., *Escherichia coli*, and non-pathogenic *Clostridia*, including *Clostridium innocuum, Clostridium bifermentans* and *Clostridium ramosum.*
(77) In another embodiment, the bacterial composition does not comprise at least one of more than one *Bacteroides* species, *Escherichia coli* and non-pathogenic *Clostridia*, such as *Clostridium butyricum, Clostridium bifermentans* and *Clostridium innocuum.*
(78) In another embodiment, the bacterial composition does not comprise at least one of *Bacteroides caccae, Bacteroides capillosus, Bacteroides coagulans, Bacteroides distasonis, Bacteroides eggerthii, Bacteroides forsythus, Bacteroides fragilis, Bacteroides fragilis-ryhm, Bacteroides gracilis, Bacteroides levii, Bacteroides macacae, Bacteroides merdae, Bacteroides ovatus, Bacteroides pneumosintes, Bacteroides putredinis, Bacteroides pyogenes, Bacteroides splanchnicus, Bacteroides stercoris, Bacteroides tectum, Bacteroides thetaiotaomicron, Bacteroides uniformis, Bacteroides ureolyticus*, and *Bacteroides vulgatus.*
(79) In another embodiment, the bacterial composition does not comprise at least one of *Bacteroides, Eubacteria, Fusobacteria, Propionibacteria, Lactobacilli,* anaerobic cocci, *Ruminococcus, Escherichia coli, Gemmiger, Desulfomonas*, and *Peptostreptococcus.*
(80) In another embodiment, the bacterial composition does not comprise at least one of *Bacteroides fragilis* ss. *Vulgatus, Eubacterium aerofaciens, Bacteroides fragilis* ss. *Thetaiotaomicron, Blautia producta* (previously known as *Peptostreptococcus productus* II), *Bacteroides fragilis* ss. *Distasonis, Fusobacterium prausnitzii, Coprococcus eutactus, Eubacterium aerofaciens* III, *Blautia producta* (previously known as *Peptostreptococcus productus* I), *Ruminococcus bromii, Bifidobacterium adolescentis, Gemmiger formicilis, Bifidobacterium longum, Eubacterium siraeum, Ruminococcus torques, Eubacterium rectale* III-H, *Eubacterium rectale* IV. *Eubacterium eligens, Bacteroides eggerthii, Clostridium leptum, Bacteroides fragilis* ss. A, *Eubacterium biforme, Bifidobacterium infantis, Eubacterium rectale* III-F, *Coprococcus comes, Bacteroides capillosus, Ruminococcus albus, Eubacterium formicigenerans, Eubacterium hallii, Eubacterium ventriosum* I, *Fusobacterium russii, Ruminococcus obeum, Eubacterium rectale* II, *Clostridium ramosum* I, *Lactobacillus leichmanii, Ruminococcus cailidus, Butyrivibrio crossotus, Acidaminococcus fermentans, Eubacterium ventriosum, Bacteroides fragilis* ss. *fragilis, Bacteroides* AR, *Coprococcus catus, Eubacterium hadrum, Eubacterium cylindroides, Eubacterium ruminantium, Eubacterium* CH-1, *Staphylococcus epidermidis, Peptostreptococcus* BL, *Eubacterium limosum, Bacteroides praeacutus, Bacteroides* L. *Fusobacterium mortiferum* I, *Fusobacterium naviforme, Clostridium innocuum, Clostridium ramosum, Propionibacterium acnes, Ruminococcus flavefaciens, Ruminococcus* AT, *Peptococcus* AU-1, *Eubacterium* AG, -AK, -AL, -AL-1, -AN; *Bacteroides fragilis* ss. *ovatus*, -ss. d, -ss. f; *Bacteroides* L-1, L-5; *Fusobacterium micleatum, Fusobacterium mortiferum, Escherichia coli, Streptococcus morbiliorum, Peptococcus magmis, Peptococcus* G, AU-2; *Streptococcus intermedius, Ruminococcus lactaris, Ruminococcus* CO *Gemmiger* X, *Coprococcus* BH, -CC; *Eubacterium tenue, Eubacterium ramulus, Eubacterium* AE, -AG-H, -AG-M, -AJ, -BN-1; *Bacteroides clostridiiformis* ss. *clostridliformis, Bacteroides coagulans, Bacteroides orails, Bacteroides ruminicola* ss. *brevis*, -ss. *ruminicola, Bacteroides splanchnicus, Desuifomonas pigra, Bacteroides* L-4, -N-i; *Fusobacterium* H, *Lactobacillus* G, and *Succinivibrio* A.
(81) In another embodiment, the bacterial composition does not comprise at least one of *Bifidobacterium bifidum* W23, *Bifidobacterium lactis* W18, *Bifidobacterium longum* W51, *Enterococcus faecium* W54, *Lactobacillus plantarum* W62, *Lactobacillus rhamnosus* W71, *Lactobacillus acidophilus* W37, *Lactobacillus acidophilus* W55, *Lactobacillus paracasei* W20, and *Lactobacillus salivarius* W24.
(82) In another embodiment, the bacterial composition does not comprise at least one of *Anaerotruncus colihominis* DSM 17241, *Blautia producta* JCM 1471, *Clostridiales bacterium* 1 7 47FAA, *Clostridium asparagiforme* DSM 15981, *Clostridium bacterium* JC13, *Clostridium bolteae* ATCC BAA 613, *Clostridium hathewayi* DSM 13479. *Clostridium indolis* CM971, *Clostridium ramosum* DSM 1402, *Clostridium saccharogumia* SDG Mt85 3 Db, *Clostridium scindens* VP 12708, *Clostridium* sp 7 3 54FAA, *Eubacterium contortum* DSM 3982, *Lachnospiraceae bacterium* 3 1 57FAA CT1, *Lachnospiraceae bacterium* 7 1 58FAA, and *Ruminococcus* sp ID8.
(83) In another embodiment, the bacterial composition does not comprise at least one of *Anaerotruncus colihominis* DSM 17241, *Blautia producta* JCM 1471, *Clostridiales bacterium* 1 7 47FAA, *Clostridium asparagiforme* DSM 15981, *Clostridium bacterium* JC13, *Clostridium bolteae* ATCC BAA 613, *Clostridium hathewayi* DSM 13479, *Clostridium indolis* CM971, *Clostridium ramosum* DSM 1402, *Clostridium saccharogumia* SDG M185 3 Db, *Clostridium scindens* VP 12708, *Clostridium* sp 7 3 54FAA, *Eubacterium contortum* DSM 3982, *Lachnospiraceae bacterium* 3 1 57FAA CT1, *Lachnospiraceae bacterium* 7 1 58FAA, *Oscillospiraceae bacterium* NML 061048, and *Ruminococcus* sp ID8.
(84) In another embodiment, the bacterial composition does not comprise at least one of *Clostridium ramosum* DSM 1402, *Clostridium saccharogumia* SDG Mt85 3 Db, and *Lachnospiraceae bacterium* 7 1 58FAA.
(85) In another embodiment, the bacterial composition does not comprise at least one of *Clostridium hathewayi* DSM 13479, *Clostridium saccharogumia* SDG Mt85 3 Db, *Clostridium* sp 7 3 54FAA, and *Lachnospiraceae bacterium* 3 1 57FAA CT1.
(86) In another embodiment, the bacterial composition does not comprise at least one of *Anaerotruncus colihominis* DSM 17241, *Blautia producta* JCM 1471, *Clostridium bacterium* JC13, *Clostridium scindens* VP 12708, and *Ruminococcus* sp ID8.
(87) In another embodiment, the bacterial composition does not comprise at least one of *Clostridiales bacterium* 1 7 47FAA, *Clostridium asparagiforme* DSM 15981, *Clostridium bolteae* ATCC BAA 613, *Clostridium indolis* CM971, and *Lachnospiraceae bacterium* 7 1 58FAA.
(88) Inhibition of Bacterial Pathogens
(89) The bacterial compositions offer a protective or therapeutic effect against infection by one or more GI pathogens of interest, some of which

are listed in Table 3.
(90) In some embodiments, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Neisseria, Mycoplasma, Mycobacterium, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, extended spectrum beta-lactam resistant Enterococci (ESBL), carbapenem-resistant Enterobacteriaceae (CRE), and vancomycin-resistant Enterococci (VRE).
(91) In some embodiments, these pathogens include, but are not limited to, *Aeromonas hydrophila, Campylobacter fetus, Plesiomonas shigelloides, Bacillus cereus, Campylobacter jejuni, Clostridium botulinum, Clostridium difficile, Clostridium perfringens*, enteroaggregative *Escherichia coli*, enterohemorrhagic *Escherichia coli*, enteroinvasive *Escherichia coli*, enterotoxigenic *Escherichia coli* (such as, but not limited to, LT and/or ST), *Escherichia coli* 0157: H7, *Fusarium* spp., *Helicobacter pylori, Klebsiella pneumonia, Klebsiella oxytoca, Lysteria monocytogenes, Morganella* spp., *Plesiomonas shigelloides, Proteus* spp., *Providencia* spp., *Salmonella* spp., *Salmonella typhi, Salmonella paratyphi, Shigella* spp., *Staphylococcus* spp., *Staphylococcus aureus*, vancomycin-resistant *enterococcus* spp., *Vibrio* spp., *Vibrio cholerae, Vibrio parahaemolyticus, Vibrio vulnificus*, and *Yersinia enterocolitica*.
(92) In one embodiment, the pathogen of interest is at least one pathogen chosen from *Clostridium difficile, Salmonella* spp., pathogenic *Escherichia coli*, vancomycin-resistant *Enterococcus* spp., and extended spectrum beta-lactam resistant Enterococci (ESBL).
(93) In Vitro Assays Substantiating Protective Effect of Bacterial Compositions
(94) In one embodiment, provided is an in vitro assay utilizing competition between the bacterial compositions or subsets thereof and *C. difficile*. Exemplary embodiments of the assay are provided herein and in the Examples.
(95) In another embodiment, provided is an in vitro assay utilizing 10% (wt/vol) Sterile-Filtered Stool (SFS). Provided is an in vitro assay to test for the protective effect of the bacterial compositions and to screen in vitro for combinations of microbes that inhibit the growth of a pathogen. The assay can operate in automated high-throughput or manual modes. Under either system, human or animal stool may be re-suspended in an anaerobic buffer solution, such as pre-reduced PBS or other suitable buffer, the particulate removed by centrifugation, and filter sterilized. This 10% sterile-filtered stool material serves as the base media for the in vitro assay. To test a bacterial composition, an investigator may add it to the sterile-filtered stool material for a first incubation period and then may inoculate the incubated microbial solution with the pathogen of interest for a second incubation period. The resulting titer of the pathogen may be quantified by any number of methods such as those described below, and the change in the amount of pathogen is compared to standard controls including the pathogen cultivated in the absence of the bacterial composition. The assay is conducted using at least one control. Stool from a healthy subject may be used as a positive control. As a negative control, antibiotic-treated stool or heat-treated stool may be used. Various bacterial compositions may be tested in this material and the bacterial compositions optionally compared to the positive and/or negative controls. The ability to inhibit the growth of the pathogen may be measured by plating the incubated material on *C. difficile* selective media and counting colonies. After competition between the bacterial composition and *C. difficile*, each well of the in vitro assay plate is serially diluted ten-fold six times, and plated on selective media, such as but not limited to cycloserine cefoxitin mannitol agar (CCMA) or cycloserine cefoxitin fructose agar (CCFA), and incubated. Colonies of *C. difficile* are then counted to calculate the concentration of viable cells in each well at the end of the competition. Colonies of *C. difficile* are confirmed by their characteristic diffuse colony edge morphology as well as fluorescence under UV light.
(96) In another embodiment, the in vitro assay utilizes Antibiotic-Treated Stool. In an alternative embodiment, and instead of using 10% sterile-filtered stool, human or animal stool may be resuspended in an anaerobic buffer solution, such as pre-reduced PBS or other suitable buffer. The resuspended stool is treated with an antibiotic, such as clindamycin, or a cocktail of several antibiotics in order to reduce the ability of stool from a healthy subject to inhibit the growth of *C. difficile*; this material is termed the antibiotic-treated matrix. While not being bound by any mechanism, it is believed that beneficial bacteria in healthy subjects protects them from infection by competing out *C. difficile*. Treating stool with antibiotics kills or reduces the population of those beneficial bacteria, allowing *C. difficile* to grow in this assay matrix. Antibiotics in addition to clindamycin that inhibit the normal flora include ceftriaxone and piperacillin-tazobactam and may be substituted for the clindamycin. The antibiotic-treated matrix is centrifuged, the supernatant removed, and the pelleted material resuspended in filter-sterilized, diluted stool in order to remove any residual antibiotic. This washed antibiotic-treated matrix may be used in the in vitro assay described above in lieu of the 10% sterile-filtered stool.
(97) Also provided is an In Vitro Assay utilizing competition between the bacterial compositions or subsets thereof and Vancomycin-resistant *Enterococcus faecium*. Exemplary embodiments of this Assay are provided herein and in the Examples.
(98) Also provided is an in vitro assay utilizing competition between the bacterial compositions or subsets thereof and *Morganella morganii*. Exemplary embodiments of this Assay are provided herein and in the Examples.
(99) Also provided is an in vitro assay utilizing competition between the bacterial compositions or subsets thereof and *Klebsiella pneumoniae*. Exemplary embodiments of this Assay are provided herein and in the Examples.
(100) Alternatively, the ability to inhibit the growth of the pathogen may be measured by quantitative PCR (qPCR). Standard techniques may be followed to generate a standard curve for the pathogen of interest. Genomic DNA may be extracted from samples using commercially available kits, such as the Mo Bio Powersoil®-htp 96 Well Soil DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA), the Mo Bio Powersoil® DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA), or the QIAamp DNA Stool Mini Kit (QIAGEN, Valencia, CA) according to the manufacturer's instructions. The qPCR may be conducted using HotMasterMix (5PRIME, Gaithersburg, MD) and primers specific for the pathogen of interest, and may be conducted on a MicroAmp® Fast Optical 96-well Reaction Plate with Barcode (0.1 mL) (Life Technologies, Grand Island, NY) and performed on a BioRad C1000™ Thermal Cycler equipped with a CFX96™ Real-Time System (BioRad, Hercules, CA), with fluorescent readings of the FAM and ROX channels. The Cq value for each well on the FAM channel is determined by the CFX Manager™ software version 2.1. The log.sub.10 (cfu/ml) of each experimental sample is calculated by inputting a given sample's Cq value into linear regression model generated from the standard curve comparing the Cq values of the standard curve wells to the known log.sub.10 (cfu/ml) of those samples. The skilled artisan may employ alternative qPCR modes.
(101) In Vivo Assay Establishing Protective Effect of Bacterial Compositions.
(102) Provided is an in vivo mouse model to test for the protective effect of the bacterial compositions against *C. difficile*. In this model (based on Chen, et al., A mouse model of *Clostridium difficile* associated disease, Gastroenterology 135 (6): 1984-1992 (2008)), mice are made susceptible to *C. difficile* by a 7 day treatment (days −12 to −5 of experiment) with 5 to 7 antibiotics (including kanamycin, colistin, gentamycin, metronidazole and vancomycin and optionally including ampicillin and ciprofloxacin) delivered via their drinking water, followed by a single dose with Clindamycin on day −3, then challenged three days later on day 0 with 10.sup.4 spores of *C. difficile* via oral gavage (i.e., oro-gastric lavage). Bacterial compositions may be given either before (prophylactic treatment) or after (therapeutic treatment) *C. difficile* gavage. Further, bacterial compositions may be given after (optional) vancomycin treatment (see below) to assess their ability to prevent recurrence and thus suppress the pathogen in vivo. The outcomes assessed each day from day −1 to day 6 (or beyond, for prevention of recurrence) are weight, clinical signs, mortality and shedding of *C. difficile* in the stool. Weight loss, clinical signs of disease, and *C. difficile* shedding are typically observed without treatment. Vancomycin provided by oral gavage on days −1 to 4 protects against these outcomes and serves as a positive

control. Clinical signs are subjective, and scored each day by the same experienced observer. Animals that lose greater than or equal to 25% of their body weight are euthanized and counted as infection-related mortalities. Stool are gathered from mouse cages (5 mice per cage) each day, and the shedding of *C. difficile* spores is detected in the stool using a selective plating assay as described for the in vitro assay above, or via qPCR for the toxin gene as described herein. The effects of test materials including 10% suspension of human stool (as a positive control), bacterial compositions, or PBS (as a negative vehicle control), are determined by introducing the test article in a 0.2 mL volume into the mice via oral gavage on day −1, one day prior to *C. difficile* challenge, on day 1, 2 and 3 as treatment or post-vancomycin treatment on days 5, 6, 7 and 8. Vancomycin, as discussed above, is given on days 1 to 4 as another positive control. Alternative dosing schedules and routes of administration (e.g., rectal) may be employed, including multiple doses of test article, and 10.sup.3 to 10.sup.10 of a given organism or composition may be delivered.
(103) Methods for Preparing a Bacterial Composition for Administration to a Subject
(104) Methods for producing bacterial compositions may include three main processing steps, combined with one or more mixing steps. The steps are: organism banking, organism production, and preservation.
(105) For banking, the strains included in the bacterial composition may be (1) isolated directly from a specimen or taken from a banked stock, (2) optionally cultured on a nutrient agar or broth that supports growth to generate viable biomass, and (3) the biomass optionally preserved in multiple aliquots in long-term storage.
(106) In embodiments using a culturing step, the agar or broth may contain nutrients that provide essential elements and specific factors that enable growth. An example would be a medium composed of 20 g/L glucose, 10 g/L yeast extract, 10 g/L soy peptone, 2 g/L citric acid, 1.5 g/L sodium phosphate monobasic, 100 mg/L ferric ammonium citrate, 80 mg/L magnesium sulfate, 10 mg/L hemin chloride, 2 mg/L calcium chloride, 1 mg/L menadione. A variety of microbiological media and variations are well known in the art (e.g., R. M. Atlas, *Handbook of Microbiological Media* (2010) CRC Press). Medium can be added to the culture at the start, may be added during the culture, or may be intermittently/continuously flowed through the culture. The strains in the bacterial composition may be cultivated alone, as a subset of the bacterial composition, or as an entire collection comprising the bacterial composition. As an example, a first strain may be cultivated together with a second strain in a mixed continuous culture, at a dilution rate lower than the maximum growth rate of either cell to prevent the culture from washing out of the cultivation.
(107) The inoculated culture is incubated under favorable conditions for a time sufficient to build biomass. For bacterial compositions for human use this is often at 37° C. temperature, pH, and other parameter with values similar to the normal human niche. The environment may be actively controlled, passively controlled (e.g., via buffers), or allowed to drift. For example, for anaerobic bacterial compositions (e.g., gut microbiota), an anoxic/reducing environment may be employed. This can be accomplished by addition of reducing agents such as cysteine to the broth, and/or stripping it of oxygen. As an example, a culture of a bacterial composition may be grown at 37° C., pH 7, in the medium above, pre-reduced with 1 g/L cysteine HCl.
(108) When the culture has generated sufficient biomass, it may be preserved for banking. The organisms may be placed into a chemical milieu that protects from freezing (adding ‘cryoprotectants’), drying (‘lyoprotectants’), and/or osmotic shock (‘osmoprotectants’), dispensing into multiple (optionally identical) containers to create a uniform bank, and then treating the culture for preservation. Containers are generally impermeable and have closures that assure isolation from the environment. Cryopreservation treatment is accomplished by freezing a liquid at ultra-low temperatures (e.g., at or below −80° C.). Dried preservation removes water from the culture by evaporation (in the case of spray drying or ‘cool drying’) or by sublimation (e.g., for freeze drying, spray freeze drying). Removal of water improves long-term bacterial composition storage stability at temperatures elevated above cryogenic. If the bacterial composition comprises spore forming species and results in the production of spores, the final composition may be purified by additional means such as density gradient centrifugation preserved using the techniques described above. Bacterial composition banking may be done by culturing and preserving the strains individually, or by mixing the strains together to create a combined bank. As an example of cryopreservation, a bacterial composition culture may be harvested by centrifugation to pellet the cells from the culture medium, the supernatant decanted and replaced with fresh culture broth containing 15% glycerol. The culture can then be aliquoted into 1 mL cryotubes, sealed, and placed at −80° C. for long-term viability retention. This procedure achieves acceptable viability upon recovery from frozen storage.
(109) Organism production may be conducted using similar culture steps to banking, including medium composition and culture conditions. It may be conducted at larger scales of operation, especially for clinical development or commercial production. At larger scales, there may be several subcultivations of the bacterial composition prior to the final cultivation. At the end of cultivation, the culture is harvested to enable further formulation into a dosage form for administration. This can involve concentration, removal of undesirable medium components, and/or introduction into a chemical milieu that preserves the bacterial composition and renders it acceptable for administration via the chosen route. For example, a bacterial composition may be cultivated to a concentration of 10.sup.10 CFU/mL, then concentrated 20-fold by tangential flow microfiltration; the spent medium may be exchanged by diafiltering with a preservative medium consisting of 2% gelatin, 100 mM trehalose, and 10 mM sodium phosphate buffer. The suspension can then be freeze-dried to a powder and titrated.
(110) After drying, the powder may be blended to an appropriate potency, and mixed with other cultures and/or a filler such as microcrystalline cellulose for consistency and ease of handling, and the bacterial composition formulated as provided herein.
(111) Formulations
(112) Provided are formulations for administration to humans and other subjects in need thereof. Generally the bacterial compositions are combined with additional active and/or inactive materials to produce a final product, which may be in single dosage unit or in a multi-dose format.
(113) In some embodiments the composition comprises at least one carbohydrate. A “carbohydrate” refers to a sugar or polymer of sugars. The terms “saccharide,” “polysaccharide,” “carbohydrate,” and “oligosaccharide” may be used interchangeably. Most carbohydrates are aldehydes or ketones with many hydroxyl groups, usually one on each carbon atom of the molecule. Carbohydrates generally have the molecular formula C.sub.nH.sub.2nO.sub.n. A carbohydrate may be a monosaccharide, a disaccharide, trisaccharide, oligosaccharide, or polysaccharide. The most basic carbohydrate is a monosaccharide, such as glucose, sucrose, galactose, mannose, ribose, arabinose, xylose, and fructose. Disaccharides are two joined monosaccharides. Exemplary disaccharides include sucrose, maltose, cellobiose, and lactose. Typically, an oligosaccharide includes between three and six monosaccharide units (e.g., raffinose, stachyose), and polysaccharides include six or more monosaccharide units. Exemplary polysaccharides include starch, glycogen, and cellulose. Carbohydrates may contain modified saccharide units such as 2′-deoxyribose wherein a hydroxyl group is removed, 2′-fluororibose wherein a hydroxyl group is replace with a fluorine, or N-acetylglucosamine, a nitrogen-containing form of glucose (e.g., 2′-fluororibose, deoxyribose, and hexose). Carbohydrates may exist in many different forms, for example, conformers, cyclic forms, acyclic forms, stereoisomers, tautomers, anomers, and isomers.
(114) In some embodiments the composition comprises at least one lipid. As used herein a “lipid” includes fats, oils, triglycerides, cholesterol, phospholipids, fatty acids in any form including free fatty acids. Fats, oils and fatty acids can be saturated, unsaturated (cis or trans) or partially unsaturated (cis or trans). In some embodiments the lipid comprises at least one fatty acid selected from lauric acid (12:0), myristic acid (14:0), palmitic acid (16:0), palmitoleic acid (16:1), margaric acid (17:0), heptadecenoic acid (17:1), stearic acid (18:0), oleic acid (18:1), linoleic acid (18:2), linolenic acid (18:3), octadecatetraenoic acid (18:4), arachidic acid (20:0), eicosenoic acid (20:1), eicosadienoic acid (20:2),

eicosatetraenoic acid (20:4), eicosapentaenoic acid (20:5) (EPA), docosanoic acid (22:0), docosenoic acid (22:1), docosapentaenoic acid (22:5), docosahexaenoic acid (22:6) (DHA), and tetracosanoic acid (24:0). In some embodiments the composition comprises at least one modified lipid, for example a lipid that has been modified by cooking.

(115) In some embodiments the composition comprises at least one supplemental mineral or mineral source. Examples of minerals include, without limitation: chloride, sodium, calcium, iron, chromium, copper, iodine, zinc, magnesium, manganese, molybdenum, phosphorus, potassium, and selenium. Suitable forms of any of the foregoing minerals include soluble mineral salts, slightly soluble mineral salts, insoluble mineral salts, chelated minerals, mineral complexes, non-reactive minerals such as carbonyl minerals, and reduced minerals, and combinations thereof.

(116) In some embodiments the composition comprises at least one supplemental vitamin. The at least one vitamin can be fat-soluble or water soluble vitamins. Suitable vitamins include but are not limited to vitamin C, vitamin A, vitamin E, vitamin B12, vitamin K, riboflavin, niacin, vitamin D, vitamin B6, folic acid, pyridoxine, thiamine, pantothenic acid, and biotin. Suitable forms of any of the foregoing are salts of the vitamin, derivatives of the vitamin, compounds having the same or similar activity of the vitamin, and metabolites of the vitamin.

(117) In some embodiments the composition comprises an excipient. Non-limiting examples of suitable excipients include a buffering agent, a preservative, a stabilizer, a binder, a compaction agent, a lubricant, a dispersion enhancer, a disintegration agent, a flavoring agent, a sweetener, and a coloring agent.

(118) In some embodiments the excipient is a buffering agent. Non-limiting examples of suitable buffering agents include sodium citrate, magnesium carbonate, magnesium bicarbonate, calcium carbonate, and calcium bicarbonate.

(119) In some embodiments the excipient comprises a preservative. Non-limiting examples of suitable preservatives include antioxidants, such as alpha-tocopherol and ascorbate, and antimicrobials, such as parabens, chlorobutanol, and phenol.

(120) In some embodiments the composition comprises a binder as an excipient. Non-limiting examples of suitable binders include starches, pregelatinized starches, gelatin, polyvinylpyrolidone, cellulose, methylcellulose, sodium carboxymethylcellulose, ethylcellulose, polyacrylamides, polyvinyloxoazolidone, polyvinylalcohols, C.sub.12-C.sub.18 fatty acid alcohol, polyethylene glycol, polyols, saccharides, oligosaccharides, and combinations thereof.

(121) In some embodiments the composition comprises a lubricant as an excipient. Non-limiting examples of suitable lubricants include magnesium stearate, calcium stearate, zinc stearate, hydrogenated vegetable oils, sterotex, polyoxyethylene monostearate, talc, polyethyleneglycol, sodium benzoate, sodium lauryl sulfate, magnesium lauryl sulfate, and light mineral oil.

(122) In some embodiments the composition comprises a dispersion enhancer as an excipient. Non-limiting examples of suitable dispersants include starch, alginic acid, polyvinylpyrrolidones, guar gum, kaolin, bentonite, purified wood cellulose, sodium starch glycolate, isoamorphous silicate, and microcrystalline cellulose as high HLB emulsifier surfactants.

(123) In some embodiments the composition comprises a disintegrant as an excipient. In some embodiments the disintegrant is a non-effervescent disintegrant. Non-limiting examples of suitable non-effervescent disintegrants include starches such as corn starch, potato starch, pregelatinized and modified starches thereof, sweeteners, clays, such as bentonite, micro-crystalline cellulose, alginates, sodium starch glycolate, gums such as agar, guar, locust bean, karaya, pecitin, and tragacanth. In some embodiments the disintegrant is an effervescent disintegrant. Non-limiting examples of suitable effervescent disintegrants include sodium bicarbonate in combination with citric acid, and sodium bicarbonate in combination with tartaric acid.

(124) In some embodiments the excipient comprises a flavoring agent. Flavoring agents can be chosen from synthetic flavor oils and flavoring aromatics; natural oils; extracts from plants, leaves, flowers, and fruits; and combinations thereof. In some embodiments the flavoring agent is selected from cinnamon oils; oil of wintergreen; peppermint oils; clover oil; hay oil; anise oil; eucalyptus; vanilla; citrus oil such as lemon oil, orange oil, grape and grapefruit oil; and fruit essences including apple, peach, pear, strawberry, raspberry, cherry, plum, pineapple, and apricot.

(125) In some embodiments the excipient comprises a sweetener. Non-limiting examples of suitable sweeteners include glucose (corn syrup), dextrose, invert sugar, fructose, and mixtures thereof (when not used as a carrier); saccharin and its various salts such as the sodium salt; dipeptide sweeteners such as aspartame; dihydrochalcone compounds, glycyrrhizin; *Stevia rebaudiana* (Stevioside); chloro derivatives of sucrose such as sucralose; and sugar alcohols such as sorbitol, mannitol, sylitol, and the like. Also contemplated are hydrogenated starch hydrolysates and the synthetic sweetener 3,6-dihydro-6-methyl-1,2,3-oxathiazin-4-one-2,2-dioxide, particularly the potassium salt (acesulfame-K), and sodium and calcium salts thereof.

(126) In some embodiments the composition comprises a coloring agent. Non-limiting examples of suitable color agents include food, drug and cosmetic colors (FD&C), drug and cosmetic colors (D&C), and external drug and cosmetic colors (Ext. D&C). The coloring agents can be used as dyes or their corresponding lakes.

(127) The weight fraction of the excipient or combination of excipients in the formulation is usually about 99% or less, such as about 95% or less, about 90% or less, about 85% or less, about 80% or less, about 75% or less, about 70% or less, about 65% or less, about 60% or less, about 55% or less, 50% or less, about 45% or less, about 40% or less, about 35% or less, about 30% or less, about 25% or less, about 20% or less, about 15% or less, about 10% or less, about 5% or less, about 2% or less, or about 1% or less of the total weight of the composition.

(128) The bacterial compositions disclosed herein can be formulated into a variety of forms and administered by a number of different means. The compositions can be administered orally, rectally, or parenterally, in formulations containing conventionally acceptable carriers, adjuvants, and vehicles as desired. The term “parenteral” as used herein includes subcutaneous, intravenous, intramuscular, or intrasternal injection and infusion techniques. In an exemplary embodiment, the bacterial composition is administered orally.

(129) Solid dosage forms for oral administration include capsules, tablets, caplets, pills, troches, lozenges, powders, and granules. A capsule typically comprises a core material comprising a bacterial composition and a shell wall that encapsulates the core material. In some embodiments the core material comprises at least one of a solid, a liquid, and an emulsion. In some embodiments the shell wall material comprises at least one of a soft gelatin, a hard gelatin, and a polymer. Suitable polymers include, but are not limited to: cellulosic polymers such as hydroxypropyl cellulose, hydroxyethyl cellulose, hydroxypropyl methyl cellulose (HPMC), methyl cellulose, ethyl cellulose, cellulose acetate, cellulose acetate phthalate, cellulose acetate trimellitate, hydroxypropylmethyl cellulose phthalate, hydroxypropylmethyl cellulose succinate and carboxymethylcellulose sodium; acrylic acid polymers and copolymers, such as those formed from acrylic acid, methacrylic acid, methyl acrylate, ammonio methylacrylate, ethyl acrylate, methyl methacrylate and/or ethyl methacrylate (e.g., those copolymers sold under the trade name “Eudragit”); vinyl polymers and copolymers such as polyvinyl pyrrolidone, polyvinyl acetate, polyvinylacetate phthalate, vinylacetate crotonic acid copolymer, and ethylene-vinyl acetate copolymers; and shellac (purified lac). In some embodiments at least one polymer functions as taste-masking agents.

(130) Tablets, pills, and the like can be compressed, multiply compressed, multiply layered, and/or coated. The coating can be single or multiple. In one embodiment, the coating material comprises at least one of a saccharide, a polysaccharide, and glycoproteins extracted from at least one of a plant, a fungus, and a microbe. Non-limiting examples include corn starch, wheat starch, potato starch, tapioca starch, cellulose, hemicellulose, dextrans, maltodextrin, cyclodextrins, inulins, pectin, mannans, gum arabic, locust bean gum, mesquite gum, guar gum, gum karaya, gum ghatti, tragacanth gum, funori, carrageenans, agar, alginates, chitosans, or gellan gum. In some embodiments the coating material comprises a protein. In some embodiments the coating material comprises at least one of a fat and an oil. In some embodiments the at least one

of a fat and an oil is high temperature melting. In some embodiments the at least one of a fat and an oil is hydrogenated or partially hydrogenated. In some embodiments the at least one of a fat and an oil is derived from a plant. In some embodiments the at least one of a fat and an oil comprises at least one of glycerides, free fatty acids, and fatty acid esters. In some embodiments the coating material comprises at least one edible wax. The edible wax can be derived from animals, insects, or plants. Non-limiting examples include beeswax, lanolin, bayberry wax, carnauba wax, and rice bran wax. Tablets and pills can additionally be prepared with enteric coatings.

(131) Alternatively, powders or granules embodying the bacterial compositions disclosed herein can be incorporated into a food product. In some embodiments the food product is a drink for oral administration. Non-limiting examples of a suitable drink include fruit juice, a fruit drink, an artificially flavored drink, an artificially sweetened drink, a carbonated beverage, a sports drink, a liquid diary product, a shake, an alcoholic beverage, a caffeinated beverage, infant formula and so forth. Other suitable means for oral administration include aqueous and nonaqueous solutions, emulsions, suspensions and solutions and/or suspensions reconstituted from non-effervescent granules, containing at least one of suitable solvents, preservatives, emulsifying agents, suspending agents, diluents, sweeteners, coloring agents, and flavoring agents.

(132) In some embodiments the food product is a solid foodstuff. Suitable examples of a solid foodstuff include without limitation a food bar, a snack bar, a cookie, a brownie, a muffin, a cracker, an ice cream bar, a frozen yogurt bar, and the like.

(133) In some embodiments, the compositions disclosed herein are incorporated into a therapeutic food. In some embodiments, the therapeutic food is a ready-to-use food that optionally contains some or all essential macronutrients and micronutrients. In some embodiments, the compositions disclosed herein are incorporated into a supplementary food that is designed to be blended into an existing meal. In some embodiments, the supplemental food contains some or all essential macronutrients and micronutrients. In some embodiments, the bacterial compositions disclosed herein are blended with or added to an existing food to fortify the food's protein nutrition. Examples include food staples (grain, salt, sugar, cooking oil, margarine), beverages (coffee, tea, soda, beer, liquor, sports drinks), snacks, sweets and other foods.

(134) In one embodiment, the formulations are filled into gelatin capsules for oral administration. An example of an appropriate capsule is a 250 mg gelatin capsule containing from 10 (up to 100 mg) of lyophilized powder (10.sup.8 to 10.sup.11 bacteria), 160 mg microcrystalline cellulose, 77.5 mg gelatin, and 2.5 mg magnesium stearate. In an alternative embodiment, from 10.sup.5 to 10.sup.12 bacteria may be used, 10.sup.5 to 10.sup.7, 10.sup.6 to 10.sup.7, or 10.sup.8 to 10.sup.10, with attendant adjustments of the excipients if necessary. In an alternative embodiment an enteric-coated capsule or tablet or with a buffering or protective composition may be used.

(135) In one embodiment, the number of bacteria of each type may be present in the same amount or in different amounts. For example, in a bacterial composition with two types of bacteria, the bacteria may be present in from a 1:10,000 ratio to a 1:1 ratio, from a 1:10,000 ratio to a 1:1,000 ratio, from a 1:1,000 ratio to a 1:100 ratio, from a 1:100 ratio to a 1:50 ratio, from a 1:50 ratio to a 1:20 ratio, from a 1:20 ratio to a 1:10 ratio, from a 1:10 ratio to a 1:1 ratio. For bacterial compositions comprising at least three types of bacteria, the ratio of type of bacteria may be chosen pairwise from ratios for bacterial compositions with two types of bacteria. For example, in a bacterial composition comprising bacteria A, B, and C, at least one of the ratio between bacteria A and B, the ratio between bacteria B and C, and the ratio between bacteria A and C may be chosen, independently, from the pairwise combinations above.

(136) Methods of Treating a Subject

(137) In some embodiments the proteins and compositions disclosed herein are administered to a subject or a user (sometimes collectively referred to as a “subject”). As used herein “administer” and “administration” encompasses embodiments in which one person directs another to consume a bacterial composition in a certain manner and/or for a certain purpose, and also situations in which a user uses a bacteria composition in a certain manner and/or for a certain purpose independently of or in variance to any instructions received from a second person. Non-limiting examples of embodiments in which one person directs another to consume a bacterial composition in a certain manner and/or for a certain purpose include when a physician prescribes a course of conduct and/or treatment to a subject, when a parent commands a minor user (such as a child) to consume a bacterial composition, when a trainer advises a user (such as an athlete) to follow a particular course of conduct and/or treatment, and when a manufacturer, distributer, or marketer recommends conditions of use to an end user, for example through advertisements or labeling on packaging or on other materials provided in association with the sale or marketing of a product.

(138) The bacterial compositions offer a protective and/or therapeutic effect against infection by one or more GI pathogens of interest and thus may be administered after an acute case of infection has been resolved in order to prevent relapse, during an acute case of infection as a complement to antibiotic therapy if the bacterial composition is not sensitive to the same antibiotics as the GI pathogen, or to prevent infection or reduce transmission from disease carriers. These pathogens include, but are not limited to, *Aeromonas hydrophila, Campylobacter fetus, Plesiomonas shigelloides, Bacillus cereus, Campylobacter jejuni, Clostridium botulinum, Clostridium difficile, Clostridium perfringens*, enteroaggregative *Escherichia coli*, enterohemorrhagic *Escherichia coli*, enteroinvasive *Escherichia coli*, enterotoxigenic *Escherichia coli* (LT and/or ST), *Escherichia coli* 0157: H7, *Helicobacter pylori, Klebsiella pneumonia, Lysteria monocytogenes, Plesiomonas shigelloides, Salmonella* spp., *Salmonella typhi, Shigella* spp., *Staphylococcus*, *Staphylococcus aureus*, vancomycin-resistant *Enterococcus* spp., *Vibrio* spp., *Vibrio cholerae, Vibrio parahaemolyticus, Vibrio vulnificus*, and *Yersinia enterocolitica.*

(139) In one embodiment, the pathogen may be *Clostridium difficile, Salmonella* spp., pathogenic *Escherichia coli,* carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL) and vancomycin-resistant Enterococci (VRE). In yet another embodiment, the pathogen may be *Clostridium difficile.*

(140) The present bacterial compositions may be useful in a variety of clinical situations. For example, the bacterial compositions may be administered alone, as a complementary treatment to antibiotics (e.g., when a subject is suffering from an acute infection, to reduce the risk of recurrence after an acute infection has subsided or, or when a subject will be in close proximity to others with or at risk of serious gastrointestinal infections (physicians, nurses, hospital workers, family members of those who are ill or hospitalized).

(141) The present bacterial compositions may be administered to animals, including humans, laboratory animals (e.g., primates, rats, mice), livestock (e.g., cows, sheep, goats, pigs, turkeys, chickens), and household pets (e.g., dogs, cats, rodents).

(142) In the present method, the bacterial composition is administered enterically, in other words by a route of access to the gastrointestinal tract. This includes oral administration, rectal administration (including enema, suppository, or colonoscopy), by an oral or nasal tube (nasogastric, nasojejunal, oral gastric, or oral jejunal), as detailed more fully herein.

(143) It has been reported that a GI dysbiosis is associated with diabetes (Qin et al., 2012. Nature 490:55). In some embodiments, a composition provided herein can be used to alter the microbiota of a subject having or susceptible diabetes. Typically, such a composition provides at least one, two, or three OTUs identified in the art as associated with an improvement in insulin sensitivity or other sign or symptom associated with diabetes, e.g., Type 2 or Type 1 diabetes. In some embodiments, the composition is associated with an increase in engraftment and/or augmentation of at least one, two, or three OTUs associated with an improvement in at least one sign or symptom of diabetes.

(144) Pretreatment Protocols

(145) Prior to administration of the bacterial composition, the subject may optionally have a pretreatment protocol to prepare the gastrointestinal tract to receive the bacterial composition. In certain embodiments, the pretreatment protocol is advisable, such as when a subject has an acute infection with a highly resilient pathogen. In other embodiments, the pretreatment protocol is entirely optional, such as when the pathogen causing the infection is not resilient, or the subject has had an acute infection that has been successfully treated but where the physician is concerned that the infection may recur. In these instances, the pretreatment protocol may enhance the ability of the bacterial composition to

affect the subject's microbiome.
(146) As one way of preparing the subject for administration of the microbial ecosystem, at least one antibiotic may be administered to alter the bacteria in the subject. As another way of preparing the subject for administration of the microbial ecosystem, a standard colon-cleansing preparation may be administered to the subject to substantially empty the contents of the colon, such as used to prepare a subject for a colonscopy. By "substantially emptying the contents of the colon," this application means removing at least 75%, at least 80%, at least 90%, at least 95%, or about 100% of the contents of the ordinary volume of colon contents. Antibiotic treatment may precede the colon-cleansing protocol.
(147) If a subject has received an antibiotic for treatment of an infection, or if a subject has received an antibiotic as part of a specific pretreatment protocol, in one embodiment the antibiotic should be stopped in sufficient time to allow the antibiotic to be substantially reduced in concentration in the gut before the bacterial composition is administered. In one embodiment, the antibiotic may be discontinued 1, 2, or 3 days before the administration of the bacterial composition. In one embodiment, the antibiotic may be discontinued 3, 4, 5, 6, or 7 antibiotic half-lives before administration of the bacterial composition. In another embodiment, the antibiotic may be chosen so the constituents in the bacterial composition have an MIC50 that is higher than the concentration of the antibiotic in the gut.
(148) MIC50 of a bacterial composition or the elements in the composition may be determined by methods well known in the art. Reller et al., Antimicrobial Susceptibility Testing: A Review of General Principles and Contemporary Practices, Clinical Infectious Diseases 49 (11): 1749-1755 (2009). In such an embodiment, the additional time between antibiotic administration and administration of the bacterial composition is not necessary. If the pretreatment protocol is part of treatment of an acute infection, the antibiotic may be chosen so that the infection is sensitive to the antibiotic, but the constituents in the bacterial composition are not sensitive to the antibiotic.
(149) Routes of Administration
(150) The bacterial compositions of the invention are suitable for administration to mammals and non-mammalian animals in need thereof. In certain embodiments, the mammalian subject is a human subject who has one or more symptoms of a dysbiosis.
(151) When the mammalian subject is suffering from a disease, disorder or condition characterized by an aberrant microbiota, the bacterial compositions described herein are suitable for treatment thereof. In some embodiments, the mammalian subject has not received antibiotics in advance of treatment with the bacterial compositions. For example, the mammalian subject has not been administered at least two doses of vancomycin, metronidazole and/or or similar antibiotic compound within one week prior to administration of the therapeutic composition. In other embodiments, the mammalian subject has not previously received an antibiotic compound in the one month prior to administration of the therapeutic composition. In other embodiments, the mammalian subject has received one or more treatments with one or more different antibiotic compounds and such treatment(s) resulted in no improvement or a worsening of symptoms.
(152) In some embodiments, the gastrointestinal disease, disorder or condition is diarrhea caused by *C. difficile* including recurrent *C. difficile* infection, ulcerative colitis, colitis, Crohn's disease, or irritable bowel disease. Beneficially, the therapeutic composition is administered only once prior to improvement of the disease, disorder or condition. In some embodiments the therapeutic composition is administered at intervals greater than two days, such as once every three, four, five or six days, or every week or less frequently than every week. Or the preparation may be administered intermittently according to a set schedule, e.g., once a day, once weekly, or once monthly, or when the subject relapses from the primary illness. In another embodiment, the preparation may be administered on a long-term basis to subjects who are at risk for infection with or who may be carriers of these pathogens, including subjects who will have an invasive medical procedure (such as surgery), who will be hospitalized, who live in a long-term care or rehabilitation facility, who are exposed to pathogens by virtue of their profession (livestock and animal processing workers), or who could be carriers of pathogens (including hospital workers such as physicians, nurses, and other health care professionals).
(153) In embodiments, the bacterial composition is administered enterically. This preferentially includes oral administration, or by an oral or nasal tube (including nasogastric, nasojejunal, oral gastric, or oral jejunal). In other embodiments, administration includes rectal administration (including enema, suppository, or colonoscopy). The bacterial composition may be administered to at least one region of the gastrointestinal tract, including the mouth, esophagus, stomach, small intestine, large intestine, and rectum. In some embodiments it is administered to all regions of the gastrointestinal tract. The bacterial compositions may be administered orally in the form of medicaments such as powders, capsules, tablets, gels or liquids. The bacterial compositions may also be administered in gel or liquid form by the oral route or through a nasogastric tube, or by the rectal route in a gel or liquid form, by enema or instillation through a colonoscope or by a suppository.
(154) If the composition is administered colonoscopically and, optionally, if the bacterial composition is administered by other rectal routes (such as an enema or suppository) or even if the subject has an oral administration, the subject may have a colon-cleansing preparation. The colon-cleansing preparation can facilitate proper use of the colonoscope or other administration devices, but even when it does not serve a mechanical purpose it can also maximize the proportion of the bacterial composition relative to the other organisms previously residing in the gastrointestinal tract of the subject. Any ordinarily acceptable colon-cleansing preparation may be used such as those typically provided when a subject undergoes a colonoscopy.
(155) Dosages and Schedule for Administration
(156) In some embodiments the bacteria and bacterial compositions are provided in a dosage form. In some embodiments the dosage form is designed for administration of at least one OTU or combination thereof disclosed herein, wherein the total amount of bacterial composition administered is selected from 0.1 ng to 10 g, 10 ng to 1 g, 100 ng to 0.1 g, 0.1 mg to 500 mg, 1 mg to 100 mg, or from 10-15 mg. In some embodiments the bacterial composition is consumed at a rate of from 0.1 ng to 10 g a day, 10 ng to 1 g a day, 100 ng to 0.1 g a day, 0.1 mg to 500 mg a day, 1 mg to 100 mg a day, or from 10-15 mg a day, or more.
(157) In some embodiments the treatment period is at least 1 day, at least 2 days, at least 3 days, at least 4 days, at least 5 days, at least 6 days, at least 1 week, at least 2 weeks, at least 3 weeks, at least 4 weeks, at least 1 month, at least 2 months, at least 3 months, at least 4 months, at least 5 months, at least 6 months, or at least 1 year. In some embodiments the treatment period is from 1 day to 1 week, from 1 week to 4 weeks, from 1 month, to 3 months, from 3 months to 6 months, from 6 months to 1 year, or for over a year.
(158) In one embodiment, from 10.sup.5 and 10.sup.12 microorganisms total may be administered to the subject in a given dosage form. In one mode, an effective amount may be provided in from 1 to 500 ml or from 1 to 500 grams of the bacterial composition having from 10.sup.7 to 10.sup.11 bacteria per ml or per gram, or a capsule, tablet or suppository having from 1 mg to 1000 mg lyophilized powder having from 10.sup.7 to 10.sup.11 bacteria. Those receiving acute treatment may receive higher doses than those who are receiving chronic administration (such as hospital workers or those admitted into long-term care facilities).
(159) Any of the preparations described herein may be administered once on a single occasion or on multiple occasions, such as once a day for several days or more than once a day on the day of administration (including twice daily, three times daily, or up to five times daily). Or the preparation may be administered intermittently according to a set schedule, e.g., once weekly, once monthly, or when the subject relapses from the primary illness. In another embodiment, the preparation may be administered on a long-term basis to individuals who are at risk for infection with or who may be carriers of these pathogens, including individuals who will have an invasive medical procedure (such as surgery), who will be hospitalized, who live in a long-term care or rehabilitation facility, who are exposed to pathogens by virtue of their profession (livestock and animal processing workers), or who could be carriers of pathogens (including hospital workers such as physicians, nurses, and other health care

professionals).
(160) Subject Selection
(161) Particular bacterial compositions may be selected for individual subjects or for subjects with particular profiles. For example, 16S sequencing may be performed for a given subject to identify the bacteria present in his or her microbiota. The sequencing may either profile the subject's entire microbiome using 16S sequencing (to the family, genera, or species level), a portion of the subject's microbiome using 16S sequencing, or it may be used to detect the presence or absence of specific candidate bacteria that are biomarkers for health or a particular disease state, such as markers of multi-drug resistant organisms or specific genera of concern such as *Escherichia*. Based on the biomarker data, a particular composition may be selected for administration to a subject to supplement or complement a subject's microbiota in order to restore health or treat or prevent disease. In another embodiment, subjects may be screened to determine the composition of their microbiota to determine the likelihood of successful treatment.
(162) Combination Therapy
(163) The bacterial compositions may be administered with other agents in a combination therapy mode, including anti-microbial agents and prebiotics. Administration may be sequential, over a period of hours or days, or simultaneous.
(164) In one embodiment, the bacterial compositions are included in combination therapy with one or more anti-microbial agents, which include anti-bacterial agents, anti-fungal agents, anti-viral agents and anti-parasitic agents.
(165) Anti-bacterial agents include cephalosporin antibiotics (cephalexin, cefuroxime, cefadroxil, cefazolin, cephalothin, cefaclor, cefamandole, cefoxitin, cefprozil, and ceftobiprole); fluoroquinolone antibiotics (cipro, Levaquin, floxin, tequin, avelox, and norflox); tetracycline antibiotics (tetracycline, minocycline, oxytetracycline, and doxycycline); penicillin antibiotics (amoxicillin, ampicillin, penicillin V, dicloxacillin, carbenicillin, vancomycin, and methicillin); and carbapenem antibiotics (ertapenem, doripenem, imipenem/cilastatin, and meropenem).
(166) Anti-viral agents include Abacavir, Acyclovir, Adefovir, Amprenavir, Atazanavir, Cidofovir, Darunavir, Delavirdine, Didanosine, Docosanol, Efavirenz, Elvitegravir, Emtricitabine, Enfuvirtide, Etravirine, Famciclovir, Foscarnet, Fomivirsen, Ganciclovir, Indinavir, Idoxuridine, Lamivudine, Lopinavir Maraviroc, MK-2048, Nelfinavir, Nevirapine, Penciclovir, Raltegravir, Rilpivirine, Ritonavir, Saquinavir, Stavudine, Tenofovir Trifluridine, Valaciclovir, Valganciclovir, Vidarabine, Ibacitabine, Amantadine, Oseltamivir, Rimantidine, Tipranavir, Zalcitabine, Zanamivir and Zidovudine.
(167) Examples of antifungal compounds include, but are not limited to polyene antifungals such as natamycin, rimocidin, filipin, nystatin, amphotericin B, candicin, and hamycin; imidazole antifungals such as miconazole, ketoconazole, clotrimazole, econazole, omoconazole, bifonazole, butoconazole, fenticonazole, isoconazole, oxiconazole, sertaconazole, sulconazole, and tioconazole; triazole antifungals such as fluconazole, itraconazole, isavuconazole, ravuconazole, posaconazole, voriconazole, terconazole, and albaconazole; thiazole antifungals such as abafungin; allylamine antifungals such as terbinafine, naftifine, and butenafine; and echinocandin antifungals such as anidulafungin, caspofungin, and micafungin. Other compounds that have antifungal properties include, but are not limited to polygodial, benzoic acid, ciclopirox, tolnaftate, undecylenic acid, flucytosine or 5-fluorocytosine, griseofulvin, and haloprogin.
(168) In one embodiment, the bacterial compositions are included in combination therapy with one or more corticosteroids, mesalazine, mesalamine, sulfasalazine, sulfasalazine derivatives, immunosuppressive drugs, cyclosporin A, mercaptopurine, azathiopurine, prednisone, methotrexate, antihistamines, glucocorticoids, epinephrine, theophylline, cromolyn sodium, anti-leukotrienes, anti-cholinergic drugs for rhinitis, anti-cholinergic decongestants, mast-cell stabilizers, monoclonal anti-IgE antibodies, vaccines, and combinations thereof.
(169) A prebiotic is a selectively fermented ingredient that allows specific changes, both in the composition and/or activity in the gastrointestinal microbiota that confers benefits upon host well being and health. Prebiotics may include complex carbohydrates, amino acids, peptides, or other essential nutritional components for the survival of the bacterial composition. Prebiotics include, but are not limited to, amino acids, biotin, fructooligosaccharide, galactooligosaccharides, inulin, lactulose, mannan oligosaccharides, oligofructose-enriched inulin, oligofructose, oligodextrose, tagatose, trans-galactooligosaccharide, and xylooligosaccharides.
(170) Methods for Characterization of Bacterial Compositions
(171) In certain embodiments, provided are methods for testing certain characteristics of bacterial compositions. For example, the sensitivity of bacterial compositions to certain environmental variables is determined, e.g., in order to select for particular desirable characteristics in a given composition, formulation and/or use. For example, the constituents in the bacterial composition may be tested for pH resistance, bile acid resistance, and/or antibiotic sensitivity, either individually on a constituent-by-constituent basis or collectively as a bacterial composition comprised of multiple bacterial constituents (collectively referred to in this section as bacterial composition).
(172) pH Sensitivity Testing. If a bacterial composition will be administered other than to the colon or rectum (i.e., through, for example, but not limited to, an oral route), optionally testing for pH resistance enhances the selection of bacterial compositions that will survive at the highest yield possible through the varying pH environments of the distinct regions of the GI tract. Understanding how the bacterial compositions react to the pH of the GI tract also assists in formulation, so that the number of bacteria in a dosage form can be increased if beneficial and/or so that the composition may be administered in an enteric-coated capsule or tablet or with a buffering or protective composition. As the pH of the stomach can drop to a pH of 1 to 2 after a high-protein meal for a short time before physiological mechanisms adjust it to a pH of 3 to 4 and often resides at a resting pH of 4 to 5, and as the pH of the small intestine can range from a pH of 6 to 7.4, bacterial compositions can be prepared that survive these varying pH ranges (specifically wherein at least 1%, 5%, 10%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or as much as 100% of the bacteria can survive gut transit times through various pH ranges). This may be tested by exposing the bacterial composition to varying pH ranges for the expected gut transit times through those pH ranges. Therefore, as a nonlimiting example only, 18-hour cultures of bacterial compositions may be grown in standard media, such as gut microbiota medium (“GMM”, see Goodman et al., Extensive personal human gut microbiota culture collections characterized and manipulated in gnotobiotic mice, PNAS 108 (15): 6252-6257 (2011)) or another animal-products-free medium, with the addition of pH adjusting agents for a pH of 1 to 2 for 30 minutes, a pH of 3 to 4 for 1 hour, a pH of 4 to 5 for 1 to 2 hours, and a pH of 6 to 7.4 for 2.5 to 3 hours. An alternative method for testing stability to acid is described in U.S. Pat. No. 4,839,281. Survival of bacteria may be determined by culturing the bacteria and counting colonies on appropriate selective or non-selective media.
(173) Bile Acid Sensitivity Testing. Additionally, in some embodiments, testing for bile-acid resistance enhances the selection of bacterial compositions that will survive exposures to bile acid during transit through the GI tract. Bile acids are secreted into the small intestine and can, like pH, affect the survival of bacterial compositions. This may be tested by exposing the bacterial compositions to bile acids for the expected gut exposure time to bile acids. For example, bile acid solutions may be prepared at desired concentrations using 0.05 mM Tris at pH 9 as the solvent. After the bile acid is dissolved, the pH of the solution may be adjusted to 7.2 with 10% HCl. Bacterial compositions may be cultured in 2.2 ml of a bile acid composition mimicking the concentration and type of bile acids in the subject, 1.0 ml of 10% sterile-filtered stool media and 0.1 ml of an 18-hour culture of the given strain of bacteria. Incubations may be conducted for from 2.5 to 3 hours or longer. An alternative method for testing stability to bile acid is described in U.S. Pat. No. 4,839,281. Survival of bacteria may be determined by culturing the bacteria and counting colonies on appropriate selective or non-selective media.
(174) Antibiotic Sensitivity Testing. As a further optional sensitivity test, bacterial compositions may be tested for sensitivity to antibiotics. In one embodiment, bacterial compositions may be chosen so that the bacterial constituents are sensitive to antibiotics such that if necessary they can be eliminated or substantially reduced from the subject's gastrointestinal tract by at least one antibiotic targeting the bacterial composition.

(175) Adherence to Gastrointestinal Cells. The bacterial compositions may optionally be tested for the ability to adhere to gastrointestinal cells. A method for testing adherence to gastrointestinal cells is described in U.S. Pat. No. 4,839,281.

(176) The specification is most thoroughly understood in light of the teachings of the references cited within the specification. The embodiments within the specification provide an illustration of embodiments and should not be construed to limit the scope. The skilled artisan readily recognizes that many other embodiments are encompassed. All publications and patents cited in this disclosure are incorporated by reference in their entirety. To the extent the material incorporated by reference contradicts or is inconsistent with this specification, the specification will supersede any such material. The citation of any references herein is not an admission that such references are prior art.

(177) Unless otherwise indicated, all numbers expressing quantities of ingredients, reaction conditions, and so forth used in the specification, including claims, are to be understood as being modified in all instances by the term "about." Accordingly, unless otherwise indicated to the contrary, the numerical parameters are approximations and may vary depending upon the desired properties sought to be obtained. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should be construed in light of the number of significant digits and ordinary rounding approaches.

(178) Unless otherwise indicated, the term "at least" preceding a series of elements is to be understood to refer to every element in the series.

EXAMPLES

(179) Below are examples of specific embodiments for carrying out the present invention. The examples are offered for illustrative purposes only, and are not intended to limit the scope of the present invention in any way. Efforts have been made to ensure accuracy with respect to numbers used (e.g., amounts, temperatures, etc.), but some experimental error and deviation should, of course, be allowed for. Examples of the techniques and protocols described herein with regard to therapeutic compositions can be found in, e.g., Remington's Pharmaceutical Sciences, 16th edition, Osol, A. (ed), 1980.

Example 1. Construction of Binary Pairs in a High-Throughput 96-Well Format/Plate Preparation

(180) Pairs of bacteria were used to identify binary pairs useful for inhibition of *C. difficile*. To prepare plates for the high-throughput screening assay of binary pairs, vials of −80° C. glycerol stock bacterial banks were thawed and diluted to 1e8 CFU/mL. Each bacterial strain was then diluted 10× (to a final concentration of 1e7 CFU/mL of each strain) into 200 uL of PBS+15% glycerol in the wells of a 96-well plate. Plates were then frozen at −80° C. When needed, plates were removed from −80° C. and thawed at room temperature under anaerobic conditions for testing in a CivSim assay with *C. difficile*.

Example 2. Construction of Ternary Combinations in a High-Throughput 96-Well Format

(181) Triplet combinations of bacteria were used to identify ternary combinations useful for inhibition of *C. difficile*. To prepare plates for high-throughput screening of ternary combinations, vials of −80° C. glycerol bacterial stock banks were thawed and diluted to 1e8 CFU/mL. Each bacterial strain was then diluted 10× (to a final concentration of 1e7 CFU/mL of each strain) into 200 uL of PBS+15% glycerol in the wells of a 96-well plate. Plates were then frozen at −80° C. When needed for the assay, plates were removed from −80° C. and thawed at room temperature under anaerobic conditions when testing in a CivSim assay with *Clostridium difficile*.

Example 3. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of *Clostridium difficile*

(182) A competition assay (CivSim assay) was used to identify compositions that can inhibit the growth of *C. difficile*. Briefly, an overnight culture of *C. difficile* was grown under anaerobic conditions in SweetB-FosIn for the growth of *C. difficile*. In some cases, other suitable media can be used. SweetB-FosIn is a version of BHI (Remel R452472) supplemented with several components as follows: Components per liter: 37 g BHI powder (Remel R452472), supplemented with 5 g yeast extract UF (Difco 210929), 1 g cysteine-HCl (Spectrum C1473), 1 g cellobiose (Sigma C7252), 1 g maltose (Spectrum MA155), 1.5 ml hemin solution, 1 g soluble starch (Sigma-Aldrich S9765), 1 g fructooligosaccharides/inulin (Jarrow Formulas 103025) and 50 mL 1 M MOPS/KOH PH 7. To prepare the hemin solution, hemin (Sigma 51280) was dissolved in 0.1 M NaOH to make a 10 mg/mL stock.

(183) After 24 hours of growth the culture was diluted 100,000 fold into a complex medium SweetB-FosIn. In some embodiments a medium is selected for use in which all desired organisms can grow, i.e., which is suitable for the growth of a wide variety of anaerobic, and, in some cases facultative anaerobic bacterial species. The diluted *C. difficile* mixture was then aliquoted to wells of a 96-well plate (180 uL to each well). 20 uL of a bacterial composition was then added to each well at a final concentration of 1e6 CFU/mL of each or two or three species. Alternatively the assay can be tested with binary pairs at different initial concentrations (1e9 CFU/mL, 1e8 CFU/mL, 1e7 CFU/mL, 1e5 CFU/mL, 1e4 CFU/mL, 1e3 CFU/mL, 1e2 CFU/mL). Control wells only inoculated with *C. difficile* were included for a comparison to the growth of *C. difficile* without inhibition. Additional wells were used for controls that either inhibit or do not inhibit the growth of *C. difficile*. One example of a positive control that inhibits growth was a combination of *Blautia producta*, *Clostridium bifermentans* and *Escherichia coli*. One example of a control that shows reduced inhibition of *C. difficile* growth as a combination of *Bacteroides thetaiotaomicron*, *Bacteroides ovatus* and *Bacteroides vulgatus*. Plates were wrapped with parafilm and incubated for 24 hours at 37° C. under anaerobic conditions. After 24 hours, the wells containing *C. difficile* alone were serially diluted and plated to determine titer. The 96-well plate was then frozen at −80 C before quantifying *C. difficile* by qPCR assay (see Example 6). Experimental combinations that inhibit *C. difficile* in this assay are useful in compositions for prevention or treatment of *C. difficile* infection.

Example 4. Construction of a CivSim Assay to Screen for Bacterial Compositions that Produce Diffusible Products Inhibitory to the Growth of *Clostridium difficile* Using a Filter Insert

(184) To identify bacterial compositions that can produce diffusible products that inhibit *C. difficile* a modified CivSim assay was designed. In this experiment, the CivSim assay described above was modified by using a 0.22 uM filter insert (Millipore™ MultiScreen™ 96-Well Assay Plates—Item MAGVS2210) in 96-well format to physically separate *C. difficile* from the bacterial compositions. The *C. difficile* was aliquoted into the 96-well plate while the bacterial compositions were aliquoted into media on the filter overlay. The nutrient/growth medium is in contact on both sides of the 0.22 uM filter, allowing exchange of nutrients, small molecules and many macromolecules (e.g., bacteriocins, cell-surface proteins, or polysaccharides) by diffusion. In this embodiment, after a 24 hour incubation, the filter insert containing the bacterial compositions was removed. The plate containing *C. difficile* was then transferred to a 96-well plate reader suitable for measuring optical density (OD) at 600 nm. The growth of *C. difficile* in the presence of different bacterial compositions was compared based on the OD measurement. The results of these experiments demonstrated that compositions that can inhibit *C. difficile* when grown in shared medium under conditions that do not permit contact between the bacteria in the composition and *C. difficile* can be identified. Such compositions are candidates for producing diffusible products that are effective for treating *C. difficile* infection and can serve as part of a process for isolating such diffusible products, e.g., for use in treating infection.

Example 5. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of *Clostridium difficile* Using *Clostridium difficile* Selective Media for Quantification

(185) The CivSim assay described above can be modified to determine final *C. difficile* titer by serially diluting and plating to *C. difficile* selective media (Bloedt et al. 2009) such as CCFA (cycloserine cefoxitin fructose agar, Anaerobe Systems), CDSA (*Clostridium difficile* selective agar, which is cycloserine cefoxitin mannitol agar, Becton Dickinson).

Example 6. Quantification of *C. difficile* Using Quantitative PCR (qPCR)

(186) A. Standard Curve Preparation

(187) To quantitate *C. difficile*, a standard curve was generated from a well on each assay plate in, e.g., a CivSim assay, containing only pathogenic *C. difficile* grown in SweetB+FosIn media as provided herein and quantified by selective spot plating. Serial dilutions of the culture were performed in sterile phosphate-buffered saline. Genomic DNA was extracted from the standard curve samples along with the other wells.

(188) B. Genomic DNA Extraction

(189) Genomic DNA was extracted from 5 μl of each sample using a dilution, freeze/thaw, and heat lysis protocol. 5 μL of thawed samples were added to 45 μL of UltraPure water (Life Technologies, Carlsbad, CA) and mixed by pipetting. The plates with diluted samples were frozen at −20° C. until use for qPCR which included a heated lysis step prior to amplification. Alternatively the genomic DNA could be isolated using the Mo Bio Powersoil®-htp 96 Well Soil DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA), Mo Bio Powersoil® DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA), or the QIAamp DNA Stool Mini Kit (QIAGEN, Valencia, CA) according to the manufacturer's instructions.

(190) C. qPCR Composition and Conditions

(191) The qPCR reaction mixture contained 1×SsoAdvanced Universal Probes Supermix, 900 nM of Wr-tcdB-F primer (AGCAGTTGAATATAGTGGTTTAGTTAGAGTTG (SEQ ID NO: 2033), IDT, Coralville, IA), 900 nM of Wr-tcdB-R primer (CATGCTTTTTTAGTTTCTGGATTGAA (SEQ ID NO: 2034), IDT, Coralville, IA), 250 nM of Wr-tcdB-P probe (6FAM-CATCCAGTCTCAATTGTATATGTTTCTCCA (SEQ ID NO: 2035)—MGB, Life Technologies, Grand Island, NY), and Molecular Biology Grade Water (Mo Bio Laboratories, Carlsbad, CA) to 18 μl (Primers adapted from: Wroblewski, D. et al., Rapid Molecular Characterization of *Clostridium difficile* and Assessment of Populations of *C. difficile* in Stool Specimens, Journal of Clinical Microbiology 47:2142-2148 (2009)). This reaction mixture was aliquoted to wells of a Hard-shell Low-Profile Thin Wall 96-well Skirted PCR Plate (BioRad, Hercules, CA). To this reaction mixture, 2 μl of diluted, frozen, and thawed samples were added and the plate sealed with a Microseal ‘B’ Adhesive Seal (BioRad, Hercules, CA). The qPCR was performed on a BioRad C1000™ Thermal Cycler equipped with a CFX96™ Real-Time System (BioRad, Hercules, CA). The thermocycling conditions were 95° C. for 15 minutes followed by 45 cycles of 95° C. for 5 seconds, 60° C. for 30 seconds, and fluorescent readings of the FAM channel. Alternatively, the qPCR could be performed with other standard methods known to those skilled in the art.

(192) D. Data Analysis

(193) The Cq value for each well on the FAM channel was determined by the CFX Manager™ 3.0 software. The log.sub.10 (cfu/mL) of *C. difficile* each experimental sample was calculated by inputting a given sample's Cq value into a linear regression model generated from the standard curve comparing the Cq values of the standard curve wells to the known log.sub.10 (cfu/mL) of those samples. The log inhibition was calculated for each sample by subtracting the log.sub.10 (cfu/mL) of *C. difficile* in the sample from the log.sub.10 (cfu/mL) of *C. difficile* in the sample on each assay plate used for the generation of the standard curve that has no additional bacteria added. The mean log inhibition was calculated for all replicates for each composition.

(194) A histogram of the range and standard deviation of each composition was plotted. Ranges or standard deviations of the log inhibitions that were distinct from the overall distribution were examined as possible outliers. If the removal of a single log inhibition datum from one of the binary pairs that were identified in the histograms would bring the range or standard deviation in line with those from the majority of the samples, that datum was removed as an outlier, and the mean log inhibition was recalculated.

(195) The pooled variance of all samples evaluated in the assay was estimated as the average of the sample variances weighted by the sample's degrees of freedom. The pooled standard error was then calculated as the square root of the pooled variance divided by the square root of the number of samples. Confidence intervals for the null hypothesis were determined by multiplying the pooled standard error to the z score corresponding to a given percentage threshold. Mean log inhibitions outside the confidence interval were considered to be inhibitory if positive or stimulatory if negative with the percent confidence corresponding to the interval used. Samples with mean log inhibition greater than the 99% confidence interval (C.I) of the null hypothesis are reported as ++++, those with a 95%<C.I.<99% as +++, those with a 90%<C.I.<95% as ++, those with a 80%<C.I.<90% as + while samples with mean log inhibition less than the 99% confidence interval (C.I) of the null hypothesis are reported as −−−, those with a 95%<C.I.<99% as −−−, those with a 90%<C.I.<95% as −−, those with a 80%<C.I.<90% as −.

Example 7. Inhibition of *C. difficile* Growth by Bacterial Compositions

(196) Using methods described herein, binary pairs were identified that can inhibit *C. difficile* (see Table 4). 622 of 989 combinations showed inhibition with a confidence interval >80%; 545 of 989 with a C.I. >90%; 507 of 989 with a C.I. >95%; 430 of 989 with a C.I. of >99%. Non-limiting but exemplary binary pairs include those with mean log reduction greater than 0.366, e.g., *Allistipes shahii* paired with *Blautia producta, Clostridium hathaweyi*, or *Collinsella aerofaciens*, or *Clostridium mayombei* paired with *C. innocuum, C. tertium, Collinsella aerofaciens*, or any of the other 424 combinations shown in Table 4. Equally important, the CivSim assay describes binary pairs that do not effectively inhibit *C. difficile.* 188 of 989 combinations promote growth with >80% confidence; 52 of 989 show a lack of inhibition with >90% confidence; 22 of 989 show a lack of inhibition with >95% confidence; 3 of 989, including *B. producta* combined with *Coprococcus catus*, *Alistipes shahii* combined with *Dorea formicigenerans*, and *Eubacterium rectale* combined with *Roseburia intestinalis*, show a lack of inhibition with >99% confidence. 249 of 989 combinations are neutral in the assay, meaning they neither promote nor inhibit *C. difficile* growth to the limit of measurement.

(197) Ternary combinations with mean log inhibition greater than the 99% confidence interval (C.I) of the null hypothesis are reported as ++++, those with a 95%<C.I.<99% as +++, those with a 90%<C.I.<95% as ++, those with a 80%<C.I.<90% as + while samples with mean log inhibition less than the 99% confidence interval (C.I) of the null hypothesis are reported as −−−, those with a 95%<C.I.<99% as −−−, those with a 90%<C.I.<95% as −−, those with a 80%<C.I.<90% as −.

(198) The CivSim assay results demonstrate that many ternary combinations can inhibit *C. difficile* (Table 4). 516 of 632 ternary combinations show inhibition with a confidence interval >80%; 507 of 632 with a C.I. >90%; 496 of 632 with a C.I. >95%; 469 of 632 with a C.I. of >99%. Non-limiting but exemplary ternary combinations include those with a score of ++++, such as *Colinsella aerofaciens*, *Coprococcus comes*, and *Blautia producta*. The CivSim assay also describes ternary combinations that do not effectively inhibit *C. difficile.* 76 of 632 combinations promote growth with >80% confidence; 67 of 632 promote growth with >90% confidence; 61 of 632, promote growth with >95% confidence; and 49 of 632 combinations such as, but not limited to, *Clostridium orbiscendens, Coprococcus comes*, and *Faecalibacterium prausnitzii* promote growth with >99% confidence. 40 of 632 combinations are neutral in the assay, meaning they neither promote nor inhibit *C. difficile* growth to the limit of confidence.

(199) Of the ternary combinations that inhibit *C. difficile* with >99% confidence, those that strongly inhibit *C. difficile* can be identified by comparing their mean log inhibition to the distribution of all results for all ternary combinations tested. Those above the 75th percentile can be considered to strongly inhibit *C. difficile*. Alternatively, those above the 50th, 60th, 70th, 80th, 90th, 95th, or 99th percentile can be considered to strongly inhibit *C. difficile*. Non-limiting but exemplary ternary combinations above the 75.sup.th percentile include *Blautia producta*, *Clostridium tertium*, and *Ruminococcus gnavus* and *Eubacterium rectale, Clostridium mayombei*, and *Ruminococcus bromii.*

(200) In addition to the demonstration that many binary and ternary combinations inhibit *C. difficile*, the CivSim demonstrates that many of these combinations synergistically inhibit *C. difficile*. Exemplary ternary combinations that demonstrate synergy in the inhibition of *C. difficile* growth include, but are not limited to, *Blautia producta, Clostridium innocuum*, *Clostridium orbiscendens* and *Colinsella aerofaciens*, *Blautia producta*, and *Eubacterium rectale*. Additional useful combinations are provided throughout, e.g., in Tables 4a, 4b, and 14-21.

(201) Two higher-order bacterial compositions were tested in the CivSim assay for inhibition of *C. difficile*. N1962 (a.k.a. S030 and N1952), a

15 member composition, inhibited *C. difficile* by an average of 2.73 log 10 CFU/mL with a standard deviation of 0.58 log 10 CFU/mL while N1984 (a.k.a. S075), a 9 member composition, inhibited *C. difficile* by an average of 1.42 log 10 CFU/mL with a standard deviation of 0.45 log 10 CFU/mL.

(202) These data collectively demonstrate that the CivSim assay can be used to identify compositions containing multiple species that are effective at inhibiting growth, that promote growth, or do not have an effect on growth of an organism, e.g., a pathogenic organism such as *C. difficile*.

Example 8. In Vivo Validation of Ternary Combinations' Efficacy in a Murine Model of *Clostridium difficile* Infection

(203) To test the therapeutic potential of a bacterial composition such as but not limited to a spore population, a prophylactic mouse model of *C. difficile* infection was used (model based on Chen et al. 2008. A mouse model of *Clostridium difficile*-associated disease. Gastroenterology 135:1984-1992). Briefly, two cages of five mice each were tested for each arm of the experiment. All mice received an antibiotic cocktail consisting of 10% glucose, kanamycin (0.5 mg/ml), gentamicin (0.044 mg/ml), colistin (1062.5 U/ml), metronidazole (0.269 mg/ml), ciprofloxacin (0.156 mg/ml), ampicillin (0.1 mg/ml) and vancomycin (0.056 mg/ml) in their drinking water on days −14 through −5 and a dose of 10 mg/kg clindamycin by oral gavage on day −3. On day −1, test compositions were spun for 5 minutes at 12,100 rcf, their supernatants' removed, and the remaining pellets were resuspended in sterile PBS, prereduced if bacterial composition was not in spore form, and delivered via oral gavage. On day 0 the mice were challenged by administration of approximately 4.5 log 10 cfu of *C. difficile* (ATCC 43255) or sterile PBS (for the naive arm) via oral gavage. Mortality, weight and clinical scoring of *C. difficile* symptoms based upon a 0-4 scale by combining scores for appearance (0-2 points based on normal, hunched, piloerection, or lethargic), and clinical signs (0-2 points based on normal, wet tail, cold-to-the-touch, or isolation from other animals), with a score of 4 in the case of death, were assessed every day from day −2 through day 6. Mean minimum weight relative to day −1 and mean maximum clinical score as well as average cumulative mortality were calculated. Reduced mortality, increased mean minimum weight relative to day −1, and reduced mean maximum clinical score with death assigned to a score of 4 relative to the vehicle control were used to assess the ability of the test composition to inhibit infection by *C. difficile*.

(204) Ternary combinations were tested in the murine model described above at 1e9 CFU/mL per strain. The results are shown in Table 5. The data demonstrate that the CivSim assay results are highly predictive of the ability of a combination to inhibit weight loss in *C. difficile* infection. Weight loss in this model is generally considered to be indicative of disease.

(205) In one embodiment, compositions to screen for efficacy in vivo can be selected by ranking the compositions based on a functional metric such as but not limited to in vitro growth inhibition scores; compositions that are ranked ≥ the 75th percentile can be considered to strongly inhibit growth and be selected for in vivo validation of the functional phenotype. In other embodiments, compositions above the 50th, 60th, 70th, 80th, 90th, 95th, or 99th percentile can be considered to be the optimal candidates. In another embodiment, combinations with mean log inhibition greater than the 99% confidence interval (C.I) of the null hypothesis are selected. In other embodiments, compositions greater than the 95%, 90%, 85%, or 80% confidence interval (C.I.) are selected. In another embodiment, compositions demonstrated to have synergistic inhibition are selected (see Example 7) for testing in an in vivo model such as that described above.

(206) Compositions selected to screen for efficacy in in vivo models can also be selected using a combination of growth inhibition metrics. In a non-limiting example: (i) compositions are selected based on their log inhibition being greater than the 99% confidence interval (C.I.) of the null hypothesis, (ii) the selected subset of compositions is further selected to represent those that are ranked ≥ the 75th percentile in the distribution of all inhibition scores, (iii) the subset of (ii) is then further selected based on compositions that demonstrate synergistic inhibition. In some embodiments, different confidence intervals (C.I.) and percentiles are used to create the composition subsets, e.g., see Table 4b.

(207) Of the twelve exemplary ternary combinations selected, all were demonstrated to inhibit *C. difficile* in the CivSim assay (see Example 6) with >99% confidence. Ten of the twelve compositions demonstrated a protective effect when compared to a vehicle control with respect to the Mean Minimum Relative Weight. All twelve compositions outperformed vehicle with respect to Mean Maximum Clinical Score while eleven of twelve compositions surpassed the vehicle control by Cumulative Mortality. A non-limiting, but exemplary ternary combination, *Collinsella aerofaciens*, *Clostridium buytricum*, and *Ruminococcus gnavus*, was protective against symptoms of *C. difficile* infection, producing a Mean Minimum Relative Weight of 0.96, a Mean Maximum Clinical Score of 0.2, and Cumulative Mortality of 0% compared to the vehicle control of 0.82, 2.6, and 30%, respectively. These results demonstrate that the in vitro CivSim assay can be used to identify compositions that are protective in an in vivo murine model. This is surprising given the inherent dynamic nature of in vivo biological systems and the inherent simplification of the in vitro assays; it would not be expected that there is a direct correlation of in vitro in in vivo measures of inhibition and efficacy. This is in part because of the complexity of the in vivo system into which a composition is administered for treatment in which it might have been expected that confounding factors would obscure or affect the ability of a composition deemed effective in vitro to be effective in vivo.

Example 9. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of Vancomycin-Resistant *Enterococcus* (VRE) Using Vancomycin-Resistant *Enterococcus* Selective Medium for Quantification and Composition Screening

(208) To determine the ability of a composition to compete with a pathogenic bacterium, e.g., vancomycin-resistant *Enterococcus*, a competition assay was developed. In these experiments, an overnight culture of a vancomycin-resistant strain of *Enterococcus faecium* was grown anaerobically in SweetB-FosIn for 24 hours. A glycerol stock of a bacterial composition was thawed from −80° C. and diluted to 1e6 CFU/mL per strain in SweetB-FosIn in the appropriate wells of a 96-well plate. The plate was incubated anaerobically at 37° C. for 1 hour to allow the previously frozen bacteria to revive. After the 1 hour initial incubation, VRE was inoculated into appropriate wells at target concentrations of 1e2 or 1e3 CFU/mL. Wells were also inoculated with VRE alone, without a bacterial composition. The plate was incubated anaerobically at 37° C. for 24 hours. Aliquots were removed at 15 hours and 24 hours and the VRE titers determined. At each time-point, well contents were serially diluted and plated to agar plates selective for VRE (Enterococcosel Agar+8 ug/mL vancomycin hydrochloride) (Enterococcosel Agar from BBL 212205, vancomycin hydrochloride from Sigma 94747). The selective plates were incubated aerobically at 37° C. for 24 hours before counting colonies to determine final titer of VRE in each well of the CivSim plate. Log Inhibition of VRE was determined by subtracting the final titer of a competition well from the final titer of a well containing VRE alone. Multiple ratios of the starting concentrations of VRE and bacterial compositions were tested to optimize for conditions resulting in the greatest signal. A competition time of 15 hours, a starting concentration of VRE at 1e2 CFU/mL and a starting concentration of N1962 (a.k.a. S030 and N1952) at 1e6 CFU/mL showed the greatest inhibition of growth compared to control.

(209) Using the conditions described above, one 15-member and 44 heterotrimeric bacterial compositions were tested in the assay, the results of which are provided in Tables 4 and 6. Of the 44 heterotrimeric compositions tested, 43 inhibited VRE with >80% confidence, 41 inhibited VRE with >95% confidence, and 39 inhibited VRE with >99% confidence. One ternary composition tested did not demonstrate inhibition or induction with >80% confidence.

(210) Of the ternary combinations that inhibit VRE with >99% confidence, those that strongly inhibit VRE can be identified by comparing their mean log inhibition to the distribution of all results for all ternary combinations tested. Those above the 75th percentile can be considered to strongly inhibit VRE. Alternatively, those above the 50th, 60th, 70th, 80th, 90th, 95th, or 99th percentile can be considered to strongly inhibit VRE. Non-limiting but exemplary ternary combinations that inhibit VRE with >99% confidence and above the 75th percentile include *Blautia producta*, *Clostridium innocuum*, and *Ruminococcus gnavus* and *Blautia producta*, *Clostridium butyricum*, and *Clostridium hylemonde*.

(211) The 15-member composition, N1962 (a.k.a. S030 and N1952), inhibited VRE by at least 0.7 log 10 CFU/mL across all of the conditions

tested and demonstrating inhibition of 5.7 log 10 CFU/mL in the optimal conditions.
(212) These data demonstrate methods of identifying compositions useful for prophylaxis and treatment of VRE infection.
Example 10. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of *Klebsiella pneumoniae* Using *Klebsiella* Selective Medium for Quantification
(213) To determine the ability of a composition to compete with a pathogenic bacterium, e.g., *Klebsiella pneumoniae*, a competition assay was developed. In these experiments, an overnight culture of a vancomycin-resistant strain of *Klebsiella pneumoniae* was grown anaerobically in SweetB-FosIn for 24 hours. A glycerol stock of a bacterial composition (N1962) was thawed from −80° C. and diluted to 1e6 CFU/mL per strain in SweetB-FosIn in the appropriate wells of a 96-well plate. The plate was incubated anaerobically at 37° C. for 1 hour to allow the previously frozen bacteria to revive. After the 1 hour initial incubation, *K. pneumoniae* was inoculated into appropriate wells at target concentrations of 1e2 or 1e3 CFU/mL. Wells were also inoculated with *K. pneumoniae* alone, without a bacterial composition. The plate was incubated anaerobically at 37° C. for 24 hours. Aliquots were removed at 15 hours and 24 hours to titer for the final concentration of *K. pneumoniae* at the end of competition. At each time-point, wells were serially diluted and plated to agar plates selective for *K. pneumoniae* (MacConkey Lactose Agar, Teknova M0149). The selective plates were incubated aerobically at 37° C. for 24 hours before counting colonies to determine final titer of *K. pneumoniae* in each well of the CivSim plate. Log Inhibition of *K. pneumoniae* was determined by subtracting the final titer of a competition well from the final titer of a well containing *K. pneumoniae* alone. Multiple ratios of the starting concentrations of *K. pneumoniae* and bacterial compositions were tested to optimize for conditions giving the greatest signal. The results of the assay are provided in Table 7. A competition time of 15 hours, a starting concentration of *K. pneumoniae* at 1e2 CFU/mL and a starting concentration of N1962 (a.k.a. S030 and N1952) at 1e6 CFU/mL showed the greatest inhibition of growth compared to control. N1962 (a.k.a. S030 and N1952) inhibited *K. pneumoniae* by 0.1-4.2 log 10 CFU/mL across the conditions tested.
Example 11. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of *Morganella morganii* Using *Morganella* Selective Media for Quantification
(214) To determine the ability of a composition to compete with a pathogenic bacterium, e.g., *Morganella morganii*, a competition assay was developed. In this experiment, an overnight culture of a vancomycin-resistant strain of *Morganella morganii* was grown anaerobically in SweetB-FosIn for 24 hours. A glycerol stock of a bacterial composition, N1962, was thawed from −80° C. and diluted to 1e6 CFU/mL per strain in SweetB-FosIn in the appropriate wells of a 96-well plate. The plate was incubated anaerobically at 37° C. for 1 hour to allow the previously frozen bacteria to revive. After the 1 hour initial incubation, *M. morganii* was inoculated into appropriate wells at target concentrations of 1e2 or 1e3 CFU/mL. Wells were also inoculated with *M. morganii* alone, without a bacterial composition. The plate was incubated anaerobically at 37° C. for 24 hours. Aliquots were removed at 15 hours and 24 hours to titer for the final concentration of *M. morganii* at the end of competition. At each time-point, wells were serially diluted and plated to agar plates selective for *M. morganii* (MacConkey Lactose Agar, Teknova M0149). The selective plates were incubated aerobically at 37° C. for 24 hours before counting colonies to determine final titer of *M. morganii* in each well of the CivSim plate. Log Inhibition of *M. morganii* was determined by subtracting the final titer of a competition well from the final titer of a well containing *M. morganii* alone. Multiple ratios of the starting concentrations of *M. morganii* and bacterial compositions were tested to optimize for conditions providing the greatest signal. A competition time of 15 hours, a starting concentration of *M. morganii* at 1e2 CFU/mL and a starting concentration of N1962 (a.k.a. S030 and N1952) at 1e6 CFU/mL showed the greatest inhibition of growth compared to control.
(215) A 15-member bacterial composition, N1962 (a.k.a. S030 and N1952), was tested in the assay, the results of which are provided in Table 8. N1962 (a.k.a. S030 and N1952) inhibited *M. morganii* by 1.4 to 5.8 log 10 CFU/mL across the conditions tested.
Example 12. Sequence-Based Genomic Characterization of Operational Taxonomic Units (OTU) and Functional Genes
(216) Method for Determining 16S rDNA Gene Sequence
(217) As described above, OTUs are defined either by full 16S sequencing of the rDNA gene, by sequencing of a specific hypervariable region of this gene (i.e., V1, V2, V3, V4, V5, V6, V7, V8, or V9), or by sequencing of any combination of hypervariable regions from this gene (e.g., V1-3 or V3-5). The bacterial 16S rDNA gene is approximately 1500 nucleotides in length and is used in reconstructing the evolutionary relationships and sequence similarity of one bacterial isolate to another using phylogenetic approaches. 16S sequences are used for phylogenetic reconstruction as they are in general highly conserved, but contain specific hypervariable regions that harbor sufficient nucleotide diversity to differentiate genera and species of most microbes. rDNA gene sequencing methods are applicable to both the analysis of non-enriched samples, but also for identification of microbes after enrichment steps that either enrich the microbes of interest from a microbial composition or a microbial sample and/or the nucleic acids that harbor the appropriate rDNA gene sequences as described below. For example, enrichment treatments prior to 16S rDNA gene characterization will increase the sensitivity of 16S as well as other molecular-based characterization nucleic acid purified from the microbes.
(218) Using techniques known in the art, to determine the full 16S sequence or the sequence of any hypervariable region of the 16S rDNA sequence, genomic DNA is extracted from a bacterial sample, the 16S rDNA (full region or specific hypervariable regions) amplified using polymerase chain reaction (PCR), the PCR products cleaned, and nucleotide sequences delineated to determine the genetic composition of 16S gene or subdomain of the gene. If full 16S sequencing is performed, the sequencing method used may be, but is not limited to, Sanger sequencing. If one or more hypervariable regions are used, such as the V4 region, the sequencing may be, but is not limited to being, performed using the Sanger method or using a next-generation sequencing method, such as an Illumina (sequencing by synthesis) method using barcoded primers allowing for multiplex reactions.
(219) Method for Determining 18S rDNA and ITS Gene Sequence
(220) Methods to assign and identify fungal OTUs by genetic means can be accomplished by analyzing 18S sequences and the internal transcribed spacer (ITS). The rRNA of fungi that forms the core of the ribosome is transcribed as a single gene and consists of the 8S, 5.8S and 28S regions with ITS4 and 5 between the 8S and 5.8S and 5.8S and 28S regions, respectively. These two intercistronic segments between the 18S and 5.8S and 5.8S and 28S regions are removed by splicing and contain significant variation between species for barcoding purposes as previously described (Schoch et al. Nuclear ribosomal internal transcribed spacer (ITS) region as a universal DNA barcode marker for Fungi. PNAS USA 109:6241-6246. 2012). 18S rDNA is typically used for phylogenetic reconstruction however the ITS can serve this function as it is generally highly conserved but contains hypervariable regions that harbor sufficient nucleotide diversity to differentiate genera and species of most fungus.
(221) Using techniques known in the art, to determine the full 18S and ITS sequences or a smaller hypervariable section of these sequences, genomic DNA is extracted from a microbial sample, the rDNA amplified using polymerase chain reaction (PCR), the PCR products cleaned, and nucleotide sequences delineated to determine the genetic composition rDNA gene or subdomain of the gene. The sequencing method used may be, but is not limited to, Sanger sequencing or using a next-generation sequencing method, such as an Illumina (sequencing by synthesis) method using barcoded primers allowing for multiplex reactions.
(222) Method for Determining Other Marker Gene Sequences
(223) In addition to the 16S and 18S rDNA gene, an OTU can be defined by sequencing a selected set of genes or portions of genes that are known marker genes for a given species or taxonomic group of OTUs. These genes may alternatively be assayed using a PCR-based screening strategy. For example, various strains of pathogenic *Escherichia coli* can be distinguished using DNAs from the genes that encode heat-labile

(LTI, LTIIa, and LTIIb) and heat-stable (STI and STII) toxins, verotoxin types 1, 2, and 2e (VT1, VT2, and VT2e, respectively), cytotoxic necrotizing factors (CNF1 and CNF2), attaching and effacing mechanisms (eaeA), enteroaggregative mechanisms (Eagg), and enteroinvasive mechanisms (Einv). The optimal genes to utilize for taxonomic assignment of OTUs by use of marker genes will be familiar to one with ordinary skill in the art of sequence based taxonomic identification.

(224) Genomic DNA Extraction

(225) Genomic DNA can be extracted from pure or enriched microbial cultures using a hot alkaline lysis method. For example, 1 μl of microbial culture is added to 9 μl of Lysis Buffer (25 mM NaOH, 0.2 mM EDTA) and the mixture is incubated at 95° C. for 30 minutes. Subsequently, the samples are cooled to 4° C. and neutralized by the addition of 10 μl of Neutralization Buffer (40 mM Tris-HCl) and then diluted 10-fold in Elution Buffer (10 mM Tris-HCl). Alternatively, genomic DNA is extracted from pure or enriched microbial cultures using commercially available kits such as the Mo Bio Ultraclean® Microbial DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA) or by methods known to those skilled in the art. For fungal samples, DNA extraction can be performed by methods described previously (e.g., see US20120135127) for producing lysates from fungal fruiting bodies by mechanical grinding methods.

(226) Amplification of 16S Sequences for Downstream Sanger Sequencing

(227) To amplify bacterial 16S rDNA (e.g., in FIG. **2** and FIG. **3**), 2 μl of extracted gDNA is added to a 20 μl final volume PCR reaction. For full-length 16 sequencing the PCR reaction also contains 1× HotMaster®Mix (5PRIME, Gaithersburg, MD), 250 nM of 27f (AGRGTTTGATCMTGGCTCAG (SEQ ID NO: 2036), IDT, Coralville, IA), and 250 nM of 1492r (TACGGYTACCTTGTTAYGACTT (SEQ ID NO: 2037), IDT, Coralville, IA), with PCR Water (Mo Bio Laboratories, Carlsbad, CA) for the balance of the volume.

(228) FIG. **2** shows the hypervariable regions mapped onto a 16s sequence and the sequence regions corresponding to these sequences on a sequence map. A schematic is shown of a 16S rDNA gene and the figure denotes the coordinates of hypervariable regions 1-9 (V1-V9), according to an embodiment of the invention. Coordinates of V1-V9 are 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294, and 1435-1465 respectively, based on numbering using *E. coli* system of nomenclature defined by Brosius et al. (Complete nucleotide sequence of a 16S ribosomal RNA gene (16S rRNA) from *Escherichia coli*, PNAS USA 75 (10): 4801-4805. 1978).

(229) Alternatively, other universal bacterial primers or thermostable polymerases known to those skilled in the art are used. For example, primers are available to those skilled in the art for the sequencing of the "V1-V9 regions" of the 16S rDNA (e.g., FIG. **2**). These regions refer to the first through ninth hypervariable regions of the 16S rDNA gene that are used for genetic typing of bacterial samples. These regions in bacteria are defined by nucleotides 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294 and 1435-1465 respectively using numbering based on the *E. coli* system of nomenclature. See Brosius et al., 1978, supra. In some embodiments, at least one of the V1, V2, V3, V4, V5, V6, V7, V8, and V9 regions are used to characterize an OTU. In one embodiment, the V1, V2, and V3 regions are used to characterize an OTU. In another embodiment, the V3, V4, and V5 regions are used to characterize an OTU. In another embodiment, the V4 region is used to characterize an OTU. A person of ordinary skill in the art can identify the specific hypervariable regions of a candidate 16S rDNA (e.g., FIG. **2**) by comparing the candidate sequence in question to the reference sequence (as in FIG. **3**) and identifying the hypervariable regions based on similarity to the reference hypervariable regions. FIG. **3** highlights in bold the nucleotide sequences for each hypervariable region in the exemplary reference *E. coli* 16S sequence described by Brosius et al., supra.

(230) The PCR is typically performed on commercially available thermocyclers such as a BioRad MyCycler™ Thermal Cycler (BioRad, Hercules, CA). The reactions are run at 94° C. for 2 minutes followed by 30 cycles of 94° C. for 30 seconds, 51° C. for 30 seconds, and 68° C. for 1 minute 30 seconds, followed by a 7 minute extension at 72° C. and an indefinite hold at 4° C. Following PCR, gel electrophoresis of a portion of the reaction products is used to confirm successful amplification of a ˜1.5 kb product.

(231) To remove nucleotides and oligonucleotides from the PCR products, 2 μl of HT ExoSap-IT® (Affymetrix, Santa Clara, CA) is added to 5 μl of PCR product followed by a 15 minute incubation at 37° C. and then a 15 minute inactivation at 80° C.

(232) Amplification of 16S Sequences for Downstream Characterization By Massively Parallel Sequencing Technologies

(233) Amplification performed for downstream sequencing by short read technologies such as Illumina require amplification using primers known to those skilled in the art that additionally include a sequence-based barcoded tag. For example, to amplify the 16s hypervariable region V4 region of bacterial 16S rDNA, 2 μl of extracted gDNA is added to a 20 μl final volume PCR reaction. The PCR reaction also contains 1× HotMasterMix (5PRIME, Gaithersburg, MD), 200 nM of V4_515f_adapt (AATGATACGGCGACCACCGAGATCTACACTATGGTAATTGTGTGCCAGCMGCCGCGGT AA (SEQ ID NO: 2038), IDT, Coralville, IA), and 200 nM of barcoded 806rbc (CAAGCAGAAGACGGCATACGAGAT_12bpGolayBarcode_AGTCAGTCAGCCGGACTACHV GGGTWTCTAAT (SEQ ID NO: 2039), IDT, Coralville, IA), with PCR Water (Mo Bio Laboratories, Carlsbad, CA) for the balance of the volume. In the preceding primer sequences non-ACTG nucleotide designations refer to conventional degenerate codes as are used in the art. These primers incorporate barcoded adapters for Illumina sequencing by synthesis. Optionally, identical replicate, triplicate, or quadruplicate reactions may be performed. Alternatively other universal bacterial primers or thermostable polymerases known to those skilled in the art are used to obtain different amplification and sequencing error rates as well as results on alternative sequencing technologies.

(234) The PCR amplification is performed on commercially available thermocyclers such as a BioRad MyCycler™ Thermal Cycler (BioRad, Hercules, CA). The reactions are run at 94° C. for 3 minutes followed by 25 cycles of 94° C. for 45 seconds, 50° C. for 1 minute, and 72° C. for 1 minute 30 seconds, followed by a 10 minute extension at 72° C. and a indefinite hold at 4° C. Following PCR, gel electrophoresis of a portion of the reaction products is used to confirm successful amplification of a ˜1.5 kb product. PCR cleanup is performed as described above.

(235) Sanger Sequencing of Target Amplicons from Pure Homogeneous Samples

(236) To detect nucleic acids for each sample, two sequencing reactions are performed to generate a forward and reverse sequencing read. For full-length 16s sequencing primers 27f and 1492r are used. 40 ng of ExoSap-IT-cleaned PCR products are mixed with 25 pmol of sequencing primer and Mo Bio Molecular Biology Grade Water (Mo Bio Laboratories, Carlsbad, CA) to 15 μl total volume. This reaction is submitted to a commercial sequencing organization such as Genewiz (South Plainfield, NJ) for Sanger sequencing.

(237) Amplification of 18S and ITS Regions for Downstream Sequencing

(238) To amplify the 18S or ITS regions, 2 μL fungal DNA were amplified in a final volume of 30 μL with 15 μL AmpliTaq Gold 360 Mastermix, PCR primers, and water. The forward and reverse primers for PCR of the ITS region are 5′-TCCTCCGCTTATTGATATGC-3′ (SEQ ID NO: 2040) and 5′-GGAAGTAAAAGTCGTAACAAGG-3′ (SEQ ID NO: 2041) and are added at 0.2 uM concentration each. The forward and reverse primers for the 18s region are 5′-GTAGTCATATGCTTGTCTC-3′ (SEQ ID NO: 2042) and 5′-CTTCCGTCAATTCCTTTAAG-3′ (SEQ ID NO: 2043) and are added at 0.4 uM concentration each. PCR is performed with the following protocol: 95° C. for 10 minutes, 35 cycles of 95° C. for 15 seconds, 52° C. for 30 seconds, 72° C. for 1.5 seconds; and finally 72° C. for 7 minutes followed by storage at 4° C. All forward primers contained the M13F-20 sequencing primer, and reverse primers included the M13R-27 sequencing primer. PCR products (3 μL) were enzymatically cleaned before cycle sequencing with 1 μL ExoSap-IT and 1 μL Tris EDTA and incubated at 37° C. for 20 minutes followed by 80° C. for 15 minutes. Cycle sequencing reactions contained 5 μL cleaned PCR product, 2 μL BigDye® Terminator v3.1 Ready Reaction Mix, 1 μL 5× Sequencing Buffer, 1.6 pmol of appropriate sequencing primers designed by one skilled in the art, and water in a final volume of 10 μL. The standard cycle sequencing protocol is 27 cycles of 10 seconds at 96° C., 5 seconds at 50° C., 4 minutes at 60° C., and hold at 4° C. Sequencing cleaning is performed with the BigDye XTerminator Purification Kit as recommended by the manufacturer for 10 μL volumes. The

genetic sequence of the resulting 18S and ITS sequences is performed using methods familiar to one with ordinary skill in the art using either Sanger sequencing technology or next-generation sequencing technologies such as but not limited to Illumina.
(239) Preparation of Extracted Nucleic Acids for Metagenomic Characterization by Massively Parallel Sequencing Technologies
(240) Extracted nucleic acids (DNA or RNA) are purified and prepared by downstream sequencing using standard methods familiar to one with ordinary skill in the art and as described by the sequencing technology's manufactures instructions for library preparation. In short, RNA or DNA are purified using standard purification kits such as but not limited to Qiagen's RNeasy® Kit or Promega's Genomic DNA purification kit. For RNA, the RNA is converted to cDNA prior to sequence library construction. Following purification of nucleic acids, RNA is converted to cDNA using reverse transcription technology such as but not limited to Nugen Ovation® RNA-Seq System or Illumina Truseq as per the manufacturer's instructions. Extracted DNA or transcribed cDNA are sheared using physical (e.g., Hydroshear), acoustic (e.g., Covaris), or molecular (e.g., Nextera) technologies and then size selected as per the sequencing technologies manufacturer's recommendations. Following size selection, nucleic acids are prepared for sequencing as per the manufacturer's instructions for sample indexing and sequencing adapter ligation using methods familiar to one with ordinary skill in the art of genomic sequencing.
(241) Massively Parallel Sequencing of Target Amplicons from Heterogeneous Samples
(242) DNA Quantification & Library Construction
(243) The cleaned PCR amplification products are quantified using the Quant-iT™ PicoGreen® dsDNA Assay Kit (Life Technologies, Grand Island, NY) according to the manufacturer's instructions. Following quantification, the barcoded cleaned PCR products are combined such that each distinct PCR product is at an equimolar ratio to create a prepared Illumina library.
(244) Nucleic Acid Detection
(245) The prepared library is sequenced on Illumina HiSeq or MiSeq sequencers (Illumina, San Diego, CA) with cluster generation, template hybridization, isothermal amplification, linearization, blocking and denaturation and hybridization of the sequencing primers performed according to the manufacturer's instructions. 16SV4SeqFw (TATGGTAATTGTGTGCCAGCMGCCGCGGTAA (SEQ ID NO: 2044)), 16SV4SeqRev (AGTCAGTCAGCCGGACTACHVGGGTWTCTAAT (SEQ ID NO: 2045)), and 16SV4Index (ATTAGAWACCCBDGTAGTCCGGCTGACTGACT (SEQ ID NO: 2046)) (IDT, Coralville, IA) are used for sequencing. Other sequencing technologies can be used such as but not limited to 454, Pacific Biosciences, Helicos, Ion Torrent, and Nanopore using protocols that are standard to someone skilled in the art of genomic sequencing.
Example 13. Sequence Read Annotation
(246) Primary Read Annotation
(247) Nucleic acid sequences are analyzed and annotated to define taxonomic assignments using sequence similarity and phylogenetic placement methods or a combination of the two strategies. A similar approach can be used to annotate protein names, protein function, transcription factor names, and any other classification schema for nucleic acid sequences. Sequence similarity based methods include those familiar to individuals skilled in the art including, but not limited to BLAST, BLASTx, BLASTn, tBLASTx, RDP-classifier, DNAclust, and various implementations of these algorithms such as Qiime or Mothur. These methods rely on mapping a sequence read to a reference database and selecting the match with the best score and e-value. Common databases include, but are not limited to the Human Microbiome Project, NCBI non-redundant database, Greengenes, RDP, and Silva for taxonomic assignments. For functional assignments reads are mapped to various functional databases such as but not limited to COG, KEGG, BioCyc, and MetaCyc. Further functional annotations can be derived from 16S taxonomic annotations using programs such as PICRUST (M. Langille, et al. 2013. Nature Biotechnology 31, 814-821). Phylogenetic methods can be used in combination with sequence similarity methods to improve the calling accuracy of an annotation or taxonomic assignment. Tree topologies and nodal structure are used to refine the resolution of the analysis. In this approach we analyze nucleic acid sequences using one of numerous sequence similarity approaches and leverage phylogenetic methods that are known to those skilled in the art, including but not limited to maximum likelihood phylogenetic reconstruction (see e.g., Liu et al., 2011. RAxML and FastTree: Comparing Two Methods for Large-Scale Maximum Likelihood Phylogeny Estimation. PLOS ONE 6: e27731; McGuire et al., 2001. Models of sequence evolution for DNA sequences containing gaps. Mol. Biol. Evol 18:481-490; Wróbel B. 2008. Statistical measures of uncertainty for branches in phylogenetic trees inferred from molecular sequences by using model-based methods. J. Appl. Genet. 49:49-67). Sequence reads (e.g., 16S, 18S, or ITS) are placed into a reference phylogeny comprised of appropriate reference sequences. Annotations are made based on the placement of the read in the phylogenetic tree. The certainty or significance of the OTU annotation is defined based on the OTU's sequence similarity to a reference nucleic acid sequence and the proximity of the OTU sequence relative to one or more reference sequences in the phylogeny. As an example, the specificity of a taxonomic assignment is defined with confidence at the level of Family, Genus, Species, or Strain with the confidence determined based on the position of bootstrap supported branches in the reference phylogenetic tree relative to the placement of the OTU sequence being interrogated. Nucleic acid sequences can be assigned functional annotations using the methods described above.
(248) Clade Assignments
(249) Clade assignments were generally made using full-length sequences of 16S rDNA and of V4. The ability of 16S-V4 OTU identification to assign an OTU as a specific species depends in part on the resolving power of the 16S-V4 region of the 16S gene for a particular species or group of species. Both the density of available reference 16S sequences for different regions of the tree as well as the inherent variability in the 16S gene between different species will determine the definitiveness of a taxonomic annotation. Given the topological nature of a phylogenetic tree and the fact that tree represents hierarchical relationships of OTUs to one another based on their sequence similarity and an underlying evolutionary model, taxonomic annotations of a read can be rolled up to a higher level using a clade-based assignment procedure. Using this approach, clades are defined based on the topology of a phylogenetic tree that is constructed from full-length 16S sequences using maximum likelihood or other phylogenetic models familiar to individuals with ordinary skill in the art of phylogenetics. Clades are constructed to ensure that all OTUs in a given clade are: (i) within a specified number of bootstrap supported nodes from one another (generally, 1-5 bootstraps), and (ii) share a defined percent similarity (for 16S molecular data typically set to 95%-97% sequence similarity). OTUs that are within the same clade can be distinguished as genetically and phylogenetically distinct from OTUs in a different clade based on 16S-V4 sequence data. OTUs falling within the same clade are evolutionarily closely related and may or may not be distinguishable from one another using 16S-V4 sequence data. The power of clade based analysis is that members of the same clade, due to their evolutionary relatedness, are likely to play similar functional roles in a microbial ecology such as that found in the human gut. Compositions substituting one species with another from the same clade are likely to have conserved ecological function and therefore are useful in the present invention. Notably in addition to 16S-V4 sequences, clade-based analysis can be used to analyze 18S, ITS, and other genetic sequences.
(250) Notably, 16S sequences of isolates of a given OTU are phylogenetically placed within their respective clades, sometimes in conflict with the microbiological-based assignment of species and genus that may have preceded 16S-based assignment. Discrepancies between taxonomic assignments based on microbiological characteristics versus genetic sequencing are known to exist from the literature.
(251) For a given network ecology or functional network ecology one can define a set of OTUs from the network's representative clades. As example, if a network was comprised of clade_100 and clade_102 it can be said to be comprised of at least one OTU from the group consisting of *Corynebacterium coyleae*, *Corynebacterium mucifaciens*, and *Corynebacterium ureicelerivorans*, and at least one OTU from the group consisting of *Corynebacterium appendicis*, *Corynebacterium genitalium*, *Corynebacterium glaucum*, *Corynebacterium imitans*,

*Corynebacterium riegelii*, *Corynebacterium* sp. L_2012475, *Corynebacterium* sp. NML 93_0481, *Corynebacterium sundsvallense*, and *Corynebacterium tuscaniae* (see Table 1). Conversely as example, if a network was said to consist of *Corynebacterium coyleae* and/or *Corynebacterium mucifaciens* and/or *Corynebacterium ureicelerivorans*, and also consisted of *Corynebacterium appendicis* and/or *Corynebacterium genitalium* and/or *Corynebacterium glaucum* and/or *Corynebacterium imitans* and/or *Corynebacterium riegelii* and/or *Corynebacterium* sp. L_2012475 and/or *Corynebacterium* sp. NML 93_0481 and/or *Corynebacterium sundsvallense* and/or *Corynebacterium tuscaniae* it can be said to be comprised of clade_100 and clade_102.

(252) The applicants made clade assignments to all OTUs disclosed herein using the above described method and these assignments are reported in Table 1. Results of the network analysis provides, in some embodiments, e.g., of compositions, substitution of clade_172 by clade_172i. In another embodiment, the network analysis provides substitution of clade_198 by clade_198i. In another embodiment, the network analysis permits substitution of clade_260 by clade_260c, clade_260g or clade_260h. In another embodiment, the network analysis permits substitution of clade_262 by clade_262i. In another embodiment, the network analysis permits substitution of clade_309 by clade_309c, clade_309e, clade_309g, clade_309h or clade 309i. In another embodiment, the network analysis permits substitution of clade 313 by clade 313f. In another embodiment, the network analysis permits substitution of clade_325 by clade 325f. In another embodiment, the network analysis permits substitution of clade_335 by clade 335i. In another embodiment, the network analysis permits substitution of clade_351 by clade_351e. In another embodiment, the network analysis permits substitution of clade_354 by clade_354e. In another embodiment, the network analysis permits substitution of clade_360 by clade_360c, clade_360g, clade_360h, or clade_360i. In another embodiment, the network analysis permits substitution of clade_378 by clade_378e. In another embodiment, the network analysis permits substitution of clade_38 by clade_38e or clade_38i. In another embodiment, the network analysis permits substitution of clade_408 by clade_408b, clade_408d, clade_408f, clade_408g or clade_408h. In another embodiment, the network analysis permits substitution of clade_420 by clade_420f. In another embodiment, the network analysis permits substitution of clade_444 by clade_444i. In another embodiment, the network analysis permits substitution of clade_478 by clade_478i. In another embodiment, the network analysis permits substitution of clade_479 by clade_479c, by clade_479g or by clade_479h. In another embodiment, the network analysis permits substitution of clade_481 by clade_481a, clade_481b, clade_481e, clade_481g, clade_481h or by clade_481i. In another embodiment, the network analysis substitution of clade_497 by clade_497e or by clade_497f. In another embodiment, the network analysis permits substitution of clade_512 by clade_512i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_516 by clade_516c, by clade_516g or by clade_516h. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_522 by clade_522i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_553 by clade_553i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_566 by clade_566f. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_572 by clade_572i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_65 by clade_65e. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_92 by clade_92e or by clade_92i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_96 by clade_96g or by clade_96h. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_98 by clade_98i. These permitted clade substitutions are described in Table 2.

(253) Metagenomic Read Annotation

(254) Metagenomic or whole genome shotgun sequence data is annotated as described above, with the additional step that sequences are either clustered or assembled prior to annotation. Following sequence characterization as described above, sequence reads are demultiplexed using the indexing (i.e. barcodes). Following demultiplexing sequence reads are either: (i) clustered using a rapid clustering algorithm such as but not limited to UCLUST (http://drive5.com/usearch/manual/uclust_algo.html) or hash methods such VICUNA (Xiao Yang, Patrick Charlebois, Sante Gnerre, Matthew G Coole, Niall J. Lennon, Joshua Z. Levin, James Qu, Elizabeth M. Ryan, Michael C. Zody, and Matthew R. Henn. 2012. De novo assembly of highly diverse viral populations. BMC Genomics 13:475). Following clustering a representative read for each cluster is identified based and analyzed as described above in “Primary Read Annotation”. The result of the primary annotation is then applied to all reads in a given cluster. (ii) A second strategy for metagenomic sequence analysis is genome assembly followed by annotation of genomic assemblies using a platform such as but not limited to MetAMOS (Treangen et al. 2013 Genome Biology 14: R2), HUMAaN (Abubucker et al. 2012. Metabolic Reconstruction for Metagenomic Data and Its Application to the Human Microbiome ed. J. A. Eisen. PLOS Computational Biology 8: e1002358) and other methods familiar to one of skill in the art.

Example 14. OTU Identification Using Microbial Culturing Techniques

(255) The identity of the bacterial species that grow up from a complex fraction can be determined in multiple ways. For example, individual colonies can be picked into liquid media in a 96 well format, grown up and saved as 15% glycerol stocks at −80° C. Aliquots of the cultures can be placed into cell lysis buffer and colony PCR methods can be used to amplify and sequence the 16S rDNA gene (Example 1). Alternatively, colonies may be streaked to purity in several passages on solid media. Well-separated colonies are streaked onto the fresh plates of the same kind and incubated for 48-72 hours at 37° C. The process is repeated multiple times to ensure purity. Pure cultures can be analyzed by phenotypic- or sequence-based methods, including 16S rDNA amplification and sequencing as described in Example 1. Sequence characterization of pure isolates or mixed communities e.g., plate scrapes and spore fractions can also include whole genome shotgun sequencing. The latter is valuable to determine the presence of genes associated with sporulation, antibiotic resistance, pathogenicity, and virulence. Colonies can also be scraped from plates en masse and sequenced using a massively parallel sequencing method as described in Example 1 such that individual 16S signatures can be identified in a complex mixture. Optionally, the sample can be sequenced prior to germination (if appropriate DNA isolation procedures are used to lyse and release the DNA from spores) in order to compare the diversity of germinable species with the total number of species in a spore sample. As an alternative or complementary approach to 16S analysis, MALDI-TOF-mass spec can also be used for species identification (Barreau et al., 2013. Improving the identification of anaerobes in the clinical microbiology laboratory through MALDI-TOF mass spectrometry. Anaerobe 22:123-125).

Example 15. Microbiological Strain Identification Approaches

(256) Pure bacterial isolates can be identified using microbiological methods as described in Wadsworth-KTL Anaerobic Microbiology Manual (Jouseimies-Somer et al., 2002. Wadsworth-KTL Anaerobic Bacteriology Manual), and The Manual of Clinical Microbiology (ASM Press, 10th Edition). These methods rely on phenotypes of strains and include Gram-staining to confirm Gram positive or negative staining behavior of the cell envelope, observance of colony morphologies on solid media, motility, cell morphology observed microscopically at 60× or 100× magnification including the presence of bacterial endospores and flagella. Biochemical tests that discriminate between genera and species are performed using appropriate selective and differential agars and/or commercially available kits for identification of Gram-negative and Gram-positive bacteria and yeast, for example, RapID tests (Remel) or API tests (bioMerieux). Similar identification tests can also be performed using instrumentation such as the Vitek 2 system (bioMerieux). Phenotypic tests that discriminate between genera and species and strains (for example the ability to use various carbon and nitrogen sources) can also be performed using growth and metabolic activity detection methods, for example the Biolog Microbial identification microplates. The profile of short chain fatty acid production during fermentation of particular carbon sources can also be used as a way to discriminate between species (Wadsworth-KTL Anaerobic Microbiology Manual, Jousimies-Somer, et al 2002). MALDI-TOF-mass spectrometry can also be used for species identification (as reviewed in Anaerobe 22:123).

Example 16. Construction of an In Vitro Assay to Screen for Combinations of Microbes Inhibitory to the Growth of Pathogenic *E. coli*
(257) A modification of the in vitro assay described herein is used to screen for combinations of bacteria inhibitory to the growth of *E. coli*. In general, the assay is modified by using a medium suitable for growth of the pathogen inoculum. For example, suitable media include Reinforced Clostridial Media (RCM), Brain Heart Infusion Broth (BHI) or Luria Bertani Broth (LB) (also known as Lysogeny Broth). *E. coli* is quantified by using alternative selective media specific for *E. coli* or using qPCR probes specific for the pathogen. For example, aerobic growth on MacConkey lactose medium selects for enteric Gram-negative bacteria, including *E. coli*. qPCR is conducted using probes specific for the shiga toxin of pathogenic *E. coli*.
(258) In general, the method can be used to test compositions in vitro for their ability to inhibit growth of any pathogen that can be cultured.
Example 17. Construction of an In Vitro Assay to Screen for Combinations of Microbes Inhibitory to the Growth of Vancomycin-Resistant *Enterococcus* (VRE)
(259) The in vitro assay can be used to screen for combinations of bacteria inhibitory to the growth of vancomycin-resistant *Enterococcus* spp. (VRE) by modifying the media used for growth of the pathogen inoculum. Several choices of media can be used for growth of the pathogen such as Reinforced Clostridial Media (RCM), Brain Heart Infusion Broth (BHI) or Luria Bertani Broth (LB). VRE is quantified by using alternative selective media specific for VRE or using qPCR probes specific for the pathogen. For example, m-*Enterococcus* agar containing sodium azide is selective for *Enterococcus* spp. and a small number of other species. Probes known in the art that are specific to the van genes conferring vancomycin resistance are used in the qPCR or such probes can be designed using methods known in the art.
Example 18. In Vitro Assay Screening Bacterial Compositions for Inhibition of *Salmonella*
(260) The in vitro assay described herein is used to screen for combinations of bacteria inhibitory to the growth of *Salmonella* spp. by modifying the media used for growth of the pathogen inoculum. Several choices of media are used for growth of the pathogen such as Reinforced Clostridial Media (RCM), Brain Heart Infusion Broth (BHI) or Luria Bertani Broth (LB). *Salmonella* spp. are quantified by using alternative selective media specific for *Salmonella* spp. or using qPCR probes specific for the pathogen. For example, MacConkey agar is used to select for *Salmonella* spp. and the invA gene is targeted with qPCR probes; this gene encodes an invasion protein carried by many pathogenic *Salmonella* spp. and is used in invading eukaryotic cells.
Example 19. In Vivo Validation of the Efficacy of Network Ecology Bacterial Compositions for Prevention of *Clostridium difficile* Infection in a Murine Model
(261) To test the therapeutic potential of the bacterial composition, a prophylactic mouse model of *C. difficile* infection was used (model based on Chen et al., 2008. A mouse model of *Clostridium difficile*-associated disease. Gastroenterology 135:1984-1992). Two cages of five mice each were tested for each arm of the experiment. All mice received an antibiotic cocktail consisting of 10% glucose, kanamycin (0.5 mg/ml), gentamicin (0.044 mg/ml), colistin (1062.5 U/ml), metronidazole (0.269 mg/ml), ciprofloxacin (0.156 mg/ml), ampicillin (0.1 mg/ml) and vancomycin (0.056 mg/ml) in their drinking water on days −14 through −5 and a dose of 10 mg/kg clindamycin by oral gavage on day −3. On day −1, test articles were spun for 5 minutes at 12,100 rcf, their supernatants' removed, and the remaining pellets were resuspended in sterile PBS, prereduced if bacterial composition was not in spore form, and delivered via oral gavage. On day 0 they were challenged by administration of approximately 4.5 log 10 cfu of *C. difficile* (ATCC 43255) or sterile PBS (for the naive arm) via oral gavage. Optionally a positive control group received vancomycin from day −1 through day 3 in addition to the antibiotic protocol and *C. difficile* challenge specified above. Stool were collected from the cages for analysis of bacterial carriage. Mortality, weight and clinical scoring of *C. difficile* symptoms based upon a 0-4 scale by combining scores for appearance (0-2 points based on normal, hunched, piloerection, or lethargic), and clinical signs (0-2 points based on normal, wet tail, cold-to-the-touch, or isolation from other animals) are assessed every day from day −2 through day 6. Mean minimum weight relative to day −1 and mean maximum clinical score where a death was assigned a clinical score of 4 as well as average cumulative mortality are calculated. Reduced mortality, increased mean minimum weight relative to day −1, and reduced mean maximum clinical score with death assigned to a score of 4 relative to the vehicle control are used to assess the success of the test article.
(262) Table 9 and Table 10 report results for 14 experiments in the prophylactic mouse model of *C. difficile* infection where treatment was with a bacterial composition. In the 14 experiments, 157 of the arms tested network ecologies, with 86 distinct networks ecologies tested (Table 10). Indicia of efficacy of a composition (test article) in these experiments is a low cumulative mortality for the test composition relative to the vehicle control, a mean minimum relative weight of at least 0.85 (e.g., at least 0.90, at least 0.95, or at least 0.97), and a mean maximum clinical score less than 1, e.g., 0.9, 0.8, 0.7, 0.5, 0.2, or 0. Of the 157 arms of the experiment, 136 of the arms and 73 of the networks performed better than the respective experiment's vehicle control arm by at least one of the following metrics: cumulative mortality, mean minimum relative weight, and mean maximum clinical score. Examples of efficacious networks include but are not limited to networks N1979 as tested in SP-361 which had 0% cumulative mortality, 0.97 mean minimum relative weight, and 0 mean maximum clinical score or N2007 which had 10% cumulative mortality, 0.91 mean minimum relative weight, and 0.9 mean maximum clinical score with both networks compared to the vehicle control in SP-361 which had 30% cumulative mortality, 0.88 mean minimum relative weight, and 2.4 mean maximum clinical score. In SP-376, N1962 had no cumulative mortality, mean maximum clinical scores of 0 at both target doses tested with mean minimum relative weights of 0.98 and 0.95 for target doses of 1e8 and 1e7 CFU/OTU/mouse respectively. These results confirm that bacterial compositions comprised of binary and ternary and combinations thereof are efficacious as demonstrated using the mouse model.
Example 20. In Vivo Validation of Network Ecology Bacterial Composition Efficacy in Prophylactic and Relapse Prevention Hamster Model
(263) Previous studies with hamsters using toxigenic and nontoxigenic strains of *C. difficile* demonstrated the utility of the hamster model in examining relapse post antibiotic treatment and the effects of prophylaxis treatments with cecal flora in *C. difficile* infection (Wilson et al., 1981. Infect Immun 34:626-628), Wilson et al., 1983. J Infect Dis 147:733, Borriello et al., 1985. J Med Microbiol 19:339-350) and more broadly in gastrointestinal infectious disease. Accordingly, to demonstrate prophylactic use of bacterial compositions comprising specific operational taxonomic units to ameliorate *C. difficile* infection, the following hamster model was used. Clindamycin (10 mg/kg s.c.) was administered to animals on day −5, the test composition or control was administered on day −3, and *C. difficile* challenge occurred on day 0. In the positive control arm, vancomycin was then administered on days 1-5 (and vehicle control was delivered on day −3). Stool were collected on days −5, −4, −1, 1, 3, 5, 7, 9 and fecal samples were assessed for pathogen carriage and reduction by microbiological methods. 16S sequencing approaches or other methods could also be utilized by one skilled in the art. Mortality was assessed multiple times per day through 21 days post *C. difficile* challenge. The percentage survival curves showed that a bacterial composition (N1962) comprised of OTUs that were shown to be inhibitory against *C. difficile* in an in vitro inhibition assay (see above examples) better protected the hamsters compared to the vancomycin control, and vehicle control (FIG. **5**).
(264) These data demonstrate the efficacy of a composition in vivo, as well as the utility of using an in vitro inhibition method as described herein to predict compositions that have activity in vivo.
Example 21. Method of Preparing a Bacterial Composition for Administration to a Subject
(265) Two or more strains that comprise the bacterial composition are independently cultured and mixed together before administration. Both strains are independently be grown at 37° C., pH 7, in a GMM or other animal-products-free medium, pre-reduced with 1 g/L cysteine HCl. After each strain reaches a sufficient biomass, it is preserved for banking by adding 15% glycerol and then frozen at −80° C. in 1 ml cryotubes.
(266) Each strain is then be cultivated to a concentration of 10.sup.10 CFU/mL, then concentrated 20-fold by tangential flow microfiltration; the

spent medium is exchanged by diafiltering with a preservative medium consisting of 2% gelatin, 100 mM trehalose, and 10 mM sodium phosphate buffer, or other suitable preservative medium. The suspension is freeze-dried to a powder and titrated.
(267) After drying, the powder is blended with microcrystalline cellulose and magnesium stearate and formulated into a 250 mg gelatin capsule containing 10 mg of lyophilized powder (10.sup.8 to 10.sup.11 bacteria), 160 mg microcrystalline cellulose, 77.5 mg gelatin, and 2.5 mg magnesium stearate.
(268) A bacterial composition can be derived by selectively fractionating the desired bacterial OTUs from a raw material such as but not limited to stool. As an example, a 10% w/v suspension of human stool material in PBS was prepared that was filtered, centrifuged at low speed, and then the supernatant containing spores was mixed with absolute ethanol in a 1:1 ratio and vortexed to mix. The suspension was incubated at room temperature for 1 hour. After incubation the suspension was centrifuged at high speed to concentrate spores into a pellet containing a purified spore-containing preparation. The supernatant was discarded and the pellet resuspended in an equal mass of glycerol, and the purified spore preparation was placed into capsules and stored at −80° C.; this preparation is referred to as an ethanol-treated spore population.
Example 22. Method of Treating a Subject with Recurrent *C. difficile* Infection with a Bacterial Composition
(269) In one example, a subject has suffered from recurrent bouts of *C. difficile*. In the most recent acute phase of the illness, the subject is treated with an antibiotic sufficient to ameliorate the symptoms of the illness. To prevent another relapse of *C. difficile* infection, a bacterial composition described herein is administered to the subject. For example, the subject is administered one of the present bacterial compositions at a dose in the range of 1e10.sup.7 to 1e10.sup.12 in, e.g., a lyophilized form, in one or more gelatin capsules (e.g., 2, 3, 4, 5, 10, 15 or more capsules) containing 10 mg of lyophilized bacteria and stabilizing components. The capsule is administered by mouth and the subject resumes a normal diet after 4, 8, 12, or 24 hours. In another embodiment, the subject may take the capsule by mouth before, during, or immediately after a meal. In a further embodiment, the subject takes the dose daily for a specified period of time.
(270) Stool is collected from the subject before and after treatment. In one embodiment stool is collected at 1 day, 3 days, 1 week, and 1 month after administration. The presence of *C. difficile* is found in the stool before administration of the bacterial composition, but stool collections after administration show a reduction in the level of *C. difficile* in the stool (for example, at least 50% less, 60%, 70%, 80%, 90%, or 95%) to no detectable levels of *C. difficile*, as measured by qPCR and if appropriate, compared to a healthy reference subject microbiome, as described above. Typically, the quantitation is performed using material extracted from the same amounts of starting material, e.g., stool. ELISA for toxin protein or traditional microbiological identification techniques may also be used. Effective treatment is defined as a reduction in the amount of *C. difficile* present after treatment.
(271) In some cases, effective treatment, i.e., a positive response to treatment with a composition disclosed herein is defined as absence of diarrhea, which itself is defined as 3 or more loose or watery stools per day for at least 2 consecutive days or 8 or more loose or watery stools in 48 hours, or persisting diarrhea (due to other causes) with repeating (three times) negative stool tests for toxins of *C. difficile*.
(272) Treatment failure is defined as persisting diarrhea with a positive *C. difficile* toxin stool test or no reduction in levels of *C. difficile*, as measured by qPCR sequencing. ELISA or traditional microbiological identification techniques may also be used.
(273) In some cases, effective treatment is determined by the lack of recurrence of signs or symptoms of *C. difficile* infection within, e.g., 2 weeks, 3 weeks, 4 weeks, 5 weeks, 10 weeks, 12 weeks, 16 weeks, 20 weeks, or 24 weeks after the treatment.
Example 23. Treatment of Subjects with *Clostridium difficile* Associated Diarrheal Disease with a Bacterial Composition
(274) Microbial Population Engraftment, Augmentation, and Reduction of Pathogen Carriage in Patients Treated with Spore Compositions
(275) Complementary genomic and microbiological methods were used to characterize the composition of the microbiota of 15 subjects with recurrent *C. difficile* associated disease (CDAD) that were treated with a bacterial composition. The microbiome of these subjects was characterized pretreatment and initially up to 4 weeks post-treatment and further to 24 weeks. An additional 15 subjects were treated and data for those subjects was collected to at least 8 weeks post-treatment and up to 24 weeks post-treatment. The bacterial compositions used for treatment were comprised of spore forming bacteria and constitute a microbial spore ecology derived from healthy human stool. Methods for preparing such compositions can be found in PCT/US2014/014715.
(276) Non-limiting exemplary OTUs and clades of the spore forming microbes identified in the initial compositions are provided in Table 11. OTUs and clades in the spore ecology treatment were observed in 1 to 15 of the initial 15 subjects treated (Table 11) and in subsequently treated subjects. Treatment of the subjects with the microbial spore ecology resolved *C. difficile* associated disease (CDAD) in all subjects treated. In addition, treatment with the microbial spore composition led to the reduction or removal of Gram (−) and Gram (+) pathobionts including but not limited to pathobionts with multi-drug resistance such as but not limited to vancomycin-resistant Enterococci (VRE) and carbapenem- or imipenem resistant bacteria. Additionally, treatment led to an increase in the total microbial diversity of the subjects gut microbiome (FIG. **6**) and the resulting microbial community that established as the result of treatment with the microbial spore ecology was different from the microbiome pretreatment and more closely represented that of a healthy individual than that of an individual with CDAD (FIG. **7**).
(277) Using novel computational approaches, applicants delineated bacterial OTUs associated with engraftment and ecological augmentation and establishment of a more diverse microbial ecology in patients treated with an ethanol-treated spore preparation (Table 11). OTUs that comprise an augmented ecology are those below the limit of detection in the patient prior to treatment and/or exist at extremely low frequencies such that they do not comprise a significant fraction of the total microbial carriage and are not detectable by genomic and/or microbiological assay methods in the bacterial composition. OTUs that are members of the engrafting and augmented ecologies were identified by characterizing the OTUs that increase in their relative abundance post treatment and that respectively are: (i) present in the ethanol-treated spore preparation and not detectable in the patient pretreatment (engrafting OTUs), or (ii) absent in the ethanol-treated spore preparation, but increase in their relative abundance in the patient through time post treatment with the preparation due to the formation of favorable growth conditions by the treatment (augmenting OTUs). Augmenting OTUs can grow from low frequency reservoirs in the patient, or can be introduced from exogenous sources such as diet.
(278) Notably, 16S sequences of isolates of a given OTU are phylogenetically placed within their respective clades despite that the actual taxonomic assignment of species and genus may suggest they are taxonomically distinct from other members of the clades in which they fall. Discrepancies between taxonomic names given to an OTU is based on microbiological characteristics versus genetic sequencing are known to exist from the literature. The OTUs footnoted in this table are known to be discrepant between the different methods for assigning a taxonomic name.
(279) Rational Design of Therapeutic Compositions from Core Ecologies
(280) To define the Core Ecology underlying the remarkable clinical efficacy of the microbial spore bacterial the following analysis was carried out. The OTU composition of the microbial spore ecology was determined by 16S-V4 rDNA sequencing and computational assignment of OTUs per Example 13. A requirement to detect at least ten sequence reads in the microbial spore ecology was set as a conservative threshold to define only OTUs that were highly unlikely to arise from errors during amplification or sequencing. Methods routinely employed by those familiar to the art of genomic-based microbiome characterization use a read relative abundance threshold of 0.005% (see e.g., Bokulich et al. 2013. Quality-filtering vastly improves diversity estimates from Illumina amplicon sequencing. Nature Methods 10:57-59), which would equate to ≥2 reads given the sequencing depth obtained for the samples analyzed in this example, as cut-off which is substantially lower than the ≥10 reads used in this analysis. All taxonomic and clade assignments were made for each OTU as described in Example 13. The resulting list of OTUs, clade

assignments, and frequency of detection in the spore preparations are shown in Table 11.

(281) In one embodiment, OTUs that comprise a "core" bacterial composition of a microbial spore ecology, augmented ecology or engrafted ecology can be defined by the percentage of total subjects in which they are observed; the greater this percentage the more likely they are to be part of a core ecology responsible for catalyzing a shift away from a dysbiotic ecology. In one embodiment, therapeutic bacterial compositions are rationally designed by identifying the OTUs that occur in the greatest number of subjects evaluated. In one embodiment OTUs that occur in 100% of subjects define a therapeutic bacterial composition. In other embodiments, OTUs that are defined to occur in ≥90%, ≥80%, ≥70%, ≥60%, or ≥50% of the subjects evaluated comprise the therapeutic bacterial composition. In a further embodiment, OTUs that are in either 100%, ≥90%, ≥80%, ≥70%, ≥60%, or ≥50% are further refined to rationally design a therapeutic bacterial composition using phylogenetic parameters or other features such as but not limited to their capacity to metabolize secondary bile acids, illicit TH17 immune signaling, or produce short-chain fatty acids.

(282) In an additional embodiment, the dominant OTUs in an ecology can be identified using several methods including but not limited to defining the OTUs that have the greatest relative abundance in either the augmented or engrafted ecologies and defining a total relative abundance threshold. As example, the dominant OTUs in the augmented ecology of Patient-1 were identified by defining the OTUs with the greatest relative abundance, which together comprise 60% of the microbial carriage in this patient's augmented ecology by day 25 post-treatment.

(283) In a further embodiment, an OTU is assigned to be a member of the Core Ecology of the bacterial composition, that OTU must be shown to engraft in a patient. Engraftment is important for at least two reasons. First, engraftment is believed to be a sine qua non of the mechanism to reshape the microbiome and eliminate *C. difficile* colonization. OTUs that engraft with higher frequency are highly likely to be a component of the Core Ecology of the spore preparation or broadly speaking a set bacterial composition. Second, OTUs detected by sequencing a bacterial composition may include non-viable cells or other contaminant DNA molecules not associated with the composition. The requirement that an OTU must be shown to engraft in the patient eliminates OTUs that represent non-viable cells or contaminating sequences. OTUs that are present in a large percentage of the bacterial composition, e.g., ethanol spore preparations analyzed and that engraft in a large number of patients represent a subset of the Core Ecology that are highly likely to catalyze the shift from a dysbiotic disease ecology to a healthy microbiome. OTUs from which to define such therapeutic bacterial compositions derived of OTUs that engraft are denoted in Table 11.

(284) A third lens was applied to further refine discoveries into the Core Ecology of the bacterial composition (e.g., microbial spore ecology). Computational-based, network analysis has enabled the description of microbial ecologies that are present in the microbiota of a broad population of healthy individuals. These network ecologies are comprised of multiple OTUs, some of which are defined as Keystone OTUs. Keystone OTUs are computationally defined OTUs that occur in a large percentage of computed networks and meet the networks in which they occur are highly prevalent in the population of subjects evaluated. Keystone OTUs form a foundation to the microbially ecologies in that they are found and as such are central to the function of network ecologies in healthy subjects. Keystone OTUs associated with microbial ecologies associated with healthy subjects are often are missing or exist at reduced levels in subjects with disease. Keystone OTUs may exist in low, moderate, or high abundance in subjects.

(285) There are several important findings from these data. A relatively small number of species, 11 in total, are detected in all of the spore preparations from 6 donors and 10 donations. This is surprising because the HMP database (www.hmpdacc.org) describes the enormous variability of commensal species across healthy individuals. The presence of a small number of consistent OTUs lends support to the concept of a Core Ecology and Backbone Networks. The engraftment data further supports this conclusion.

(286) In another embodiment, three factors—prevalence in the bacterial composition such as but not limited to a spore preparation, frequency of engraftment, and designation as a Keystone OTUs—enabled the creation of a "Core Ecology Score" (CES) to rank individual OTUs. CES was defined as follows: 40% weighting for presence of OTU in spore preparation multiplier of 1 for presence in 1-3 spore preparations multiplier of 2.5 for presence in 4-8 spore preparations multiplier of 5 for presences in ≥9 spore preparations 40% weighting for engraftment in a patient multiplier of 1 for engraftment in 1-4 patients multiplier of 2.5 for engraftment in 5-6 patients multiplier of 5 for engraftment in ≥7 patients 20% weighting to Keystone OTUs multiplier of 1 for a Keystone OTU multiplier of 0 for a non-Keystone OTU

(287) Using this guide, the CES has a maximum possible score of 5 and a minimum possible score of 0.8. As an example, an OTU found in 8 of the 10 bacterial composition such as but not limited to a spore preparations that engrafted in 3 patients and was a Keystone OTU would be assigned the follow CES:

$CES=(0.4\times2.5)+(0.4\times1)+(0.2\times1)=1.6$

(288) Table 11 provides a rank of OTUs by CES. Bacterial compositions rationally designed using a CES score are highly likely to catalyze the shift from a dysbiotic disease ecology to a healthy microbiome. In additional embodiments, the CES score can be combined with other factors to refine the rational design of a therapeutic bacterial composition. Such factors include but are not limited to: using phylogenetic parameters or other features such as but not limited to their capacity to metabolize secondary bile acids, illicit TH17 immune signaling, or produce short-chain fatty acids. In an additional embodiment, refinement can be done by identifying the OTUs that have the greatest relative abundance in either the augmented or engrafted ecologies and defining a total relative abundance threshold.

(289) The number of organisms in the human gastrointestinal tract, as well as the diversity between healthy individuals, is indicative of the functional redundancy of a healthy gut microbiome ecology (see The Human Microbiome Consortia. 2012. Structure, function and diversity of the healthy human microbiome. Nature 486:207-214). This redundancy makes it highly likely that subsets of the Core Ecology describe therapeutically beneficial components of the bacterial composition such as but not limited to an ethanol-treated spore preparation and that such subsets may themselves be useful compositions for populating the GI tract and for the treatment of *C. difficile* infection given the ecologies functional characteristics. Using the CES, as well as other key metrics as defined above, individual OTUs can be prioritized for evaluation as an efficacious subset of the Core Ecology.

(290) Another aspect of functional redundancy is that evolutionarily related organisms (i.e., those close to one another on the phylogenetic tree, e.g., those grouped into a single clade) will also be effective substitutes in the Core Ecology or a subset thereof for treating *C. difficile.*

(291) To one skilled in the art, the selection of appropriate OTU subsets for testing in vitro or in vivo is straightforward. Subsets may be selected by picking any 2, 3, 4, 5, 6, 7, 8, 9, 10, or more than 10 OTUs from Table 11, typically selecting those with higher CES. In addition, using the clade relationships defined in Example 13 above and Table 11, related OTUs can be selected as substitutes for OTUs with acceptable CES values. These organisms can be cultured anaerobically in vitro using the appropriate media, and then combined in a desired ratio. A typical experiment in the mouse *C. difficile* model utilizes at least 104 and preferably at least 105, 106, 107, 108, 109 or more than 109 colony forming units of a each microbe in the composition. In some compositions, organisms are combined in unequal ratios, for example, due to variations in culture yields, e.g., 1:10, 1:100, 1:1,000, 1:10,000, 1:100,000, or greater than 1:100,000. What is important in these compositions is that each strain be provided in a minimum amount so that the strain's contribution to the efficacy of the Core Ecology subset can be therapeutically effective, and in some cases, measured. Using the principles and instructions described here, one of skill in the art can make clade-based substitutions to test the efficacy of subsets of the Core Ecology. Table 11 and Table 2 describe the clades for each OTU from which such substitutions can be derived.

(292) Rational Design of Therapeutic Compositions by Integration of In Vitro and Clinical Microbiome Data

(293) In one embodiment, efficacious subsets of the treatment microbial spore ecology as well as subsets of the microbial ecology of the subject post-treatment are defined by rationally interrogating and the composition of these ecologies with respect to compositions comprising 2, 3, 4, 5, 6, 7, 8, 9, 10, or some larger number of OTUs. In one embodiment, the bacterial compositions that have demonstrated efficacy in an in vitro pathogen inhibition assay and that are additionally identified as constituents of the ecology of the treatment itself and/or the microbial ecology of 100%, ≥90%, >80%, ≥70%, ≥60%, or ≥50% of the subject's can by an individual with ordinary skill in the art be prioritize for functional screening. Functional screens can include but are not limited to in vivo screens using various pathogen or non-pathogen models (as example, murine models, hamster models, primate models, or human). Table 12 provides bacterial compositions that exhibited inhibition against *C. difficile* as measured by a mean log inhibition greater than the 99% confidence interval (C.I) of the null hypothesis (see Example 6, ++++) and that are identified in at least one spore ecology treatment or subject post-treatment. In another embodiment compositions found in the 95%, 90%, or 80% confidence intervals (C.I.) and occurring in the treatment and post-treatment ecologies are selected. In other embodiments, bacterial compositions are selected for screening for therapeutic potential by selecting OTUs that occur in the treatment or post-treatment ecologies and the measured growth inhibition of the composition is ranked ≥ the 75.sup.th percentile of all growth inhibition scores. In other embodiments, compositions ranked ≥ the 50th, 60th, 70th, 80th, 90th, 95th, or 99th percentiles are selected. In another embodiment, compositions demonstrated to have synergistic inhibition are selected (see Example 7). In yet a further embodiment, compositions selected to screen for efficacy in in vivo models are selected using a combination of growth inhibition metrics. As non-limiting example: (i) compositions are first selected based on their log inhibition being greater than the 99% confidence interval (C.I.) of the null hypothesis, (ii) then this subset of compositions further selected to represent those that are ranked ≥ the 75.sup.th percentile in the distribution of all inhibition scores, (iii) this subset is then further selected based on compositions that demonstrate synergistic inhibition. In some embodiments, different confidence intervals (C.I.) and percentiles are used to subset and rationally select the compositions. In yet another embodiment, bacterial compositions are further rationally defined for their therapeutic potential using phylogenetic criteria, such as but not limited to, the presence of particular phylogenetic clade, or other features such as but not limited to their capacity to metabolize secondary bile acids, illicit TH17 immune signaling, or produce short-chain fatty acids.
(294) In a related embodiment, all unique bacterial compositions that can be delineated in silico using the OTUs that occur in 100% of the dose spore ecologies are defined; exemplary bacterial compositions are denoted in Table 13. In other embodiments, compositions are derived form OTUs that occur in ≥90%, ≥80%, ≥70%, ≥60%, or ≥50% of the dose spore ecology or the subject's post-treatment ecologies. One with ordinary skill in the art can interrogate the resulting bacterial compositions and using various metrics including, but not limited to the percentage of spore formers, the presence of keystone OTUs, phylogenetic composition, or the OTUs' ability to metabolize secondary bile acids or the ability to produce short-chain fatty acids to rationally define bacterial compositions with suspected efficacy and suitability for further screening.
Example 24. Computational Analysis of Administered Spore Ecology Dose Compositions, and Augmentation and Engraftment Following Administration of Spore Ecology Doses
(295) The clinical trial described in Example 23 enrolled 15 additional subjects. Further analyses were carried out on information combining data from all subjects responding to treatment in the trial (29 of 30 subjects). The treatment was with a complex formulation of microbes derived from human stool. Analyses of these results are provided in Tables 14-21. Table 22 is provided for convenience, and lists alternative names for certain organisms. Typically, the presence of an OTU is made using a method known in the art, for example, using qPCR under conditions known in the art and described herein.
(296) The set of doses used in the trial is the collection of doses that was provided to at least one patient. Thus, a dose is implicitly a member of the set of doses. Consequently, the set of all OTUs in doses is defined as the unique set of OTUs such that each OTU is present in at least one dose.
(297) As described herein, an engrafting OTU is an OTU that is not detectable in a patient, e.g., in their stool, pre-treatment, but is present in the composition delivered to the subject and is detected in the subject, (e.g., in the subject's stool) in at least one post-treatment sample from the subject. The set of all engrafting OTUs is defined as the unique set of engrafting OTUs found in at least one subject. An augmenting OTU is an OTU detected in a subject that is not engrafting and has an abundance ten times greater than the pre-treatment abundance at some post-treatment time point. The set of all augmenting OTUs is the unique set of augmenting OTUs found in at least one subject. The set of all augmenting and engrafting OTUs is defined as the unique set of OTUs that either augment or engraft in at least one subject.
(298) The set of all unique ternary combinations can be generated from the experimentally derived set of OTUs by considering the all combinations of OTUs such that 1) each OTU of the ternary is different and 2) the three OTUs were not used together previously. A computer program can be used to generate such combinations.
(299) Table 14 is generated from the set of all augmenting and engrafting OTUs and provides the OTUs that either were found to engraft or augment in at least one subject after they were treated with the composition. Each listed ternary combination is either in all doses provided to subjects or were detected together in all patients for at least one post-treatment time point. Typically, a useful composition includes at least one of the ternary compositions. In some embodiments, all three members of the ternary composition either engraft or augment in at least, e.g., 68%, 70%, 71%, 75%, 79%, 86%, 89%, 93%, or 100% of subjects. Because all subjects analyzed responded to treatment, the ternaries listed in the Table are useful in compositions for treatment of a dysbiosis.
(300) Table 15 provides the list of unique ternary combinations of OTUs that were present in at least 95% of doses (rounding to the nearest integer) and that engrafted in at least one subject. Note that ternary combinations that were present in 100% of doses are listed in Table 14. Compositions that include a ternary combination are useful in compositions for treating a dysbiosis.
(301) Table 16 provides the set of all unique ternary combinations of augmenting OTUs such that each ternary combination was detected in at least 75% of the subjects at a post-treatment time point.
(302) Table 17 provides the set of all unique ternary combinations that were present in at least 75% of doses and for which the subject receiving the dose containing the ternary combination had *Clostridiales* sp. SM4/1 present as either an engrafting or augmenting OTU. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 17 is useful for increasing *Clostridiales* sp. SM4/1 in a subject.
(303) Table 18 provides the set of all unique ternary combinations generated from the set of all OTUs in doses such that each ternary is present at least 75% of the doses and for which the subject receiving the dose containing the ternary combination had *Clostridiales* sp. SSC/2 present as either an engrafting or augmenting OTU after treatment. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 18 is useful for increasing *Clostridiales* sp. SSC/2 in a subject.
(304) Table 19 provides the set of all unique ternary combinations generated from the set of all OTUs present in doses such that each ternary is present at least 75% of the doses and for which the subject to whom the doses containing the ternary combination was administered had *Clostridium* sp. NML 04A032 present as either an engrafting or augmenting OTU after treatment. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 19 is useful for increasing *Clostridium* sp. NML 04A032 in a subject.
(305) Table 20 provides the set of all unique ternary combinations generated from the set of all OTUs in doses such that the ternary is present at least 75% of the doses and for which the subject to whom the dose containing the ternary was administered had *Clostridium* sp. NML 04A032,

*Ruminococcus lactaris*, and *Ruminococcus torques* present as either an engrafting or augmenting OTUs. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 20 is useful for increasing *Clostridium* sp. NML 04A032, *Ruminococcus lactaris*, and *Ruminococcus torques* in a subject.

(306) Table 21 shows the set of all unique ternary combinations generated from the set of all OTUs in doses such that each ternary is present at least 75% of the doses and for which the subject to whom the dose containing the ternary combination was administered has *Eubacterium* rectale, *Faecalibacterium prausnitzii*, *Oscillibacter* sp. G2, *Ruminococcus lactaris*, and *Ruminococcus torques* present as either an engrafting or augmenting OTU. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 21 is useful for increasing *Eubacterium* rectale, *Faecalibacterium prausnitzii*, *Oscillibacter* sp. G2, *Ruminococcus lactaris*, and *Ruminococcus torques* in a subject.

(307) Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the embodiments. Consider the specification and examples as exemplary only, with a true scope and spirit being indicated by the following claims.

(308) Tables

(309) TABLE-US-00001 TABLE 1 SEQ ID Public DB Phylogenetic Spore Pathogen OTU Number Accession Clade Former Status *Corynebacterium coyleae* 697 X96497 clade_100 N N *Corynebacterium mucifaciens* 711 NR_026396 clade_100 N N *Corynebacterium ureicelerivorans* 733 AM397636 clade_100 N N *Corynebacterium appendicis* 684 NR_028951 clade_102 N N *Corynebacterium genitalium* 698 ACLJ01000031 clade_102 N N *Corynebacterium glaucum* 699 NR_028971 clade_102 N N *Corynebacterium imitans* 703 AF537597 clade_102 N N *Corynebacterium riegelii* 719 EU848548 clade_102 N N *Corynebacterium* sp. L_2012475 723 HE575405 clade_102 N N *Corynebacterium* sp. NML 93_0481 724 GU238409 clade_102 N N *Corynebacterium sundsvallense* 728 Y09655 clade_102 N N *Corynebacterium tuscaniae* 730 AY677186 clade_102 N N *Prevotella maculosa* 1504 AGEK01000035 clade_104 N N *Prevotella oris* 1513 ADDV01000091 clade_104 N N *Prevotella salivae* 1517 AB108826 clade_104 N N *Prevotella* sp. ICM55 1521 HQ616399 clade_104 N N *Prevotella* sp. oral clone AA020 1528 AY005057 clade_104 N N *Prevotella* sp. oral clone GI032 1538 AY349396 clade_104 N N *Prevotella* sp. oral taxon G70 1558 GU432179 clade_104 N N *Prevotella corporis* 1491 L16465 clade_105 N N *Bacteroides* sp. 4_1_36 312 ACTC01000133 clade_110 N N *Bacteroides* sp. AR20 315 AF139524 clade_110 N N *Bacteroides* sp. D20 319 ACPT01000052 clade_110 N N *Bacteroides* sp. F_4 322 AB470322 clade_110 N N *Bacteroides uniformis* 329 AB050110 clade_110 N N *Prevotella nanceiensis* 1510 JN867228 clade_127 N N *Prevotella* sp. oral taxon 299 1548 ACWZ01000026 clade_127 N N *Prevotella bergensis* 1485 ACKS01000100 clade_128 N N *Prevotella buccalis* 1489 JN867261 clade_129 N N *Prevotella timonensis* 1564 ADEF01000012 clade_129 N N *Prevotella oralis* 1512 AEPE01000021 clade_130 N N *Prevotella* sp. SEQ072 1525 JN867238 clade_130 N N *Leuconostoc carnosum* 1177 NR_040811 clade_135 N N *Leuconostoc gasicomitatum* 1179 FN822744 clade_135 N N *Leuconostoc inhae* 1180 NR_025204 clade_135 N N *Leuconostoc kimchii* 1181 NR_075014 clade_135 N N *Edwardsiella tarda* 777 CP002154 clade_139 N N *Photorhabdus asymbiotica* 1466 Z76752 clade_139 N N *Psychrobacter arcticus* 1607 CP000082 clade_141 N N *Psychrobacter cibarius* 1608 HQ698586 clade_141 N N *Psychrobacter cryohalolentis* 1609 CP000323 clade_141 N N *Psychrobacter faecalis* 1610 HQ698566 clade_141 N N *Psychrobacter nivimaris* 1611 HQ698587 clade_141 N N *Psychrobacter pulmonis* 1612 HQ698582 clade_141 N N *Pseudomonas aeruginosa* 1592 AABQ07000001 clade_154 N N *Pseudomonas* sp. 2_1_26 1600 ACWU01000257 clade_154 N N *Corynebacterium confusum* 691 Y15886 clade_158 N N *Corynebacterium propinquum* 712 NR_037038 clade_158 N N *Corynebacterium pseudodiphtheriticum* 713 X84258 clade_158 N N *Bartonella bacilliformis* 338 NC_008783 clade_159 N N *Bartonella grahamii* 339 CP001562 clade_159 N N *Bartonella henselae* 340 NC_005956 clade_159 N N *Bartonella quintana* 341 BX897700 clade_159 N N *Bartonella tamiae* 342 EF672728 clade_159 N N *Bartonella washoensis* 343 FJ719017 clade_159 N N *Brucella abortus* 430 ACBJ01000075 clade_159 N Category-B *Brucella canis* 431 NR_044652 clade_159 N Category-B *Brucella ceti* 432 ACJD01000006 clade_159 N Category-B *Brucella melitensis* 433 AE009462 clade_159 N Category-B *Brucella microti* 434 NR_042549 clade_159 N Category-B *Brucella ovis* 435 NC_009504 clade_159 N Category-B *Brucella* sp. 83_13 436 ACBQ01000040 clade_159 N Category-B *Brucella* sp. BO1 437 EU053207 clade_159 N Category-B *Brucella suis* 438 ACBK01000034 clade_159 N Category-B *Ochrobactrum anthropi* 1360 NC_009667 clade_159 N N *Ochrobactrum intermedium* 1361 ACQA01000001 clade_159 N N *Ochrobactrum pseudintermedium* 1362 DQ365921 clade_159 N N *Prevotella* genomosp. C2 1496 AY278625 clade_164 N N *Prevotella multisaccharivorax* 1509 AFJE01000016 clade_164 N N *Prevotella* sp. oral clone IDR_CEC_0055 1543 AY550997 clade_164 N N *Prevotella* sp. oral taxon 292 1547 GQ422735 clade_164 N N *Prevotella* sp. oral taxon 300 1549 GU409549 clade_164 N N *Prevotella marshii* 1505 AEEI01000070 clade_166 N N *Prevotella* sp. oral clone IK053 1544 AY349401 clade_166 N N *Prevotella* sp. oral taxon 781 1554 GQ422744 clade_166 N N *Prevotella stercorea* 1562 AB244774 clade_166 N N *Prevotella brevis* 1487 NR_041954 clade_167 N N *Prevotella ruminicola* 1516 CP002006 clade_167 N N *Prevotella* sp. sp24 1560 AB003384 clade_167 N N *Prevotella* sp. sp34 1561 AB003385 clade_167 N N *Prevotella albensis* 1483 NR_025300 clade_168 N N *Prevotella copri* 1490 ACBX02000014 clade_168 N N *Prevotella oulorum* 1514 L16472 clade_168 N N *Prevotella* sp. BI_42 1518 AJ581354 clade_168 N N *Prevotella* sp. oral clone P4PB_83 P2 1546 AY207050 clade_168 N N *Prevotella* sp. oral taxon G60 1557 GU432133 clade_168 N N *Prevotella amnii* 1484 AB547670 clade_169 N N *Bacteroides caccae* 268 EU136686 clade_170 N N *Bacteroides finegoldii* 277 AB222699 clade_170 N N *Bacteroides intestinalis* 283 ABJL02000006 clade_171 N N *Bacteroides* sp. XB44A 326 AM230649 clade_171 N N Bifidobacteriaceae genomosp. C1 345 AY278612 clade_172 N N *Bifidobacterium adolescentis* 346 AAXD02000018 clade_172 N N *Bifidobacterium angulatum* 347 ABYS02000004 clade_172 N N *Bifidobacterium animalis* 348 CP001606 clade_172 N N *Bifidobacterium breve* 350 CP002743 clade_172 N N *Bifidobacterium catenulatum* 351 ABXY01000019 clade_172 N N *Bifidobacterium dentium* 352 CP001750 clade_172 N OP *Bifidobacterium gallicum* 353 ABXB03000004 clade_172 N N *Bifidobacterium infantis* 354 AY151398 clade_172 N N *Bifidobacterium kashiwanohense* 355 AB491757 clade_172 N N *Bifidobacterium longum* 356 ABQQ01000041 clade_172 N N *Bifidobacterium pseudocatenulatum* 357 ABXX02000002 clade_172 N N *Bifidobacterium pseudolongum* 358 NR_043442 clade_172 N N *Bifidobacterium scardovii* 359 AJ307005 clade_172 N N *Bifidobacterium* sp. HM2 360 AB425276 clade_172 N N *Bifidobacterium* sp. HMLN12 361 JF519685 clade_172 N N *Bifidobacterium* sp. M45 362 HM626176 clade_172 N N *Bifidobacterium* sp. MSX5B 363 HQ616382 clade_172 N N *Bifidobacterium* sp. TM_7 364 AB218972 clade_172 N N *Bifidobacterium thermophilum* 365 DQ340557 clade_172 N N *Leuconostoc citreum* 1178 AM157444 clade_175 N N *Leuconostoc lactis* 1182 NR_040823 clade_175 N N *Eubacterium saburreum* 858 AB525414 clade_178 Y N *Eubacterium* sp. oral clone IR009 866 AY349376 clade_178 Y N Lachnospiraceae bacterium ICM62 1061 HQ616401 clade_178 Y N Lachnospiraceae bacterium MSX33 1062 HQ616384 clade_178 Y N Lachnospiraceae bacterium oral taxon 107 1063 ADDS01000069 clade_178 Y N *Alicyclobacillus acidocaldarius* 122 NR_074721 clade_179 Y N *Alicyclobacillus acidoterrestris* 123 NR_040844 clade_179 N N *Alicyclobacillus cycloheptanicus* 125 NR_024754 clade_179 N N *Acinetobacter baumannii* 27 ACYQ01000014 clade_181 N N *Acinetobacter calcoaceticus* 28 AM157426 clade_181 N N *Acinetobacter* genomosp. C1 29 AY278636 clade_181 N N *Acinetobacter haemolyticus* 30 ADMT01000017 clade_181 N N Acinetobacter johnsonii 31 ACPL01000162 clade_181 N N *Acinetobacter junii* 32 ACPM01000135 clade_181 N N *Acinetobacter lwoffii* 33 ACPN01000204 clade_181 N N *Acinetobacter parvus* 34 AIEB01000124 clade_181 N N *Acinetobacter schindleri* 36 NR_025412 clade_181 N N *Acinetobacter* sp. 56A1 37 GQ178049 clade_181 N N *Acinetobacter* sp. CIP 101934 38 JQ638573 clade_181 N N *Acinetobacter* sp. CIP 102143 39 JQ638578 clade_181 N N *Acinetobacter* sp. M16_22 41 HM366447 clade_181 N N *Acinetobacter* sp. RUH2624 42 ACQF01000094 clade_181 N N *Acinetobacter* sp. SH024 43 ADCH01000068 clade_181 N N *Lactobacillus jensenii* 1092 ACQD01000066 clade_182 N N *Alcaligenes faecalis* 119 AB680368 clade_183 N N *Alcaligenes* sp. CO14 120 DQ643040 clade_183 N N *Alcaligenes* sp. S3 121 HQ262549 clade_183 N N *Oligella ureolytica* 1366 NR_041998 clade_183 N N *Oligella urethralis* 1367 NR_041753

clade_183 N N *Eikenella corrodens* 784 ACEA01000028 clade_185 N N *Kingella denitrificans* 1019 AEWV01000047 clade_185 N N *Kingella* genomosp. P1 oral cone 1020 DQ003616 clade_185 N N MB2_C20 *Kingella kingae* 1021 AFHS01000073 clade_185 N N *Kingella oralis* 1022 ACJW02000005 clade_185 N N *Kingella* sp. oral clone ID059 1023 AY349381 clade_185 N N *Neisseria elongate* 1330 ADBF01000003 clade_185 N N *Neisseria* genomosp. P2 oral clone 1332 DQ003630 clade_185 N N MB5_P15 *Neisseria* sp. oral clone JC012 1345 AY349388 clade_185 N N *Neisseria* sp. SMC_A9199 1342 FJ763637 clade_185 N N *Simonsiella muelleri* 1731 ADCY01000105 clade_185 N N *Corynebacterium glucuronolyticum* 700 ABYP01000081 clade_193 N N *Corynebacterium pyruviciproducens* 716 FJ185225 clade_193 N N *Rothia aeria* 1649 DQ673320 clade_194 N N *Rothia dentocariosa* 1650 ADDW01000024 clade_194 N N *Rothia* sp. oral taxon 188 1653 GU470892 clade_194 N N *Corynebacterium accolens* 681 ACGD01000048 clade_195 N N *Corynebacterium macginleyi* 707 AB359393 clade_195 N N *Corynebacterium pseudogenitalium* 714 ABYQ01000237 clade_195 N N *Corynebacterium tuberculostearicum* 729 ACVP01000009 clade_195 N N *Lactobacillus casei* 1074 CP000423 clade_198 N N *Lactobacillus paracasei* 1106 ABQV01000067 clade_198 N N *Lactobacillus zeae* 1143 NR_037122 clade_198 N N *Prevotella dentalis* 1492 AB547678 clade_205 N N *Prevotella* sp. oral clone ASCG10 1529 AY923148 clade_206 N N *Prevotella* sp. oral clone HF050 1541 AY349399 clade_206 N N *Prevotella* sp. oral clone ID019 1542 AY349400 clade_206 N N *Prevotella* sp. oral clone IK062 1545 AY349402 clade_206 N N *Prevotella* genomosp. P9 oral clone MB7_G16 1499 DQ003633 clade_207 N N *Prevotella* sp. oral clone AU069 1531 AY005062 clade_207 N N *Prevotella* sp. oral clone CY006 1532 AY005063 clade_207 N N *Prevotella* sp. oral clone FL019 1534 AY349392 clade_207 N N *Actinomyces* genomosp. C1 56 AY278610 clade_212 N N *Actinomyces* genomosp. C2 57 AY278611 clade_212 N N *Actinomyces* genomosp. P1 oral clone MB6_C03 58 DQ003632 clade_212 N N *Actinomyces georgiae* 59 GU561319 clade_212 N N *Actinomyces israelii* 60 AF479270 clade_212 N N *Actinomyces massiliensis* 61 AB545934 clade_212 N N *Actinomyces meyeri* 62 GU561321 clade_212 N N *Actinomyces odontolyticus* 66 ACYT01000123 clade_212 N N *Actinomyces orihominis* 68 AJ575186 clade_212 N N *Actinomyces* sp. CCUG 37290 71 AJ234058 clade_212 N N *Actinomyces* sp. ICM34 75 HQ616391 clade_212 N N *Actinomyces* sp. ICM41 76 HQ616392 clade_212 N N *Actinomyces* sp. ICM47 77 HQ616395 clade_212 N N *Actinomyces* sp. ICM54 78 HQ616398 clade_212 N N *Actinomyces* sp. oral clone IP081 87 AY349366 clade_212 N N *Actinomyces* sp. oral taxon 178 91 AEUH01000060 clade_212 N N *Actinomyces* sp. oral taxon 180 92 AEPP01000041 clade_212 N N *Actinomyces* sp. TeJ5 80 GU561315 clade_212 N N *Haematobacter* sp. BC14248 968 GU396991 clade_213 N N *Paracoccus denitrificans* 1424 CP000490 clade_213 N N *Paracoccus marcusii* 1425 NR_044922 clade_213 N N *Grimontia hollisae* 967 ADAQ01000013 clade_216 N N *Shewanella putrefaciens* 1723 CP002457 clade_216 N N *Afipia* genomosp. 4 111 EU117385 clade_217 N N *Rhodopseudomonas palustris* 1626 CP000301 clade_217 N N *Methylobacterium extorquens* 1223 NC_010172 clade_218 N N *Methylobacterium podarium* 1224 AY468363 clade_218 N N *Methylobacterium radiotolerans* 1225 GU294320 clade_218 N N *Methylobacterium* sp. 1sub 1226 AY468371 clade_218 N N *Methylobacterium* sp. MM4 1227 AY468370 clade_218 N N *Clostridium baratii* 555 NR_029229 clade_223 Y N *Clostridium colicanis* 576 FJ957863 clade_223 Y N *Clostridium paraputrificum* 611 AB536771 clade_223 Y N *Clostridium sardiniense* 621 NR_041006 clade_223 Y N *Eubacterium budayi* 837 NR_024682 clade_223 Y N *Eubacterium moniliforme* 851 HF558373 clade_223 Y N *Eubacterium multiforme* 852 NR_024683 clade_223 Y N *Eubacterium nitritogenes* 853 NR_024684 clade_223 Y N *Achromobacter denitrificans* 18 NR_042021 clade_224 N N *Achromobacter piechaudii* 19 ADMS01000149 clade_224 N N *Achromobacter xylosoxidans* 20 ACRC01000072 clade_224 N N *Bordetella bronchiseptica* 384 NR_025949 clade_224 N OP *Bordetella holmesii* 385 AB683187 clade_224 N OP *Bordetella parapertussis* 386 NR_025950 clade_224 N OP *Bordetella pertussis* 387 BX640418 clade_224 N OP *Microbacterium chocolatum* 1230 NR_037045 clade_225 N N *Microbacterium flavescens* 1231 EU714363 clade_225 N N *Microbacterium lacticum* 1233 EU714351 clade_225 N N *Microbacterium oleivorans* 1234 EU714381 clade_225 N N *Microbacterium oxydans* 1235 EU714348 clade_225 N N *Microbacterium paraoxydans* 1236 AJ491806 clade_225 N N *Microbacterium phyllosphaerae* 1237 EU714359 clade_225 N N *Microbacterium schleiferi* 1238 NR_044936 clade_225 N N *Microbacterium* sp. 768 1239 EU714378 clade_225 N N *Microbacterium* sp. oral strain C24KA 1240 AF287752 clade_225 N N *Microbacterium testaceum* 1241 EU714365 clade_225 N N *Corynebacterium atypicum* 686 NR_025540 clade_229 N N *Corynebacterium mastitidis* 708 AB359395 clade_229 N N *Corynebacterium* sp. NML 97_0186 725 GU238411 clade_229 N N *Mycobacterium elephantis* 1275 AF385898 clade_237 N OP *Mycobacterium paraterrae* 1288 EU919229 clade_237 N OP *Mycobacterium phlei* 1289 GU142920 clade_237 N OP *Mycobacterium* sp. 1776 1293 EU703152 clade_237 N N *Mycobacterium* sp. 1781 1294 EU703147 clade_237 N N *Mycobacterium* sp. AQ1GA4 1297 HM210417 clade_237 N N *Mycobacterium* sp. GN_10546 1299 FJ497243 clade_237 N N *Mycobacterium* sp. GN_10827 1300 FJ497247 clade_237 N N *Mycobacterium* sp. GN_11124 1301 FJ652846 clade_237 N N *Mycobacterium* sp. GN_9188 1302 FJ497240 clade_237 N N *Mycobacterium* sp. GR_2007_210 1303 FJ555538 clade_237 N N *Anoxybacillus contaminans* 172 NR_029006 clade_238 N N *Anoxybacillus flavithermus* 173 NR_074667 clade_238 Y N *Bacillus aeolius* 195 NR_025557 clade_238 N N *Bacillus aerophilus* 196 NR_042339 clade_238 Y N *Bacillus aestuarii* 197 GQ980243 clade_238 Y N *Bacillus amyloliquefaciens* 199 NR_075005 clade_238 Y N *Bacillus anthracis* 200 AAEN01000020 clade_238 Y Category-A *Bacillus atrophaeus* 201 NR_075016 clade_238 Y OP *Bacillus badius* 202 NR_036893 clade_238 Y OP Bacillus cereus 203 ABDJ01000015 clade_238 Y OP *Bacillus circulans* 204 AB271747 clade_238 Y OP *Bacillus firmus* 207 NR_025842 clade_238 Y OP *Bacillus flexus* 208 NR_024691 clade_238 Y OP *Bacillus fordii* 209 NR_025786 clade_238 Y OP *Bacillus halmapalus* 211 NR_026144 clade_238 Y OP *Bacillus herbersteinensis* 213 NR_042286 clade_238 Y OP *Bacillus idriensis* 215 NR_043268 clade_238 Y OP *Bacillus lentus* 216 NR_040792 clade_238 Y OP *Bacillus licheniformis* 217 NC_006270 clade_238 Y OP *Bacillus megaterium* 218 GU252124 clade_238 Y OP *Bacillus nealsonii* 219 NR_044546 clade_238 Y OP *Bacillus niabensis* 220 NR_043334 clade_238 Y OP *Bacillus niacini* 221 NR_024695 clade_238 Y OP *Bacillus pocheonensis* 222 NR_041377 clade_238 Y OP *Bacillus pumilus* 223 NR_074977 clade_238 Y OP *Bacillus safensis* 224 JQ624766 clade_238 Y OP *Bacillus simplex* 225 NR_042136 clade_238 Y OP *Bacillus sonorensis* 226 NR_025130 clade_238 Y OP *Bacillus* sp. 10403023 MM10403188 227 CAET01000089 clade_238 Y OP *Bacillus* sp. 2_A_57_CT2 230 ACWD01000095 clade_238 Y OP *Bacillus* sp. 2008724126 228 GU252108 clade_238 Y OP *Bacillus* sp. 2008724139 229 GU252111 clade_238 Y OP *Bacillus* sp. 7_16AIA 231 FN397518 clade_238 Y OP *Bacillus* sp. AP8 233 JX101689 clade_238 Y OP *Bacillus* sp. B27(2008) 234 EU362173 clade_238 Y OP *Bacillus* sp. BT1B_CT2 235 ACWC01000034 clade_238 Y OP *Bacillus* sp. GB1.1 236 FJ897765 clade_238 Y OP *Bacillus* sp. GB9 237 FJ897766 clade_238 Y OP *Bacillus* sp. HU19.1 238 FJ897769 clade_238 Y OP *Bacillus* sp. HU29 239 FJ897771 clade_238 Y OP *Bacillus* sp. HU33.1 240 FJ897772 clade_238 Y OP *Bacillus* sp. JC6 241 JF824800 clade_238 Y OP *Bacillus* sp. oral taxon F79 248 HM099654 clade_238 Y OP *Bacillus* sp. SRC_DSF1 243 GU797283 clade_238 Y OP *Bacillus* sp. SRC_DSF10 242 GU797292 clade_238 Y OP *Bacillus* sp. SRC_DSF2 244 GU797284 clade_238 Y OP *Bacillus* sp. SRC_DSF6 245 GU797288 clade_238 Y OP *Bacillus* sp. tc09 249 HQ844242 clade_238 Y OP *Bacillus* sp. zh168 250 FJ851424 clade_238 Y OP *Bacillus sphaericus* 251 DQ286318 clade_238 Y OP *Bacillus sporothermodurans* 252 NR_026010 clade_238 Y OP *Bacillus subtilis* 253 EU627588 clade_238 Y OP *Bacillus thermoamylovorans* 254 NR_029151 clade_238 Y OP *Bacillus thuringiensis* 255 NC_008600 clade_238 Y OP *Bacillus weihenstephanensis* 256 NR_074926 clade_238 Y OP *Brevibacterium frigoritolerans* 422 NR_042639 clade_238 N N *Geobacillus kaustophilus* 933 NR_074989 clade_238 Y N *Geobacillus* sp. E263 934 DQ647387 clade_238 N N *Geobacillus* sp. WCH70 935 CP001638 clade_238 N N *Geobacillus stearothermophilus* 936 NR_040794 clade_238 Y N *Geobacillus thermocatenulatus* 937 NR_043020 clade_238 N N *Geobacillus thermodenitrificans* 938 NR_074976 clade_238 Y N *Geobacillus thermoglucosidasius* 939 NR_043022 clade_238 Y N *Geobacillus thermoleovorans* 940 NR_074931 clade_238 N N *Lysinibacillus fusiformis* 1192 FN397522 clade_238 N N *Lysinibacillus sphaericus* 1193 NR_074883 clade_238 Y N *Planomicrobium koreense* 1468 NR_025011 clade_238 N N *Sporosarcina newyorkensis* 1754 AFPZ01000142 clade_238 N N *Sporosarcina* sp. 2681 1755 GU994081 clade_238 N N *Ureibacillus composti* 1968 NR_043746 clade_238 N N *Ureibacillus suwonensis* 1969 NR_043232 clade_238 N N *Ureibacillus terrenus* 1970

NR_025394 clade_238 N N *Ureibacillus thermophilus* 1971 NR_043747 clade_238 N N *Ureibacillus thermosphaericus* 1972 NR_040961 clade_238 N N *Prevotella micans* 1507 AGWK01000061 clade_239 N N *Prevotella* sp. oral clone DA058 1533 AY005065 clade_239 N N *Prevotella* sp. SEQ053 1523 JN867222 clade_239 N N *Treponema socranskii* 1937 NR_024868 clade_240 N OP *Treponema* sp. 6:H:D15A_4 1938 AY005083 clade_240 N N *Treponema* sp. oral taxon 265 1953 GU408850 clade_240 N N *Treponema* sp. oral taxon G85 1958 GU432215 clade_240 N N *Porphyromonas endodontalis* 1472 ACNN01000021 clade_241 N N *Porphyromonas* sp. oral clone BB134 1478 AY005068 clade_241 N N *Porphyromonas* sp. oral clone F016 1479 AY005069 clade_241 N N *Porphyromonas* sp. oral clone P2PB_52 P1 1480 AY207054 clade_241 N N *Porphyromonas* sp. oral clone P4GB_100 P2 1481 AY207057 clade_241 N N *Acidovorax* sp. 98_63833 26 AY258065 clade_245 N N Comamonadaceae bacterium NML000135 663 JN585335 clade_245 N N Comamonadaceae bacterium NML790751 664 JN585331 clade_245 N N Comamonadaceae bacterium NML910035 665 JN585332 clade_245 N N Comamonadaceae bacterium NML910036 666 JN585333 clade_245 N N *Comamonas* sp. NSP5 668 AB076850 clade_245 N N *Delftia acidovorans* 748 CP000884 clade_245 N N *Xenophilus aerolatus* 2018 JN585329 clade_245 N N Clostridiales sp. SS3/4 543 AY305316 clade_246 Y N *Oribacfcerium* sp. oral taxon 078 1380 ACIQ02000009 clade_246 N N *Oribacterium* sp. oral taxon 102 1381 GQ422713 clade_246 N N *Weissella cibaria* 2007 NR_036924 clade_247 N N *Weissella confusa* 2008 NR_040816 clade_247 N N *Weissella hellenica* 2009 AB680902 clade_247 N N *Weissella kandleri* 2010 NR_044659 clade_247 N N *Weissella koreensis* 2011 NR_075058 clade_247 N N *Weissella paramesenteroides* 2012 ACKU01000017 clade_247 N N *Weissella* sp. KLDS 7.0701 2013 EU600924 clade_247 N N *Mobiluncus curtisii* 1251 AEPZ01000013 clade_249 N N *Clostridium beijerinckii* 557 NR_074434 clade_252 Y N *Clostridium botulinum* 560 NC_010723 clade_252 Y Category-A *Clostridium butyricum* 561 ABDT01000017 clade_252 Y N *Clostridium chauvoei* 568 EU106372 clade_252 Y N *Clostridium favososporum* 582 X76749 clade_252 Y N *Clostridium histolyticum* 592 HF558362 clade_252 Y N *Clostridium isatidis* 597 NR_026347 clade_252 Y N *Clostridium limosum* 602 FR870444 clade_252 Y N *Clostridium sartagoforme* 622 NR_026490 clade_252 Y N *Clostridium septicum* 624 NR_026020 clade_252 Y N *Clostridium* sp. 7_2_43FAA 626 ACDK01000101 clade_252 Y N *Clostridium sporogenes* 645 ABKW02000003 clade_252 Y N *Clostridium tertium* 653 Y18174 clade_252 Y N *Clostridium carnis* 564 NR_044716 clade_253 Y N *Clostridium celatum* 565 X77844 clade_253 Y N *Clostridium disporicum* 579 NR_026491 clade_253 Y N *Clostridium gasigenes* 585 NR_024945 clade_253 Y N *Clostridium quinii* 616 NR_026149 clade_253 Y N *Enhydrobacter aerosaccus* 785 ACYI01000081 clade_256 N N *Moraxella osloensis* 1262 JN175341 clade_256 N N *Moraxella* sp. GM2 1264 JF837191 clade_256 N N *Brevibacterium casei* 420 JF951998 clade_257 N N *Brevibacterium epidermidis* 421 NR_029262 clade_257 N N *Brevibacterium sanguinis* 426 NR_028016 clade_257 N N *Brevibacterium* sp. H15 427 AB177640 clade_257 N N *Clostridium hylemonae* 593 AB023973 clade_260 Y N *Clostridium scindens* 623 AF262238 clade_260 Y N Lachnospiraceae bacterium 5_1_57FAA 1054 ACTR01000020 clade_260 Y N *Acinetobacter radioresistens* 35 ACVR01000010 clade_261 N N *Clostridium glycyrrhizinilyticum* 588 AB233029 clade_262 Y N *Clostridium nexile* 607 X73443 clade_262 Y N *Coprococcus comes* 674 ABVR01000038 clade_262 Y N Lachnospiraceae bacterium 1_1_57FAA 1048 ACTM01000065 clade_262 Y N Lachnospiraceae bacterium 1_4_56FAA 1049 ACTN01000028 clade_262 Y N Lachnospiraceae bacterium 8_1_57FAA 1057 ACWQ01000079 clade_262 Y N *Ruminococcus lactaris* 1663 ABOU02000049 clade_262 Y N *Ruminococcus torques* 1670 AAVP02000002 clade_262 Y N *Lactobacillus alimentarius* 1068 NR_044701 clade_263 N N *Lactobacillus farciminis* 1082 NR_044707 clade_263 N N *Lactobacillus kimchii* 1097 NR_025045 clade_263 N N *Lactobacillus nodensis* 1101 NR_041629 clade_263 N N *Lactobacillus tucceti* 1138 NR_042194 clade_263 N N *Pseudomonas mendocina* 1595 AAUL01000021 clade_265 N N *Pseudomonas pseudoalcaligenes* 1598 NR_037000 clade_265 N N *Pseudomonas* sp. NP522b 1602 EU723211 clade_265 N N *Pseudomonas stutzeri* 1603 AM905854 clade_265 N N *Paenibacillus barcinonensis* 1390 NR_042272 clade_270 N N *Paenibacillus barengoltzii* 1391 NR_042756 clade_270 N N *Paenibacillus chibensis* 1392 NR_040885 clade_270 N N *Paenibacillus cookii* 1393 NR_025372 clade_270 N N *Paenibacillus durus* 1394 NR_037017 clade_270 N N *Paenibacillus glucanolyticus* 1395 D78470 clade_270 N N *Paenibacillus lactis* 1396 NR_025739 clade_270 N N *Paenibacillus lautus* 1397 NR_040882 clade_270 Y N *Paenibacillus pabuli* 1398 NR_040853 clade_270 N N *Paenibacillus polymyxa* 1399 NR_037006 clade_270 Y N *Paenibacillus popilliae* 1400 NR_040888 clade_270 N N *Paenibacillus* sp. CIP 101062 1401 HM212646 clade_270 N N *Paenibacillus* sp. HGF5 1402 AEXS01000095 clade_270 Y N *Paenibacillus* sp. HGF7 1403 AFDH01000147 clade_270 Y N *Paenibacillus* sp. JC66 1404 JF824808 clade_270 N N *Paenibacillus* sp. R_27413 1405 HE586333 clade_270 N N *Paenibacillus* sp. 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HKU31 74 HQ335393 clade_303 N N *Actinomyces* sp. oral taxon C55 94 HM099646 clade_303 N N *Kerstersia gyiorum* 1018 NR_025669 clade_307 N N *Pigmentiphaga daeguensis* 1467 JN585327 clade_307 N N *Aeromonas allosaccharophila* 104 S39232 clade_308 N N *Aeromonas enteropelogenes* 105 X71121 clade_308 N N *Aeromonas hydrophila* 106 NC_008570 clade_308 N N *Aeromonas jandaei* 107 X60413 clade_308 N N *Aeromonas salmonicida* 108 NC_009348 clade_308 N N *Aeromonas trota* 109 X60415 clade_308 N N *Aeromonas veronii* 110 NR_044845 clade_308 N N *Blautia coccoides* 373 AB571656 clade_309 Y N *Blautia glucerasea* 374 AB588023 clade_309 Y N *Blautia glucerasei* 375 AB439724 clade_309 Y N *Blautia hansenii* 376 ABYU02000037 clade_309 Y N *Blautia luti* 378 AB691576 clade_309 Y N *Blautia producta* 379 AB600998 clade_309 Y N *Blautia schinkii* 380 NR_026312 clade_309 Y N *Blautia* sp. M25 381 HM626178 clade_309 Y N *Blautia stercoris* 382 HM626177 clade_309 Y N *Blautia wexlerae* 383 EF036467 clade_309 Y N *Bryantella formatexigens* 439

ACCL02000018 clade_309 Y N *Clostridium coccoides* 573 EF025906 clade_309 Y N *Eubacterium cellulosolvens* 839 AY178842 clade_309 Y N Lachnospiraceae bacterium 6_1_63FAA 1056 ACTV01000014 clade_309 Y N *Marvinbryantia formatexigens* 1196 AJ505973 clade_309 N N *Ruminococcus hansenii* 1662 M59114 clade_309 Y N *Ruminococcus obeum* 1664 AY169419 clade_309 Y N *Ruminococcus* sp. 5_1_39BFAA 1666 ACII01000172 clade_309 Y N *Ruminococcus* sp. K_1 1669 AB222208 clade_309 Y N *Syntrophococcus sucromutans* 1911 NR_036869 clade_309 Y N *Rhodobacter* sp. oral taxon C30 1620 HM099648 clade_310 N N *Rhodobacter sphaeroides* 1621 CP000144 clade_310 N N *Lactobacillus antri* 1071 ACLL01000037 clade_313 N N *Lactobacillus coleohominis* 1076 ACOH01000030 clade_313 N N *Lactobacillus fermentum* 1083 CP002033 clade_313 N N *Lactobacillus gastricus* 1085 AICN01000060 clade_313 N N *Lactobacillus mucosae* 1099 FR693800 clade_313 N N *Lactobacillus oris* 1103 AEKL01000077 clade_313 N N *Lactobacillus pontis* 1111 HM218420 clade_313 N N *Lactobacillus reuteri* 1112 ACGW02000012 clade_313 N N *Lactobacillus* sp. KLDS 1.0707 1127 EU600911 clade_313 N N *Lactobacillus* sp. KLDS 1.0709 1128 EU600913 clade_313 N N *Lactobacillus* sp. KLDS 1.0711 1129 EU600915 clade_313 N N *Lactobacillus* sp. KLDS 1.0713 1131 EU600917 clade_313 N N *Lactobacillus* sp. KLDS 1.0716 1132 EU600921 clade_313 N N *Lactobacillus* sp. 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KLDS 1.0705 1126 EU600909 clade_315 N N *Agrobacterium radiobacter* 115 CP000628 clade_316 N N *Agrobacterium tumefaciens* 116 AJ389893 clade_316 N N *Corynebacterium argentoratense* 685 EF463055 clade_317 N N *Corynebacterium diphtheriae* 693 NC_002935 clade_317 N OP *Corynebacterium pseudotuberculosis* 715 NR_037070 clade_317 N N *Corynebacterium renale* 717 NR_037069 clade_317 N N *Corynebacterium ulcerans* 731 NR_074467 clade_317 N N *Aurantimonas coralicida* 191 AY065627 clade_318 N N *Aureimonas altamirensis* 192 FN658986 clade_318 N N *Lactobacillus acidipiscis* 1066 NR_024718 clade_320 N N *Lactobacillus salivarius* 1117 AEBA01000145 clade_320 N N *Lactobacillus* sp. KLDS 1.0719 1134 EU600923 clade_320 N N *Lactobacillus buchneri* 1073 ACGH01000101 clade_321 N N *Lactobacillus* genomosp. C1 1086 AY278619 clade_321 N N *Lactobacillus* genomosp. 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C1 1166 AY278621 clade_334 N N *Leptotrichia shahii* 1169 AY029806 clade_334 N N *Leptotrichia* sp. neutropenicPatient 1170 AF189244 clade_334 N N *Leptotrichia* sp. oral clone GT018 1171 AY349384 clade_334 N N *Leptotrichia* sp. oral clone GT020 1172 AY349385 clade_334 N N *Bacteroides* sp. 20_3 296 ACRQ01000064 clade_335 N N *Bacteroides* sp. 3_1_19 307 ADCJ01000062 clade_335 N N *Bacteroides* sp. 3_2_5 311 ACIB01000079 clade_335 N N *Parabacteroides distasonis* 1416 CP000140 clade_335 N N *Parabacteroides goldsteinii* 1417 AY974070 clade_335 N N *Parabacteroides gordonii* 1418 AB470344 clade_335 N N *Parabacteroides* sp. D13 1421 ACPW01000017 clade_335 N N *Capnocytophaga* genomosp. C1 477 AY278613 clade_336 N N *Capnocytophaga ochracea* 480 AEOH01000054 clade_336 N N *Capnocytophaga* sp. GEJ8 481 GU561335 clade_336 N N *Capnocytophaga* sp. oral strain A47ROY 486 AY005077 clade_336 N N *Capnocytophaga* sp. 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P6 oral clone MB4_G11 1930 DQ003625 clade_339 N N *Treponema* sp. oral taxon 254 1952 GU408803 clade_339 N N *Treponema* sp. oral taxon 508 1956 GU413616 clade_339 N N *Treponema* sp. oral taxon 518 1957 GU413640 clade_339 N N *Chlamydia muridarum* 502 AE002160 clade_341 N OP *Chlamydia trachomatis* 504 U68443 clade_341 N OP *Chlamydia psittaci* 503 NR_036864 clade_342 N Category-B *Chiamydophila pneumoniae* 509 NC_002179 clade_342 N OP *Chlamydophila psittaci* 510 D85712 clade_342 N OP *Anaerococcus octavius* 146 NR_026360 clade_343 N N *Anaerococcus* sp. 8405254 149 HM587319 clade_343 N N *Anaerococcus* sp. 9401487 150 HM587322 clade_343 N N *Anaerococcus* sp. 9403502 151 HM587325 clade_343 N N *Gardnerella vaginalis* 923 CP001849 clade_344 N N *Campylobacter lari* 466 CP000932 clade_346 N OP *Anaerobiospirillum succiniciproducens* 142 NR_026075 clade_347 N N *Anaerobiospirillum thomasii* 143 AJ420985 clade_347 N N *Ruminobacter amylophilus* 1654 NR_026450 clade_347 N N *Succinatimonas hippei* 1897 AEVO01000027 clade_347 N N *Actinomyces europaeus* 54 NR_026363 clade_348 N N *Actinomyces* sp. oral clone GU009 82 AY349361 clade_348 N N *Moraxella catarrhalis* 1260 CP002005 clade_349 N N *Moraxella lincolnii* 1261 FR822735 clade_349 N N *Moraxella* sp. 16285 1263 JF682466 clade_349 N N *Psychrobacter* sp. 13983 1613 HM212668 clade_349 N N *Actinobaculum massiliae* 49 AF487679 clade_350 N N *Actinobaculum schaalii* 50 AY957507 clade_350 N N *Actinobaculum* sp. 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FJ159523 clade_354 Y N *Eubacterium tenue* 872 M59118 clade_354 Y N *Clostridium argentinense* 553 NR_029232 clade_355 Y N *Clostridium* sp. JC122 630 CAEV01000127 clade_355 Y N *Clostridium* sp. NMBHI_1 636 JN093130 clade_355 Y N *Clostridium subterminale* 650 NR_041795 clade_355 Y N *Clostridium sulfidigenes* 651 NR_044161 clade_355 Y N *Blastomonas natatoria* 372 NR_040824 clade_356 N N *Novospbingobium aromaticivorans* 1357 AAAV03000008 clade_356 N N *Sphingomonas* sp. oral clone FI012 1745 AY349411 clade_356 N N *Sphingopyxis alaskensis* 1749 CP000356 clade_356 N N *Oxalobacter formigenes* 1389 ACDQ01000020 clade_357 N N *Veillonella atypica* 1974 AEDS01000059 clade_358 N N *Veillonella dispar* 1975 ACIK02000021 clade_358 N N *Veillonella* genomosp. 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BT6 1121 HQ616370 clade_373 N N *Mycobacterium mageritense* 1282 FR798914 clade_374 N OP *Mycobacterium neoaurum* 1286 AF268445 clade_374 N OP *Mycobacterium smegmatis* 1291 CP000480 clade_374 N OP *Mycobacterium* sp. 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NB_8 323 AB117565 clade_378 N N *Bacteroides vulgatus* 331 CP000139 clade_378 N N *Bacteroides ovatus* 287 ACWH01000036 clade_38 N N *Bacteroides* sp. 1_1_30 294 ADCL01000128 clade_38 N N *Bacteroides* sp. 2_1_22 297 ACPQ01000117 clade_38 N N *Bacteroides* sp. 2_2_4 299 ABZZ01000168 clade_38 N N *Bacteroides* sp. 3_1_23 308 ACRS01000081 clade_38 N N *Bacteroides* sp. D1 318 ACAB02000030 clade_38 N N *Bacteroides* sp. D2 321 ACGA01000077 clade_38 N N *Bacteroides* sp. D22 320 ADCK01000151 clade_38 N N *Bacteroides xylanisolvens* 332 ADKP01000087 clade_38 N N *Treponema lecithinolyticum* 1931 NR_026247 clade_380 N OP *Treponema parvum* 1933 AF302937 clade_380 N OP *Treponema* sp. oral clone JU025 1940 AY349417 clade_380 N N *Treponema* sp. oral taxon 270 1954 GQ422733 clade_380 N N *Parascardovia denticolens* 1428 ADEB01000020 clade_381 N N *Scardovia inopinata* 1688 AB029087 clade_381 N N *Scardovia wiggsiae* 1689 AY278626 clade_381 N N Clostridiales bacterium 9400853 533 HM587320 clade_384 N N *Eubacterium infirmum* 849 U13039 clade_384 Y N *Eubacterium* sp. WAL 14571 864 FJ687606 clade_384 Y N *Mogibacterium diversum* 1254 NR_027191 clade_384 N N *Mogibacterium neglectum* 1255 NR_027203 clade_384 N N *Mogibacterium pumilum* 1256 NR_028608 clade_384 N N *Mogibacterium timidum* 1257 Z36296 clade_384 N N Erysipeiotrichaceae bacterium 5_2_54FAA 823 ACZW01000054 clade_385 Y N *Eubacterium biforme* 835 ABYT01000002 clade_385 Y N *Eubacterium cylindroides* 842 FP929041 clade_385 Y N *Eubacterium dolichum* 844 L34682 clade_385 Y N *Eubacterium* sp. 3_1_31 861 ACTL01000045 clade_385 Y N *Eubacterium tortuosum* 873 NR_044648 clade_385 Y N *Borrelia burgdorferi* 389 ABGI01000001 clade_386 N OP *Borrelia garinii* 392 ABJV01000001 clade_386 N OP *Borrelia* sp. NE49 397 AJ224142 clade_386 N OP *Caldimonas manganoxidans* 457 NR_040787 clade_387 N N Comamonadaceae bacterium oral taxon F47 667 HM099651 clade_387 N N *Lautropia mirabilis* 1149 AEQP01000026 clade_387 N N *Lautropia* sp. oral clone AP009 1150 AY005030 clade_387 N N *Bulleidia extructa* 441 ADFR01000011 clade_388 Y N *Solobacterium moorei* 1739 AECQ01000039 clade_388 Y N *Peptoniphilus asaccharolyticus* 1441 D14145 clade_389 N N *Peptoniphilus duerdenii* 1442 EU526290 clade_389 N N *Peptoniphilus harei* 1443 NR_026358 clade_389 N N *Peptoniphilus indolicus* 1444 AY153431 clade_389 N N *Peptoniphilus lacrimalis* 1446 ADDO01000050 clade_389 N N Peptoniphilus sp. gpac077 1450 AM176527 clade_389 N N *Peptoniphilus* sp. JC140 1447 JF824803 clade_389 N N *Peptoniphilus* sp. oral taxon 386 1452 ADCS01000031 clade_389 N N *Peptoniphilus* sp. oral taxon 836 1453 AEAA01000090 clade_389 N N Peptostreptococcaceae bacterium ph1 1454 JN837495 clade_389 N N *Dialister pneumosintes* 765 HM596297 clade_390 N N *Dialister* sp. oral taxon 502 767 GQ422739 clade_390 N N *Cupriavidus metallidurans* 741 GU230889 clade_391 N N *Herbaspirillum seropedicae* 1001 CP002039 clade_391 N N *Herbaspirillum* sp. JC206 1002 JN657219 clade_391 N N *Janthinobacterium* sp. SY12 1015 EF455530 clade_391 N N *Massilia* sp. CCUG 43427A 1197 FR773700 clade_391 N N *Ralstonia pickettii* 1615 NC_010682 clade_391 N N *Ralstonia* sp. 5_7_47FAA 1616 ACUF01000076 clade_391 N N *Francisella novicida* 889 ABSS01000002 clade_392 N N *Francisella philomiragia* 890 AY928394 clade_392 N N *Francisella tularensis* 891 ABAZ01000082 clade_392 N Category-A *Ignatzschineria indica* 1009 HQ823562 clade_392 N N *Ignatzschineria* sp. NML 95_0260 1010 HQ823559 clade_392 N N *Coprococcus catus* 673 EU266552 clade_393 Y N Lachnospiraceae bacterium oral taxon F15 1064 HM099641 clade_393 Y N *Streptococcus mutans* 1814 AP010655 clade_394 N N *Clostridium cochlearium* 574 NR_044717 clade_395 Y N *Clostridium malenominatum* 604 FR749893 clade_395 Y N *Clostridium tetani* 654 NC_004557 clade_395 Y N *Acetivibrio ethanolgignens* 6 FR749897 clade_396 Y N *Anaerosporobacter mobilis* 161 NR_042953 clade_396 Y N *Bacteroides pectinophilus* 288 ABVQ01000036 clade_396 Y N *Clostridium aminovalericum* 551 NR_029245 clade_396 Y N *Clostridium phytofermentans* 613 NR_074652 clade_396 Y N *Eubacterium hallii* 848 L34621 clade_396 Y N *Eubacterium xylanophilum* 875 L34628 clade_396 Y N *Lactobacillus gasseri* 1084 ACOZ01000018 clade_398 N N *Lactobacillus hominis* 1090 FR681902 clade_398 N N *Lactobacillus iners* 1091 AEKJ01000002 clade_398 N N *Lactobacillus johnsonii* 1093 AE017198 clade_398 N N *Lactobacillus senioris* 1119 AB602570 clade_398 N N *Lactobacillus* sp. oral clone HT002 1135 AY349382 clade_398 N N *Weissella beninensis* 2006 EU439435 clade_398 N N *Sphingomonas echinoides* 1744 NR_024700 clade_399 N N *Sphingomonas* sp. oral taxon A09 1747 HM099639 clade_399 N N *Sphingomonas* sp. oral taxon F71 1748 HM099645 clade_399 N N *Zymomonas mobilis* 2032 NR_074274 clade_399 N N *Arcanobacterium haemolyticum* 174 NR_025347 clade_400 N N *Arcanobacterium pyogenes* 175 GU585578 clade_400 N N *Trueperella pyogenes* 1962 NR_044858 clade_400 N N *Lactococcus garvieae* 1144 AF061005 clade_401 N N *Lactococcus lactis* 1145 CP002365 clade_401 N N *Brevibacterium mcbrellneri* 424 ADNU01000076 clade_402 N N *Brevibacterium paucivorans* 425 EU086796 clade_402 N N *Brevibacterium* sp. JC43 428 JF824806 clade_402 N N

*Selenomonas artemidis* 1692 HM596274 clade_403 N N *Selenomonas* sp. FOBRC9 1704 HQ616378 clade_403 N N *Selenomonas* sp. oral taxon 137 1715 AENV01000007 clade_403 N N *Desmospora activa* 751 AM940019 clade_404 N N *Desmospora* sp. 8437 752 AFHT01000143 clade_404 N N *Paenibacillus* sp. oral taxon F45 1407 HM099647 clade_404 N N *Corynebacterium ammoniagenes* 682 ADNS01000011 clade_405 N N *Corynebacterium aurimucosum* 687 ACLH01000041 clade_405 N N *Corynebacterium bovis* 688 AF537590 clade_405 N N *Corynebacterium canis* 689 GQ871934 clade_405 N N *Corynebacterium casei* 690 NR_025101 clade_405 N N *Corynebacterium durum* 694 Z97069 clade_405 N N *Corynebacterium efficiens* 695 ACLI01000121 clade_405 N N *Corynebacterium falsenii* 696 Y13024 clade_405 N N *Corynebacterium flavescens* 697 NR_037040 clade_405 N N *Corynebacterium glutamicum* 701 BA000036 clade_405 N N *Corynebacterium jeikeium* 704 ACYW01000001 clade_405 N OP *Corynebacterium kroppenstedtii* 705 NR_026380 clade_405 N N *Corynebacterium lipophiloflavum* 706 ACHJ01000075 clade_405 N N *Corynebacterium matruchotii* 709 ACSH02000003 clade_405 N N *Corynebacterium minutissimum* 710 X82064 clade_405 N N *Corynebacterium resistens* 718 ADGN01000058 clade_405 N N *Corynebacterium simulans* 720 AF537604 clade_405 N N *Corynebacterium singulare* 721 NR_026394 clade_405 N N *Corynebacterium* sp. 1 ex sheep 722 Y13427 clade_405 N N *Corynebacterium* sp. NML 99_0018 726 GU238413 clade_405 N N *Corynebacterium striatum* 727 ACGE01000001 clade_405 N OP *Corynebacterium urealyticum* 732 X81913 clade_405 N OP *Corynebacterium variabile* 734 NR_025314 clade_405 N N *Ruminococcus callidus* 1658 NR_029160 clade_406 Y N *Ruminococcus champanellensis* 1659 FP929052 clade_406 Y N *Ruminococcus* sp. 18P13 1665 AJ515913 clade_406 Y N *Ruminococcus* sp. 9SE51 1667 FM954974 clade_406 Y N *Aerococcus sanguinicola* 98 AY837833 clade_407 N N *Aerococcus urinae* 99 CP002512 clade_407 N N *Aerococcus urinaeequi* 100 NR_043443 clade_407 N N *Aerococcus viridans* 101 ADNT01000041 clade_407 N N *Anaerostipes caccae* 162 ABAX03000023 clade_408 Y N *Anaerostipes* sp. 3_2_56FAA 163 ACWB01000002 clade_408 Y N Clostridiales bacterium 1_7_47FAA 541 ABQR01000074 clade_408 Y N Clostridiales sp. SM4_1 542 FP929060 clade_408 Y N Clostridiales sp. SSC_2 544 FP929061 clade_408 Y N *Clostridium aerotolerans* 546 X76163 clade_408 Y N *Clostridium aldenense* 547 NR_043680 clade_408 Y N *Clostridium algidixylanolyticum* 550 NR_028726 clade_408 Y N *Clostridium amygdalinum* 552 AY353957 clade_408 Y N *Clostridium asparagiforme* 554 ACCJ01000522 clade_408 Y N *Clostridium bolteae* 559 ABCC02000039 clade_408 Y N *Clostridium celerecrescens* 566 JQ246092 clade_408 Y N *Clostridium citroniae* 569 ADLJ01000059 clade_408 Y N *Clostridium clostridiiformes* 571 M59089 clade_408 Y N *Clostridium clostridioforme* 572 NR_044715 clade_408 Y N *Clostridium hathewayi* 590 AY552788 clade_408 Y N *Clostridium indolis* 594 AF028351 clade_408 Y N *Clostridium lavalense* 600 EF564277 clade_408 Y N *Clostridium saccharolyticum* 620 CP002109 clade_408 Y N *Clostridium* sp. M62_1 633 ACFX02000046 clade_408 Y N *Clostridium* sp. SS2_1 638 ABGC03000041 clade_408 Y N *Clostridium sphenoides* 643 X73449 clade_408 Y N *Clostridium symbiosum* 652 ADLQ01000114 clade_408 Y N *Clostridium xylanolyticum* 658 NR_037068 clade_408 Y N *Eubacterium hadrum* 847 FR749933 clade_408 Y N *Fusobacterium naviforme* 898 HQ223106 clade_408 N N Lachnospiraceae bacterium 3_1_57FAA 1052 ACTP01000124 clade_408 Y N Lachnospiraceae bacterium 5_1_63FAA 1055 ACTS01000081 clade_408 Y N Lachnospiraceae bacterium A4 1059 DQ789118 clade_408 Y N Lachnospiraceae bacterium DJF VP30 1060 EU728771 clade_408 Y N Lachnospiraceae genomosp. C1 1065 AY278618 clade_408 Y N *Moryella indoligenes* 1268 AF527773 clade_408 N N *Clostridium difficile* 578 NC_013315 clade_409 Y OP *Selenomonas* genomosp. P5 1697 AY341820 clade_410 N N *Selenomonas* sp. oral clone IQ048 1710 AY349408 clade_410 N N *Selenomonas sputigena* 1717 ACKP02000033 clade_410 N N *Hyphomicrobium sulfonivorans* 1007 AY468372 clade_411 N N *Methylocella silvestris* 1228 NR_074237 clade_411 N N *Legionella pneumophila* 1153 NC_002942 clade_412 N OP *Lactobacillus coryniformis* 1077 NR_044705 clade_413 N N *Arthrobacter agilis* 178 NR_026198 clade_414 N N *Arthrobacter arilaitensis* 179 NR_074608 clade_414 N N Arthrobacter bergerei 180 NR_025612 clade_414 N N *Arthrobacter globiformis* 181 NR_026187 clade_414 N N *Arthrobacter nicotianae* 182 NR_026190 clade_414 N N *Mycobacterium abscessus* 1269 AGQU01000002 clade_418 N OP *Mycobacterium chelonae* 1273 AB548610 clade_418 N OP *Bacteroides salanitronis* 291 CP002530 clade_419 N N *Paraprevotella xylaniphila* 1427 AFBR01000011 clade_419 N N *Barnesiella intestinihominis* 336 AB370251 clade_420 N N *Barnesiella viscericola* 337 NR_041508 clade_420 N N *Parabacteroides* sp. NS31_3 1422 JN029805 clade_420 N N Porphyromonadaceae bacterium NML 060648 1470 EF184292 clade_420 N N *Tannerella forsythia* 1913 CP003191 clade_420 N N *Tannerella* sp. 6_1_58FAA_CT1 1914 ACWX01000068 clade_420 N N *Mycoplasma amphoriforme* 1311 AY531656 clade_421 N N *Mycoplasma genitalium* 1317 L43967 clade_421 N N *Mycoplasma pneumoniae* 1322 NC_000912 clade_421 N N *Mycoplasma penetrans* 1321 NC_004432 clade_422 N N *Ureaplasma parvum* 1966 AE002127 clade_422 N N *Ureaplasma urealyticum* 1967 AAYN01000002 clade_422 N N *Treponema* genomosp. P1 1927 AY341822 clade_425 N N *Treponema* sp. oral taxon 228 1943 GU408580 clade_425 N N *Treponema* sp. oral taxon 230 1944 GU408603 clade_425 N N *Treponema* sp. oral taxon 231 1945 GU408631 clade_425 N N *Treponema* sp. oral taxon 232 1946 GU408646 clade_425 N N *Treponema* sp. oral taxon 235 1947 GU408673 clade_425 N N *Treponema* sp. ovine footrot 1959 AJ010951 clade_425 N N *Treponema vincentii* 1960 ACYH01000036 clade_425 N OP *Eubacterium* sp. AS15b 862 HQ616364 clade_428 Y N *Eubacterium* sp. OBRC9 863 HQ616354 clade_428 Y N *Eubacterium* sp. oral clone OH3A 871 AY947497 clade_428 Y N *Eubacterium yurii* 876 AEES01000073 clade_428 Y N *Clostridium acetobutylicum* 545 NR_074511 clade_430 Y N *Clostridium algidicarnis* 549 NR_041746 clade_430 Y N *Clostridium cadaveris* 562 AB542932 clade_430 Y N *Clostridium carboxidivorans* 563 FR733710 clade_430 Y N *Clostridium estertheticum* 580 NR_042153 clade_430 Y N *Clostridium fallax* 581 NR_044714 clade_430 Y N *Clostridium felsineum* 583 AF270502 clade_430 Y N *Clostridium frigidicarnis* 584 NR_024919 clade_430 Y N *Clostridium kluyveri* 598 NR_074165 clade_430 Y N *Clostridium magnum* 603 X77835 clade_430 Y N *Clostridium putrefaciens* 615 NR_024995 clade_430 Y N *Clostridium* sp. HPB_46 629 AY862516 clade_430 Y N *Clostridium tyrobutyricum* 656 NR_044718 clade_430 Y N *Burkholderiales bacterium* 1_1_47 452 ADCQ01000066 clade_432 N OP *Parasutterella excrementihominis* 1429 AFBP01000029 clade_432 N N *Parasutterella secunda* 1430 AB491209 clade_432 N N *Sutterella morbirenis* 1898 AJ832129 clade_432 N N *Sutterella parvirubra* 1899 AB300989 clade_432 Y N *Sutterella sanguinus* 1900 AJ748647 clade_432 N N *Sutterella* sp. YIT 12072 1901 AB491210 clade_432 N N *Sutterella stercoricanis* 1902 NR_025600 clade_432 N N *Sutterella wadsworthensis* 1903 ADMF01000048 clade_432 N N *Propionibacterium freudenreichii* 1572 NR_036972 clade_433 N N *Propionibacterium* sp. oral taxon 192 1580 GQ422728 clade_433 N N *Tessaracoccus* sp. oral taxon F04 1917 HM099640 clade_433 N N *Peptoniphilus ivorii* 1445 Y07840 clade_434 N N *Peptoniphilus* sp. gpac007 1448 AM176517 clade_434 N N *Peptoniphilus* sp. gpac018A 1449 AM176519 clade_434 N N *Peptoniphilus* sp. gpac148 1451 AM176535 clade_434 N N *Flexispira rappini* 887 AY126479 clade_436 N N *Helicobacter bilis* 993 ACDN01000023 clade_436 N N *Helicobacter cinaedi* 995 ABQT01000054 clade_436 N N *Helicobacter* sp. None 998 U44756 clade_436 N N *Brevundimonas subvibrioides* 429 CP002102 clade_438 N N *Hyphomonas neptunium* 1008 NR_074092 clade_438 N N *Phenylobacterium zucineum* 1465 AY628697 clade_438 N N *Acetanaerobaeterium elongatum* 4 NR_042930 clade_439 Y N *Clostridium cellulosi* 567 NR_044624 clade_439 Y N *Ethanoligenens harbinense* 832 AY675965 clade_439 Y N *Streptococcus downei* 1793 AEKN01000002 clade_441 N N *Streptococcus* sp. SHV515 1848 Y07601 clade_441 N N *Acinetobacter* sp. CIP 53.82 40 JQ638584 clade_443 N N *Halomonas elongata* 990 NR_074782 clade_443 N N *Halomonas johnsoniae* 991 FR775979 clade_443 N N *Butyrivibrio fibrisolvens* 456 U41172 clade_444 N N *Eubacterium rectale* 856 FP929042 clade_444 Y N *Eubacterium* sp. oral clone GI038 865 AY349374 clade_444 Y N *Lachnobacterium bovis* 1045 GU324407 clade_444 Y N *Roseburia cecicola* 1634 GU233441 clade_444 Y N *Roseburia faecalis* 1635 AY804149 clade_444 Y N *Roseburia faecis* 1636 AY305310 clade_444 Y N *Roseburia hominis* 1637 AJ270482 clade_444 Y N *Roseburia intestinalis* 1638 FP929050 clade_444 Y N *Roseburia inulinivorans* 1639 AJ270473 clade_444 Y N *Roseburia* sp. 11SE37 1640 FM954975 clade_444 N N *Roseburia* sp. 11SE38 1641 FM954976 clade_444 N N *Shuttleworthia satelles* 1728 ACIP02000004 clade_444 N N *Shuttleworthia* sp. MSX8B 1729 HQ616383 clade_444 N N *Shuttleworthia* sp. oral taxon G69 1730 GU432167 clade_444 N N *Bdellovibrio* sp. MPA 344 AY294215 clade_445 N N *Desulfobulbus* sp. oral clone CH031 755 AY005036 clade_445 N N *Desulfovibrio* *desulfuricans* 757 DQ092636 clade_445 N N *Desulfovibrio fairfieldensis* 758 U42221 clade_445 N N *Desulfovibrio piger* 759 AF192152 clade_445 N N *Desulfovibrio* sp. 3_1_syn3 760 ADDR01000239 clade_445 N N *Geobacter bemidjiensis* 941 CP001124 clade_445 N N *Brachybacterium alimentarium* 401 NR_026269 clade_446 N N *Brachybacterium conglomeratum* 402 AB537169 clade_446 N N *Brachybacterium tyrofermentans* 403 NR_026272 clade_446 N N *Dermabacter hominis* 749 FJ263375 clade_446 N N *Aneurinibacillus thermoaerophilus* 171 NR_029303 clade_448 N N *Brevibacillus agri* 409 NR_040983 clade_448 N N *Brevibacillus brevis* 410 NR_041524 clade_448 Y N *Brevibacillus centrosporus* 411 NR_043414 clade_448 N N *Brevibacillus choshinensis* 412 NR_040980 clade_448 N N *Brevibacillus invocatus* 413 NR_041836 clade_448 N N *Brevibacillus laterosporus* 414 NR_037005 clade_448 Y N *Brevibacillus parabrevis* 415 NR_040981 clade_448 N N *Brevibacillus reuszeri* 416 NR_040982 clade_448 N N *Brevibacillus* sp. phR 417 JN837488 clade_448 N N *Brevibacillus thermoruber* 418 NR_026514 clade_448 N N *Lactobacillus murinus* 1100 NR_042231 clade_449 N N *Lactobacillus oeni* 1102 NR_043095 clade_449 N N *Lactobacillus ruminis* 1115 ACGS02000043 clade_449 N N *Lactobacillus vini* 1141 NR_042196 clade_449 N N *Gemella haemolysans* 924 ACDZ02000012 clade_450 N N *Gemella morbillorum* 925 NR_025904 clade_450 N N *Gemella morbillorum* 926 ACRX01000010 clade_450 N N *Gemella sanguinis* 927 ACRY01000057 clade_450 N N *Gemella* sp. oral clone ASCE02 929 AY923133 clade_450 N N *Gemella* sp. oral clone ASCF04 930 AY923139 clade_450 N N *Gemella* sp. oral clone ASCF12 931 AY923143 clade_450 N N *Gemella* sp. WAL 1945J 928 EU427463 clade_450 N N *Bacillus coagulans* 206 DQ297928 clade_451 Y OP *Sporolactobacillus inulinus* 1752 NR_040962 clade_451 Y N *Sporolactobacillus nakayamae* 1753 NR_042247 clade_451 N N *Gluconacetobacter entanii* 945 NR_028909 clade_452 N N *Gluconacetobacter europaeus* 946 NR_026513 clade_452 N N *Gluconacetobacter hansenii* 947 NR_026133 clade_452 N N *Gluconacetobacter oboediens* 949 NR_041295 clade_452 N N Gluconacetobacter xylinus 950 NR_074338 clade_452 N N *Auritibacter ignavus* 193 FN554542 clade_453 N N *Dermacoccus* sp. Ellin185 750 AEIQ01000090 clade_453 N N *Janibacter limosus* 1013 NR_026362 clade_453 N N *Janibacter melonis* 1014 EF063716 clade_453 N N *Kocuria palustris* 1041 EU333884 clade_453 Y N *Acetobacter aceti* 7 NR_026121 clade_454 N N *Acetobacter fabarum* 8 NR_042678 clade_454 N N *Acetobacter lovaniensis* 9 NR_040832 clade_454 N N *Acetobacter malorum* 10 NR_025513 clade_454 N N *Acetobacter orientalis* 11 NR_028625 clade_454 N N *Acetobacter pasteurianus* 12 NR_026107 clade_454 N N *Acetobacter pomorum* 13 NR_042112 clade_454 N N *Acetobacter syzygii* 14 NR_040868 clade_454 N N *Acetobacter tropicalis* 15 NR_036881 clade_454 N N *Gluconacetobacter azotocaptans* 943 NR_028767 clade_454 N N *Gluconacetobacter diazotrophicus* 944 NR_074292 clade_454 N N *Gluconacetobacter johannae* 948 NR_024959 clade_454 N N *Nocardia brasiliensis* 1351 AIHV01000038 clade_455 N N *Nocardia cyriacigeorgica* 1352 HQ009486 clade_455 N N *Nocardia farcinica* 1353 NC_006361 clade_455 Y N *Nocardia puris* 1354 NR_028994 clade_455 N N *Nocardia* sp. 01_Je_025 1355 GU574059 clade_455 N N *Rhodococcus equi* 1623 ADNW01000058 clade_455 N N *Bacillus* sp. oral taxon F28 247 HM099650 clade_456 Y OP *Oceanobacillus caeni* 1358 NR_041533 clade_456 N N *Oceanobacillus* sr. Ndiop 1359 CAER01000083 clade_456 N N *Ornithinibacillus bavariensis* 1384 NR_044923 clade_456 N N *Ornithinibacillus* sp. 7_10AIA 1385 FN397526 clade_456 N N *Virgibacillus proomii* 2005 NR_025308 clade_456 N N *Corynebacterium amycolatum* 683 ABZU01000033 clade_457 N OP *Corynebacterium hansenii* 702 AM946639 clade_457 N N *Corynebacterium xerosis* 735 FN179330 clade_457 N OP Staphylococcaceae bacterium NML 92 0017 1756 AY841362 clade_458 N N *Staphylococcus fleurettii* 1766 NR_041326 clade_458 N N *Staphylococcus sciuri* 1774 NR_025520 clade_458 N N *Staphylococcus vitulinus* 1779 NR_024670 clade_458 N N *Stenotrophomonas maltophilia* 1782 AAVZ01000005 clade_459 N N *Stenotrophomonas* sp. FG_6 1783 EF017810 clade_459 N N *Mycobacterium africanum* 1270 AF480605 clade_46 N OP *Mycobacterium alsiensis* 1271 AJ938169 clade_46 N OP *Mycobacterium avium* 1272 CP000479 clade_46 N OP *Mycobacterium colombiense* 1274 AM062764 clade_46 N OP *Mycobacterium gordonae* 1276 GU142930 clade_46 N OP *Mycobacterium intracellulare* 1277 GQ153276 clade_46 N OP *Mycobacterium kansasii* 1278 AF480601 clade_46 N OP *Mycobacterium lacus* 1279 NR_025175 clade_46 N OP *Mycobacterium leprae* 1280 FM211192 clade_46 N OP *Mycobacterium lepromatosis* 1281 EU203590 clade_46 N OP *Mycobacterium mantenii* 1283 FJ042897 clade_46 N OP *Mycobacterium marinum* 1284 NC_010612 clade_46 N OP *Mycobacterium microti* 1285 NR_025234 clade_46 N OP *Mycobacterium parascrofulaceum* 1287 ADNV01000350 clade_46 N OP *Mycobacterium seoulense* 1290 DQ536403 clade_46 N OP *Mycobacterium* sp. 1761 1292 EU703150 clade_46 N N *Mycobacterium* sp. 1791 1295 EU703148 clade_46 N N *Mycobacterium* sp. 1797 1296 EU703149 clade_46 N N *Mycobacterium* sp. 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KTR9 953 DQ068383 clade_463 N N *Gordonia sputi* 954 FJ536304 clade_463 N N *Gordonia terrae* 955 GQ848239 clade_463 N N *Leptotrichia goodfellowii* 1167 ADAD01000110 clade_465 N N *Leptotrichia* sp. oral clone IK040 1174 AY349387 clade_465 N N *Leptotrichia* sp. oral clone P2PB_51 P1 1175 AY207053 clade_465 N N Bacteroidales genomosp. P7 oral clone MB3_P19 264 DQ003623 clade_466 N N *Butyricimonas virosa* 454 AB443949 clade_466 N N *Odoribacter laneus* 1363 AB490805 clade_466 N N *Odoribacter splanchnicus* 1364 CP002544 clade_466 N N *Capnocytophaga gingivalis* 478 ACLQ01000011 clade_467 N N *Capnocytophaga granulosa* 479 X97248 clade_467 N N *Capnocytophaga* sp. oral clone AH015 483 AY005074 clade_467 N N *Copnocytophoga* sp. oral strain S3 487 AY005073 clade_467 N N *Copnocytophaga* sp. oral taxon 338 488 AEXX01000050 clade_467 N N *Capnocytophaga canimorsus* 476 CP002113 clade_468 N N *Copnocytophoga* sp. oral clone ID062 485 AY349368 clade_468 N N *Catenibacterium mitsuokai* 495 AB030224 clade_469 Y N *Clostridium* sp. 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OB7196 140 AB425070 clade_475 N N *Bifidobacterium urinalis* 366 AJ278695 clade_475 N N *Eubacterium nodatum* 854 U13041 clade_476 Y N *Eubacterium saphenum* 859 NR_026031 clade_476 Y N *Eubacterium* sp. oral clone JH012 867 AY349373 clade_476 Y N *Eubacterium* sp. oral clone JS001 870 AY349378 clade_476 Y N *Faecalibacterium prausnitzii* 880 ACOP02000011 clade_478 Y N *Gemmiger formicilis* 932 GU562446 clade_478 Y N *Subdoligranulum variabile* 1896 AJ518869 clade_478 Y N Clostridiaceae bacterium JC13 532 JF824807 clade_479 Y N *Clostridium* sp. MLG055 634 AF304435 clade_479 Y N Erysipelotrichaceae bacterium 3_1_53 822 ACTJ01000113 clade_479 Y N Prevotella loescheii 1503 JN867231 clade_48 N N *Prevotella* sp. oral clone ASCG12

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D7 672 ACDT01000199 clade_481 Y N Clostridiales bacterium SY8519 535 AB477431 clade_482 Y N *Clostridium* sp. SY8519 639 AP012212 clade_482 Y N *Eubacterium ramulus* 855 AJ011522 clade_482 Y N *Agrococcus jenensis* 117 NR_026275 clade_484 Y N *Microbacterium gubbeenense* 1232 NR_025098 clade_484 N N *Pseudoclavibacter* sp. Timone 1590 FJ375951 clade_484 N N *Tropheryma whipplei* 1961 BX251412 clade_484 N N *Zimmermannella bifida* 2031 AB012592 clade_484 N N *Erysipelothrix inopinata* 819 NR_025594 clade_485 Y N *Erysipelothrix rhusiopathiae* 820 ACLK01000021 clade_485 Y N *Erysipelothrix tonsillarum* 821 NR_040871 clade_485 Y N *Holdemania filiformis* 1004 Y11466 clade_485 Y N *Mollicutes bacterium pACH93* 1258 AY297808 clade_485 Y N *Coxiella burnetii* 736 CP000890 clade_486 Y Category-B *Legionella hackeliae* 1151 M36028 clade_486 N OP *Legionella longbeachae* 1152 M36029 clade_486 N OP *Legionella* sp. D3923 1154 JN380999 clade_486 N OP *Legionella* sp. 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NML98_0157 1648 AF533360 clade_490 N N *Rickettsia akari* 1627 CP000847 clade_492 N OP *Rickettsia conorii* 1628 AE008647 clade_492 N OP *Rickettsia prowazekii* 1629 M21789 clade_492 N Category-B *Rickettsia rickettsii* 1630 NC_010263 clade_492 N OP *Rickettsia slovaca* 1631 L36224 clade_492 N OP *Rickettsia typhi* 1632 AE017197 clade_492 N OP *Anaeroglobus geminatus* 160 AGCJ01000054 clade_493 N N *Megasphaera* genomosp. C1 1201 AY278622 clade_493 N N *Megasphaera micronuciformis* 1203 AECS01000020 clade_493 N N *Clostridium orbiscindens* 609 Y18187 clade_494 Y N *Clostridium* sp. NML 04A032 637 EU815224 clade_494 Y N *Flavonifractor plautii* 886 AY724678 clade_494 Y N *Pseudoflavonifractor capillosus* 1591 AY136666 clade_494 Y N Ruminococcaceae bacterium D16 1655 ADDX01000083 clade_494 Y N *Acetivibrio cellulolyticus* 5 NR_025917 clade_495 Y N Clostridiales genomosp. 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FOBRC15 470 HQ616380 clade_502 N OP *Campylobacter* sp. oral clone BB120 471 AY005038 clade_502 N OP *Campylobacter sputorum* 472 NR_044839 clade_502 N OP *Bacteroides ureolyticus* 330 GQ167666 clade_504 N N *Campylobacter gracilis* 463 ACYG01000026 clade_504 N OP *Campylobacter hominis* 464 NC_009714 clade_504 N OP *Dialister invisus* 762 ACIM02000001 clade_506 N N *Dialister micraerophilus* 763 AFBB01000028 clade_506 N N *Dialister microaerophilus* 764 AENT01000008 clade_506 N N *Dialister propionicifaciens* 766 NR_043231 clade_506 N N *Dialister succinatiphilus* 768 AB370249 clade_506 N N *Megasphaera elsdenii* 1200 AY038996 clade_506 N N *Megasphaera* genomosp. type_1 1202 ADGP01000010 clade_506 N N *Megasphaera* sp. BLPYG_07 1204 HM990964 clade_506 N N *Megasphaera* sp. UPII 199_6 1205

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P4P_31 P3 1458 AY207059 clade_509 N N *Helicobacter pylori* 997 CP00001.2 clade_510 N OP *Anaplasma marginale* 165 ABOR01000019 clade_511 N N *Anaplasma phagocytophilum* 166 NC_007797 clade_511 N N *Ehrlichia chaffeensis* 783 AAIF01000035 clade_511 N OP *Neorickettsia risticii* 1349 CP001431 clade_511 N N *Neorickettsia sennetsu* 1350 NC_007798 clade_511 N N *Eubacterium barkeri* 834 NR_044661 clade_512 Y N *Eubacterium callanderi* 838 NR_026310 clade_512 N N *Eubacterium limosum* 850 CP002273 clade_512 Y N *Pseudoramibacter alactolyticus* 1606 AB036759 clade_512 N N *Veillonella montpellierensis* 1977 AF473836 clade_513 N N *Veillonella* sp. oral clone ASCA08 1988 AY923118 clade_513 N N *Veillonella* sp. oral clone ASCB03 1989 AY923122 clade_513 N N *Inquilinus limosus* 1012 NR_029046 clade_514 N N *Sphingomonas* sp. oral clone FZ016 1746 AY349412 clade_514 N N *Anaerococcus lactolyticus* 145 ABYO01000217 clade_515 N N *Anaerococcus prevotii* 147 CP001708 clade_515 N N *Anaerococcus* sp. gpac104 152 AM176528 clade_515 N N *Anaerococcus* sp. gpac126 153 AM176530 clade_515 N N *Anaerococcus* sp. gpac155 154 AM176536 clade_515 N N *Anaerococcus* sp. gpac199 155 AM176539 clade_515 N N *Anaerococcus tetradius* 157 ACGC01000107 clade_515 N N *Bacteroides coagulans* 271 AB547639 clade_515 N N Clostridiales bacterium 9403326 534 HM587324 clade_515 N N Clostridiales bacterium ph2 539 JN837487 clade_515 N N *Peptostreptococcus* sp. 9succ1 1457 X90471 clade_515 N N *Peptostreptococcus* sp. oral clone AP24 1459 AB175072 clade_515 N N *Tissierella praeacuta* 1924 NR_044860 clade_515 N N *Anaerotruncus colihominis* 164 ABGD02000021 clade_516 Y N *Clostridium methylpentosum* 606 ACEC01000059 clade_516 Y N *Clostridium* sp. 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P2 oral clone MB1_G13 259 DQ003613 clade_529 N N Bacteroidales genomosp. P3 oral clone MB1_G34 260 DQ003615 clade_529 N N Bacteroidales genomosp. P4 oral clone MB2_G17 261 DQ003617 clade_529 N N Bacteroidales genomosp. P5 oral clone MB2_P04 262 DQ003619 clade_529 N N Bacteroidales genomosp. P6 oral clone MB3_C19 263 DQ003634 clade_529 N N Bacteroidales genomosp. P8 oral clone MB4_G15 265 DQ003626 clade_529 N N Bacteroidetes bacterium oral taxon D27 333 HM099638 clade_530 N N Bacteroidetes bacterium oral taxon F31 334 HM099643 clade_530 N N Bacteroidetes bacterium oral taxon F44 335 HM099649 clade_530 N N *Flavobacterium* sp. NF2_1 885 FJ195988 clade_530 N N *Myroides odoratimimus* 1326 NR_042354 clade_530 N N *Myroides* sp. MY15 1327 GU253339 clade_530 N N Chlamydiales bacterium NS16 507 JN606076 clade_531 N N *Chlamydophila pecorum* 508 D88317 clade_531 N OP *Parachlamydia* sp. UWE25 1423 BX908798 clade_531 N N *Fusobacterium russii* 903 NR_044687 clade_532 N N *Streptobacillus moniliformis* 1784 NR_027615 clade_532 N N Eubacteriaceae bacterium P4P_50 P4 833 AY207060 clade_533 N N *Eubacterium brachy* 836 U13038 clade_533 Y N *Filifactor alocis* 881 CP002390 clade_533 Y N *Filifactor villosus* 882 NR_041928 clade_533 Y N *Abiotrophia defectiva* 1 ACIN02000016 clade_534 N N *Abiotrophia* sp. oral clone P4PA_155 P1 3 AY207063 clade_534 N N *Catonella* genomosp. P1 oral clone MB5_P12 496 DQ003629 clade_534 N N *Catonella morbi* 497 ACIL02000016 clade_534 N N *Catonella* sp. oral clone FL037 498 AY349369 clade_534 N N *Eremococcus coleocola* 818 AENN01000008 clade_534 N N Facklamia hominis 879 Y10772 clade_534 N N *Granulicatella* sp. 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ICM57 189 HQ616400 clade_539 N N *Atopobium vaginae* 190 AEDQ01000024 clade_539 N N Coriobacteriaceae bacterium BV3Ac1 677 JN809768 clade_539 N N *Actinomyces naeslundii* 63 X81062 clade_54 N N *Actinomyces oricola* 67 NR_025559 clade_54 N N *Actinomyces oris* 69 BABV01000070 clade_54 N N *Actinomyces* sp. 7400942 70 EU484334 clade_54 N N *Actinomyces* sp. ChDC B197 72 AF543275 clade_54 N N *Actinomyces* sp. GEJ15 73 GU561313 clade_54 N N *Actinomyces* sp. 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L2_50 631 AAYW02000018 clade_543 Y N *Coprococcus eutactus* 675 EF031543 clade_543 Y N *Coprococcus* sp. ART55_1 676 AY350746 clade_543 Y N *Eubacterium ruminantium* 857 NR_024661 clade_543 Y N *Aeromicrobium marinum* 102 NR_025681 clade_544 N N *Aeromicrobium* sp. JC14 103 JF824798 clade_544 N N *Luteococcus sanguinis* 1190 NR_025507 clade_544 N N Propionibacteriaceae bacterium NML 02_0265 1568 EF599122 clade_544 N N *Rhodococcus corynebacterioides* 1622 X80615 clade_546 N N *Rhodococcus erythropolis* 1624 ACNO01000030 clade_546 N N *Rhodococcus fascians* 1625 NR_037021 clade_546 N N *Segniliparus rotundus* 1690 CP001958 clade_546 N N *Segniliparus rugosus* 1691 ACZI01000025 clade_546 N N *Exiguobacterium acetylicum* 878 FJ970034 clade_547 N N *Micrococcus caseolyticus* 1194 NR_074941 clade_547 N N *Streptomyces* sp. 1 AIP_2009 1890 FJ176782 clade_548 N N *Streptomyces* sp. SD 524 1892 EU544234 clade_548 N N *Streptomyces* sp. SD 528 1893 EU544233 clade_548 N N *Streptomyces thermoviolaceus* 1895 NR_027616 clade_548 N N *Borrelia afzelii* 388 ABCU01000001 clade_549 N OP *Borrelia crocidurae* 390 DQ057990 clade_549 N OP *Borrelia duttonii* 391 NC_011229 clade_549 N OP *Borrelia hermsii* 393 AY597657 clade_549 N OP *Borrelia hispanica* 394 DQ057988 clade_549 N OP *Borrelia persica* 395 HM161645 clade_549 N OP *Borrelia recurrentis* 396 AF107367 clade_549 N OP *Borrelia spielmanii* 398 ABKB01000002 clade_549 N OP *Borrelia turicatae* 399 NC_008710 clade_549 N OP *Borrelia valaisiana* 400 ABCY01000002 clade_549 N OP *Providencia alcalifaciens* 1586 ABXW01000071 clade_55 N N *Providencia rettgeri* 1587 AM040492 clade_55 N N *Providencia rustigianii* 1588 AM040489 clade_55 N N *Providencia stuartii* 1589 AF008581 clade_55 N N *Treponema pallidum* 1932 CP001752 clade_550 N OP *Treponema phagedenis* 1934 AEFH01000172 clade_550 N N *Treponema* sp. clone DDKL_4 1939 Y08894 clade_550 N N

*Acholeplasma laidlawii* 17 NR_074448 clade_551 N N *Mycoplasma putrefaciens* 1323 U26055 clade_551 N N Mycoplasmataceae genomosp P1 oral clone 1325 DQ003614 clade_551 N N *Spiroplasma insolitum* 1750 NR_025705 clade_551 N N *Collinsella aerofaciens* 659 AAVN02000007 clade_553 Y N *Collinsella intestinalis* 660 ABXH02000037 clade_553 N N *Collinsella stercoris* 661 ABXJ01000150 clade_553 N N *Collinsella tanakaei* 662 AB490807 clade_553 N N *Alkaliphilus metalliredigenes* 137 AY137848 clade_554 Y N *Alkaliphilus oremlandii* 138 NR_043674 clade_554 Y N *Caminicella sporogenes* 458 NR_025485 clade_554 N N *Clostridium sticklandii* 648 L04167 clade_554 Y N *Turicibacter sanguinis* 1965 AF349724 clade_555 Y N *Acidaminococcus fermentans* 21 CP001859 clade_556 N N *Acidaminococcus intestini* 22 CP003058 clade_556 N N *Acidaminococcus* sp. D21 23 ACGB01000071 clade_556 N N *Phascolarctobacterium faecium* 1462 NR_026111 clade_556 N N *Phascolarctobacterium* sp. YIT 12068 1463 AB490812 clade_556 N N *Phascolarctobacterium succinatutens* 1464 AB490811 clade_556 N N *Acidithiobacillus ferrivorans* 25 NR_074660 clade_557 N N *Fulvimonas* sp. NML 060897 892 EF589680 clade_557 Y N Xanthomonadaceae bacterium NML 03_0222 2015 EU313791 clade_557 N N *Catabacter hongkongensis* 494 AB671763 clade_558 N N *Christensenella minuta* 512 AB490809 clade_558 N N Clostridiales bacterium oral clone P4PA 536 AY207065 clade_558 N N Clostridiales bacterium oral taxon 093 537 GQ422712 clade_558 N N *Desulfitobacterium frappieri* 753 AJ276701 clade_560 Y N *Desulfitobacterium hafniense* 754 NR_074996 clade_560 Y N *Desulfotomaculum nigrificans* 756 NR_044832 clade_560 Y N *Heliobacterium modesticaldum* 1000 NR_074517 clade_560 N N *Alistipes indistinctus* 130 AB490804 clade_561 N N Bacteroidales bacterium ph8 257 JN837494 clade_561 N N *Candidatus Sulcia muelleri* 475 CP002163 clade_561 N N *Cytophaga xylanolytica* 742 FR733683 clade_561 N N Flavobacteriaceae genomosp. C1 884 AY278614 clade_561 N N *Gramella forsetii* 958 NR_074707 clade_561 N N *Sphingobacterium faecium* 1740 NR_025537 clade_562 N N *Sphingobacterium mizutaii* 1741 JF708889 clade_562 N N *Sphingobacterium multivorum* 1742 NR_040953 clade_562 N N *Sphingobacterium spiritivorum* 1743 ACHA02000013 clade_562 N N *Jonquetella anthropi* 1017 ACOO02000004 clade_563 N N *Pyramidobacter piscolens* 1614 AY207056 clade_563 N N Synergistes genomosp. C1 1904 AY278615 clade_563 N N Synergistes sp. RMA 14551 1905 DQ412722 clade_563 N N Synergistetes bacterium ADV897 1906 GQ258968 clade_563 N N *Candidatus Arthromitus* sp. SFB_mouse_Yit 474 NR_074460 clade_564 N N *Gracilibacter thermotolerans* 957 NR_043559 clade_564 N N *Lutispora thermophila* 1191 NR_041236 clade_564 Y N *Brachyspira aalborgi* 404 FM178386 clade_565 N N *Brachyspira pilosicoli* 405 NR_075069 clade_565 Y N Brachyspira sp. HIS3 406 FM178387 clade_565 N N *Brachyspira* sp. HIS4 407 FM178388 clade_565 N N *Brachyspira* sp. HIS5 408 FM178389 clade_565 N N *Adlercreutzia equolifaciens* 97 AB306661 clade_566 N N Coriobacteriaceae bacterium JC110 678 CAEM01000062 clade_566 N N Coriobacteriaceae bacterium phI 679 JN837493 clade_566 N N *Cryptobacterium curtum* 740 GQ422741 clade_566 N N *Eggerthella lenta* 778 AF292375 clade_566 Y N *Eggerthella sinensis* 779 AY321958 clade_566 N N *Eggerthella* sp. 1_3_56FAA 780 ACWN01000099 clade_566 N N *Eggerthella* sp. HGA1 781 AEXR01000021 clade_566 N N *Eggerthella* sp. YY7918 782 AP012211 clade_566 N N *Gordonibacter pamelaeae* 680 AM886059 clade_566 N N *Gordonibacter pamelaeae* 956 FP929047 clade_566 N N *Slackia equolifaciens* 1732 EU377663 clade_566 N N *Slackia exigua* 1733 ACUX01000029 clade_566 N N *Slackia faecicanis* 1734 NR_042220 clade_566 N N *Slackia heliotrinireducens* 1735 NR_074439 clade_566 N N *Slackia isoflavoniconvertens* 1736 AB566418 clade_566 N N *Slackia piriformis* 1737 AB490806 clade_566 N N *Slackia* sp. NATTS 1738 AB505075 clade_566 N N *Streptomyces albus* 1888 AJ697941 clade_566 Y N Chlamydiales bacterium NS11 505 JN606074 clade_567 Y N Chlamydiales bacterium NS13 506 JN606075 clade_567 N N Victivallaceae bacterium NML 080035 2003 FJ394915 clade_567 N N *Victivallis vadensis* 2004 ABDE02000010 clade_567 N N *Anaerofustis stercorihominis* 159 ABIL02000005 clade_570 Y N *Butyricicoccus pullicaecorum* 453 HH793440 clade_572 Y N *Eubacterium desmolans* 843 NR_044644 clade_572 Y N *Papillibacter cinnamivorans* 1415 NR_025025 clade_572 Y N *Sporobacter termitidis* 1751 NR_044972 clade_572 Y N *Streptomyces griseus* 1889 NR_074787 clade_573 N N *Streptomyces* sp. SD 511 1891 EU544231 clade_573 N N *Streptomyces* sp. SD 534 1894 EU544232 clade_573 N N *Cloacibacillus evryensis* 530 GQ258966 clade_575 N N *Deferribacteres* sp. oral clone JV001 743 AY349370 clade_575 N N *Deferribacteres* sp. oral clone JV006 744 AY349371 clade_575 Y N *Deferribacteres* sp. oral clone JV023 745 AY349372 clade_575 N N Synergistetes bacterium LBVCM1157 1907 GQ258969 clade_575 N N Synergistetes bacterium oral taxon 362 1909 GU410752 clade_575 N N Synergistetes bacterium oral taxon D48 1910 GU430992 clade_575 N N *Clostridium colinum* 577 NR_026151 clade_576 Y N *Clostridium lactatifermentans* 599 NR_025651 clade_576 Y N *Clostridium piliforme* 614 D14639 clade_576 Y N *Peptococcus* sp. oral clone JM048 1439 AY349389 clade_576 N N *Helicobacter winghamensis* 999 ACDO01000013 clade_577 N N *Wolinella succinogenes* 2014 BX571657 clade_577 N N *Olsenella* genomosp. C1 1368 AY278623 clade_578 N N *Olsenella profusa* 1369 FN178466 clade_578 N N *Olsenella* sp. F0004 1370 EU592964 clade_578 N N *Olsenella* sp. oral taxon 809 1371 ACVE01000002 clade_578 N N *Olsenella uli* 1372 CP002106 clade_578 N N *Nocardiopsis dassonvillei* 1356 CP002041 clade_579 N N *Saccharomonospora viridis* 1671 X54286 clade_579 Y N *Thermobifida fusca* 1921 NC_007333 clade_579 Y N *Peptococcus niger* 1438 NR_029221 clade_580 N N *Peptococcus* sp. oral taxon 167 1440 GQ422727 clade_580 N N *Akkermansia muciniphila* 118 CP001071 clade_583 N N *Opitutus terrae* 1373 NR_074978 clade_583 N N Clostridiales bacterium oral taxon F32 538 HM099644 clade_584 N N *Leptospira borgpetersenii* 1161 NC_008508 clade_585 N OP *Leptospira broomii* 1162 NR_043200 clade_585 N OP *Leptospira interrogans* 3163 NC_005823 clade_585 N OP *Leptospira licerasiae* 1164 EF612284 clade_585 Y OP *Methanobrevibacter gottschalkii* 1213 NR_044789 clade_587 N N *Methanobrevibacter millerae* 1214 NR_042785 clade_587 N N *Methanobrevibacter oralis* 1216 HE654003 clade_587 N N *Methanobrevibacter thaueri* 1219 NR_044787 clade_587 N N *Methanobrevibacter smithii* 1218 ABYV02000002 clade_588 N N *Deinococcus radiodurans* 746 AE000513 clade_589 N N *Deinococcus* sp. R_43890 747 FR682752 clade_589 N N *Thermus aquaticus* 1923 NR_025900 clade_589 N N *Actinomyces* sp. c109 81 AB167239 clade_590 N N *Moorella thermoacetica* 1259 NR_075001 clade_590 Y N Syntrophomonadaceae genomosp. P1 1912 AY341821 clade_590 N N *Thermoanaerobacter pseudethanolicus* 1920 CP000924 clade_590 Y N *Anaerobaculum hydrogeniformans* 141 ACJX02000009 clade_591 N N *Flexistipes sinusarabici* 888 NR_074881 clade_591 Y N *Microcystis aeruginosa* 1246 NC_010296 clade_592 N N *Prochlorococcus marinus* 1567 CP000551 clade_592 N N *Methanobrevibacter acididurans* 1208 NR_028779 clade_593 N N *Methanobrevibacter arboriphilus* 1209 NR_042783 clade_593 N N *Methanobrevibacter curvatus* 1210 NR_044796 clade_593 N N *Methanobrevibacter cuticularis* 1211 NR_044776 clade_593 N N *Methanobrevibacter filiformis* 1212 NR_044801 clade_593 N N *Methanobrevibacter woesei* 1220 NR_044788 clade_593 N N *Roseiflexus castenholzii* 1642 CP000804 clade_594 N N *Methanobrevibacter olleyae* 1215 NR_043024 clade_595 N N *Methanobrevibacter ruminantium* 1217 NR_042784 clade_595 N N *Methanobrevibacter wolinii* 1221 NR_044790 clade_595 N N *Methanosphaera stadtmanae* 1222 AY196684 clade_595 N N Chloroflexi genomosp. P1 511 AY331414 clade_596 N N *Gloeobacter violaceus* 942 NR_074282 clade_596 Y N *Halorubrum lipolyticum* 992 AB477978 clade_597 N N *Methanobacterium formicicum* 1207 NR_025028 clade_597 N N *Acidilobus saccharovorans* 24 AY350586 clade_598 N N *Hyperthermus butylicus* 1006 CP000493 clade_598 N N *Ignicoccus islandicus* 1011 X99562 clade_598 N N *Metallosphaera sedula* 1206 D26491 clade_598 N N *Thermofilum pendens* 1922 X14835 clade_598 N N *Prevotella melaninogenica* 1506 CP002122 clade_6 N N *Prevotella* sp. ICM1 1520 HQ616385 clade_6 N N *Prevotella* sp. oral clone FU048 1535 AY349393 clade_6 N N *Prevotella* sp. oral done GI030 1537 AY349395 clade_6 N N *Prevotella* sp. SEQ116 1526 JN867246 clade_6 N N *Streptococcus anginosus* 1787 AECT01000011 clade_60 N N *Streptococcus milleri* 1812 X81023 clade_60 N N *Streptococcus* sp. 16362 1829 JN590019 clade_60 N N *Streptococcus* sp. 69130 1832 X78825 clade_60 N N *Streptococcus* sp. AC15 1833 HQ616356 clade_60 N N *Streptococcus* sp. CM7 1839 HQ616373 clade_60 N N *Streptococcus* sp. OBRC6 1847 HQ616352 clade_60 N N *Burkholderia ambifaria* 442 AAUZ01000009 clade_61 N OP *Burkholderia cenocepacia* 443 AAHI01000060 clade_61 N OP *Burkholderia cepacia* 444 NR_041719 clade_61 N OP *Burkholderia mallei* 445 CP000547 clade_61 N Category-B *Burkholderia multivorans* 446 NC_010086 clade_61 N OP *Burkholderia oklahomensis* 447 DQ108388 clade_61 N OP *Burkholderia pseudomallei* 448 CP001408 clade_61 N Category-B *Burkholderia* *rhizoxinica* 449 HQ005410 clade_61 N OP *Burkholderia* sp. 383 450 CP000151 clade_61 N OP *Burkholderia xenovorans* 451 U86373 clade_61 N OP *Prevotella buccae* 1488 ACRB01000001 clade_62 N N *Prevotella* genomosp. P8 oral clone MB3_P13 1498 DQ003622 clade_62 N N *Prevotella* sp. oral clone FW035 1536 AY349394 clade_62 N N *Prevotella bivia* 1486 ADFO01000096 clade_63 N N *Prevotella disiens* 1494 AEDO01000026 clade_64 N N *Bacteroides faecis* 276 GQ496624 clade_65 N N *Bacteroides fragilis* 279 AP006841 clade_65 N N *Bacteroides nordii* 285 NR_043017 clade_65 N N *Bacteroides salyersiae* 292 EU136690 clade_65 N N Bacteroides sp. 1_1_14 293 ACRP01000155 clade_65 N N *Bacteroides* sp. 1_1_6 295 ACIC01000215 clade_65 N N *Bacteroides* sp. 2_1_56FAA 298 ACWI01000065 clade_65 N N *Bacteroides* sp. AR29 316 AF139525 clade_65 N N *Bacteroides* sp. B2 317 EU722733 clade_65 N N *Bacteroides thetaiotaomicron* 328 NR_074277 clade_65 N N *Actinobacillus minor* 45 ACFT01000025 clade_69 N N *Haemophilus parasuis* 978 GU226366 clade_69 N N *Vibrio cholerae* 1996 AAUR01000095 clade_71 N Category-B *Vibrio fluvialis* 1997 X76335 clade_71 N Category-B *Vibrio furnissii* 1998 CP002377 clade_71 N Category-B *Vibrio mimicus* 1999 ADAF01000001 clade_71 N Category-B *Vibrio parahaemolyticus* 2000 AAWQ01000116 clade_71 N Category-B *Vibrio* sp. RC341 2001 ACZT01000024 clade_71 N Category-B *Vibrio vulnificus* 2002 AE016796 clade_71 N Category-B *Lactobacillus acidophilus* 1067 CP000033 clade_72 N N *Lactobacillus amylolyticus* 1069 ADNY01000006 clade_72 N N *Lactobacillus amylovorus* 1070 CP002338 clade_72 N N *Lactobacillus crispatus* 1078 ACOG01000151 clade_72 N N *Lactobacillus delbrueckii* 1080 CP002341 clade_72 N N *Lactobacillus helveticus* 1088 ACLM01000202 clade_72 N N *Lactobacillus kalixensis* 1094 NR_029083 clade_72 N N *Lactobacillus kefiranofaciens* 1095 NR_042440 clade_72 N N *Lactobacillus leichmannii* 1098 JX986966 clade_72 N N *Lactobacillus* sp. 66c 1120 FR681900 clade_72 N N *Lactobacillus* sp. KLDS 1.0701 1122 EU600905 clade_72 N N *Lactobacillus* sp. KLDS 1.0712 1130 EU600916 clade_72 N N *Lactobacillus* sp. oral clone HT070 1136 AY349383 clade_72 N N *Lactobacillus ultunensis* 1139 ACGU01000081 clade_72 N N *Prevotella intermedia* 1502 AF414829 clade_81 N N *Prevotella nigrescens* 1511 AFPX01000069 clade_81 N N *Prevotella pallens* 1515 AFPY01000135 clade_81 N N *Prevotella* sp. oral taxon 310 1551 GQ422737 clade_81 N N *Prevotella* genomosp. C1 1495 AY278624 clade_82 N N *Prevotella* sp. CM38 1519 HQ610181 clade_82 N N *Prevotella* sp. oral taxon 317 1552 ACQH01000158 clade_82 N N *Prevotella* sp. SG12 1527 GU561343 clade_82 N N *Prevotella denticola* 1493 CP002589 clade_83 N N *Prevotella* genomosp. P7 oral clone MB2_P31 1497 DQ003620 clade_83 N N *Prevotella histicola* 1501 JN867315 clade_83 N N *Prevotella multiformis* 1508 AEWX01000054 clade_83 N N *Prevotella* sp. JCM 6330 1522 AB547699 clade_83 N N *Prevotella* sp. oral clone GI059 1539 AY349397 clade_83 N N *Prevotella* sp. oral taxon 782 1555 GQ422745 clade_83 N N *Prevotella* sp. oral taxon G71 1559 GU432180 clade_83 N N *Prevotella* sp. SEQ065 1524 JN867234 clade_83 N N *Prevotella veroralis* 1565 ACVA01000027 clade_83 N N *Bacteroides acidifaciens* 266 NR_028607 clade_85 N N *Bacteroides cellulosilyticus* 269 ACCH01000108 clade_85 N N *Bacteroides clarus* 270 AFBM01000011 clade_85 N N *Bacteroides eggerthii* 275 ACWG01000065 clade_85 N N *Bacteroides oleiciplenus* 286 AB547644 clade_85 N N *Bacteroides pyogenes* 290 NR_041280 clade_85 N N *Bacteroides* sp. 315_5 300 FJ848547 clade_85 N N *Bacteroides* sp. 31SF15 301 AJ583248 clade_85 N N *Bacteroides* sp. 31SF18 302 AJ583249 clade_85 N N *Bacteroides* sp. 35AE31 303 AJ583244 clade_85 N N *Bacteroides* sp. 35AE37 304 AJ583245 clade_85 N N *Bacteroides* sp. 35BE34 305 AJ583246 clade_85 N N *Bacteroides* sp. 35BE35 306 AJ583247 clade_85 N N *Bacteroides* sp. WH2 324 AY895180 clade_85 N N *Bacteroides* sp. XB12B 325 AM230648 clade_85 N N *Bacteroides stercoris* 327 ABFZ02000022 clade_85 N N *Actinobacillus pleuropneumoniae* 46 NR_074857 clade_88 N N *Actinobacillus ureae* 48 AEVG01000167 clade_88 N N *Haemophilus aegyptius* 969 AFBC01000053 clade_88 N N *Haemophilus ducreyi* 970 AE017143 clade_88 N OP *Haemophilus haemolyticus* 973 JN175335 clade_88 N N *Haemophilus influenzae* 974 AADP01000001 clade_88 N OP *Haemophilus parahaemolyticus* 975 GU561425 clade_88 N N *Haemophilus parainfluenzae* 976 AEWU01000024 clade_88 N N *Haemophilus paraphrophaemolyticus* 977 M75076 clade_88 N N *Haemophilus somnus* 979 NC_008309 clade_88 N N *Haemophilus* sp. 70334 980 HQ680854 clade_88 N N *Haemophilus* sp. HK445 981 FJ685624 clade_88 N N *Haemophilus* sp. oral clone ASCA07 982 AY923117 clade_88 N N *Haemophilus* sp. oral clone ASCG06 983 AY923147 clade_88 N N *Haemophilus* sp. oral clone BJ021 984 AY005034 clade_88 N N *Haemophilus* sp. oral clone BJ095 985 AY005033 clade_88 N N *Haemophilus* sp. oral taxon 851 987 AGRK01000004 clade_88 N N *Haemophilus sputorum* 988 AFNK01000005 clade_88 N N *Histophilus somni* 1003 AF549387 clade_88 N N *Mannheimia haemolytica* 1195 ACZX01000102 clade_88 N N *Pasteurella bettyae* 1433 L06088 clade_88 N N *Moellerella wisconsensis* 1253 JN175344 clade_89 N N *Morganella morganii* 1265 AJ301681 clade_89 N N *Morganella* sp. JB_T16 1266 AJ781005 clade_89 N N *Proteus mirabilis* 1582 ACLE01000013 clade_89 N N *Proteus penneri* 1583 ABVP01000020 clade_89 N N *Proteus* sp. HS7514 1584 DQ512963 clade_89 N N *Proteus vulgaris* 1585 AJ233425 clade_89 N N *Eubacterium* sp. oral clone JN088 869 AY349377 clade_90 Y N *Oribacterium sinus* 1374 ACKX01000142 clade_90 N N *Oribacterium* sp. ACB1 1375 HM120210 clade_90 N N *Oribacterium* sp. ACB7 1376 HM120211 clade_90 N N *Oribacterium* sp. CM12 1377 HQ616374 clade_90 N N *Oribacterium* sp. ICM51 1378 HQ616397 clade_90 N N *Oribacterium* sp. OBRC12 1379 HQ616355 clade_90 N N *Oribacterium* sp. oral taxon 108 1382 AFIH01000001 clade_90 N N *Actinobacillus actinomycetemcomitans* 44 AY362885 clade_92 N N *Actinobacillus succinogenes* 47 CP000746 clade_92 N N *Aggregatibacter actinomycetemcomitans* 112 CP001733 clade_92 N N *Aggregatibacter aphrophilus* 113 CP001607 clade_92 N N *Aggregatibacter segnis* 114 AEPS01000017 clade_92 N N *Averyella dalhousiensis* 194 DQ481464 clade_92 N N Bisgaard Taxon 368 AY683487 clade_92 N N Bisgaard Taxon 369 AY683489 clade_92 N N Bisgaard Taxon 370 AY683491 clade_92 N N Bisgaard Taxon 371 AY683492 clade_92 N N *Buchnera aphidicola* 440 NR_074609 clade_92 N N *Cedecea davisae* 499 AF493976 clade_92 N N *Citrobacter amalonaticus* 517 FR870441 clade_92 N N *Citrobacter braakii* 518 NR_028687 clade_92 N N *Citrobacter farmeri* 519 AF025371 clade_92 N N *Citrobacter freundii* 520 NR_028894 clade_92 N N *Citrobacter gillenii* 521 AF025367 clade_92 N N *Citrobacter koseri* 522 NC_009792 clade_92 N N *Citrobacter murliniae* 523 AF025369 clade_92 N N *Citrobacter rodentium* 524 NR_074903 clade_92 N N *Citrobacter sedlakii* 525 AF025364 clade_92 N N *Citrobacter* sp. 30_2 526 ACDJ01000053 clade_92 N N *Citrobacter* sp. KMSI_3 527 GQ468398 clade_92 N N *Citrobacter werkmanii* 528 AF025373 clade_92 N N *Citrobacter youngae* 529 ABWL02000011 clade_92 N N *Cronobacter malonaticus* 737 GU122174 clade_92 N N *Cronobacter sakazakii* 738 NC_009778 clade_92 N N *Cronobacter turicensis* 739 FN543093 clade_92 N N *Enterobacter aerogenes* 786 AJ251468 clade_92 N N *Enterobacter asburiae* 787 NR_024640 clade_92 N N *Enterobacter cancerogenus* 788 Z96078 clade_92 N N *Enterobacter cloacae* 789 FP929040 clade_92 N N Enterobacter cowanii 790 NR_025566 clade_92 N N *Enterobacter hormaechei* 791 AFHR01000079 clade_92 N N *Enterobacter* sp. 247BMC 792 HQ122932 clade_92 N N *Enterobacter* sp. 638 793 NR_074777 clade_92 N N *Enterobacter* sp. JC163 794 JN657217 clade_92 N N *Enterobacter* sp. SCSS 795 HM007811 clade_92 N N *Enterobacter* sp. TSE38 796 HM156134 clade_92 N N Enterobacteriaceae bacterium 9_2_54FAA 797 ADCU01000033 clade_92 N N Enterobacteriaceae bacterium CF01Ent_1 798 AJ489826 clade_92 N N Enterobacteriaceae bacterium Smarlab 3302238 799 AY538694 clade_92 N N *Escherichia albertii* 824 ABKX01000012 clade_92 N N *Escherichia coli* 825 NC_008563 clade_92 N Category-B *Escherichia fergusonii* 826 CU928158 clade_92 N N *Escherichia hermannii* 827 HQ407266 clade_92 N N *Escherichia* sp. 1_1_43 828 ACID01000033 clade_92 N N *Escherichia* sp. 4_1_40B 829 ACDM02000056 clade_92 N N *Escherichia* sp. B4 830 EU722735 clade_92 N N *Escherichia vulneris* 831 NR_041927 clade_92 N N *Ewingella americana* 877 JN175329 clade_92 N N *Haemophilus* genomosp. P2 oral clone MB3_C24 971 DQ003621 clade_92 N N *Haemophilus* genomosp. P3 oral clone MB3_C38 972 DQ003635 clade_92 N N *Haemophilus* sp. oral clone JM053 986 AY349380 clade_92 N N *Hafnia alvei* 989 DQ412565 clade_92 N N *Klebsiella oxytoca* 1024 AY292871 clade_92 N OP *Klebsiella pneumoniae* 1025 CP000647 clade_92 N OP *Klebsiella* sp. AS10 1026 HQ616362 clade_92 N N *Klebsiella* sp. Co9935 1027 DQ068764 clade_92 N N *Klebsiella* sp. enrichment culture clone *SRC* _DSD25 1036 HM195210 clade_92 N N *Klebsiella* sp. OBRC7 1028 HQ616353 clade_92 N N *Klebsiella* sp. SP_BA 1029 FJ999767 clade_92 N N *Klebsiella* sp. SRC_DSD1 1033 GU797254 clade_92 N N *Klebsiella* sp. SRC_DSD11 1030 GU797263 clade_92 N N *Klebsiella* sp. SRC_DSD12 1031 GU797264 clade_92 N N *Klebsiella* sp. SRC_DSD15 1032 GU797267 clade_92 N N *Klebsiella* sp. SRC_DSD2 1034 GU797253 clade_92 N N *Klebsiella* sp. SRC_DSD6 1035 GU797258 clade_92 N N *Klebsiella variicola* 1037 CP001891 clade_92 N N *Kluyvera ascorbata* 1038 NR_028677 clade_92 N N *Kluyvera cryocrescens* 1039 NR_028803 clade_92 N N *Leminorella grimontii* 1159 AJ233421 clade_92 N N *Leminorella richardii* 1160 HF558368 clade_92 N N *Pantoea agglomerans* 1409 AY335552 clade_92 N N *Pantoea ananatis* 1410 CP001875 clade_92 N N *Pantoea brenneri* 1411 EU216735 clade_92 N N *Pantoea citrea* 1412 EF688008 clade_92 N N *Pantoea conspicua* 1413 EU216737 clade_92 N N *Pantoea septica* 1414 EU216734 clade_92 N N *Pasteurella dagmatis* 1434 ACZR01000003 clade_92 N N *Pasteurella multocida* 1435 NC_002663 clade_92 N N *Plesiomonas shigelloides* 1469 X60418 clade_92 N N *Raoultella ornithinolytica* 1617 AB364958 clade_92 N N *Raoultella planticola* 1618 AF129443 clade_92 N N *Raoultella terrigena* 1619 NR_037085 clade_92 N N *Salmonella bongori* 1683 NR_041699 clade_92 N Category-B *Salmonella enterica* 1672 NC_011149 clade_92 N Category-B *Salmonella enterica* 1673 NC_011205 clade_92 N Category-B *Salmonella enterica* 1674 DQ344532 clade_92 N Category-B *Salmonella enterica* 1675 ABEH02000004 clade_92 N Category-B *Salmonella enterica* 1676 ABAK02000001 clade_92 N Category-B *Salmonella enterica* 1677 NC_011080 clade_92 N Category-B *Salmonella enterica* 1678 EU118094 clade_92 N Category-B *Salmonella enterica* 1679 NC_011094 clade_92 N Category-B *Salmonella enterica* 1680 AE014613 clade_92 N Category-B *Salmonella enterica* 1682 ABFH02000001 clade_92 N Category-B *Salmonella enterica* 1684 ABEM01000001 clade_92 N Category-B *Salmonella enterica* 1685 ABAM02000001 clade_92 N Category-B *Salmonella typhimurium* 1681 DQ344533 clade_92 N Category-B *Salmonella typhimurium* 1686 AF170176 clade_92 N Category-B *Serratia fonticola* 1718 NR_025339 clade_92 N N *Serratia liquefaciens* 1719 NR_042062 clade_92 N N *Serratia marcescens* 1720 GU826157 clade_92 N N *Serratia odorifera* 1721 ADBY01000001 clade_92 N N *Serratia proteamaculans* 1722 AAUN01000015 clade_92 N N *Shigella boydii* 1724 AAKA01000007 clade_92 N Category-B *Shigella dysenteriae* 1725 NC_007606 clade_92 N Category-B *Shigella flexneri* 1726 AE005674 clade_92 N Category-B *Shigella sonnei* 1727 NC_007384 clade_92 N Category-B *Tatumella ptyseos* 1916 NR_025342 clade_92 N N *Trabulsiella guamensis* 1925 AY373830 clade_92 N N *Yersinia aldovae* 2019 AJ871363 clade_92 N OP *Yersinia aleksiciae* 2020 AJ627597 clade_92 N OP *Yersinia bercovieri* 2021 AF366377 clade_92 N OP *Yersinia enterocolitica* 2022 FR729477 clade_92 N Category-B *Yersinia frederiksenii* 2023 AF366379 clade_92 N OP *Yersinia intermedia* 2024 AF366380 clade_92 N OP *Yersinia kristensenii* 2025 ACCA01000078 clade_92 N OP *Yersinia mollaretii* 2026 NR_027546 clade_92 N OP *Yersinia pestis* 2027 AE013632 clade_92 N Category-A *Yersinia pseudotuberculosis* 2028 NC_009708 clade_92 N OP *Yersinia rohdei* 2029 ACCD01000071 clade_92 N OP *Yokenella regensburgei* 2030 AB273739 clade_92 N N *Conchiformibius kuhniae* 669 NR_041821 clade_94 N N *Morococcus cerebrosus* 1267 JN175352 clade_94 N N *Neisseria bacilliformis* 1328 AFAY01000058 clade_94 N N *Neisseria cinerea* 1329 ACDY01000037 clade_94 N N *Neisseria flavescens* 1331 ACQV01000025 clade_94 N N *Neisseria gonorrhoeae* 1333 CP002440 clade_94 N OP *Neisseria lactamica* 1334 ACEQ01000095 clade_94 N N *Neisseria macacae* 1335 AFQE01000146 clade_94 N N *Neisseria meningitidis* 1336 NC_003112 clade_94 N OP *Neisseria mucosa* 1337 ACDX01000110 clade_94 N N *Neisseria pharyngis* 1338 AJ239281 clade_94 N N *Neisseria polysaccharea* 1339 ADBE01000137 clade_94 N N *Neisseria sicca* 1340 ACKO02000016 clade_94 N N *Neisseria* sp. KEM232 1341 GQ203291 clade_94 N N *Neisseria* sp. oral clone AP132 1344 AY005027 clade_94 N N *Neisseria* sp. oral strain B33KA 1346 AY005028 clade_94 N N *Neisseria* sp. oral taxon 014 1347 ADEA01000039 clade_94 N N *Neisseria* sp. TM10_1 1343 DQ279352 clade_94 N N *Neisseria subflava* 1348 ACEO01000067 clade_94 N N *Clostridium oroticum* 610 FR749922 clade_96 Y N *Clostridium* sp. D5 627 ADBG01000142 clade_96 Y N *Eubacterium contortum* 840 FR749946 clade_96 Y N *Eubacterium fissicatena* 846 FR749935 clade_96 Y N *Okadaella gastrococcus* 1365 HQ699465 clade_98 N N *Streptococcus agalactiae* 1785 AAJO01000130 clade_98 N N *Streptococcus alactolyticus* 1786 NR_041781 clade_98 N N *Streptococcus australis* 1788 AEQR01000024 clade_98 N N *Streptococcus bovis* 1789 AEEL01000030 clade_98 N N *Streptococcus canis* 1790 AJ413203 clade_98 N N *Streptococcus constellatus* 1791 AY277942 clade_98 N N *Streptococcus cristatus* 1792 AEVC01000028 clade_98 N N *Streptococcus dysgalactiae* 1794 AP010935 clade_98 N N *Streptococcus equi* 1795 CP001129 clade_98 N N *Streptococcus equinus* 1796 AEVB01000043 clade_98 N N *Streptococcus gallolyticus* 1797 FR824043 clade_98 N N *Streptococcus* genomosp. C1 1798 AY278629 clade_98 N N *Streptococcus* genomosp. C2 1799 AY278630 clade_98 N N *Streptococcus* genomosp. C3 1800 AY278631 clade_98 N N Streptococcus genomosp. C4 1801 AY278632 clade_98 N N *Streptococcus* genomosp. C5 1802 AY278633 clade_98 N N *Streptococcus* genomosp. C6 1803 AY278634 clade_98 N N *Streptococcus* genomosp. C7 1804 AY278635 clade_98 N N *Streptococcus* genomosp. C8 1805 AY278609 clade_98 N N *Streptococcus gordonii* 1806 NC_009785 clade_98 N N *Streptococcus infantarius* 1807 ABJK02000017 clade_98 N N *Streptococcus infantis* 1808 AFNN01000024 clade_98 N N *Streptococcus intermedius* 1809 NR_028736 clade_98 N N *Streptococcus lutetiensis* 1810 NR_037096 clade_98 N N *Streptococcus massiliensis* 1811 AY769997 clade_98 N N *Streptococcus mitis* 1813 AM157420 clade_98 N N *Streptococcus oligofermentans* 1815 AY099095 clade_98 N N *Streptococcus oralis* 1816 ADMV01000001 clade_98 N N *Streptococcus parasanguinis* 1817 AEKM01000012 clade_98 N N *Streptococcus pasteurianus* 1818 AP012054 clade_98 N N *Streptococcus peroris* 1819 AEVF01000016 clade_98 N N *Streptococcus pneumoniae* 1820 AE008537 clade_98 N N *Streptococcus porcinus* 1821 EF121439 clade_98 N N *Streptococcus pseudopneumoniae* 1822 FJ827123 clade_98 N N *Streptococcus pseudoporcinus* 1823 AENS01000003 clade_98 N N *Streptococcus pyogenes* 1824 AE006496 clade_98 N OP *Streptococcus ratti* 1825 X58304 clade_98 N N *Streptococcus salivarius* 1826 AGBV01000001 clade_98 N N *Streptococcus sanguinis* 1827 NR_074974 clade_98 N N *Streptococcus sinensis* 1828 AF432857 clade_98 N N *Streptococcus* sp. 2_1_36FAA 1831 ACOI01000028 clade_98 N N *Streptococcus* sp. 2285_97 1830 AJ131965 clade_98 N N *Streptococcus* sp. ACS2 1834 HQ616360 clade_98 N N *Streptococcus* sp. AS20 1835 HQ616366 clade_98 N N *Streptococcus* sp. BS35a 1836 HQ616369 clade_98 N N *Streptococcus* sp. C150 1837 ACRI01000045 clade_98 N N *Streptococcus* sp. CM6 1838 HQ616372 clade_98 N N *Streptococcus* sp. ICM10 1840 HQ616389 clade_98 N N *Streptococcus* sp. ICM12 1841 HQ616390 clade_98 N N *Streptococcus* sp. ICM2 1842 HQ616386 clade_98 N N *Streptococcus* sp. ICM4 1844 HQ616387 clade_98 N N *Streptococcus* sp. ICM45 1843 HQ616394 clade_98 N N *Streptococcus* sp. M143 1845 ACRK01000025 clade_98 N N *Streptococcus* sp. M334 1846 ACRL01000052 clade_98 N N *Streptococcus* sp. oral clone ASB02 1849 AY923121 clade_98 N N *Streptococcus* sp. oral clone ASCA03 1850 DQ272504 clade_98 N N *Streptococcus* sp. oral clone ASCA04 1851 AY923116 clade_98 N N *Streptococcus* sp. oral clone ASCA09 1852 AY923119 clade_98 N N *Streptococcus* sp. oral clone ASCB04 1853 AY923123 clade_98 N N *Streptococcus* sp. oral clone ASCB06 1854 AY923124 clade_98 N N *Streptococcus* sp. oral clone ASCC04 1855 AY923127 clade_98 N N *Streptococcus* sp. oral clone ASCC05 1856 AY923128 clade_98 N N *Streptococcus* sp. oral clone ASCC12 1857 DQ272507 clade_98 N N *Streptococcus* sp. oral clone ASCD01 1858 AY923129 clade_98 N N *Streptococcus* sp. oral clone ASCD09 1859 AY923130 clade_98 N N *Streptococcus* sp. oral clone ASCD10 1860 DQ272509 clade_98 N N *Streptococcus* sp. oral clone ASCE03 1861 AY923134 clade_98 N N *Streptococcus* sp. oral clone ASCE04 1862 AY953253 clade_98 N N *Streptococcus* sp. oral clone ASCE05 1863 DQ272510 clade_98 N N *Streptococcus* sp. oral clone ASCE06 1864 AY923135 clade_98 N N *Streptococcus* sp. oral clone ASCE09 1865 AY923136 clade_98 N N *Streptococcus* sp. oral clone ASCE10 1866 AY923137 clade_98 N N *Streptococcus* sp. oral clone ASCE12 1867 AY923138 clade_98 N N *Streptococcus* sp. oral clone ASCF05 1868 AY923140 clade_98 N N *Streptococcus* sp. oral clone ASCF07 1869 AY953255 clade_98 N N *Streptococcus* sp. oral clone ASCF09 1870 AY923142 clade_98 N N *Streptococcus* sp. oral clone ASCG04 1871 AY923145 clade_98 N N *Streptococcus* sp. oral clone BW009 1872 AY005042 clade_98 N N *Streptococcus* sp. oral clone CH016 1873 AY005044 clade_98 N N *Streptococcus* sp. oral clone GK051 1874 AY349413 clade_98 N N *Streptococcus* sp. oral clone GM006 1875 AY349414 clade_98 N N *Streptococcus* sp. oral clone P2PA_41 P2 1876 AY207051 clade_98 N N *Streptococcus* sp. oral clone P4PA_30 P4 1877 AY207064 clade_98 N N *Streptococcus* sp. oral taxon 071 1878 AEEP01000019 clade_98 N N *Streptococcus* sp. oral taxon G59 1879 GU432132 clade_98 N N *Streptococcus* sp. oral taxon G62 1880 GU432146 clade_98 N N *Streptococcus* sp. oral taxon G63 1881 GU432150 clade_98 N N *Streptococcus suis* 1882 FM252032 clade_98 N N *Streptococcus thermophilus* 1883 CP000419 clade_98 N N *Streptococcus* *uberis* 1884 HQ391900 clade_98 N N *Streptococcus urinalis* 1885 DQ303194 clade_98 N N *Streptococcus vestibularis* 1886 AEKO01000008 clade_98 N N *Streptococcus viridans* 1887 AF076036 clade_98 N N *Synergistetes bacterium* oral clone 03 5 D05 1908 GU227192 clade_98 N N

(310) TABLE-US-00002 TABLE 2 Phylogenetic Clade OTUs in clade clade_172 Bifidobacteriaceae genomosp. C1, *Bifidobacterium adolescentis*, *Bifidobacterium angulatum*, *Bifidobacterium animalis*, *Bifidobacterium breve*, *Bifidobacterium catenulatum*, *Bifidobacterium dentium*, *Bifidobacterium gallicuin*, *Bifidobacterium infantis*, *Bifidobacterium kashiwanohense*, *Bifidobacterium longum*, *Bifidobacterium pseudocatenulatum*, *Bifidobacterium pseudolongum*, *Bifidobacterium scardovii*, *Bifidobacterium* sp. HM2, *Bifidobacterium* sp. HMLN12, *Bifidobacterium* sp. M45, *Bifidobacterium* sp. MSX5B, *Bifidobacterium* sp. TM_7, *Bifidobacterium thermophilum* clade_172i Bifidobacteriaceae genomosp. C1, *Bifidobacterium angulatum*, *Bifidobacterium animalis*, *Bifidobacterium breve*, *Bifidobacterium catenulatum*, *Bifidobacterium dentium*, *Bifidobacterium gallicum*, *Bifidobacterium infantis*, *Bifidobacterium kashiwanohense*, *Bifidobacterium pseudocatenulatum*, *Bifidobacterium pseudolongum*, *Bifidobacterium scardovii*, *Bifidobacterium* sp. HM2, *Bifidobacterium* sp. HMLN12, *Bifidobacterium* sp. M45, *Bifidobacterium* sp. MSX5B, *Bifidobacterium* sp. TM 7, *Bifidobacterium thermophilum* clade_198 *Lactobacillus casei*, *Lactobacillus paracasei*, *Lactobacillus zeae* clade_198i *Lactobacillus zeae* clade_260 *Clostridium hylemonae*, *Clostridium scindens*, Lachnospiraceae bacterium 5_1_57FAA clade_260c *Clostridium hylemonae*, Lachnospiraceae bacterium 5_1_57FAA clade_260g *Clostridium hylemonae*, Lachnospiraceae bacterium 5_1_57FAA clade_260h *Clostridium hylemonae*, Lachnospiraceae bacterium 5_1_57FAA clade_262 *Clostridium glycyrrhizinilyticum*, *Clostridium nexile*, *Coprococcus comes*, Lachnospiraceae bacterium 1_1_57FAA, Lachnospiraceae bacterium 1_4_56FAA, Lachnospiraceae bacterium 8_1_57FAA, *Ruminococcus lactaris*, *Rummococcus torques* clade_262i *Clostridium glycyrrhizinilyticum*, *Clostridium nexile*, *Coprococcus comes*, Lachnospiraceae bacterium 1_1_57FAA, Lachnospiraceae bacterium 1_4_56FAA, Lachnospiraceae bacterium 8_1_57FAA, *Ruminococcus lactaris* clade_309 *Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia producta*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolveris*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans* clade_309c *Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans* clade_309e *Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans* clade_309g *Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans* clade_309h *Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans* clade_309i *Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans* clade_313 *Lactobacillus antri*, *Lactobacillus coleohominis*, *Lactobacillus fermentum*, *Lactobacillus gastricus*, *Lactobacillus mucosae*, *Lactobacillus oris*, *Lactobacillus pontis*, *Lactobacillus reuteri*, *Lactobacillus* sp. KLDS 1.0707, *Lactobacillus* sp. KLDS 1.0709, *Lactobacillus* sp. KLDS 1.0711, *Lactobacillus* sp. KLDS 1.0713, *Lactobacillus* sp. KLDS 1.0716, *Lactobacillus* sp. KLDS 1.0718, *Lactobacillus* sp. oral taxon 052, *Lactobacillus vaginalis* clade_313f *Lactobacillus antri*, *Lactobacillus coleohominis*, *Lactobacillus fermentum*, *Lactobacillus gastricus*, *Lactobacillus mucosae*, *Lactobacillus oris*, *Lactobacillus pontis*, *Lactobacillus* sp. KLDS 1.0707, *Lactobacillus* sp. KLDS 1.0709, *Lactobacillus* sp. KLDS 1.0711, *Lactobacillus* sp. KLDS 1.0713, *Lactobacillus* sp. KLDS 1.0716, *Lactobacillus* sp. KLDS 1.0718, *Lactobacillus* sp. oral taxon 052, *Lactobacillus vaginalis* clade_325 *Staphylococcus aureus*, *Staphylococcus auricularis*, *Staphylococcus capitis*, *Staphylococcus caprae*, *Staphylococcus carnosus*, *Staphylococcus cohnii*, *Staphylococcus condimenti*, *Staphylococcus epidermidis*, *Staphylococcus equorum*, *Staphylococcus haemolyticus*, *Staphylococcus hominis*, *Staphylococcus lugdunensis*, *Staphylococcus pasteuri*, *Staphylococcus pseudintermedius*, *Staphylococcus saccharolyticus*, *Staphylococcus saprophyticus*, *Staphylococcus* sp. H292, *Staphylococcus* sp. H780, *Staphylococcus* sp. clone bottae7, *Staphylococcus succinus*, *Staphylococcus warneri*, *Staphylococcus xylosus* clade_325f *Staphylococcus aureus*, *Staphylococcus auricularis*, *Staphylococcus capitis*, *Staphylococcus caprae*, *Staphylococcus carnosus*, *Staphylococcus cohnii*, *Staphylococcus condimenti*, *Staphylococcus epidermidis*, *Staphylococcus equorum*, *Staphylococcus haemolyticus*, *Staphylococcus hominis*, *Staphylococcus lugdunensis*, *Staphylococcus pseudintermedius*, *Staphylococcus saccharolyticus*, *Staphylococcus saprophyticus*, *Staphylococcus* sp. H292, *Staphylococcus* sp. H780, *Staphylococcus* sp. clone bottae7, *Staphylococcus succinus*, *Staphylococcus xylosus* clade_335 *Bacteroides* sp. 20_3, *Bacteroides* sp. 3_1_19, *Bacteroides* sp. 3_2_5, *Parabacteroides distasonis*, *Parabacteroides goldsteinii*, *Parabacteroides gordonii*, *Parabacteroides* sp. D13 clade_335i *Bacteroides* sp. 20_3, *Bacteroides* sp. 3_1_19, *Bacteroides* sp. 3_2_5, *Parabacteroides goldsteinii*, *Parabacteroides gordonii*, *Parabacteroides* sp. D13 clade_351 *Clostridium innocuum*, *Clostridium* sp. HGF2 clade_351e *Clostridium* sp. HGF2 clade_354 *Clostridium bartlettii*, *Clostridium bifermentans*, *Clostridium ghonii*, *Clostridium glycolicum*, *Clostridium mayombei*, *Clostridium sordellii*, *Clostridium* sp. MT4 E, *Eubacterium tenue* clade_354e *Clostridium bartlettii*, *Clostridium ghonii*, *Clostridium glycolicum*, *Clostridium mayombei*, *Clostridium sordellii*, *Clostridium* sp. MT4 E, *Eubacterium tenue* clade_360 *Dorea formicigenerans*, *Dorea longicatena*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus*, *Ruminococcus* sp. ID8 clade_360c *Dorea formicigenerans*, *Dorea longicatena*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus* clade_360g *Dorea formicigenerans*, *Dorea longicatena*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus* clade_360h *Dorea formicigenerans*, *Dorea longicatena*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus* clade_360i *Dorea formicigenerans*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus*, *Ruminococcus* sp. ID8 clade_378 *Bacteroides barnesiae*, *Bacteroides coprocola*, *Bacteroides coprophilus*, *Bacteroides dorei*, *Bacteroides massiliensis*, *Bacteroides plebeius*, *Bacteroides* sp. 3_1_33FAA, *Bacteroides* sp. 3_1_40A, *Bacteroides* sp. 4_3_47FAA, *Bacteroides* sp. 9_1_42FAA, *Bacteroides* sp. NB_8, *Bacteroides vulgatus* clade_378e *Bacteroides barnesiae*, *Bacteroides coprocola*, *Bacteroides coprophilus*, *Bacteroides dorei*, *Bacteroides massiliensis*, *Bacteroides plebeius*, *Bacteroides* sp. 3_1_33FAA, *Bacteroides* sp. 3_1_40A, *Bacteroides* sp. 4_3_47FAA, *Bacteroides* sp. 9_1_42FAA, *Bacteroides* sp. NB_8 clade_38 *Bacteroides ovatus*, *Bacteroides* sp. 1_1_30, *Bacteroides* sp. 2_1_22, *Bacteroides* sp. 2_2_4, *Bacteroides* sp. 3_1_23, *Bacteroides* sp.

D1, *Bacteroides* sp. D2, *Bacteroides* sp. D22, *Bacteroides xylanisolvens* clade_38e *Bacteroides* sp. 1_1_30, *Bacteroides* sp. 2_1_22, *Bacteroides* sp. 2_2_4, *Bacteroides* sp. 3_1_23, *Bacteroides* sp. D1, *Bacteroides* sp. D2, *Bacteroides* sp. D22, *Bacteroides xylanisolvens* clade_38i *Bacteroides* sp. 1_1_30, *Bacteroides* sp. 2_1_22, *Bacteroides* sp. 2_2_4, *Bacteroides* sp. 3_1_23, *Bacteroides* sp. D1, *Bacteroides* sp. D2, *Bacteroides* sp. D22, *Bacteroides xylanisolvens* clade_408 *Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales bacterium 1_7_47FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium asparagiforme*, *Clostridium bolteae*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium hathewayi*, *Clostridium indolis*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 3_1_57FAA, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes* clade_408b *Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales bacterium 1_7_47FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium asparagiforme*, *Clostridium bolteae*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium indolis*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes* clade_408d *Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium hathewayi*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 3_1_57FAA, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes* clade_408f *Anaerostipes* sp. 3_2_56FAA, Clostridiales bacterium 1_7_47FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium asparagiforme*, *Clostridium bolteae*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium hathewayi*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 3_1_57FAA, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes* clade_408g *Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes* clade_408h *Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes* clade_420 *Barnesiella intestinihominis*, *Barnesiella viscericola*, *Parabacteroides* sp. NS31_3, Porphyromonadaceae bacterium NML 060648, *Tannerella forsythia*, *Tannerella* sp. 6_1_58FAA_CT1 clade_420f *Barnesiella viscericola*, *Parabacteroides* sp. NS31_3. Porphyromonadaceae bacterium NML 060648, *Tannerella forsythia*, *Tannerella* sp. 6_1_58FAA_CT1 clade_444 *Butyrivibrio fibrisolvens*, *Eubacterium rectale*, *Eubacterium* sp. oral clone GI038, *Lachnobacterium bovis*, *Roseburia cecicola*, *Roseburia faecalis*, *Roseburia faecis*, *Roseburia hominis*, *Roseburia intestinalis*, *Roseburia inulinivorans*, *Roseburia* sp. 11SE37, *Roseburia* sp. 11SE38, *Shuttleworthia satelles*, *Shuttleworthia* sp. MSX8B, *Shuttleworthia* sp. oral taxon G69 clade_444i *Butyrivibrio fibrisolvens*, *Eubacterium* sp. oral clone GI038, *Lachnobacterium bovis*, *Roseburia cecicola*, *Roseburia faecis*, *Roseburia hominis*, *Roseburia inulinivorans*, *Roseburia* sp. 11SE37, *Roseburia* sp. 11SE38, *Shuttleworthia satelles*, *Shuttleworthia* sp. MSX8B, *Shuttleworthia* sp. oral taxon G69 clade_478 *Faecalibacterium prausnitzii*, *Gemmiger formicilis*, *Subdoligranulum variabile* clade_478i *Gemmiger formicilis*, *Subdoligranulum variabile* clade_479 Clostridiaceae bacterium JC13, *Clostridium* sp. MLG055, Erysipelotrichaceae bacterium 3_1_53 clade_479c *Clostridium* sp. MLG055, Erysipelotrichaceae bacterium 3_1_53 clade_479g *Clostridium* sp. MLG055, Erysipelotrichaceae bacterium 3_1_53 clade_479h *Clostridium* sp. MLG055, Erysipelotrichaceae bacterium 3_1_53 clade_481 *Clostridium cocleatum*, *Clostridium ramosum*, *Clostridium saccharogumia*, *Clostridium spiroforme*, *Coprobacillus* sp. D7 clade_481a *Clostridium cocleatum*, *Clostridium spiroforme*, *Coprobacillus* sp. D7 clade_481b *Clostridium cocleatum*, *Clostridium ramosum*, *Clostridium spiroforme*, *Coprobacillus* sp. D7 clade_481e *Clostridium cocleatum*, *Clostridium saccharogumia*, *Clostridium spiroforme*, *Coprobacillus* sp. D7 clade_481g *Clostridium cocleatum*, *Clostridium spiroforme*, *Coprobacillus* sp. D7 clade_481h *Clostridium cocleatum*, *Clostridium spiroforme*, *Coprobacillus* sp. D7 clade_481i *Clostridium ramosum*, *Clostridium saccharogumia*, *Clostridium spiroforme*, *Coprobacillus* sp. D7 clade_497 *Abiotrophia para_adiacens*, *Carnobacterium divergens*, *Carnobacterium maltaromaticum*, *Enterococcus avium*, *Enterococcus caccae*, *Enterococcus casseliflavus*, *Enterococcus durans*, *Enterococcus faecalis*, *Enterococcus faecium*, *Enterococcus gallinarum*, *Enterococcus gilvus*, *Enterococcus hawaiiensis*, *Enterococcus hirae*, *Enterococcus italicus*, *Enterococcus mundtii*, *Enterococcus raffinosus*, *Enterococcus* sp. BV2CASA2, *Enterococcus* sp. CCRI 16620, *Enterococcus* sp. F95, *Enterococcus* sp. RfL6, *Enterococcus thailandieus*, *Fusobacterium canifelinum*, *Fusobacterium* genomosp. C1, *Fusobacterium* genomosp. C2, *Fusobacterium nucleatum*, *Fusobacterium periodonticum*, *Fusobacterium* sp. 11_3_2, *Fusobacterium* sp. 1_1_41FAA, *Fusobacterium* sp. 2_1_31, *Fusobacterium* sp. 3_1_27, *Fusobacterium* sp. 3_1_33, *Fusobacterium* sp. 3_1_36A2, *Fusobacterium* sp. AC18, *Fusobacterium* sp. ACB2, *Fusobacterium* sp. AS2, *Fusobacterium* sp. CM1, *Fusobacterium* sp. CM21, *Fusobacterium* sp. CM22, *Fusobacterium* sp. oral clone ASCF06, *Fusobacterium* sp. oral clone ASCF11, *Granulicatella adiacens*, *Granulicatella elegans*, *Granulicatella paradiacens*, *Granulicatella* sp. oral clone ASC02, *Granulicatella* sp. oral clone ASCA05, *Granulicatella* sp. oral clone ASCB09, *Granulicatella* sp. oral clone ASCG05, *Tetragenococcus halophilus*, *Tetragenococcus koreensis*, *Vagococcus fluvialis* clade_497e *Abiotrophia para_adiacens*, *Carnobacterium divergens*, *Carnobacterium maltaromaticum*, *Enterococcus avium*, *Enterococcus caccae*, *Enterococcus casseliflavus*, *Enterococcus durans*, *Enterococcus faecium*, *Enterococcus gallinarum*, *Enterococcus gilvus*, *Enterococcus hawaiiensis*, *Enterococcus hirae*, *Enterococcus italicus*, *Enterococcus mundtii*, *Enterococcus raffinosus*, *Enterococcus* sp. BV2CASA2, *Enterococcus* sp. CCRI 16620, *Enterococcus* sp. F95, *Enterococcus* sp. RfL6, *Enterococcus thailandicus*, *Fusobacterium canifelinum*, *Fusobacterium* genomosp. C1, *Fusobacterium* genomosp. C2, *Fusobacterium nucleatum*, *Fusobacterium periodonticum*, *Fusobacterium* sp. 11_3_2, *Fusobacterium* sp. 1_1_41FAA, *Fusobacterium* sp. 2_1_31, *Fusobacterium* sp. 3_1_27, *Fusobacterium* sp. 3_1_33, *Fusobacterium* sp. 3_1_36A2, *Fusobacterium* sp. AC18, *Fusobacterium* sp. ACB2, *Fusobacterium* sp. AS2, *Fusobacterium* sp. CM1, *Fusobacterium* sp. CM21, *Fusobacterium* sp. CM22, *Fusobacterium* sp. oral clone ASCF06, *Fusobacterium* sp. oral clone ASCF11, *Granulicatella adiacens*, *Granulicatella elegans*, *Granulicatella paradiacens*, *Granulicatella* sp. oral clone ASC02, *Granulicatella* sp. oral clone ASCA05, *Granulicatella* sp. oral clone ASCB09, *Granulicatella* sp. oral clone ASCG05, *Tetragenococcus halophilus*, *Tetragenococcus koreensis*, *Vagococcus fluvialis* clade_497f *Abiotrophia para_adiacens*, *Carnobacterium divergens*, *Carnobacterium maltaromaticum*, *Enterococcus avium*, *Enterococcus caccae*, *Enterococcus casseliflavus*, *Enterococcus faecalis*, *Enterococcus gallinarum*, *Enterococcus gilvus*, *Enterococcus hawaiiensis*, *Enterococcus italicus*, *Enterococcus mundtii*, *Enterococcus raffinosus*, *Enterococcus* sp. BV2CASA2, *Enterococcus* sp. CCRI 16620, *Enterococcus* sp. F95, *Enterococcus* sp. RfL6, *Enterococcus thailandicus*, *Fusobacterium canifelinum*, *Fusobacterium* genomosp. C1, *Fusobacterium* genomosp. C2, *Fusobacterium nucleatum*, *Fusobacterium periodonticum*, *Fusobacterium* sp. 11_3_2, *Fusobacterium* sp. 1_1_41FAA, *Fusobacterium* sp. 2_1_31, *Fusobacterium* sp. 3_1_27, *Fusobacterium* sp. 3_1_33, *Fusobacterium* sp. 3_1_36A2, *Fusobacterium* sp. AC18, *Fusobacterium* sp. ACB2, *Fusobacterium* sp. AS2, *Fusobacterium* sp. CM1, *Fusobacterium* sp. CM21, *Fusobacterium* sp. CM22, *Fusobacterium* sp. oral clone ASCF06, *Fusobacterium* sp. oral clone ASCF11, *Granulicatella adiacens*, *Granulicatella elegans*, *Granulicatella paradiacens*, *Granulicatella* sp. oral clone ASC02, *Granulicatella* sp. oral clone ASCA05, *Granulicatella* sp. oral clone ASCB09, *Granulicatella* sp. oral clone ASCG05, *Tetragenococcus halophilus*, *Tetragenococcus koreensis*, *Vagococcus fluvialis* clade_512 *Eubacterium barkeri*, *Eubacterium callanderi*, *Eubacterium limosum*, *Pseudoramibacter alactolyticus* clade_512i *Eubacterium barkeri*, *Eubacterium callanderi*, *Pseudoramibacter alactolyticus* clade_516 *Anaerotruncus colihominis*, *Clostridium methylpentosum*, *Clostridium* sp. YIT 12070, *Hydrogenoanaerobacterium saccharovorans*, *Ruminococcus albus*, *Ruminococcus flavefaciens* clade_516c *Clostridium methylpentosum*, *Clostridium* sp. YIT 12070, *Hydrogenoanaerobacterium saccharovorans*, *Ruminococcus albus*, *Ruminococcus flavefaciens* clade_516g *Clostridium methylpentosum*, *Clostridium* sp. YIT 12070, *Hydrogenoanaerobacterium saccharovorans*, *Ruminococcus albus*, *Ruminococcus flavefaciens* clade_516h *Clostridium methylpentosum*, *Clostridium* sp. YIT 12070, *Hydrogenoanaerobacterium saccharovorans*, *Ruminococcus albus*, *Ruminococcus flavefaciens* clade_519 *Eubacterium ventriosum* clade_522 *Bacteroides galacturonicus*, *Eubacterium eligens*, *Lachnospira multipara*, *Lachnospira pectinoschiza*, *Lactobacillus rogosae* clade_522i *Bacteroides galacturonicus*, *Lachnospira multipara*, *Lachnospira pectinoschiza*, *Lactobacillus rogosae* clade_553 *Collinsella aerofaciens*, *Collinsella intestinalis*, *Collinsella stercoris*, *Collinsella tanakaei* clade_553i *Collinsella intestinalis*, *Collinsella stercoris*, *Collinsella tanakaei* clade_566 *Adlercreutzia equolifaciens*, Coriobacteriaceae bacterium JC110, Coriobacteriaceae bacterium phI, *Cryptobacterium curtum*, *Eggerthella lenta*, *Eggerthella sinensis*, *Eggerthella* sp. 1_3_56FAA, *Eggerthella* sp. HGA1, *Eggerthella* sp. YY7918, *Gordonibacter pamelaeae*, *Slackia equolifaciens*, *Slackia exigua*, *Slackia faecicanis*, *Slackia heliotrinireducens*, *Slackia isoflavoniconvertens*, *Slackia piriformis*, *Slackia* sp. NATTS, *Streptomyces albus* clade_566f Coriobacteriaceae bacterium JC110, Coriobacteriaceae bacterium phI, *Cryptobacterium curtum*, *Eggerthella lenta*, *Eggerthella sinensis*, *Eggerthella* sp. 1_3_56FAA, *Eggerthella* sp. HGA1, *Eggerthella* sp. YY7918, *Gordonibacter pamelaeae*, *Slackia equolifaciens*, *Slackia exigua*, *Slackia faecicanis*, *Slackia heliotrinireducens*, *Slackia isoflavoniconvertens*, *Slackia piriformis*, *Slackia* sp. NATTS, *Streptomyces albus* clade_572 *Butyricicoccus pullicaecorum*, *Eubacterium desmolans*, *Papillibacter cinnamivorans*, *Sporobacter termitidis* clade_572i *Butyricicoccus pullicaecorum*, *Papillibacter cinnamivorans*, *Sporobacter termitidis* clade_65 *Bacteroides faecis*, *Bacteroides fragilis*, *Bacteroides nordii*, *Bacteroides salyersiae*, *Bacteroides* sp. 1_1_14, *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 2_1_56FAA, *Bacteroides* sp. AR29, *Bacteroides* sp. B2, *Bacteroides thetaiotaomicron* clade_65e *Bacteroides faecis*, *Bacteroides fragilis*, *Bacteroides nordii*, *Bacteroides salyersiae*, *Bacteroides* sp. 1_1_14, *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 2_1_56FAA, *Bacteroides* sp. AR29, *Bacteroides* sp. B2 clade_92 *Actinobacillus actinomycetemcomitans*, *Actinobacillus succinogenes*, *Aggregatibacter actinomycetemcomitans*, *Aggregatibacter aphrophilus*, *Aggregatibacter segnis*, *Averyella dalhousiensis*, Bisgaard Taxon, *Buchnera aphidicola*, *Cedecea davisae*, *Citrobacter amalonaticus*, *Citrobacter braakii*, *Citrobacter farmeri*, *Citrobacter freundii*, *Citrobacter gillenii*, *Citrobacter koseri*, *Citrobacter murliniae*, *Citrobacter rodentium*, *Citrobacter sedlakii*, *Citrobacter* sp. 30_2, *Citrobacter* sp. KMSI_3, *Citrobacter werkmanii*, *Citrobacter youngae*, *Cronobacter malonaticus*, *Cronobacter sakazakii*, *Cronobacter turicensis*, *Enterobacter aerogenes*, *Enterobacter asburiae*, *Enterobacter cancerogenus*, *Enterobacter cloacae*, *Enterobacter cowanii*, *Enterobacter hormaechei*, *Enterobacter* sp. 247BMC, *Enterobacter* sp. 638, *Enterobacter* sp. JC163, *Enterobacter* sp. SCSS, *Enterobacter* sp. TSE38, Enterobacteriaceae bacterium 9_2_54FAA, Enterobacteriaceae bacterium CF01Ent_1, Enterobacteriaceae bacterium Smarlab 3302238, *Escherichia albertii*, *Escherichia coli*, *Escherichia fergusonii*, *Escherichia hermannii*, *Escherichia* sp. 1_1_43, *Escherichia* sp. 4_1_40B, *Escherichia* sp. B4, *Escherichia vulneris*, *Ewingella americana*, *Haemophilus* genomosp. P2 oral clone MB3_C24, *Haemophilus* genomosp. P3 oral clone MB3_C38, *Haemophilus* sp. oral clone JM053, *Hafnia alvei*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Klebsiella* sp. AS10, *Klebsiella* sp. Co9935, *Klebsiella* sp. OBRC7, *Klebsiella* sp. SP_BA, *Klebsiella* sp. SRC_DSD1, *Klebsiella* sp. SRC_DSD11, *Klebsiella* sp. SRC_DSD12, *Klebsiella* sp. SRC_DSD15, *Klebsiella* sp. SRC_DSD2, *Klebsiella* sp. SRC_DSD6, *Klebsiella* sp. enrichment culture clone SRC_DSD25, *Klebsiella variicola*, *Kluyvera ascorbata*, *Kluyvera cryocrescens*, *Leminorella grimontii*, *Leminorella richardii*, *Pantoea agglomerans*, *Pantoea ananatis*, *Pantoea brenneri*, *Pantoea citrea*, *Pantoea conspicua*, *Pantoea septica*, *Pasteurella dagmatis*, *Pasteurella multocida*, *Plesiomonas shigelloides*, *Raoultella ornithinolytica*, *Raoultella planticola*, *Raoultella terrigena*, *Salmonella bongori*, *Salmonella enterica*, *Salmonella typhimurium*, *Serratia fonticola*, *Serratia liquefaciens*, *Serratia marcescens*, *Serratia odorifera*, *Serratia proteamaculans*, *Shigella boydii*, *Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Tatumella ptyseos*, *Trabulsiella guamensis*, *Yersinia aldovae*, *Yersinia aleksiciae*, *Yersinia bercovieri*, *Yersinia enterocolitica*, *Yersinia frederiksenii*, *Yersinia intermedia*, *Yersinia kristensenii*, *Yersinia mollaretii*, *Yersinia pestis*, *Yersinia pseudotuberculosis*, *Yersinia rohdei*, *Yokenella regensburgei* clade_92e *Actinobacillus actinomycetemcomitans*, *Actinobacillus succinogenes*, *Aggregatibacter actinomycetemcomitans*, *Aggregatibacter aphrophilus*, *Aggregatibacter segnis*, *Averyella dalhousiensis*, Bisgaard Taxon, *Buchnera aphidicola*, *Cedecea davisae*, *Citrobacter amalonaticus*, *Citrobacter braakii*, *Citrobacter farmeri*, *Citrobacter freundii*, *Citrobacter gillenii*, *Citrobacter koseri*, *Citrobacter murliniae*, *Citrobacter rodentium*, *Citrobacter sedlakii*, *Citrobacter* sp. 30_2, *Citrobacter* sp. KMSI_3, *Citrobacter werkmanii*, *Citrobacter youngae*, *Cronobacter malonaticus*, *Cronobacter sakazakii*, *Cronobacter turicensis*, *Enterobacter aerogenes*, *Enterobacter asburiae*, *Enterobacter cancerogenus*, *Enterobacter cloacae*, *Enterobacter cowanii*, *Enterobacter hormaechei*, *Enterobacter* sp. 247BMC, *Enterobacter* sp. 638, *Enterobacter* sp. JC163, *Enterobacter* sp. SCSS, *Enterobacter* sp. TSE38, Enterobacteriaceae bacterium 9_2_54FAA, Enterobacteriaceae bacterium CF01Ent_1, Enterobacteriaceae bacterium Smarlab 3302238, *Escherichia albertii*, *Escherichia fergusonii*, *Escherichia hermannii*, *Escherichia* sp. 1_1_43, *Escherichia* sp. 4_1_40B, *Escherichia* sp. B4, *Escherichia vulneris*, *Ewingella americana*, *Haemophilus* genomosp. P2 oral clone MB3_C24, *Haemophilus* genomosp. P3 oral clone MB3_C38, *Haemophilus* sp. oral clone JM053, *Hafnia alvei*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Klebsiella* sp. AS10, *Klebsiella* sp. Co9935, *Klebsiella* sp. OBRC7, *Klebsiella* sp. SP_BA, *Klebsiella* sp. SRC_DSD1, *Klebsiella* sp. SRC_DSD11, *Klebsiella* sp. SRC_DSD12, *Klebsiella* sp. SRC_DSD15, *Klebsiella* sp. SRC_DSD2, *Klebsiella* sp. SRC_DSD6, *Klebsiella* sp. enrichment culture clone SRC_DSD25, *Klebsiella variicola*, *Kluyvera ascorbata*, *Kluyvera cryocrescens*, *Leminorella grimontii*, *Leminorella richardii*, *Pantoea agglomerans*, *Pantoea ananatis*, *Pantoea brenneri*, *Pantoea citrea*, *Pantoea conspicua*, *Pantoea septica*, *Pasteurella dagmatis*, *Pasteurella multocida*, *Plesiomonas shigelloides*, *Raoultella ornithinolytica*, *Raoultella planticola*, *Raoultella terrigena*, *Salmonella bongori*, *Salmonella enterica*, *Salmonella typhimurium*, *Serratia fonticola*, *Serratia liquefaciens*, *Serratia marcescens*, *Serratia odorifera*, *Serratia proteamaculans*, *Shigella boydii*, *Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Tatumella ptyseos*, *Trabulsiella guamensis*, *Yersinia aldovae*, *Yersinia aleksiciae*, *Yersinia bercovieri*, *Yersinia enterocolitica*, *Yersinia frederiksenii*, *Yersinia intermedia*, *Yersinia kristensenii*, *Yersinia mollaretii*, *Yersinia pestis*, *Yersinia pseudotuberculosis*, *Yersinia rohdei*, *Yokenella regensburgei* clade_92i *Actinobacillus actinomycetemcomitans*, *Actinobacillus succinogenes*, *Aggregatibacter actinomycetemcomitans*, *Aggregatibacter aphrophilus*, *Aggregatibacter segnis*, *Averyella dalhousiensis*, Bisgaard Taxon, *Buchnera aphidicola*, *Cedecea davisae*, *Citrobacter amalonaticus*, *Citrobacter braakii*, *Citrobacter farmeri*, *Citrobacter freundii*, *Citrobacter gillenii*, *Citrobacter koseri*, *Citrobacter murliniae*, *Citrobacter rodentium*, *Citrobacter sedlakii*, *Citrobacter* sp. 30_2, *Citrobacter* sp. KMSI_3, *Citrobacter werkmanii*, *Citrobacter youngae*, *Cronobacter malonaticus*, *Cronobacter sakazakii*, *Cronobacter turicensis*, *Enterobacter aerogenes*, *Enterobacter asburiae*, *Enterobacter cancerogenus*, *Enterobacter cloacae*, *Enterobacter cowanii*, *Enterobacter hormaechei*, *Enterobacter* sp. 247BMC, *Enterobacter* sp. 638, *Enterobacter* sp. JC163, *Enterobacter* sp. SCSS, *Enterobacter* sp. TSE38, Enterobacteriaceae bacterium 9_2_54FAA, Enterobacteriaceae bacterium CF01Ent_1, Enterobacteriaceae bacterium Smarlab 3302238, *Escherichia albertii*, *Escherichia fergusonii*, *Escherichia hermannii*, *Escherichia* sp. 1_1_43, *Escherichia* sp. 4_1_40B, *Escherichia* sp. B4, *Escherichia vulneris*, *Ewingella americana*, *Haemophilus* genomosp. P2 oral clone MB3_C24, *Haemophilus* genomosp. P3 oral clone MB3_C38, *Haemophilus* sp. oral clone JM053, *Hafnia alvei*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Klebsiella* sp. AS10, *Klebsiella* sp. Co9935, *Klebsiella* sp. OBRC7, *Klebsiella* sp. SP_BA, *Klebsiella* sp. SRC_DSD1, *Klebsiella* sp. SRC_DSD11, *Klebsiella* sp. SRC_DSD12, *Klebsiella* sp. SRC_DSD15, *Klebsiella* sp. SRC_DSD2, *Klebsiella* sp. SRC_DSD6, *Klebsiella* sp. enrichment culture clone SRC_DSD25, *Klebsiella variicola*, *Kluyvera ascorbata*, *Kluyvera cryocrescens*, *Leminorella grimontii*, *Leminorella richardii*, *Pantoea agglomerans*, *Pantoea ananatis*, *Pantoea brenneri*, *Pantoea citrea*, *Pantoea conspicua*, *Pantoea septica*, *Pasteurella dagmatis*, *Pasteurella multocida*, *Plesiomonas shigelloides*, *Raoultella ornithinolytica*, *Raoultella planticola*, *Raoultella terrigena*, *Salmonella bongori*, *Salmonella enterica*, *Salmonella typhimurium*, *Serratia fonticola*, *Serratia liquefaciens*, *Serratia marcescens*, *Serratia odorifera*, *Serratia proteamaculans*, *Shigella boydii*, *Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Tatumella ptyseos*, *Trabulsiella guamensis*, *Yersinia aldovae*, *Yersinia aleksiciae*, *Yersinia bercovieri*, *Yersinia enterocolitica*, *Yersinia frederiksenii*, *Yersinia intermedia*, *Yersinia kristensenii*, *Yersinia mollaretii*, *Yersinia pestis*, *Yersinia pseudotuberculosis*, *Yersinia rohdei*, *Yokenella regensburgei* clade_96 *Clostridium oroticum*, *Clostridium* sp. D5, *Eubacterium contortum*, *Eubacterium fissicatena* clade_96g *Clostridium oroticum*, *Clostridium* sp. D5, *Eubacterium fissicatena* clade_96h *Clostridium oroticum*, *Clostridium* sp. D5, *Eubacterium fissicatena* clade_98 *Okadaella gastrococcus*, *Streptococcus agalactiae*, *Streptococcus alactolyticus*, *Streptococcus australis*, *Streptococcus bovis*, *Streptococcus canis*, *Streptococcus constellatus*, *Streptococcus cristatus*, *Streptococcus dysgalactiae*, *Streptococcus equi*, *Streptococcus equinus*, *Streptococcus gallolyticus*, *Streptococcus* genomosp. C1, *Streptococcus* genomosp. C2, *Streptococcus* genomosp. C3, *Streptococcus* genomosp. C4, *Streptococcus* genomosp. C5, *Streptococcus* genomosp. C6, *Streptococcus* genomosp. C7, *Streptococcus* genomosp. C8, *Streptococcus gordonii*, *Streptococcus infantarius*, *Streptococcus infantis*, *Streptococcus intermedius*, *Streptococcus lutetiensis*, *Streptococcus massiliensis*, *Streptococcus mitis*, *Streptococcus oligofermentans*, *Streptococcus oralis*, *Streptococcus parasanguinis*, *Streptococcus pasteurianus*, *Streptococcus peroris*, *Streptococcus pneumoniae*, *Streptococcus porcinus*, *Streptococcus pseudopneumoniae*, *Streptococcus pseudoporcinus*, *Streptococcus pyogenes*, *Streptococcus ratti*, *Streptococcus salivarius*, *Streptococcus sanguinis*, *Streptococcus sinensis*, *Streptococcus* sp. 2285_97, *Streptococcus* sp. 2_1_36FAA, *Streptococcus* sp. ACS2, *Streptococcus* sp. AS20, *Streptococcus* sp. BS35a, *Streptococcus* sp. C150, *Streptococcus* sp. CM6, *Streptococcus* sp. ICM10, *Streptococcus* sp. ICM12, *Streptococcus* sp. ICM2, *Streptococcus* sp. ICM4, *Streptococcus* sp. ICM45, *Streptococcus* sp. M143, *Streptococcus* sp. M334, *Streptococcus* sp. oral clone ASB02, *Streptococcus* sp. oral clone ASCA03, *Streptococcus* sp. oral clone ASCA04, *Streptococcus* sp. oral clone ASCA09, *Streptococcus* sp. oral clone ASCB04, *Streptococcus* sp. oral clone ASCB06, *Streptococcus* sp. oral clone ASCC04, *Streptococcus* sp. oral clone ASCC05, *Streptococcus* sp. oral clone ASCC12, *Streptococcus* sp. oral clone ASCD01, *Streptococcus* sp. oral clone ASCD09, *Streptococcus* sp. oral clone ASCD10, *Streptococcus* sp. oral clone ASCE03, *Streptococcus* sp. oral clone ASCE04, *Streptococcus* sp. oral clone ASCE05, *Streptococcus* sp. oral clone ASCE06, *Streptococcus* sp. oral clone ASCE09, *Streptococcus* sp. oral clone ASCE10, *Streptococcus* sp. oral clone ASCE12, *Streptococcus* sp. oral clone ASCF05, *Streptococcus* sp. oral clone ASCF07, *Streptococcus* sp. oral clone ASCF09, *Streptococcus* sp. oral clone ASCG04, *Streptococcus* sp. oral clone RW009, *Streptococcus* sp. oral clone CH016, *Streptococcus* sp. oral clone GK051, *Streptococcus* sp. oral clone GM006, *Streptococcus* sp. oral clone P2PA_41 P2, *Streptococcus* sp. oral clone P4PA_30 P4, *Streptococcus* sp. oral taxon 071, *Streptococcus* sp. oral taxon G59, *Streptococcus* sp. oral taxon G62, *Streptococcus* sp. oral taxon G63, *Streptococcus suis*, *Streptococcus thermophilus*, *Streptococcus uberis*, *Streptococcus urinalis*, *Streptococcus vestibularis*, *Streptococcus viridans*, Synergistetes bacterium oral clone 03 5 D05 clade_98i *Okadaella gastrococcus*, *Streptococcus agalactiae*, *Streptococcus alactolyticus*, *Streptococcus australis*, *Streptococcus bovis*, *Streptococcus canis*, *Streptococcus constellatus*, *Streptococcus cristatus*, *Streptococcus dysgalactiae*, *Streptococcus equi*, *Streptococcus equinus*, *Streptococcus gallolyticus*, *Streptococcus* genomosp. C1, *Streptococcus* genomosp. C2, *Streptococcus* genomosp. C3, *Streptococcus* genomosp. C4, *Streptococcus* genomosp. C5, *Streptococcus* genomosp. C6, *Streptococcus* genomosp. C7, *Streptococcus* genomosp. C8, *Streptococcus gordonii*, *Streptococcus infantarius*, *Streptococcus infantis*, *Streptococcus intermedius*, *Streptococcus lutetiensis*, *Streptococcus massiliensis*, *Streptococcus oligofermentans*, *Streptococcus oralis*, *Streptococcus parasanguinis*, *Streptococcus pasteurianus*, *Streptococcus peroris*, *Streptococcus pneumoniae*, *Streptococcus porcinus*, *Streptococcus pseudopneumoniae*, *Streptococcus pseudoporcinus*, *Streptococcus pyogenes*, *Streptococcus ratti*, *Streptococcus salivarius*, *Streptococcus sanguinis*, *Streptococcus sinensis*, *Streptococcus* sp. 2285_97, *Streptococcus* sp. 2_1_36FAA, *Streptococcus* sp. ACS2, *Streptococcus* sp. AS20, *Streptococcus* sp. BS35a, *Streptococcus* sp. C150, *Streptococcus* sp. CM6, *Streptococcus* sp. ICM10, *Streptococcus* sp. ICM12, *Streptococcus* sp. ICM2, *Streptococcus* sp. ICM4, *Streptococcus* sp. ICM45, *Streptococcus* sp. M143, *Streptococcus* sp. M334, *Streptococcus* sp. oral clone ASB02, *Streptococcus* sp. oral clone ASCA03, *Streptococcus* sp. oral clone ASCA04, *Streptococcus* sp. oral clone ASCA09, *Streptococcus* sp. oral clone ASCB04, *Streptococcus* sp. oral clone ASCB06, *Streptococcus* sp. oral clone ASCC04, *Streptococcus* sp. oral clone ASCC05, *Streptococcus* sp. oral clone ASCC12, *Streptococcus* sp. oral clone ASCD01, *Streptococcus* sp. oral clone ASCD09, *Streptococcus* sp. oral clone ASCD10, *Streptococcus* sp. oral clone ASCE03, *Streptococcus* sp. oral clone ASCE04, *Streptococcus* sp. oral clone ASCE05, *Streptococcus* sp. oral clone ASCE06, *Streptococcus* sp. oral clone ASCE09, *Streptococcus* sp. oral clone ASCE10, *Streptococcus* sp. oral clone ASCE12, *Streptococcus* sp. oral clone ASCF05, *Streptococcus* sp. oral clone ASCF07, *Streptococcus* sp. oral clone ASCF09, *Streptococcus* sp. oral clone ASCG04, *Streptococcus* sp. oral clone BW009, *Streptococcus* sp. oral clone CH016, *Streptococcus* sp. oral clone GK051, *Streptococcus* sp. oral clone GM006, *Streptococcus* sp. oral clone P2PA_41 P2, *Streptococcus* sp. oral clone P4PA_30 P4, *Streptococcus* sp. oral taxon 071, *Streptococcus* sp. oral taxon G59, *Streptococcus* sp. oral taxon G62, *Streptococcus* sp. oral taxon G63, *Streptococcus suis*, *Streptococcus thermophilus*, *Streptococcus uberis*, *Streptococcus urinalis*, *Streptococcus vestibularis*, *Streptococcus viridans*, Synergistetes bacterium oral clone 03 5 D05

(311) TABLE-US-00003 TABLE 3 Disease Indication Abdominal cavity inflammation *Absidia* infection *Acinetobacter baumanii* infection *Acinetobacter* infection *Acinetobacter lwoffii* infection Acne *Actinomyces israelii* infection Adenovirus infection Adult varicella zoster virus infection Aging Alcoholism (and effects) Allergic conjunctivitis Allergic rhinitis Allergy ALS Alzheimers disease Amoeba infection Anal cancer Antibiotic treatment Antibiotic associated diarrhea Arteriosclerosis Arthritis *Aspergillus fumigatus* infection *Aspergillus* infection Asthma Atherosclerosis Atopic dermatitis Atopy/Allergic Sensitivity Autism Autoimmune disease *Bacillus anthracis* infection *Bacillus* infection Bacterial endocarditis Bacterial eye infection Bacterial infection Bacterial meningitis Bacterial pneumonia Bacterial respiratory tract infection Bacterial skin infection Bacterial susceptibility Bacterial urinary tract infection Bacterial vaginosis *Bacteroides caccae* infection *Bacteroides fragilis* infection *Bacteroides* infection *Bacteroides thetaiotaomicron* infection *Bacteroides uniformis* infection *Bacteroides vulgatus* infection *Bartonella bacilliformis* infection *Bartonella* infection *Bifidobacterium* infection Biliary cancer Biliary cirrhosis Biliary tract disease Biliary tract infection Biliary tumor BK virus infection *Blastomyces* infection Bone and joint infection Bone infection *Bordetella pertussis* infection *Borrelia burgdorferi* infection *Borrelia recurrentis* infection *Brucella* infection *Burkholderia* infection Cachexia *Campylobacter* fetus infection *Campylobacter* infection *Campylobacter jejuni* infection Cancer *Candida albicans* infection *Candida* infection *Candida krusei* infection Celiac

Disease Cervix infection Chemotherapy-induced diarrhea *Chlamydia* infection *Chlamydia pneumoniae* infection *Chlamydia trachomatis* infection Chlamydiae infection Chronic fatigue syndrome Chronic infection Chronic inflammatory demyelinating polyneuropathy Chronic Polio Shedders Circadian rhythm sleep disorder Cirrhosis *Citrobacter* infection *Cladophialophora* infection Clostridiaceae infection *Clostridium botulinum* infection *Clostridium difficile* infection *Clostridium* infection *Clostridium tetani* infection *Coccidioides* infection Colitis Colon cancer Colorectal cancer Common cold Compensated liver cirrhosis Complicated skin and skin structure infection Complicated urinary tract infection Constipation Constipation predominant irritable bowel syndrome *Corynebacterium diphtheriae* infection *Corynebacterium* infection *Coxiella* infection carbapenem-resistant Enterobacteriaceae (CRE) infection Crohns disease *Cryptococcus* infection *Cryptococcus neoformans* infection *Cryptosporidium* infection Cutaneous lupus erythematosus Cystic fibrosis Cystitis Cytomegalovirus infection Dementia Dengue virus infection Depression Dermatitis Diabetes mellitus Diabetic complication Diabetic foot ulcer Diarrhea Diarrhea predominant irritable bowel syndrome Discoid lupus erythematosus Diverticulitis DNA virus infection Duodenal ulcer Ebola virus infection *Entamoeba histolytica* infection *Enterobacter aerogenes* infection *Enterobacter cloacae* infection *Enterobacter* infection Enterobacteriaceae infection *Enterococcus faecalis* infection *Enterococcus faecium* infection *Enterococcus* infection Enterocolitis Enterovirus 71 infection *Epidermophyton* infection Epstein Barr virus infection ESBL (Extended Spectrum Beta Lactamase) Producing Bacterial Infection *Escherichia coli* infection Esophageal cancer *Exophiala* infection Familial cold autoinflammatory syndrome *Fasciola hepatica* infection Female genital tract infection Female genital tract tumor Female infertility Fibrosis Flavivirus infection Food Allergy *Francisella tularensis* infection Functional bowel disorder Fungal infection Fungal respiratory tract infection Fungal urinary tract infection *Fusarium* infection *Fusobacterium* infection Gastric ulcers Gastrointestinal infection Gastrointestinal pain Gastrointestinal ulcer Genital tract infection Genitourinary disease Genitourinary tract rumor Gestational diabetes *Giardia lamblia* infection Gingivitis Gram negative bacterium infection Gram positive bacterium infection *Haemophilus aegyptus* infection *Haemophilus ducreyi* infection *Haemophilus* infection *Haemophilus influenzae* infection *Haemophilus parainfluenzae* infection Hantavirus infection *Helicobacter pylori* infection Helminth infection Hepatitis A virus infection Hepatitis B virus infection Hepatitis C virus infection Hepatitis D virus infection Hepatitis E virus infection Hepatitis virus infection Herpes simplex virus infection Herpesvirus infection *Histoplasma* infection HIV infection HIV-1 infection HIV-2 infection HSV-1 infection HSV-2 infection Human T cell leukemia virus 1 infection Hypercholesterolemia Hyperoxaluria Hypertension Infectious arthritis Infectious disease Infectious endocarditis Infertility Inflammatory bowel disease Inflammatory disease Influenza virus A infection Influenza virus B infection Influenza virus infection Insomnia Insulin dependent diabetes Intestine infection Irritable bowel syndrome Japanese encephalitis virus infection Joint infection Juvenile rheumatoid arthritis *Klebsiella granulomatis* infection *Klebsiella* infection *Klebsiella pneumoniae* infection *Legionella* infection *Legionella pneumophila* infection *Leishmania braziliensis* infection *Leishmania donovani* infection *Leishmania* infection *Leishmania tropica* infection Leptospiraceae infection *Listeria monocytogenes* infection Listerosis Liver cirrhosis Liver fibrosis Lower respiratory tract infection Lung infection Lung inflammation Lupus erythematosus panniculitis Lupus nephritis Lyme disease Male infertility Marburg virus infection Measles virus infection Metabolic disorder Metabolic Syndrome Metastatic colon cancer Metastatic colorectal cancer Metastatic esophageal cancer Metastatic gastrointestinal cancer Metastatic stomach cancer Micrococcaceae infection *Micrococcus* infection Microsporidial infection *Microsporum* infection *Molluscum contagiosum* infection Monkeypox virus infection *Moraxella catarrhalis* infection *Moraxella* infection *Morganella* infection *Morganella morganii* infection MRSA infection Mucor infection Multidrug resistant infection Multiple sclerosis Mumps virus infection Musculoskeletal system inflammation *Mycobacterium* infection *Mycobacterium leprae* infection *Mycobacterium tuberculosis* infection *Mycoplasma* infection *Mycoplasma pneumoniae* infection Necrotizing enterocolitis Necrotizing Pancreatitis *Neisseria gonorrhoeae* infection *Neisseria* infection *Neisseria meningitidis* infection Nematode infection Non alcoholic fatty liver disease Non-alcoholic steatohepatitis Non-insulin dependent diabetes Obesity Ocular infection Ocular inflammation Orbital inflammatory disease Osteoarthritis Otorhinolaryngological infection Pain Papillomavirus infection Parasitic infection Parkinsons disease Pediatric varicella zoster virus infection Pelvic inflammatory disease *Peptostreptococcus* infection Perennial allergic rhinitis Periarthritis Pink eye infection *Plasmodium falciparum* infection *Plasmodium* infection *Plasmodium malariae* infection *Plasmodium vivax* infection *Pneumocystis carinii* infection Poliovirus infection Polyomavirus infection Post-surgical bacterial leakage Pouchitis *Prevotella* infection Primary biliary cirrhosis Primary sclerosing cholangitis *Propionibacterium acnes* infection *Propionibacterium* infection Prostate cancer *Proteus* infection *Proteus mirabilis* infection Protozoal infection *Providencia* infection *Pseudomonas aeruginosa* infection *Pseudomonas* infection Psoriasis Psoriatic arthritis Pulmonary fibrosis Rabies virus infection Rectal cancer Respiratory syncytial virus infection Respiratory tract infection Respiratory tract inflammation Rheumatoid arthritis Rhinitis *Rhizomucor* infection *Rhizopus* infection *Rickettsia* infection Ross River virus infection Rotavirus infection Rubella virus infection *Salmonella* infection *Salmonella typhi* infection Sarcopenia SARS coronavirus infection Scabies infection *Scedosporium* infection Scleroderma Seasonal allergic rhinitis *Serratia* infection *Serratia marcescens* infection *Shigella boydii* infection *Shigella dysenteriae* infection *Shigella flexneri* infection *Shigella* infection *Shigella sonnei* infection Short bowel syndrome Skin allergy Skin cancer Skin infection Skin Inflammatory disease Sleep disorder Spondylarthritis *Staphylococcus aureus* infection *Staphylococcus epidermidis* infection *Staphylococcus* infection *Staphylococcus saprophyticus* infection *Stenotrophomonas maltophilia* infection Stomach cancer Stomach infection Stomach ulcer *Streptococcus agalactiae* infection *Streptococcus constellatus* infection *Streptococcus* infection *Streptococcus intermedius* infection *Streptococcus mitis* infection *Streptococcus oralis* infection *Streptococcus pneumoniae* infection *Streptococcus pyogenes* infection Systemic lupus erythematosus Traveler's diarrhea Trench mouth *Treponema* infection *Treponema pallidum* infection *Trichomonas* infection *Trichophyton* infection *Trypanosoma brucei* infection *Trypanosoma cruzi* infection Type 1 Diabetes Type 2 Diabetes Ulcerative colitis Upper respiratory tract infection *Ureaplasma urealyticum* infection Urinary tract disease Urinary tract infection Urinary tract tumor Urogenital tract infection Uterus infection Vaccinia virus infection Vaginal infection Varicella zoster virus infection Variola virus infection *Vibrio cholerae* infection Viral eye infection Viral infection Viral respiratory tract infection Viridans group *Streptococcus* infection Vancomycin-Resistant *Enterococcus* infection Wasting Syndrome Weight loss West Nile virus infection Whipple's disease Xenobiotic metabolism Yellow fever virus infection *Yersinia pestis* infection Flatulence Gastrointestinal Disorder General Inflammation

(312) TABLE-US-00004 TABLE 4a Strain ID OTU1 of Composition OTU2 of Composition OTU3 of Composition OTU1 *Escherichia_coli Escherichia_coli* SPC00001 *Escherichia_coli Bacteroides_vulgatus* SPC00001 *Escherichia_coli Bacteroides*_sp_1_1_6 SPC00001 *Escherichia_coli Bacteroides*_sp_3_1_23 SPC00001 *Escherichia_coli Enterococcus_faecalis* SPC00001 *Escherichia_coli Coprobacillus*_sp_D7 SPC00001 *Escherichia_coli Streptococcus_thermophilus* SPC00001 *Escherichia_coli Dorea_formicigenerans* SPC00001 *Escherichia_coli Blautia_producta* SPC00001 *Escherichia_coli Eubacterium_eligens* SPC00001 *Escherichia_coli Clostridium_nexile* SPC00001 *Escherichia_coli Clostridium*_sp_HGF2 SPC00001 *Escherichia_coli Faecalibacterium_prausnitzii* SPC00001 *Escherichia_coli Odoribacter_splanchnicus* SPC00001 *Escherichia_coli Dorea_longicatena* SPC00001 *Escherichia_coli Roseburia_intestinalis* SPC00001 *Escherichia_coli Coprococcus_catus* SPC00001 *Escherichia_coli* Erysipelotrichaceae_*bacterium*_3_1_53 SPC00001 *Escherichia_coli Bacteroides*_sp_D20 SPC00001 *Escherichia_coli Bacteroides_ovatus* SPC00001 *Escherichia_coli Parabacteroides_merdae* SPC00001 *Escherichia_coli Bacteroides_vulgatus* SPC00001 *Escherichia_coli Collinsella_aerofaciens* SPC00001 *Escherichia_coli Escherichia_coli* SPC00001 *Escherichia_coli Ruminococcus_obeum* SPC00001 *Escherichia_coli Bacteroides_caccae* SPC00001 *Escherichia_coli Bacteroides_eggerthii* SPC00001 *Escherichia_coli Ruminococcus_torques* SPC00001 *Escherichia_coli Clostridium_hathewayi* SPC00001 *Escherichia_coli Bifidobacterium_pseudocatenulatum* SPC00001 *Escherichia_coli Bifidobacterium_adolescentis* SPC00001 *Escherichia_coli Coprococcus_comes* SPC00001 *Escherichia_coli Clostridium_symbiosum* SPC00001 *Escherichia_coli Eubacterium_rectale* SPC00001

Escherichia_coli Faecalibacterium_prausnitzii SPC00001 Escherichia_coli Odoribacter_splanchnicus SPC00001 Escherichia_coli Lachnospiraceae_bacterium_5_1_57FAA SPC00001 Escherichia_coli Blautia_schinkii SPC00001 Escherichia_coli Alistipes_shahii SPC00001 Escherichia_coli Blautia_producta SPC00001 Bacteroides_vulgatus Bacteroides_vulgatus SPC00005 Bacteroides_vulgatus Bacteroides_sp_1_1_6 SPC00005 Bacteroides_vulgatus Bacteroides_sp_3_1_23 SPC00005 Bacteroides_vulgatus Enterococcus_faecalis SPC00005 Bacteroides_vulgatus Coprobacillus_sp_D7 SPC00005 Bacteroides_vulgatus Streptococcus_thermophilus SPC00005 Bacteroides_vulgatus Dorea_formicigenerans SPC00005 Bacteroides_vulgatus Blautia_producta SPC00005 Bacteroides_vulgatus Eubacterium_eligens SPC00005 Bacteroides_vulgatus Clostridium_nexile SPC00005 Bacteroides_vulgatus Clostridium_sp_HGF2 SPC00005 Bacteroides_vulgatus Faecalibacterium_prausnitzii SPC00005 Bacteroides_vulgatus Odoribacter_splanchnicus SPC00005 Bacteroides_vulgatus Dorea_longicatena SPC00005 Bacteroides_vulgatus Roseburia_intestinalis SPC00005 Bacteroides_vulgatus Coprococcus_catus SPC00005 Bacteroides_vulgatus Erysipelotrichaceae_bacterium_3_1_53 SPC00005 Bacteroides_vulgatus Bacteroides_sp_D20 SPC00005 Bacteroides_vulgatus Bacteroides_ovatus SPC00005 Bacteroides_vulgatus Parabacteroides_merdae SPC00005 Bacteroides_vulgatus Bacteroides_vulgatus SPC00005 Bacteroides_vulgatus Collinsella_aerofaciens SPC00005 Bacteroides_vulgatus Escherichia_coli SPC00005 Bacteroides_vulgatus Ruminococcus_obeum SPC00005 Bacteroides_vulgatus Bacteroides_caccae SPC00005 Bacteroides_vulgatus Bacteroides_eggerthii SPC00005 Bacteroides_vulgatus Ruminococcus_torques SPC00005 Bacteroides_vulgatus Clostridium_hathewayi SPC00005 Bacteroides_vulgatus Bifidobacterium_pseudocatenulatum SPC00005 Bacteroides_vulgatus Bifidobacterium_adolescentis SPC00005 Bacteroides_vulgatus Coprococcus_comes SPC00005 Bacteroides_vulgatus Clostridium_symbiosum SPC00005 Bacteroides_vulgatus Eubacterium_rectale SPC00005 Bacteroides_vulgatus Faecalibacterium_prausnitzii SPC00005 Bacteroides_vulgatus Odoribacter_splanchnicus SPC00005 Bacteroides_vulgatus Lachnospiraceae_bacterium_5_1_57FAA SPC00005 Bacteroides_vulgatus Blautia_schinkii SPC00005 Bacteroides_vulgatus Alistipes_shahii SPC00005 Bacteroides_vulgatus Blautia_producta SPC00005 Bacteroides_sp_1_1_6 Bacteroides_sp_1_1_6 SPC00006 Bacteroides_sp_1_1_6 Bacteroides_sp_3_1_23 SPC00006 Bacteroides_sp_1_1_6 Enterococcus_faecalis SPC00006 Bacteroides_sp_1_1_6 Coprobacillus_sp_D7 SPC00006 Bacteroides_sp_1_1_6 Streptococcus_thermophilus SPC00006 Bacteroides_sp_1_1_6 Dorea_formicigenerans SPC00006 Bacteroides_sp_1_1_6 Blautia_producta SPC00006 Bacteroides_sp_1_1_6 Eubacterium_eligens SPC00006 Bacteroides_sp_1_1_6 Clostridium_nexile SPC00006 Bacteroides_sp_1_1_6 Clostridium_sp_HGF2 SPC00006 Bacteroides_sp_1_1_6 Faecalibacterium_prausnitzii SPC00006 Bacteroides_sp_1_1_6 Odoribacter_splanchnicus SPC00006 Bacteroides_sp_1_1_6 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*Faecalibacterium_prausnitzii* Lachnospiraceae_*bacterium*_5_1_57FAA SPC00054 *Faecalibacterium_prausnitzii Blautia_schinkii* SPC00054 *Faecalibacterium_prausnitzii Alistipes_shahii* SPC00054 *Faecalibacterium_prausnitzii Blautia_producta* SPC00054 *Odoribacter_splanchnicus Odoribacter_splanchnicus* SPC00056 *Odoribacter_splanchnicus Dorea_longicatena* SPC00056 *Odoribacter_splanchnicus Roseburia_intestinalis* SPC00056 *Odoribacter_splanchnicus Coprococcus_catus* SPC00056 *Odoribacter_splanchnicus* Erysipelotrichaceae_*bacterium*_3_1_53 SPC00056 *Odoribacter_splanchnicus Bacteroides*_sp_D20 SPC00056 *Odoribacter_splanchnicus Bacteroides_ovatus* SPC00056 *Odoribacter_splanchnicus Parabacteroides_merdae* SPC00056 *Odoribacter_splanchnicus Bacteroides_vulgatus* SPC00056 *Odoribacter_splanchnicus Collinsella_aerofaciens* SPC00056 Odoribacter_*splanchnicus Escherichia_coli* SPC00056 *Odoribacter_splanchnicus Ruminococcus_obeum* SPC00056 *Odoribacter_splanchnicus Bacteroides_caccae* SPC00056 *Odoribacter_splanchnicus Bacteroides_eggerthii* SPC00056 *Odoribacter_splanchnicus Ruminococcus_torques* SPC00056 *Odoribacter_splanchnicus Clostridium_hathewayi* SPC00056 *Odoribacter_splanchnicus Bifidobacterium_pseudocatenulatum* SPC00056 *Odoribacter_splanchnicus Bifidobacterium_adolescentis* SPC00056 *Odoribacter_splanchnicus Coprococcus_comes* SPC00056 *Odoribacter_splanchnicus Clostridium_symbiosum* SPC00056 *Odoribacter_splanchnicus Eubacterium_rectale* SPC00056 *Odoribacter_splanchnicus Faecalibacterium_prausnitzii* SPC00056 *Odoribacter_splanchnicus Odoribacter_splanchnicus* SPC00056 *Odoribacter_splanchnicus* Lachnospiraceae_*bacterium*_5_1_57FAA SPC00056 *Odoribacter_splanchnicus Blautia_schinkii* SPC00056 *Odoribacter_splanchnicus Alistipes_shahii* SPC00056 *Odoribacter_splanchnicus Blautia_producta* SPC00056 *Dorea_longicatena Dorea_longicatena* SPC00057 *Dorea_longicatena Roseburia_intestinalis* SPC00057 *Dorea_longicatena Coprococcus_catus* SPC00057 *Dorea_longicatena* Erysipelotrichaceae_*bacterium*_3_1_53 SPC00057 *Dorea_longicatena Bacteroides*_sp_D20 SPC00057 *Dorea_longicatena Bacteroides_ovatus* SPC00057 *Dorea_longicatena Parabacteroides_merdae* SPC00057 *Dorea_longicatena Bacteroides_vulgatus* SPC00057 *Dorea_longicatena Collinsella_aerofaciens* SPC00057 *Dorea_longicatena Escherichia_coli* SPC00057 *Dorea_longicatena Ruminococcus_obeum* SPC00057 *Dorea_longicatena Bacteroides_caccae* SPC00057 *Dorea_longicatena Bacteroides_eggerthii* SPC00057 *Dorea_longicatena Ruminococcus_torques* SPC00057 *Dorea_longicatena Clostridium_hathewayi* SPC00057 *Dorea_longicatena Bifidobacterium_pseudocatenulatum* SPC00057 *Dorea_longicatena Bifidobacterium_adolescentis* SPC00057 *Dorea_longicatena Coprococcus_comes* SPC00057 *Dorea_longicatena Clostridium_symbiosum* SPC00057 *Dorea_longicatena Eubacterium_rectale* SPC00057 *Dorea_longicatena Faecalibacterium_prausnitzii* SPC00057 *Dorea_longicatena Odoribacter_splanchnicus* SPC00057 *Dorea_longicatena* Lachnospiraceae_*bacterium*_5_1_57FAA SPC00057 *Dorea_longicatena Blautia_schinkii* SPC00057 *Dorea_longicatena Alistipes_shahii* SPC00057 *Dorea_longicatena Blautia_producta* SPC00057 *Roseburia_intestinalis Roseburia_intestinalis* SPC00061 *Roseburia_intestinalis Coprococcus_catus* SPC00061 *Roseburia_intestinalis* Erysipelotrichaceae_*bacterium*_3_1_53 SPC00061 *Roseburia_intestinalis Bacteroides*_sp_D20 SPC00061 *Roseburia_intestinalis Bacteroides_ovatus* SPC00061 *Roseburia_intestinalis Parabacteroides_merdae* SPC00061 *Roseburia_intestinalis Bacteroides_vulgatus* SPC00061 *Roseburia_intestinalis Collinsella_aerofaciens* SPC00061 *Roseburia_intestinalis Escherichia_coli* SPC00061 *Roseburia_intestinalis Ruminococcus_obeum* SPC00061 *Roseburia_intestinalis Bacteroides_caccae* SPC00061 *Roseburia_intestinalis Bacteroides_eggerthii* SPC00061 *Roseburia_intestinalis Ruminococcus_torques* SPC00061 *Roseburia_intestinalis Clostridium_hathewayi* SPC00061 *Roseburia_intestinalis 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*Coprococcus_catus Escherichia_coli* SPC00080 *Coprococcus_catus Ruminococcus_obeum* SPC00080 *Coprococcus_catus Bacteroides_caccae* SPC00080 *Coprococcus_catus Bacteroides_eggerthii* SPC00080 *Coprococcus_catus Ruminococcus_torques* SPC00080 *Coprococcus_catus Clostridium_hathewayi* SPC00080 *Coprococcus_catus Bifidobacterium_pseudocatenulatum* SPC00080 *Coprococcus_catus*

*Bifidobacterium_adolescentis* SPC00080 *Coprococcus_catus Coprococcus_comes* SPC00080 *Coprococcus_catus Clostridium_symbiosum*
SPC00080 *Coprococcus_catus Eubacterium_rectale* SPC00080 *Coprococcus_catus Faecalibacterium_prausnitzii* SPC00080
*Coprococcus_catus Odoribacter_splanchnicus* SPC00080 *Coprococcus_catus* Lachnospiraceae_*bacterium*_5_1_57FAA SPC00080
*Coprococcus_catus Blautia_schinkii* SPC00080 *Coprococcus_catus Alistipes_shahii* SPC00080 *Coprococcus_catus Blautia_producta*
SPC00080 Erysipelotrichaceae_*bacterium*_3_1_53 Erysipelotrichaceae_*bacterium*_3_1_53 SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53
*Bacteroides*_sp_D20 SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Bacteroides_ovatus* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53
*Parabacteroides_merdae* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Bacteroides_vulgatus* SPC10001
Erysipelotrichaceae_*bacterium*_3_1_53 *Collinsella_aerofaciens* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Escherichia_coli* SPC10001
Erysipelotrichaceae_*bacterium*_3_1_53 *Ruminococcus_obeum* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Bacteroides_caccae*
SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Bacteroides_eggerthii* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53
*Ruminococcus_torques* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Clostridium_hathewayi* SPC10001
Erysipelotrichaceae_*bacterium*_3_1_53 *Bifidobacterium_pseudocatenulatum* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53
*Bifidobacterium_adolescentis* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Coprococcus_comes* SPC10001
Erysipelotrichaceae_*bacterium*_3_1_53 *Clostridium_symbiosum* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Eubacterium_rectale*
SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Faecalibacterium_prausnitzii* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53
*Odoribacter_splanchnicus* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 Lachnospiraceae_*bacterium*_5_1_57FAA SPC10001
Erysipelotrichaceae_*bacterium*_3_1_53 *Blautia_schinkii* SPC10001 Erysipelotrichaceae_*bacterium*_3_1_53 *Alistipes_shahii* SPC10001
Erysipelotrichaceae_*bacterium*_3_1_53 *Blautia_producta* SPC10001 *Bacteroides*_sp_D20 *Bacteroides*_sp_D20 SPC10019 *Bacteroides*_sp_D20
*Bacteroides_ovatus* SPC10019 *Bacteroides*_sp_D20 *Parabacteroides_merdae* SPC10019 *Bacteroides*_sp_D20 *Bacteroides_vulgatus* SPC10019
*Bacteroides*_sp_D20 *Collinsella_aerofaciens* SPC10019 *Bacteroides*_sp_D20 *Escherichia_coli* SPC10019 *Bacteroides*_sp_D20
*Ruminococcus_obeum* SPC10019 *Bacteroides*_sp_D20 *Bacteroides_caccae* SPC10019 *Bacteroides*_sp_D20 *Bacteroides_eggerthii* SPC10019
*Bacteroides*_sp_D20 *Ruminococcus_torques* SPC10019 *Bacteroides*_sp_D20 *Clostridium_hathewayi* SPC10019 *Bacteroides*_sp_D20
*Bifidobacterium_pseudocatenulatum* SPC10019 *Bacteroides*_sp_D20 *Bifidobacterium_adolescentis* SPC10019 *Bacteroides*_sp_D20
*Coprococcus_comes* SPC10019 *Bacteroides*_sp_D20 *Clostridium_symbiosum* SPC10019 Bacteroides_sp_D20 *Eubacterium_rectale* SPC10019
*Bacteroides*_sp_D20 *Faecalibacterium_prausnitzii* SPC10019 *Bacteroides*_sp_D20 *Odoribacter_splanchnicus* SPC10019 *Bacteroides*_sp_D20
Lachnospiraceae_*bacterium*_5_1_57FAA SPC10019 *Bacteroides*_sp_D20 *Blautia_schinkii* SPC10019 *Bacteroides*_sp_D20 *Alistipes_shahii*
SPC10019 *Bacteroides*_sp_D20 *Blautia_producta* SPC10019 *Bacteroides_ovatus Bacteroides_ovatus* SPC10030 *Bacteroides_ovatus*
*Parabacteroides_merdae* SPC10030 *Bacteroides_ovatus Bacteroides_vulgatus* SPC10030 *Bacteroides_ovatus Collinsella_aerofaciens*
SPC10030 *Bacteroides_ovatus Escherichia_coli* SPC10030 *Bacteroides_ovatus Ruminococcus_obeum* SPC10030 *Bacteroides_ovatus*
*Bacteroides_caccae* SPC10030 *Bacteroides_ovatus Bacteroides_eggerthii* SPC10030 *Bacteroides_ovatus Ruminococcus_torques* SPC10030
*Bacteroides_ovatus Clostridium_hathewayi* SPC10030 *Bacteroides_ovatus Bifidobacterium_pseudocatenulatum* SPC10030 *Bacteroides_ovatus*
*Bifidobacterium_adolescentis* SPC10030 *Bacteroides_ovatus Coprococcus_comes* SPC10030 *Bacteroides_ovatus Clostridium_symbiosum*
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*Bacteroides_ovatus Odoribacter_splanchnicus* SPC10030 *Bacteroides_ovatus* Lachnospiraceae_*bacterium*_5_1_57FAA SPC10030
*Bacteroides_ovatus Blautia_schinkii* SPC10030 *Bacteroides_ovatus Alistipes_shahii* SPC10030 *Bacteroides_ovatus Blautia_producta*
SPC10030 *Parabacteroides_merdae Parabacteroides_merdae* SPC10048 *Parabacteroides_merdae Bacteroides_vulgatus* SPC10048
*Parabacteroides_merdae Collinsella_aerofaciens* SPC10048 *Parabacteroides_merdae Escherichia_coli* SPC10048 *Parabacteroides_merdae*
*Ruminococcus_obeum* SPC10048 *Parabacteroides_merdae Bacteroides_caccae* SPC10048 *Parabacteroides_merdae Bacteroides_eggerthii*
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*Parabacteroides_merdae Bifidobacterium_pseudocatenulatum* SPC10048 *Parabacteroides_merdae Bifidobacterium_adolescentis* SPC10048
*Parabacteroides_merdae Coprococcus_comes* SPC10048 *Parabacteroides_merdae Clostridium_symbiosum* SPC10048
*Parabacteroides_merdae Eubacterium_rectale* SPC10048 *Parabacteroides_merdae Faecalibacterium_prausnitzii* SPC10048
*Parabacteroides_merdae Odoribacter_splanchnicus* SPC10048 *Parabacteroides_merdae* Lachnospiraceae_*bacterium*_5_1_57FAA SPC10048
*Parabacteroides_merdae Blautia_schinkii* SPC10048 *Parabacteroides_merdae Alistipes_shahii* SPC10048 *Parabacteroides_merdae*
*Blautia_producta* SPC10048 *Bacteroides_vulgatus Bacteroides_vulgatus* SPC10081 *Bacteroides_vulgatus Collinsella_aerofaciens* SPC10081
*Bacteroides_vulgatus Escherichia_coli* SPC10081 *Bacteroides_vulgatus Ruminococcus_obeum* SPC10081 *Bacteroides_vulgatus*
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*Bacteroides_vulgatus Clostridium_hathewayi* SPC10081 *Bacteroides_vulgatus Bifidobacterium_pseudocatenulatum* SPC10081
*Bacteroides_vulgatus Bifidobacterium_adolescentis* SPC10081 *Bacteroides_vulgatus Coprococcus_comes* SPC10081 *Bacteroides_vulgatus*
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SPC10081 *Bacteroides_vulgatus Odoribacter_splanchnicus* SPC10081 *Bacteroides_vulgatus* Lachnospiraceae_*bacterium*_5_1_57FAA
SPC10081 *Bacteroides_vulgatus Blautia_schinkii* SPC10081 Bacteroides_*vulgatus Alistipes_shahii* SPC10081 *Bacteroides_vulgatus*
*Blautia_producta* SPC10081 *Collinsella_aerofaciens Collinsella_aerofaciens* SPC10097 *Collinsella_aerofaciens Escherichia_coli* SPC10097
*Collinsella_aerofaciens Ruminococcus_obeum* SPC10097 *Collinsella_aerofaciens Bacteroides_caccae* SPC10097 *Collinsella_aerofaciens*
*Bacteroides_eggerthii* SPC10097 *Collinsella_aerofaciens Ruminococcus_torques* SPC10097 *Collinsella_aerofaciens Clostridium_hathewayi*
SPC10097 *Collinsella_aerofaciens Bifidobacterium_pseudocatenulatum* SPC10097 *Collinsella_aerofaciens Bifidobacterium_adolescentis*
SPC10097 *Collinsella_aerofaciens Coprococcus_comes* SPC10097 *Collinsella_aerofaciens Clostridium_symbiosum* SPC10097
*Collinsella_aerofaciens Eubacterium_rectale* SPC10097 *Collinsella_aerofaciens Faecalibacterium_prausnitzii* SPC10097
*Collinsella_aerofaciens Odoribacter_splanchnicus* SPC10097 *Collinsella_aerofaciens* Lachnospiraceae_*bacterium*_5_1_57FAA SPC10097
*Collinsella_aerofaciens Blautia_schinkii* SPC10097 *Collinsella_aerofaciens Alistipes_shahii* SPC10097 *Collinsella_aerofaciens*
*Blautia_producta* SPC10097 *Escherichia_coli Escherichia_coli* SPC10110 *Escherichia_coli Ruminococcus_obeum* SPC10110 *Escherichia_coli*
*Bacteroides_caccae* SPC10110 *Escherichia_coli Bacteroides_eggerthii* SPC10110 *Escherichia_coli Ruminococcus_torques* SPC10110
*Escherichia_coli Clostridium_hathewayi* SPC10110 *Escherichia_coli Bifidobacterium_pseudocatenulatum* SPC10110 *Escherichia_coli*
*Bifidobacterium_adolescentis* SPC10110 *Escherichia_coli Coprococcus_comes* SPC10110 *Escherichia_coli Clostridium_symbiosum* SPC10110
*Escherichia_coli Eubacterium_rectale* SPC10110 *Escherichia_coli Faecalibacterium_prausnitzii* SPC10110 *Escherichia_coli*
*Odoribacter_splanchnicus* SPC10110 *Escherichia_coli* Lachnospiraceae_*bacterium*_5_1_57FAA SPC10110 *Escherichia_coli Blautia_schinkii*
SPC10110 *Escherichia_coli Alistipes_shahii* SPC10110 *Escherichia_coli Blautia_producta* SPC10110 *Ruminococcus_obeum*
*Ruminococcus_obeum* SPC10197 *Ruminococcus_obeum Bacteroides_caccae* SPC10197 *Ruminococcus_obeum Bacteroides_eggerthii*
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*Ruminococcus_obeum Bifidobacterium_pseudocatenulatum* SPC10197 *Ruminococcus_obeum Bifidobacterium_adolescentis* SPC10197
*Ruminococcus_obeum Coprococcus_comes* SPC10197 *Ruminococcus_obeum Clostridium_symbiosum* SPC10197 *Ruminococcus_obeum*
*Eubacterium_rectale* SPC10197 *Ruminococcus_obeum Faecalibacterium_prausnitzii* SPC10197 *Ruminococcus_obeum*
*Odoribacter_splanchnicus* SPC10197 *Ruminococcus_obeum* Lachnospiraceae_*bacterium*_5_1_57FAA SPC10197 *Ruminococcus_obeum*

*Blautia_schinkii* SPC10197 *Ruminococcus_obeum Alistipes_shahii* SPC10197 *Ruminococcus_obeum Blautia_producta* SPC10197 *Bacteroides_caccae Bacteroides_caccae* SPC10211 *Bacteroides_caccae Bacteroides_eggerthii* SPC10211 *Bacteroides_caccae Ruminococcus_torques* SPC10211 *Bacteroides_caccae Clostridium_hathewayi* SPC10211 *Bacteroides_caccae Bifidobacterium_pseudocatenulatum* SPC10211 *Bacteroides_caccae Bifidobacterium_adolescentis* SPC10211 *Bacteroides_caccae Coprococcus_comes* SPC10211 *Bacteroides_caccae Clostridium_symbiosum* SPC10211 *Bacteroides_caccae Eubacterium_rectale* SPC10211 *Bacteroides_caccae Faecalibacterium_prausnitzii* SPC10211 *Bacteroides_caccae Odoribacter_splanchnicus* SPC10211 *Bacteroides_caccae* Lachnospiraceae_*bacterium*_5_1_57FAA SPC10211 Bacteroides_*caccae Blautia_schinkii* SPC10211 *Bacteroides_caccae Alistipes_shahii* SPC10211 *Bacteroides_caccae Blautia_producta* SPC10211 *Bacteroides_eggerthii Bacteroides_eggerthii* SPC10213 *Bacteroides_eggerthii 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(313) TABLE-US-00005 TABLE 4b OTU1 ID1 OTU2 ID2 OTU3 ID3 Clade1 Clade2 Clade3 *Bacteroides* sp. 1_1_6 295 *Clostridium* sp. HGF2 628 clade_65 clade_351 *Bacteroides* sp. 1_1_6 295 *Bifidobacterium* 357 clade_65 clade_172 *pseudocatenulatum Bacteroides* sp. 1_1_6 295 *Clostridium symbiosum* 652 clade_65 clade_408 *Bacteroides* sp. 3_1_23 308 *Clostridium nexile* 607 clade_38 clade_262 *Bacteroides* sp. 3_1_23 308 *Bifidobacterium* 357 clade_38 clade_172 *pseudocatenulatum Bacteroides* sp. 3_1_23 308 *Clostridium symbiosum* 652 clade_38 clade_408 *Streptococcus thermophilus* 1883 *Bifidobacterium* 357 clade_98 clade_172 *pseudocatenulatum Clostridium nexile* 607 *Bifidobacterium* 357 clade_262 clade_172 *pseudocatenulatum Parabacteroides merdae* 1420 *Bifidobacterium* 357 clade_286 clade_172 *pseudocatenulatum Clostridium tertium* 653 *Clostridium mayombei* 605 clade_252 clade_354 *Clostridium tertium* 653 *Clostridium butyricum* 561 clade_252 clade_252 *Clostridium tertium* 653 *Coprococcus comes* 674 clade_252 clade_262 *Clostridium tertium* 653 *Clostridium hylemonae* 593

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symbiosum* 652 *Clostridium butyricum* 561 *Ruminococcus gnavus* 1661 clade_408 clade_252 clade_360 *Clostridium symbiosum* 652 *Clostridium butyricum* 561 *Ruminococcus bromii* 1657 clade_408 clade_252 clade_537 Lachnospiraceae *bacterium* 1054 *Clostridium tertium* 653 *Clostridium butyricum* 561 clade_260 clade_252 clade_252 5_1_57FAA Lachnospiraceae *bacterium* 1054 *Clostridium disporicum* 579 *Clostridium butyricum* 561 clade_260 clade_253 clade_252 5_1_57FAA Lachnospiraceae *bacterium* 1054 *Clostridium mayombei* 605 *Ruminococcus gnavus* 1661 clade_260 clade_354 clade_360 5_1_57FAA Lachnospiraceae *bacterium* 1054 *Clostridium mayombei* 605 *Ruminococcus bromii* 1657 clade_260 clade_354 clade_537 5_1_57FAA Lachnospiraceae *bacterium* 1054 *Clostridium butyricum* 561 *Clostridium hylemonae* 593 clade_260 clade_252 clade_260 5_1_57FAA Lachnospiraceae *bacterium* 1054 *Clostridium butyricum* 561 *Clostridium orbiscindens* 609 clade_260 clade_252 clade_494 5_1_57FAA Lachnospiraceae *bacterium* 1054 *Clostridium butyricum* 561 *Ruminococcus gnavus* 1661 clade_260 clade_252 clade_360 5_1_57FAA Lachnospiraceae *bacterium* 1054 *Clostridium butyricum* 561 *Ruminococcus bromii* 1657 clade_260 clade_252 clade_537 5_1_57FAA *Clostridium butyricum* 561 *Coprococcus comes* 674 *Clostridium symbiosum* 652 clade_252 clade_262 clade_408 *Clostridium butyricum* 561 *Coprococcus comes* 674 Lachnospiraceae *bacterium* 1054 clade_252 clade_262 clade_260 5_1_57FAA *Clostridium butyricum* 561 *Clostridium symbiosum* 652 Lachnospiraceae *bacterium* 1054 clade_252 clade_408 clade_260 5_1_57FAA

(314) TABLE-US-00006 TABLE 5 CivSim Mean Mean Inhibition CivSim Min. Max. SP Seres (log10 Inhibition Rel. Clin. Expt. Arm Key Arm Treatment OTU1 OTU2 OTU3 CFU/mL) (Confidence) Weight Score Cum. Mort. Clade1 Clade2 Clade3 SP427 1 PBS 1 Vehicle n/a n/a 0.82 2.6 3 SP427 14 V5F 2 FSV33_10pct n/a n/a 0.99 0 0 (EMT) SP427 6 79F 3 DE277512.1 *Collinsella Faecalibacterium Blautia* 1.84 ++++ 0.85 1.4 1 clade_553 clade_478 clade_309 *aerofaciens prausnitzii producta* SP427 12 V7B 4 DE643314.1 *Faecalibacterium Blautia producta Eubacterium* 1.27 ++++ 0.81 2 3 clade_478 clade_309 clade_444 *prausnitzii rectale* SP427 5 699 5 DE022136.1 *Clostridium* Lachnospiraceae *Blautia* 0.81 ++++ 0.93 1 0 clade_408 clade_260 clade_309 *bolteae bacterium producta* 5_1_57FAA SP427 10 E4W 6 DE061176.1 *Collinsella Clostridium Ruminococcus* 2.87 ++++ 0.96 0.2 0 clade_553 clade_252 clade_360 *aerofaciens butyricum gnavus* SP427 7 852 7 DE554703.1 *Collinsella Clostridium Clostridium* 3.09 ++++ 0.78 1.7 2 clade_553 clade_252 clade_260 *aerofaciens butyricum hylemonae* SP427 3 366 8 DE705158.1 *Coprococcus Clostridium Clostridium* 3 ++++ 0.85 1.7 2 clade_262 clade_351 clade_252 *comes innocuum butyricum* SP427 9 CE3 9 DE266960.1 *Clostridium Clostridium Ruminococcus* 3.06 ++++ 0.91 1 0 clade_408 clade_252 clade_360 *bolteae butyricum gnavus* SP427 8 BCY 10 DE897971.1 *Clostridium Collinsella Clostridium* 1.46 ++++ 0.83 1.7 1 clade_354 clade_553 clade_408 *mayombei aerofaciens symbiosum* SP427 13 Y4K 11 DE001210.1 *Clostridium Collinsella Blautia* 2 ++++ 0.88 1.1 0 clade_252 clade_553 clade_309 *tertium aerofaciens producta* SP427 11 FBK 12 DE586246.1 *Clostridium* Lachnospiraceae *Blautia* 1.88 ++++ 0.94 1 0 clade_354 clade_260 clade_309 *mayombei bacterium producta* 5_1_57FAA SP427 4 3R1 13 DE844277.1 *Coprococcus Clostridium Blautia* 2.1 ++++ 0.88 1 0 clade_262 clade_354 clade_309 *comes mayombei* sp. M25 SP427 2 1HR 14 DE208485.1 *Coprococcus Clostridium Clostridium* 1.8 ++++ 0.89 1.3 1 clade_262 clade_252 clade_494 *comes tertium orbiscindens* SP427 31 PBS 31 Naive n/a n/a 1 0 0 NoCdiff

(315) TABLE-US-00007 TABLE 6 Initial Initial Concentration of Concentration Ecobiotic ™ Incu- Inhibition of VRE (log10 Composition (log10 bation (log10 CFU/mL) CFU/mL/Strain) Time (h) CFU/mL) 3 6 15 3.1 3 4 15 1.3 2 6 15 5.2 2 4 15 1.6 3 6 24 2.7 3 4 24 0.7 2 6 24 4.6

(316) TABLE-US-00008 TABLE 7 Initial Initial Concentration of Concentration of Ecobiotic ™ Incu- Inhibition *K. pneumoniae* (log10 Composition (log10 bation (log10 CFU/mL) CFU/mL/Strain) Time (h) CFU/mL) 3 6 15 2.5 3 4 15 0.4 2 6 15 4.2 2 4 15 1.2 3 6 24 1.7 3 4 24 0.2 2 6 24 3.1 2 4 24 0.1

(317) TABLE-US-00009 TABLE 8 Initial Initial Concentration of Concentration of Ecobiotic ™ Incu- Inhibition *M. morganii* (log10 Composition (log10 bation (log10 CFU/mL) CFU/mL/Strain) Time (h) CFU/mL) 3 6 15 4.3 3 4 15 2.1 2 6 15 5.8 2 4 15 3.3 3 6 24 3.9 3 4 24 1.4 2 6 24 5.1 2 4 24 2.5

(318) TABLE-US-00010 TABLE 9 Mean Target Dose Cumulative Min. Mean Max. SP (CFU/OTU/ Mortality Rel. Clin. Score SP Expt. Arm Test Article mouse) (%) Weight (Death = 4) SP-327 3 Vehicle Control 30 0.89 2.2 SP-327 4 Vanco. Positive Control 0 0.99 1 SP-327 12 N1957 2.0E+07 0 0.87 0 SP-327 13 N1957 2.0E+06 40 0.86 2.2 SP-327 14 N1957 2.0E+05 50 0.80 2.8 SP-338 1 Vehicle Control 60 0.81 3.2 SP-338 2 Vanco. Positive Control 0 1.00 0 SP-338 3 10% fecal suspension 0 0.95 1 SP-338 5 N1957 2.0E+07 10 0.80 2 SP-338 6 N1957 2.0E+06 0 0.97 1 SP-338 7 N1957 2.0E+05 20 0.85 1.7 SP-338 11 N1957 2.0E+07 20 0.86 2 SP-338 12 N1957 2.0E+06 30 0.83 2.5 SP-338 13 N1961 2.0E+07 10 0.93 1.3 SP-338 14 N1955 2.0E+07 0 0.91 1.2 SP-338 15 N1955 2.0E+06 10 0.90 1.5 SP-338 16 N1955 2.0E+05 10 0.89 2.7 SP-338 17 N1967 2.0E+07 10 0.94 1.4 SP-338 18 N1983 2.0E+07 0 0.92 1 SP-338 19 N1989 2.0E+07 10 0.91 1.3 SP-338 20 N1996 2.0E+07 10 0.93 1.3

SP-338 21 Naïve 0 1.00 0 SP-339 1 Vehicle Control 20 0.88 2.2 SP-339 2 Vanco. Positive Control 0 0.99 0 SP-339 3 10% fecal suspension 0 0.97 0 SP-339 4 N1995 2.0E+07 20 0.83 2.1 SP-339 5 N1995 2.0E+06 10 0.91 1.5 SP-339 6 N1995 2.0E+05 0 0.96 1.2 SP-339 7 N1950 2.0E+07 0 0.94 1 SP-339 8 N1994 2.0E+07 20 0.87 1.8 SP-339 9 N1997 2.0E+07 0 0.95 1.2 SP-339 10 N1967 2.0E+07 0 0.93 1.2 SP-339 11 N1983 2.0E+07 10 0.83 2.2 SP-339 12 N1989 2.0E+07 0 0.88 1.5 SP-339 13 N1996 2.0E+07 0 0.97 1 SP-339 14 N2002 2.0E+07 20 0.92 2 SP-339 15 N2000 2.0E+07 0 0.98 1.2 SP-339 21 Naïve 0 0.98 0 SP-342 1 Vehicle Control 40 0.85 2.5 SP-342 2 Vanco. Positive Control 0 1.00 0 SP-342 5 N1957 2.0E+08 0 0.94 0.2 SP-342 6 N1957 2.0E+07 0 0.96 0 SP-342 7 N1957 2.0E+06 10 0.88 1.3 SP-342 8 N1980 2.0E+08 10 0.92 1.8 SP-342 9 N1998 2.0E+08 20 0.83 2.8 SP-342 10 N1976 2.0E+08 10 0.92 1.4 SP-342 11 N1987 2.0E+08 10 0.93 1.6 SP-342 12 N2005 2.0E+08 20 0.86 2.4 SP-342 13 N1958 2.0E+08 0 0.94 1.5 SP-342 14 N2004 2.0E+08 10 0.93 1.4 SP-342 15 N1949 2.0E+08 10 0.87 1.5 SP-342 18 N1970 2.0E+08 50 0.81 3 SP-342 21 Naïve 0 0.99 0 SP-361 1 Vehicle Control 30 0.88 2.6 SP-361 2 10% fecal suspension 0 0.99 0 SP-361 3 N435 1.0E+07 80 0.83 3.6 SP-361 4 N1979 1.0E+07 0 0.97 0 SP-361 5 N414 1.0E+07 0 0.97 0 SP-361 6 N512 1.0E+07 20 0.94 1.6 SP-361 7 N582 1.0E+07 10 0.93 0.9 SP-361 8 N571 1.0E+07 30 0.88 2.1 SP-361 9 N510 1.0E+07 0 0.93 0.3 SP-361 10 N1981 1.0E+07 40 0.83 2.8 SP-361 11 N1969 1.0E+07 80 0.82 3.6 SP-361 12 N461 1.0E+07 10 0.89 1.2 SP-361 13 N460 1.0E+07 0 0.93 1.1 SP-361 14 N1959 1.0E+07 30 0.89 1.9 SP-361 15 N2006 1.0E+07 30 0.89 1.9 SP-361 16 N1953 1.0E+07 10 0.83 2.3 SP-361 17 N1960 1.0E+07 0 0.92 1 SP-361 18 N2007 1.0E+07 10 0.91 0.9 SP-361 19 N1978 1.0E+07 10 0.91 1.3 SP-361 20 N1972 1.0E+07 30 0.83 2.6 SP-361 21 Naïve 0 1.00 0 SP-363 1 Vehicle Control 30 0.85 2.6 SP-363 2 10% fecal suspension 0 0.95 0 SP-363 8 N1974 1.0E+07 60 0.81 3.2 SP-363 9 N582 1.0E+07 60 0.81 3.2 SP-363 10 N435 1.0E+07 30 0.86 2.1 SP-363 11 N414 1.0E+07 40 0.83 2.5 SP-363 12 N457 1.0E+07 30 0.83 2.2 SP-363 13 N511 1.0E+07 20 0.87 2 SP-363 14 N513 1.0E+07 0 0.88 0.2 SP-363 15 N682 1.0E+07 30 0.82 2.6 SP-363 16 N736 1.0E+07 40 0.82 2.8 SP-363 17 N732 1.0E+07 10 0.86 1.3 SP-363 18 N1948 1.0E+07 60 0.85 3.2 SP-363 19 N853 1.0E+07 10 0.85 2.2 SP-363 20 N1979 1.0E+07 60 0.78 3.2 SP-363 21 N879 1.0E+07 40 0.83 2.8 SP-363 22 N999 1.0E+07 20 0.88 2.4 SP-363 23 N975 1.0E+07 30 0.80 2.6 SP-363 24 N861 1.0E+07 50 0.85 3 SP-363 25 N1095 1.0E+07 80 0.83 3.6 SP-363 26 Naïve 0 1.00 0 SP-364 1 Vehicle Control 40 0.83 2.8 SP-364 4 N582 1.0E+07 0 0.81 0.9 SP-364 5 N582 1.0E+06 0 0.84 0.9 SP-364 6 N582 1.0E+05 40 0.76 2.5 SP-364 13 N414 1.0E+07 0 0.84 0 SP-364 14 N414 1.0E+06 30 0.79 2.4 SP-364 15 N414 1.0E+05 10 0.76 2 SP-364 22 10% fecal suspension 0 0.97 0 SP-364 23 Nave 0 0.99 0 SP-365 1 Vehicle Control 40 0.83 2.8 SP-365 4 10% fecal suspension 0 0.98 0 SP-365 13 N582 1.0E+07 60 0.80 3.2 SP-365 14 N582 1.0E+06 10 0.89 1.5 SP-365 15 N414 1.0E+07 20 0.86 1.7 SP-365 16 N414 1.0E+06 80 0.83 3.5 SP-365 21 Naïve 0 1.00 0 SP-366 1 Vehicle Control 20 0.82 2.4 SP-366 4 10% fecal suspension 0 0.93 1 SP-366 7 N582 1.0E+07 0 0.86 1 SP-366 10 N414 1.0E+07 20 0.83 2.4 SP-366 13 N402 1.0E+07 30 0.81 2.1 SP-366 16 N1982 1.0E+07 0 0.90 1.1 SP-366 19 N460 1.0E+07 10 0.83 2.2 SP-366 22 N513 6.7E+06 40 0.82 2.8 SP-366 23 N1966 1.0E+07 0 0.90 0.5 SP-366 24 N1977 1.0E+07 20 0.83 1.9 SP-366 25 N1979 1.0E+07 20 0.83 2.4 SP-366 26 N682 1.0E+07 20 0.83 2.3 SP-366 27 N1947 1.0E+07 10 0.82 1.3 SP-366 28 N582 1.0E+07 20 0.82 1.8 SP-366 29 N414 1.0E+07 0 0.85 1.5 SP-366 30 N603 1.0E+07 30 0.82 2.2 SP-366 31 Naïve 0 0.99 0 SP-368 1 Vehicle Control 50 0.85 2.8 SP-368 2 10% fecal suspension 0 0.97 0 SP-368 7 N1966 1.0E+07 0 0.89 1 SP-368 8 N1966 1.0E+06 10 0.91 1.5 SP-368 9 N1966 1.0E+05 50 0.82 3.1 SP-368 21 Naïve 0 1.00 0 SP-374 1 Vehicle Control 100 0.83 4 SP-374 4 10% fecal suspension 10 0.89 0.5 SP-374 11 N1966 1.0E+08 0 0.87 1 SP-374 12 N1966 1.0E+08 0 0.91 0.5 SP-374 13 N1966 1.0E+07 10 0.88 1.3 SP-374 14 N1966 1.0E+06 50 0.79 3 SP-374 15 N584 1.0E+08 0 0.89 1 SP-374 16 N584 1.0E+07 30 0.84 2.4 SP-374 17 N1962 1.0E+07 0 0.93 0 SP-374 18 N382 1.0E+07 10 0.85 1.5 SP-374 19 N1964 1.0E+07 20 0.89 1.8 SP-374 20 N1965 1.0E+07 30 0.85 2.1 SP-374 21 N306 1.0E+07 10 0.90 0.4 SP-374 22 N1988 1.0E+07 0 0.89 1 SP-374 23 N2003 1.0E+07 0 0.92 1.2 SP-374 24 N1993 1.0E+07 20 0.77 2.4 SP-374 25 Naïve 0 0.99 0 SP-376 1 Vehicle Control 60 0.83 3.2 SP-376 2 10% fecal suspension 0 0.98 0 SP-376 3 N1966 1.0E+08 30 0.79 2.4 SP-376 4 N1966 1.0E+07 0 0.95 0 SP-376 5 N1966 1.0E+08 30 0.79 2.6 SP-376 6 N1966 1.0E+07 10 0.88 2.2 SP-376 7 N1986 1.0E+07 40 0.80 2.8 SP-376 8 N1962 1.0E+08 0 0.98 0 SP-376 9 N1962 1.0E+07 0 0.95 0 SP-376 10 N1963 1.0E+07 40 0.81 2.6 SP-376 11 N1984 1.0E+08 0 0.97 0 SP-376 12 N1984 1.0E+07 0 0.90 1.1 SP-376 13 N1990 1.0E+08 0 0.92 1 SP-376 14 N1990 1.0E+07 0 0.92 1 SP-376 15 N1999 1.0E+08 10 0.87 1.4 SP-376 16 N1999 1.0E+07 0 0.93 0 SP-376 17 N1968 1.0E+07 50 0.78 3 SP-376 18 N1951 1.0E+07 0 0.93 1 SP-376 19 N1991 1.0E+07 0 0.93 1.1 SP-376 20 N1975 1.0E+07 50 0.78 3 SP-376 21 Naïve 0 0.99 0 SP-383 1 Vehicle Control 100 0.83 4 SP-383 2 10% fecal suspension 0 0.92 0.1 SP-383 9 N1962 1.0E+09 10 0.95 1.3 SP-383 10 N1962 1.0E+08 10 0.93 1.3 SP-383 11 N1962 1.0E+07 0 0.92 1 SP-383 12 N1984 1.0E+09 0 0.89 1 SP-383 13 N1984 1.0E+08 10 0.94 1.3 SP-383 14 N1984 1.0E+07 10 0.90 1.3 SP-383 21 Naïve 0 1.00 0 SP-390 1 Vehicle Control 80 0.82 3.6 SP-390 2 10% fecal suspension 0 0.98 0.1 SP-390 3 N1962 2.0E+07 0 0.97 0 SP-390 4 N1962 2.0E+06 0 0.98 0 SP-390 5 N1984 2.0E+07 0 0.95 1 SP-390 6 N1984 2.0E+06 0 0.95 0.1 SP-390 9 N1962 2.0E+07 0 0.93 1 SP-390 10 N1962 2.0E+06 10 0.93 1.3 SP-390 11 N1984 2.0E+07 20 0.86 2.2 SP-390 12 N1984 2.0E+09 30 0.88 2.1 SP-390 13 N1952 2.0E+07 0 0.89 1 SP-390 14 N2001 2.0E+07 0 0.95 0.2 SP-390 15 N1973 2.0E+07 10 0.90 0.7 SP-390 16 N1954 2.0E+07 0 0.94 1.1 SP-390 17 N1985 2.0E+07 10 0.86 1.8 SP-390 18 N1971 2.0E+07 0 0.89 0.9 SP-390 19 N1956 2.0E+07 0 0.95 0 SP-390 20 N1992 2.0E+07 0 0.95 0 SP-390 31 Naïve 0 0.98 0

(319) TABLE-US-00011 TABLE 10 Network Ecology ID Exemplary Network Clades Exemplary Network OTUs N306 clade_252, (clade_260 or clade_260c or *Blautia producta*, *Clostridium hylemonae*, clade_260g or clade_260h), (clade_262 or *Clostridium innocuum*, *Clostridium orbiscindens*, clade_262i), (clade_309 or clade_309c or *Clostridium symbiosum*, *Clostridium tertium*, clade_309e or clade_309g or clade_309h or *Collinsella aerofaciens*, *Coprobacillus* sp. D7, clade_309i), (clade_351 or clade_351e), *Coprococcus comes*, *Eubacterium rectale*, (clade_38 or clade_38e or clade_38i), *Eubacterium* sp. WAL 14571, *Faecalibacterium* (clade_408 or clade_408b or clade_408d or *prausnitzii*, Lachnospiraceae bacterium 5_1_57FAA, clade_408f or clade_408g or clade_408h), *Roseburia faecalis*, *Ruminococcus obeum*, (clade_444 or clade_444i), (clade_478 or *Ruminococcus torques* clade_478i), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), clade_494, (clade_553 or clade_553i) N382 clade_252, (clade_260 or clade_260c or *Blautia producta*, *Clostridium hylemonae*, clade_260g or clade_260h), (clade_262 or *Clostridium innocuum*, *Clostridium orbiscindens*, clade_262i), (clade_309 or clade_309c or *Clostridium symbiosum*, *Clostridium tertium*, clade_309e or clade_309g or clade_309h or *Collinsella aerofaciens*, *Coprococcus comes*, clade_309i), (clade_351 or clade_351e), Lachnospiraceae bacterium 5_1_57FAA, (clade_360 or clade_360c or clade_360g or *Ruminococcus bromii*, *Ruminococcus gnavus* clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i) N402 (clade_262 or clade_262i), clade_286, *Alistipes shahii*, *Coprococcus comes*, *Dorea* (clade_309 or clade_309c or clade_309e or *formicigenerans*, *Dorea longicatena*, *Eubacterium* clade_309g or clade_309h or clade_309i), *rectale*, *Faecalibacterium prausnitzii*, *Odoribacter* (clade_360 or clade_360c or clade_360g or *splanchnicus*, *Parabacteroides merdae*, clade_360h or clade_360i), (clade_444 or *Ruminococcus obeum*, *Ruminococcus torques* clade_444i), clade_466, (clade_478 or clade_478i), clade_500 N414 (clade_262 or clade_262i), (clade_309 or *Coprococcus comes*, *Dorea formicigenerans*, *Dorea* clade_309c or clade_309e or clade_309g or *longicatena*, *Eubacterium eligens*, *Eubacterium* clade_309h or clade_309i), (clade_360 or *rectale*, *Faecalibacterium prausnitzii*, *Odoribacter* clade_360c or clade_360g or clade_360h or *splanchnicus*, *Ruminococcus obeum*, *Ruminococcus* clade_360i), (clade_444 or clade_444i), *torques* clade_466, (clade_478 or clade_478i), (clade_522 or clade_522i) N435 (clade_262 or clade_262i), (clade_309 or *Coprococcus comes*, *Dorea formicigenerans*, *Dorea* clade_309c or clade_309e or clade_309g or *longicatena*, *Eubacterium rectale*, *Faecalibacterium* clade_309h or clade_309i), (clade_360 or *prausnitzii*, *Odoribacter splanchnicus*, clade_360c or clade_360g or clade_360h or *Ruminococcus obeum*, *Ruminococcus torques* clade_360i), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i) N457 (clade_262 or clade_262i), (clade_309 or *Dorea longicatena*, *Eubacterium eligens*, clade_309c or clade_309e or clade_309g or *Eubacterium rectale*, *Faecalibacterium prausnitzii*, clade_309h or clade_309i), (clade_360 or *Roseburia intestinalis*, *Ruminococcus obeum*, clade_360c or clade_360g or clade_360h or *Ruminococcus torques* clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_522 or clade_522i) N460 (clade_262 or

clade_262i), (clade_309 or *Coprococcus comes*, *Dorea formicigenerans*, clade_309c or clade_309e or clade_309g or *Eubacterium rectale*, *Faecalibacterium prausnitzii*, clade_309h or clade_309i), (clade_360 or *Odoribacter splanchnicus*, *Ruminococcus obeum*, clade_360c or clade_360g or clade_360h or *Ruminococcus torques* clade_360i), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i) N461 (clade_262 or clade_262i), (clade_309 or *Coprococcus comes*, *Dorea formicigenerans*, *Dorea* clade_309c or clade_309e or clade_309g or *longicatena*, *Eubacterium rectale*, *Faecalibacterium* clade_309h or clade_309i), (clade_360 or *prausnitzii*, *Ruminococcus obeum*, *Ruminococcus* clade_360c or clade_360g or clade_360h or *torques* clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i) N510 (clade_262 or clade_262i), (clade_309 or *Dorea longicatena*, *Eubacterium eligens*, clade_309c or clade_309e or clade_309g or *Eubacterium rectale*, *Faecalibacterium prausnitzii*, clade_309h or clade_309i), (clade_360 or *Ruminococcus obeum*, *Ruminococcus torques* clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_522 or clade_522i) N511 (clade_262 or clade_262i), (clade_309 or *Coprococcus comes*, *Eubacterium rectale*, clade_309c or clade_309e or clade_309g or *Faecalibacterium prausnitzii*, *Roseburia intestinalis*, clade_309h or clade_309i), (clade_444 or *Ruminococcus obeum*, *Ruminococcus torques* clade_444i), (clade_478 or clade_478i) N512 (clade_262 or clade_262i), (clade_309 or *Coprococcus comes*, *Dorea formicigenerans*, *Dorea* clade_309c or clade_309e or clade_309g or *longicatena*, *Eubacterium rectale*, *Ruminococcus* clade_309h or clade_309i), (clade_360 or *obeum*, *Ruminococcus torques* clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i) N513 (clade_262 or clade_262i), (clade_309 or *Coprococcus comes*, *Dorea formicigenerans*, clade_309c or clade_309e or clade_309g or *Eubacterium rectale*, *Faecalibacterium prausnitzii*, clade_309h or clade_309i), (clade_360 or *Ruminococcus obeum*, *Ruminococcus torques* clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i) N571 (clade_262 or clade_262i), (clade_309 or *Dorea longicatena*, *Eubacterium rectale*, clade_309c or clade_309e or clade_309g or *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, clade_309h or clade_309i), (clade_360 or *Ruminococcus torques* clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i) N582 (clade_262 or clade_262i), (clade_309 or *Clostridium symbiosum*, *Eubacterium rectale*, clade_309c or clade_309e or clade_309g or *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, clade_309h or clade_309i), (clade_408 or *Ruminococcus torques* clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i) N584 (clade_260 or clade_260c or clade_260g or *Blautia producta*, *Clostridium symbiosum*, clade_260h), (clade_262 or clade_262i), *Collinsella aerofaciens*, *Coprococcus comes*, (clade_309 or clade_309c or clade_309e or Lachnospiraceae bacterium 5_1_57FAA clade_309g or clade_309h or clade_309i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_553 or clade_553i) N603 (clade_172 or clade_172i), (clade_309 or *Bifidobacterium adolescentis*, *Dorea longicatena*, clade_309c or clade_309e or clade_309g or *Eubacterium rectale*, *Faecalibacterium prausnitzii*, clade_309h or clade_309i), (clade_360 or *Ruminococcus obeum* clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i) N682 clade_170, (clade_309 or clade_309c or *Bacteroides caccae*, *Eubacterium rectale*, clade_309e or clade_309g or clade_309h or *Faecalibacterium prausnitzii*, *Ruminococcus obeum* clade_309i), (clade_444 or clade_444i), (clade_478 or clade_478i) N732 (clade_262 or clade_262i), (clade_309 or *Coprococcus comes*, *Faecalibacterium prausnitzii*, clade_309c or clade_309e or clade_309g or *Ruminococcus obeum*, *Ruminococcus torques* clade_309h or clade_309i), (clade_478 or clade_478i) N736 (clade_262 or clade_262i), (clade_309 or *Dorea longicatena*, *Faecalibacterium prausnitzii*, clade_309c or clade_309e or clade_309g or *Ruminococcus obeum*, *Ruminococcus torques* clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_478 or clade_478i) N853 (clade_262 or clade_262i), (clade_444 or *Eubacterium rectale*, *Faecalibacterium prausnitzii*, clade_444i), (clade_478 or clade_478i) *Ruminococcus torques* N861 (clade_262 or clade_262i), (clade_309 or *Clostridium hathewayi*, *Ruminococcus obeum*, clade_309c or clade_309e or clade_309g or *Ruminococcus torques* clade_309h or clade_309i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h) N879 (clade_309 or clade_309c or clade_309e or *Faecalibacterium prausnitzii*, *Roseburia intestinalis*, clade_309g or clade_309h or clade_309i), *Ruminococcus obeum* (clade_444 or clade_444i), (clade_478 or clade_478i) N975 clade_170, (clade_262 or clade_262i), *Bacteroides caccae*, *Coprococcus comes*, *Dorea* (clade_360 or clade_360c or clade_360g or *longicatena* clade_360h clade_360i) N999 (clade_309 or clade_309c or clade_309e or *Dorea formicigenerans*, *Faecalibacterium* clade_309g or clade_309h or clade_309i), *prausnitzii*, *Ruminococcus obeum* (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_478 or clade_478i) N1095 (clade_444 or clade_444i), (clade_522 or *Eubacterium eligens*, *Eubacterium rectale* clade_522i) N1947 (clade_262 or clade_262i), (clade_309 or *Bacteroides* sp. 3_1_23, *Collinsella aerofaciens*, clade_309c or clade_309e or clade_309g or *Dorea longicatena*, *Escherichia coli*, *Eubacterium* clade_309h or clade_309i), (clade_360 or *rectale*, *Faecalibacterium prausnitzii*, *Roseburia* clade_360c or clade_360g or clade_360h or *intestinalis*, *Ruminococcus obeum*, *Ruminococcus* clade_360i), (clade_38 or clade_38e or *torques* clade_38i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_553 or clade_553i), (clade_92 or clade_92e or clade_92i) N1948 (clade_262 or clade_262i), (clade_38 or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_38e or clade_38i), (clade_478 or *Faecalibacterium prausnitzii*, *Ruminococcus* clade_478i), (clade_65 or clade_65e) *torques* N1949 (clade_309 or clade_309c or clade_309e or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_309g or clade_309h or clade_309i), *Bacteroides vulgatus*, *Blautia producta*, (clade_378 or clade_378e), (clade_38 or *Enterococcus faecalis*, Erysipelotrichaceae clade_38e or clade_38i), (clade_479 or bacterium 3_1_53, *Escherichia coli* clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1950 clade_253, (clade_309 or clade_309c or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_309e or clade_309g or clade_309h or *Bacteroides vulgatus*, *Blautia producta*, *Clostridium* clade_309i), (clade_378 or clade_378e), *disporicum*, Erysipelotrichaceae bacterium 3_1_53 (clade_38 or clade_38e or clade_38i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_65 or clade_65e) N1951 (clade_260 or clade_260c or clade_260g or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260h), (clade_262 or clade_262i), *hylemonae*, *Clostridium symbiosum*, *Coprococcus* (clade_309 or clade_309c or clade_309e or *comes*, *Eubacterium rectale*, clade_309g or clade_309h or clade_309i), Lachnospiraceae bacterium 5_1_57FAA, (clade_360 or clade_360c or clade_360g or *Ruminococcus gnavus* clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i) N1952 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *hylemonae*, *Clostridium innocuum*, *Clostridium* clade_309c or clade_309e or clade_309g or *mayombei*, *Clostridium orbiscindens*, *Clostridium* clade_309h or clade_309i), (clade_351 or *symbiosum*, *Clostridium tertium*, *Collinsella* clade_351e), (clade_354 or clade_354e), *aerofaciens*, *Coprococcus comes*, (clade_360 or clade_360c or clade_360g or Lachnospiraceae bacterium 5_1_57FAA, clade_360h or clade_360i), (clade_408 or *Ruminococcus bromii*, *Ruminococcus gnavus* clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i) N1953 clade_253, (clade_262 or clade_262i), *Bacteroides* sp. 1_1_6, *Bacteroides vulgatus*, (clade_309 or clade_309c or clade_309e or *Clostridium disporicum*, *Clostridium mayombei*, clade_309g or clade_309h or clade_309i), *Clostridium symbiosum*, *Coprobacillus* sp. D7, (clade_354 or clade_354e), (clade_360 or *Coprococcus comes*, *Dorea formicigenerans*, clade_360c or clade_360g or clade_360h or *Enterococcus faecalis*, Erysipelotrichaceae clade_360i), (clade_378 or clade_378e), bacterium 3_1_53, *Escherichia coli*, *Eubacterium* (clade_408 or clade_408b or clade_408d or *rectale*, *Faecalibacterium prausnitzii*, *Odoribacter* clade_408f or clade_408g or clade_408h), *splanchnicus*, *Ruminococcus obeum* (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1954 clade_252, clade_253, (clade_260 or *Blautia* sp. M25, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *hylemonae*, *Clostridium innocuum*, *Clostridium* clade_309c or clade_309e or clade_309g or *mayombei*, *Clostridium orbiscindens*, *Clostridium* clade_309h or clade_309i), (clade_351 or *symbiosum*, *Clostridium tertium*, *Collinsella* clade_351e), (clade_354 or clade_354e), *aerofaciens*, *Coprococcus comes*, (clade_360 or clade_360c or clade_360g or Lachnospiraceae bacterium 5_1_57FAA, clade_360h or clade_360i), (clade_408 or *Ruminococcus bromii*, *Ruminococcus gnavus* clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i) N1955 (clade_309 or clade_309c or clade_309e or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 2_1_22, clade_309g or clade_309h or clade_309i), *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, (clade_354 or clade_354e), (clade_378 or *Blautia producta*, *Clostridium sordellii*, clade_378e), (clade_38 or clade_38e or *Coprobacillus* sp. D7, *Enterococcus faecalis*, clade_38i), (clade_479 or clade_479c or *Enterococcus faecium*, Erysipelotrichaceae clade_479g or clade_479h), (clade_481 or bacterium 3_1_53, *Escherichia coli* clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1956 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *hylemonae*, *Clostridium innocuum*, *Clostridium* clade_309c or clade_309e or clade_309g or *mayombei*, *Clostridium nexile*, *Clostridium* clade_309h or clade_309i), (clade_351 or *orbiscindens*, *Clostridium symbiosum*, *Clostridium* clade_351e), (clade_354 or clade_354e), *tertium*, *Collinsella aerofaciens*, (clade_360 or clade_360c or clade_360g or Lachnospiraceae bacterium 5_1_57FAA, clade_360h or clade_380i), (clade_408 or *Ruminococcus bromii*, *Ruminococcus gnavus* clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i) N1957 (clade_309 or clade_309c or clade_309e or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_309g or clade_309h or clade_309i), *Bacteroides vulgatus*, *Blautia producta*, *Clostridium* (clade_351 or clade_351e), (clade_354 or *innocuum*, *Clostridium sordellii*, *Coprobacillus* sp. clade_354e), (clade_378 or clade_378e), D7, *Enterococcus faecalis*, *Escherichia coli* (clade_38 or clade_38e or clade_38i), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1958 (clade_351 or clade_351e), (clade_354 or *Clostridium innocuum*, *Clostridium sordellii*, clade_354e), (clade_481 or clade_481a or *Coprobacillus* sp. D7 clade_481b or clade_481e or clade_481g or clade_481h or clade_481i) N1959 clade_253, (clade_262 or clade_262i), *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, (clade_309 or clade_309c or clade_309e or *Bacteroides vulgatus*, *Blautia producta*, *Clostridium* clade_309g or clade_309h or clade_309i), *disporicum*, *Coprococcus comes*, *Dorea* (clade_360 or clade_360c or clade_360g or *formicigenerans*, *Dorea longicatena*, *Enterococcus* clade_360h or clade_360i), (clade_378 or *faecalis*, Erysipelotrichaceae bacterium 3_1_53, clade_378e), (clade_38 or clade_38e or *Escherichia coli*, *Eubacterium eligens*, *Eubacterium* clade_38i), (clade_444 or clade_444i), *rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus* (clade_478 or clade_478i), (clade_479 or *obeum*, *Ruminococcus torques* clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_522 or clade_522i), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1960 clade_253, (clade_262 or clade_262i), *Clostridium disporicum*, *Clostridium mayombei*, (clade_309 or clade_309c or clade_309e or *Clostridium symbiosum*, *Coprobacillus* sp. D7, clade_309g or clade_309h or clade_309i), *Coprococcus comes*, *Dorea formicigenerans*, (clade_354 or clade_354e), (clade_360 or *Enterococcus faecalis*, Erysipelotrichaceae bacterium clade_360c or clade_360g or clade_360h or 3_1_53, *Escherichia coli*, *Eubacterium* clade_360i), (clade_408 or clade_408b or *rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus* clade_408d or clade_408f or clade_408g or *obeum* clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_92 or clade_92e or clade_92i) N1961 (clade_309 or clade_309c or clade_309e or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_309g or clade_309h or clade_309i), *Bacteroides vulgatus*, *Blautia producta*, *Clostridium* (clade_351 or clade_351e), (clade_378 or *innocuum*, *Enterococcus faecalis*, *Escherichia coli* clade_378e), (clade_38 or clade_38e or clade_38i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1962 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *hylemonae*, *Clostridium innocuum*, *Clostridium* clade_309c or clade_309e or clade_309g or *mayombei*, *Clostridium orbiscindens*, *Clostridium* clade_309h or clade_309i), (clade_351 or *symbiosum*, *Clostridium tertium*, *Collinsella* clade_351e), (clade_354 or clade_354e), *aerofaciens*, *Coprococcus comes*, (clade_360 or clade_360c or clade_360g or Lachnospiraceae bacterium 5_1_S7FAA, clade_360h or clade_360i), (clade_408 or *Ruminococcus bromii*, *Ruminococcus gnavus* clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i) N1963 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium disporicum*, clade_260c or clade_260g or clade_260h), *Clostridium hylemonae*, *Clostridium innocuum*, (clade_262 or clade_262i), (clade_309 or *Clostridium orbiscindens*,*Clostridium symbiosum*, clade_309c or clade_309e or clade_309g or *Clostridium tertium*, *Collinsella aerofaciens*, clade_309h or clade_309i), (clade_351 or *Coprococcus comes*, Lachnospiraceae bacterium clade_351e), (clade_360 or clade_360c or 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus* clade_360g or clade_360h or clade_360i), *gnavus* (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i) N1964 clade_170, (clade_260 or clade_260c or *Alistipes shahii*, *Bacteroides caccae*, *Bacteroides* clade_260g or clade_260h), (clade_262 or *stercoris*, *Blautia producta*, *Clostridium hathewayi*, clade_262i), clade_286, (clade_309 or *Clostridium symbiosum*, *Collinsella aerofaciens*, clade_309c or clade_309e or clade_309g or *Coprococcus comes*, *Dorea formicigenerans*, clade_309h or clade_309i), (clade 360 or *Eubacterium rectale*, *Holdemania filiformis*, clade_360c or clade_360g or clade_360h or Lachnospiraceae bacterium 5_1_57FAA, clade_360i), (clade_408 or clade_408b or *Parabacteroides merdae*, *Ruminococcus bromii*, clade_408d or clade_408f or clade_408g or *Ruminococcus obeum*, *Ruminococcus torques* clade_408h), (clade_444 or clade_444i), clade_485, clade_500, clade_537, (clade_553or clade_553i), clade_85 N1965 clade_252, clade_253, (clade_260 or *Blautia products*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *mayombel*, *Clostridium symbiosum*, *Collinsella* clade_309c or clade_309e or clade_309g or *aerofaciens*, *Coprococcus comes*, *Eubacterium* clade_309h or clade_309i), (clade_354 or *rectale*, *Faecalibacterium prausnitzii*, clade_354e), (clade_408 or clade_408b or Lachnospiraceae bacterium 5_1_57FAA, *Roseburia* clade_408d or clade_408f or clade_408g or *intestinalis*, *Ruminococcus bromii*, *Ruminococcus* clade_408h), (clade_444 or clade_444i), *obeum* (clade_478 or clade_478i), clade_537, (clade_553 or clade_553i) N1966 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *mayombei*, *Clostridium symbiosum*, *Collinsella* clade_309c or clade_309e or clade_309g or *aerofaciens*, *Coprococcus comes*, clade_309h or clade_309i), (clade_354 or Lachnospiraceae bacterium 5_1_57FAA clade_354e), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_553 or clade_553i) N1967 (clade_309 or clade_309c or clade_309e or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_309g or clade_309h or clade_309i), *Bacteroides vulgatus*, *Blautia producta*, (clade_378 or clade_378e), (clade_38 or *Enterococcus faecium*, *Escherichia coli* clade_38e or clade_38i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1968 clade_252, clade_253, (clade_351 or *Clostridium butyricum*, *Clostridium disporicum*, clade_351e), (clade_354 or clade_354e), *Clostridium innocuum*, *Clostridium mayombei*, (clade_478 or clade_478i), clade_494, *Clostridium orbiscindens*, *Clostridium tertium*, clade_537, (clade_553 or clade_553i) *Collinsella aerofaciens*, *Faecalibacterium prausnitzii*, *Ruminococcus bromii* N1969 clade_252, clade_253, (clade_262 or *Blautia producta*, *Clostridium butyricum*, clade_262i), (clade_309 or clade_309c or *Clostridium disporicum*, *Clostridium mayombei*, clade_309e or clade_309g or clade_309h or *Dorea formicigenerans*, Erysipelotrichaceae clade_309i), (clade_354 or clade_354e), bacterium 3_1_53, *Ruminococcus torques* (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_479 or clade_479c or clade_479g or clade_479h) N1970 (clade_351 or clade_351e), (clade_354 or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_354e), (clade_378 or clade_378e), *Bacteroides vulgatus*, *Clostridium innocuum*, (clade_38 or clade_38e or clade_38i), *Clostridium sordellii*, *Coprobacillus* sp. D7, (clade_481 or clade_481a or clade_481b or *Enterococcus faecalis*, *Escherichia coli* clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (cladc_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1971 clade_252, clade_253, (clade_260 or *Clostridium bolteae*, *Clostridium butyricum*, clade_260c or clade_260g or clade_260h), *Clostridium disporicum*, *Clostridium hylemonae*, (clade_262 or clade_262i), (clade_351 or *Clostridium innocuum*, *Clostridium mayombei*, clade_351e), (clade_354 or clade_354e), *Clostridium orbiscindens*, *Clostridium symbiosum*, (clade_360 or clade_360c or clade_360g or *Clostridium tertium*, *Collinsessa aerofaciens*, clade_360h or clade_360i), (clade_408 or *Coprococcus comes*, Lachnospiraceae bacterium clade_408b or clade_408d or clade_408f or 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus* clade_408g or clade_408h), clade_494, *gnavus* clade_537, (clade_553 or clade_553i) N1972 clade_253, (clade_262 or clade_262i), *Clostridium disporicum*, *Clostridium mayombei*, (clade_309 or clade_309c or clade_309e or *Clostridium symbiosum*, *Coprobacillus* sp. D7, clade_309g or clade_309h or clade_309i), *Coprococcus comes*, *Dorea formicigenerans*, (clade_354 or clade_354e), (clade_360 or Erysipelotrichaceae bacterium 3_1_53, clade_360c or clade_360g or clade_360h or *Eubacterium rectale*, *Faecalibacterium prausnitzii*, clade_360i), (clade_408 or clade_408b or *Ruminococcus obeum* clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i) N1973 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *hylemonae*, *Clostridium innocuum*, *Clostridium* clade_309c or clade_309e or clade_309g or *mayombei*, *Clostridium orbiscindens*, *Clostridium* clade_309h or clade_309i), (clade_351 or *symbiosum*, *Clostridium tertium*, *Coprococcus* clade_351e), (clade_354 or clade_354e), *comes*, Lachnospiraceae bacterium 5_1_57FAA, (clade_360 or clade_360c or clade_360g or *Ruminococcus bromii*, *Ruminococcus gnavus* clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537 N1974 (clade_262 or clade_262i), (clade_309 or *Coprococcus comes*, *Dorea formicigenerans*, *Dorea* clade_309c or clade_309e or clade_309g or *longicatena*, *Eubacterium rectale*, *Faecalibacterium* clade_309h or clade_309f), (clade_360 or *prausnitzii*, *Odoribacter splanchnicus*, *Roseburia* clade_360c or clade_360g or clade_360h or *intestinalis*, *Ruminococcus obeum*, *Ruminococcus* clade_360i), (clade_444 or clade_444i), *torques* clade_466, (clade_478 or clade_478i) N1975 (clade_260 or clade_260c or clade_260g or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260h), (clade_262 or clade_262i), *hylemonae*, *Clostridium innocuum*, *Clostridium* (clade_309 or clade_309c or clade_309e or *mayombei*, *Clostridium orbiscindens*, *Clostridium* clade_309g or clade_309h or clade_309i), *symbiosum*, *Collinsella aerofaciens*, *Coprococcus* (clade_351 or clade_351e), (clade_354 or *comes*, *Eubacterium rectale*, *Faecalibacterium* clade_354e), (clade_360 or clade_360c or *prausnitzii*, Lachnospiraceae bacterium 5_1_57FAA, clade_360g or clade_360h or clade_360i), *Ruminococcus bromii*, *Ruminococcus gnavus* (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), clade_494, clade_537, (clade_553 or clade_553i) N1976 (clade_351 or clade_351e), (clade_378 or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_378e), (clade_38 or clade_38e or *Bacteroides vulgatus*, *Clostridium innocuum*, clade_38i), (clade_481 or clade_481a or *Coprobacillus* sp. D7, *Enterococcus faecalis* clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e) N1977 clade_252, clade_253, (clade_262 or *Blautia producta*, *Clostridium butyricum*, clade_262i), (clade_309 or clade_309c or *Clostridium disporicum*, *Clostridium mayombei*, clade_309e or clade_309g or clade_309h or *Dorea formicigenerans*, Erysipelotrichaceae clade_309i), (clade_354 or clade_354e), bacterium 3_1_53, *Eubacterium tenue*, (clade_360 or clade_360c or clade_360g or *Ruminococcus torques* clade_360h or clade_360i), (clade_479 or clade_479c or clade_479g or clade_479h) N1978 clade_253, (clade_262 or clade_262i), *Bacteroides* sp. 1_1_6, *Bacteroides vulgatus*, (clade_309 or clade_309c or clade_309e or *Clostridium disporicum*, *Clostridium mayombei*, clade_309g or clade_309h or clade_309i), *Clostridium symbiosum*, *Coprobacillus* sp. D7, (clade_354 or clade_354e), (clade_360 or *Coprococcus comes*, *Dorea formicigenerans*, clade_360c or clade_360g or clade_360h or Erysipelotrichaceae bacterium 3_1_53, *Escherichia* clade_360i), (clade_378 or clade_378e), *coli*, *Eubacterium rectale*, *Faecalibacterium* (clade_408 or clade_408b or clade_408d or *prausnttzii*, *Odoribacter splanchnicus*, clade_408f or clade_408g or clade_408h), *Ruminococcus obeum* (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1979 (clade_262 or clade_262i), (clade_309 or *Bacteroides* sp. 1_1_6, *Coprococcus comes*, *Dorea* clade_309c or clade_309e or clade_309g or *formicigenerans*, *Dorea longicatena*, *Eubacterium* clade_309h or clade_309i), (clade_360 or *rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus* clade_360c or clade_360g or clade_360h or *obeum*, *Ruminococcus torques* clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_65 or clade_65e) N1980 (clade_309 or clade_309c or clade_309e or *Blautia producta*, *Clostridium innocuum*, clade_309g or clade_309h or clade_309i), *Clostridium sordellii*, *Coprobacillus* sp. D7, (clade_351 or clade_351e), (clade_354 or *Enterococcus faecalis*, *Escherichia coli* clade_354e), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_92 or clade_92e or clade_92i) N1981 (clade_262 or clade_262i), (clade_309 or *Bacteroides* sp. 3_1_23, *Dorea longicatena*, clade_309c or clade_309e or clade_309g or *Eubacterium eligens*, *Eubacterium rectale*, clade_309h or clade_309i), (clade_360 or *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, clade_360c or clade_360g or clade_360h or *Ruminococcus torques* clade_360i), (clade_38 or clade_38e or clade_38i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_522 or clade_522i) N1982 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *mayombei*, *Clostridium symbiosum*, *Collinsella* clade_309c or clade_309e or clade_309g or *aerofaciens*, *Coprococcus comes*, *Dorea* clade_309h or clade_309i), (clade_354 or *formicigenerans*, Erysipelotrichaceae bacterium clade_354e), (clade_360 or clade_360c or 3_1_53, *Eubacterium rectale*, clade_360g or clade_360h or clade_360i), Lachnospiraceae bacterium 5_1_57FAA, (clade_408 or clade_408b or clade_408d or *Ruminococcus obeum*, *Ruminococcus torques* clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_553 or clade_553i) N1983 clade_252, clade_253, (clade_309 or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_309c or clade_309e or clade_309g or *Bacteroides vulgatus*, *Blautia producta*, *Clostridium* clade_309h or clade_309i), (clade_354 or *butyricum*, *Clostridium disporicum*, *Clostridium* clade_354e), (clade_378 or clade_378e), *mayombei*, *Enterococcus faecium*, (clade_38 or clade_38e or clade_38i), Erysipelotrichaceae bacterium 3_1_53, *Escherichia* (clade_479 or clade_479c or clade_479g or *coli* clade_479h), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1984 clade_253, (clade_260 or clade_260c or *Blautia producta*, *Clostridium disporicum*, clade_260g or clade_260h), (clade_309 or *Clostridium innocuum*, *Clostridium mayombei*, clade_309c or clade_309e or clade_309g or *Clostridium orbiscindens*, *Clostridium symbiosum*, clade_309h or clade_309i), (clade_351 or *Collinsella aerofaciens*, *Eubacterium rectale*, clade_351e), (clade_354 or clade_354e), Lachnospiraceae bacterium 5_1_57FAA (clade_403 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), clade_494, (clade_553 or clade_553i) N1985 clade_252, clade_253, (clade_260 or *Blautia glucerasei*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *hylemonae*, *Clostridium innocuum*, *Clostridium* clade_309c or clade_309e or clade_309g or *mayombei*, *Clostridium orbiscindens*, *Clostridium* clade_309h or clade_309i), (clade_351 or *symbiosum*, *Clostridium tertiurm*, *Collinsella* clade_351e), (clade_354 or clade_354e), *aerofaciens*, *Coprococcus comes*, (clade_360 or clade_360c or clade_360g or Lachnospiraceae bacterium 5_1_57FAA, clade_360h or clade_360i), (clade_408 or *Ruminococcus bromii*, *Ruminococcus gnavus* clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i) N1986 clade_253, (clade_260 or clade_260c or *Blautia producta*, *Clostridium disporicum*, clade_260g or clade_260h), (clade_262 or *Clostridium symbiosum*, *Collinsella aerofaciens*, clade_262i), (clade_309 or clade_309c or *Coprococcus comes*, Lachnospiraceae bacterium clade_309e or clade_309g or clade_309h or 5_1_57FAA clade_309i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_553 or clade_553i) N1987 (clade_309 or clade_309c or clade_309e or *Blautia producta*, *Clostridium sordellii*, *Escherichia* clade_309g or clade_309h or clade_309i), *coli* (clade_354 or clade_354e), (clade_92 or clade_92e or clade_92i) N1988 clade_252, clade_253, (clade_260 or *Alistipes shahii*, *Blautia producta*, *Clostridium* clade_260c or clade_260g or clade_260h), *bolteae*, *Clostridium butyricum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *disporicum*, *Clostridium mayombei*, *Clostridium* clade_309c or clade_309e or clade_309g or *symbiosum*, *Collinsella aerofaciens*, *Coprococcus* clade_309h or clade_309i), (clade_354 or *comes*, *Eubacterium rectale*, *Faecalibacterium* clade_354e), (clade_408 or clade_408b or *prausnitzii*, *Holdemania filiformis*, Lachnospiraceae clade_408d or clade_408f or clade_408g or bacterium 5_1_57FAA, *Roseburia intestinalis*, clade_408h), (clade_444 or clade_444i), *Ruminococcus obeum*, *Ruminococcus torques* (clade_478 or clade_478i), clade_485, clade_500, (clade_553 or clade_553i) N1989 clade_252, clade_253, (clade_262 or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_262i), (clade_309 or clade_309c or *Bacteroides vulgatus*, *Blautia producta*, *Clostridium* clade_309e or clade_309g or clade_309h or *butyricum*, *Clostridium disporicum*, *Clostridium* clade_309i), (clade_354 or clade_354e), *mayombei*, *Dorea formicigenerans*, *Enterococcus* (clade_360 or clade_360c or clade_360g or *faecium*, Erysipelotrichaceae bacterium 3_1_53, clade_360h or clade_360i), (clade_378 or *Escherichia coli*, *Eubacterium tenue*, *Ruminococcus* clade_378e), (clade_38 or clade_38e or *torques* clade_38i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1990 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium disporicum*, clade_260c or clade_260g or clade_260h), *Clostridium hylemonae*, *Clostridium innocuum*, (clade_262 or clade_262i), (clade_309 or *Clostridium orbiscindens*, *Clostridium symbiosum*, clade_309c or clade_309e or clade_309g or *Clostridium tertium*, *Collinsella aerofaciens*, clade_309h or clade_309i), (clade_351 or *Coprococcus comes*, *Eubacterium rectale*, clade_351e), (clade_360 or clade_360c or *Faecalibacterium prausnitzii*, clade_360g or clade_360h or clade_360i), Lachnospiraceae bacterium 5_1_57FAA, (clade_408 or clade_408b or clade_408d or *Ruminococcus brornii*, *Ruminococcus gnavus* clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), clade_494, clade_537, (clade_553 or clade_553i) N1991 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostrtdium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *hylemonae*, *Clostridium innocuum*, *Clostridium* clade_309c or clade_309e or clade_309g or *mayombei*, *Clostridium symbiosum*, *Clostridium* clade_309h or clade_309i), (clade_351 or *tertium*, *Collinsella aerofaciens*, *Coprococcus* clade_351e), (clade_354 or clade_354e), *comes*, *Eubacterium rectale*, Lachnospiraceae (clade_360 or clade_360c or clade_360g or bacterium 5_1_57FAA, *Ruminococcus gnavus* clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_553 or clade_553i) N1992 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *hylemonae*, *Clostridium innocuum*, *Clostridium* clade_309c or clade_309e or clade_309g or *mayombei*, *Clostridium orbiscindens*, *Clostridium* clade_309h or clade_309i), (clade_351 or *symbiosum*, *Clostridium tertium*, *Collinsella* clade_351e), (clade_354 or clade_354e), *aerofaciens*, Lachnospiraceae bacterium (clade_360 or clade_360c or clade_360g or 1_4_56FAA, Lachnospiraceae bacterium clade_360h or clade_360i), (clade_408 or 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus* clade_408b or clade_408d or clade_408f or *gnavus* clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i) N1993 clade_110, clade_170, (clade_378 or *Bacteroides caccae*, *Bacteroides eggerthii*, clade_378e), (clade_38 or clade_38e or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_38i), (clade_65 or clade_65e), *Bacteroides stercoris*, *Bacteroides uniformis*, clade_85 *Bacteroides vulgatus* N1994 clade_253, (clade_378 or clade_378e), *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, (clade_38 or clade_38e or clade_38i), *Bacteroides vulgatus*, *Clostridium disporicum*, (clade_479 or clade_479c or clade_479g or *Enterococcus faecium*, Erysipelotrichaceae clade_479h), (clade_497 or clade_497e or bacterium 3_1_53, *Escherichia coli* clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1995 clade_253, (clade_309 or clade_309c or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_309e or clade_309g or clade_309h or *Bacteroides vulgatus*, *Blautia producta*, *Clostridium* clade_309i), (clade_378 or clade_378e), *disporicum*, *Enterococcus faecium*, (clade_38 or clade_38e or clade_38i), Erysipelotrichaceae bacterium 3_1_53, *Escherichia* (clade_479 or clade_479c or clade_479g or *coli* clade_479h), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N1996 clade_253, (clade_309 or clade_309c or *Blautia producta*, *Clostridium disporicum*, clade_309e or clade_309g or clade_309h or Erysipelotrichaceae bacterium 3_1_53 clade_309i), (clade_479 or clade_479c or clade_479g or clade_479h) N1997 clade_253, (clade_309 or clade_309c or *Blautia producta*, Clostridium disporicum, clade_309e or clade_309g or clade_309h or *Enterococcus faecium*, Erysipelotrichaceae clade_309i), (clade_479 or clade_479c or bacterium 3_1_53, *Escherichia coli* clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_92 or clade_92e or clade_92i) N1998 (clade_378 or clade_378e), (clade_38 or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_38e or clade_38i), (clade_65 or *Bacteroides vulgatus* clade_65e) N1999 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *mayombei*, *Clostridium symbiosum*, *Collinsella* clade_309c or clade_309e or clade_309g or *aerofaciens*, *Coprococcus comes*, *Eubacterium* clade_309h or clade_309i), (clade_354 or *rectale*, *Faecalibacterium prausnitzii*, clade_354e), (clade_408 or clade_408b or Lachnospiraceae bacterium 5_1_57FAA clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_553 or clade_553i) N2000 clade_252, clade_253, (clade_262 or *Blautia producta*, *Clostridium butyricum*, clade_262i), (clade_309 or clade_309c or *Clostridium disporicum*, *Clostridium mayombei*, clade_309e or clade_309g or clade_309h or *Dorea formicigenerans*, clade_309i), (clade_354 or clade_354e), Erysipelotrichaceae bacterium 3_1_53, (clade_360 or clade_360c or clade_360g or *Eubacterium tenue*, *Ruminococcus torques* clade_360h or clade_360i), (clade_479 or clade_479c or clade_479g or clade_479h) N2001 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *disporicum*, *Clostridium hylemonae*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *innocuum*, *Clostridium mayombei*, *Clostridium* clade_309c or clade_309e or clade_309g or *orbiscindens*, *Clostridium symbiosum*, *Clostridium* clade_309h or clade_309i), (clade_351 or *tertium*, *Collinsella aerofaciens*, *Coprococcus* clade_351e), (clade_354 or clade_354e), *comes*, Lachnospiraceae bacterium 5_1_57FAA, (clade_360 or clade_360c or clade_360g or *Ruminococcus bromii*, *Ruminococcus gnavus* clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i) N2002 clade_252, clade_253, (clade_309 or *Blautia producta*, *Clostridium butyricum*, clade_309c or clade_309e or clade_309g or *Clostridium disporicum*, *Clostridium mayombei*, clade_309h or clade_309i), (clade_354 or Erysipelotrichaceae bacterium 3_1_53 clade_354e), (clade_479 or clade_479c or clade_479g or clade_479h) N2003 clade_252, clade_253, (clade_260 or *Blautia producta*, *Clostridium bolteae*, *Clostridium* clade_260c or clade_260g or clade_260h), *butyricum*, *Clostridium disporicum*, *Clostridium* (clade_262 or clade_262i), (clade_309 or *hylemonae*, *Clostridium mayombei*, *Clostridium* clade_309c or clade_309e or clade_309g or *sordellii*, *Clostridium symbiosum*, *Clostridium* clade_309h or clade_309i), (clade_354 or *tertium*, *Collinsella aerofaciens*, *Coprobacillus* sp. clade_354e), (clade_360 or clade_360c or D7, *Coprococcus comes*, *Eubacterium* sp. WAL clade_360g or clade_360h or clade_360i), 14571, Lachnospiraceae bacterium 5_1_57FAA, (clade_38 or clade_38e or clade_38i), *Rumtinococcus bromii*, *Ruminococcus gnavus* (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), clade_537, (clade_553 or clade_553i) N2004 (clade_351 or clade_351e), (clade_378 or

*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_378e), (clade_38 or clade_38e or *Bacteroides vulgatus*, *Clostridium innocuum*, clade_38i), (clade_497 or clade_497e or *Enterococcus faecalis*, *Escherichia coli* clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N2005 (clade_309 or clade_309c or dade_309e or *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, clade_309g or clade_309h or clade_309i) *Bacteroides vulgatus*, *Blautia producta*, (clade_378 or clade_378e), (clade_38 or *Enterococcus faecalis*, *Escherichia coli* clade_38e or clade_38i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i) N2006 clade_170, (clade_262 or clade_262i), *Bacteroides caccae*, *Bacteroides* sp. 1_1_6, (clade_309 or clade_309c or clade_309e or *Coprococcus comes*, *Dorea formicigenerans*, *Dorea* clade_309g or clade_309h or clade_309i), *longicatena*, *Eubacterium rectale*, *Faecalibacterium* (clade_360 or clade_360c or clade_360g or *prausnitzii*, *Ruminococcus obeum*, *Ruminococcus* clade_360h or clade_360i), (clade_444 or *torques* clade_444i), (clade_478 or clade_478i), (clade_65 or dade_65e) N2007 clade_253, (clade_202 or clade_262i), *Bacteroides* sp. 1_1_6, *Bacteroides vulgatus*, (clade_309 or clade_309c or clade_309e or *Clostridium disporicurn*, *Clostridium mayombei*, clade_309g or clade_309h or clade_309i), *Clostridium symbiosum*, *Coprobacillus* sp. D7, (clade_354 or clade_354e), (clade_360 or *Coprococcus comes*, *Dorea formicigenerans*, clade_360c or clade_360g or clade_360h or *Enterococcus faecalis*, clade_360i), (clade_378 or clade_378e), Erysipelotrichaceae bacterium 3_1_53, (clade_408 or clade_408b or clade_408d or *Eubacterium rectale*, *Faecalibacterium prausnitzii*, clade_408f or clade_408g or clade_408h), *Odoribacter splanchnicus*, *Ruminococcus obeum* (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e)

(320) TABLE-US-00012 TABLE 11 Percent of Percent of Percent of Dose Engrafted Augemented Spore Ecologies Ecologies Ecologies Keystone OTU Clade Genus Family Order Former Occurs Occurs Occurs OTU CES *Blautia_luti* clade_309 *Blautia* Lachnospiraceae Clostridiales yes 100 100 0 0 4.0 *Blautia_schinkii* clade_309 *Blautia* Lachnospiraceae Clostridiales yes 100 93 7 0 4.0 *Blautia*_sp_M25 clade_309 *Blautia* Lachnospiraceae Clostridiales yes 100 93 7 1 5.0 *Subdoligranulum_variabile* clade_478 *Subdoligranulum* Ruminococcaceae Clostridiales yes 100 93 0 1 5.0 *Eubacterium_rectale* clade_444 *Eubacterium* Eubacteriaceae Clostridiales yes 100 87 13 1 5.0 Lachnospiraceae_*bacterium*_2_1_58FAA clade_360 unassigned Lachnospiraceae Clostridiales yes 100 80 7 0 4.0 *Clostridium_leptum* clade_537 *Clostridium* Clostridiaceae Clostridiales yes 100 73 13 1 4.0 *Faecalibacterium_prausnitzii* clade_478 *Faecalibacterium* Ruminococcaceae Clostridiales yes 100 73 0 1 4.0 *Ruminococcus_bromii* clade_537 *Ruminococcus* Ruminococcaceae Clostridiales yes 100 67 7 1 4.0 *Clostridium_citroniae* clade_408 *Clostridium* Clostridiaceae Clostridiales yes 100 53 27 0 3.0 *Christensenella_minuta* clade_558 *Christensenella* Christensenellaceae Clostridiales yes 100 0 27 0 2.0 *Ruminococcus_torques* clade_262 *Blautia* Lachnospiraceae Clostridiales yes 93 87 7 1 5.0 *Dorea_longicatena* clade_360 *Dorea* Lachnospiraceae Clostridiales yes 93 80 20 1 5.0 *Eubacterium_hadrum* clade_408 *Anaerostipes* Lachnospiraceae Clostridiales yes 93 80 7 1 5.0 *Blautia_hansenii* clade_309 *Blautia* Lachnospiraceae Clostridiales yes 93 73 13 0 3.0 *Clostridium_ramosum* clade_481 unassigned Erysipelotrichaceae Erysipelotrichales yes 93 73 13 0 3.0 *Ruminococcus_lactaris* clade_262 *Ruminococcus* Ruminococcaceae Clostridiales yes 93 67 33 1 4.0 Clostridiales_sp_SS3_4 clade_246 *Clostridiales* unclassified Clostridiales yes 86 73 20 0 3.0 *Dorea_formicigenerans* clade_360 *Dorea* Lachnospiraceae Clostridiales yes 86 53 7 1 4.0 *Coprococcus_comes* clade_262 *Coprococcus* Lachnospiraceae Clostridiales yes 86 47 0 1 4.0 Lachnospiraceae_*bacterium*_A4 clade_408 unassigned Lachnospiraceae Clostridiales yes 86 40 20 0 2.4 *Eubacterium_hallii* clade_396 *Eubacterium* Eubacteriaceae Clostridiales yes 86 40 7 1 3.4 *Eubacterium_brachy* clade_533 *Eubacterium* Eubacteriaceae Clostridiales yes 86 13 0 0 2.4 *Ruminococcus_callidus* clade_406 *Ruminococcus* Ruminococcaceae Clostridiales yes 79 67 13 0 2.3 *Clostridium_bartlettii* clade_354 unassigned Peptostreptococcaceae Clostridiales yes 79 67 7 0 2.3 *Clostridium_sporosphaeroides* clade_537 *Clostridium* Clostridiaceae Clostridiales yes 79 60 20 0 2.3 *Clostridium_bifermentans* clade_354 unassigned Peptostreptococcaceae Clostridiales yes 79 53 13 0 2.3 *Turicibacter_sanguinis* clade_555 *Turicibacter* Erysipelotrichaceae Erysipelotrichales yes 79 40 13 0 1.7 *Ruminococcus_albus* clade_516 *Ruminococcus* Ruminococcaceae Clostridiales yes 71 60 27 0 2.3 Eubacterium_*ramulus* clade_482 *Eubacterium* Eubacteriaceae Clostridiales yes 71 47 33 0 2.3 *Eubacterium_desmolans* clade_572 *Eubacterium* Eubacteriaceae Clostridiales yes 71 40 27 0 1.7 *Coprococcus_catus* clade_393 *Coprococcus* Lachnospiraceae Clostridiales yes 71 33 7 1 2.7 *Clostridium_oroticum* clade_96 *Clostridium* Clostridiaceae Clostridiales yes 71 27 7 0 1.7 *Blautia_glucerasea* clade_309 *Blautia* Lachnospiraceae Clostridiales yes 64 33 33 0 1.7 Lachnospiraceae_*bacterium*_3_1_57FAA clade_408 unassigned Lachnospiraceae Clostridiales yes 64 33 13 1 2.7 *Clostridium_viride* clade_540 *Clostridium* Clostridiaceae Clostridiales yes 64 27 13 0 1.7 *Ruminococcus_obeum* clade_309 *Blautia* Lachnospiraceae Clostridiales yes 64 27 7 1 2.7 *Eubacterium_ruminantium* clade_543 *Eubacterium* Eubacteriaceae Clostridiales yes 57 20 27 0 1.7 *Clostridium_thermocellum* clade_495 *Clostridium* Clostridiaceae Clostridiales yes 50 47 20 0 2.3 *Oscillibacter_valericigenes* clade_540 *Oscillibacter* Oscillospiraceae Clostridiales yes 50 40 60 1 2.7 *Eubacterium_coprostanoligenes* clade_537 *Eubacterium* Eubacteriaceae Clostridiales yes 50 27 13 1 2.7 *Clostridium_disporicum* clade_253 *Clostridium* Clostridiaceae Clostridiales yes 50 20 27 0 1.7 *Clostridium_mayombei* clade_354 *Clostridium* Clostridiaceae Clostridiales yes 50 20 0 0 1.7 *Roseburia_faecalis* clade_444 *Roseburia* Lachnospiraceae Clostridiales yes 43 27 60 0 1.7 Lachnospiraceae_*bacterium*_1_4_56FAA clade_262 unassigned Lachnospiraceae Clostridiales yes 43 13 20 1 2.7 *Clostridium_spiroforme* clade_481 unassigned Erysipelotrichaceae Erysipelotrichales yes 36 33 33 0 1.7 *Eubacterium_siraeum* clade_538 *Eubacterium* Eubacteriaceae Clostridiales yes 36 27 60 1 2.7 *Lachnospira_pectinoschiza* clade_522 *Lachnospira* Lachnospiraceae Clostridiales yes 36 27 53 0 1.7 *Papillibacter_cinnamivorans* clade_572 *Papillibacter* Ruminococcaceae Clostridiales yes 36 27 53 0 1.7 *Clostridium_tyrobutyricum* clade_430 *Clostridium* Clostridiaceae Clostridiales yes 36 27 13 0 1.7 *Roseburia_inulinivorans* clade_444 *Roseburia* Lachnospiraceae Clostridiales yes 36 20 60 1 2.7 *Ethanoligenens_harbinense* clade_439 *Ethanoligenens* Ruminococcaceae Clostridiales yes 36 20 27 0 1.7 *Eggerthella_lenta* clade_566 *Eggerthella* Coriobacteriaceae Coriobacteriales yes 36 13 40 0 1.7 *Clostridium_orbiscindens* clade_494 unassigned unassigned Clostridiales yes 29 20 60 0 1.1 *Eubacterium_ventriosum* clade_519 *Eubacterium* Eubacteriaceae Clostridiales yes 29 20 40 0 1.1 *Clostridium_paraputrificum* clade_223 *Clostridium* Clostridiaceae Clostridiales yes 29 13 33 0 1.1 *Clostridium*_sp_YIT_12069 clade_537 *Clostridium* Clostridiaceae Clostridiales yes 29 7 53 0 1.1 *Eubacterium_barkeri* clade_512 *Eubacterium* Eubacteriaceae Clostridiales yes 29 0 13 0 0.7 *Eubacterium_biforme* clade_385 unassigned Erysipelotrichaceae Erysipelotrichales yes 29 0 13 0 0.7 *Alkaliphilus_oremlandii* clade_554 *Alkaliphilus* Clostridiaceae Clostridiales yes 29 0 7 0 0.7 Lachnospiraceae_*bacterium*_5_1_57FAA clade_260 unassigned Lachnospiraceae Clostridiales yes 21 20 60 0 1.1 *Eubacterium_eligens* clade_522 *Eubacterium* Eubacteriaceae Clostridiales yes 21 20 33 0 1.1 *Bacillus*_sp_9_3AIA clade_527 *Bacillus* Bacillaceae Bacillales yes 21 13 73 0 1.1 *Eubacterium*_sp_WAL_14571 clade_384 *Eubacterium* Eubacteriaceae Clostridiales yes 21 7 60 0 1.1 *Anaerosporobacter_mobilis* clade_396 *Anaerosporobacter* Clostridiaceae Clostridiales yes 21 7 47 0 1.1 *Coprococcus_eutactus* clade_543 *Coprococcus* Lachnospiraceae Clostridiales yes 21 7 20 0 1.1 *Eubacterium*_sp_oral_clone_JH012 clade_476 *Eubacterium* Eubacteriaceae Clostridiales yes 21 7 13 0 1.1 *Lachnospira_multipara* clade_522 *Lachnospira* Lachnospiraceae Clostridiales yes 21 7 13 0 1.1 *Clostridium_carnis* clade_253 *Clostridium* Clostridiaceae Clostridiales yes 21 7 0 0 1.1 *Clostridium_colinum* clade_576 *Clostridium* Clostridiaceae Clostridiales yes 21 7 0 0 1.1 *Clostridium_hylemonae* clade_260 *Clostridium* Clostridiaceae Clostridiales yes 21 0 40 0 0.7 *Gloeobacter_violaceus* clade_596 *Gloeobacter* unassigned Gloeobacterales yes 21 0 7 0 0.7 *Clostridium_algidicarnis* clade_430 *Clostridium* Clostridiaceae Clostridiales yes 14 20 67 0 1.1 *Holdemania_filiformis* clade_485 *Holdemania* Erysipelotrichaceae Erysipelotrichales yes 14 13 73 1 2.1 *Clostridium_aldenense* clade_408 *Clostridium* Clostridiaceae Clostridiales yes 14 13 67 0 1.1 *Sporobacter_termitidis* clade_572 *Sporobacter* Ruminococcaceae Clostridiales yes 14 13 67 1 2.1 *Ruminococcus*_sp_ID8 clade_360 *Ruminococcus* Ruminococcaceae Clostridiales yes 14 13 47 0 1.1 Lachnospiraceae_*bacterium*_4_1_37FAA clade_360 unassigned Lachnospiraceae Clostridiales yes 14 0 60 0 0.7 *Ruminococcus*_sp_18P13

clade_406 *Ruminococcus* Ruminococcaceae Clostridiales yes 14 0 7 0 0.7 *Blautia_hydrogenotrophica* clade_368 *Blautia* Lachnospiraceae Clostridiales yes 14 0 0 0 0.7 *Anaerotruncus_colihominis* clade_516 *Anaerotruncus* Ruminococcaceae Clostridiales yes 7 7 93 0 1.1 *Clostridium_symbiosum* clade_408 *Clostridium* Clostridiaceae Clostridiales yes 7 7 87 0 1.1 *Clostridium_lactatifermentans* clade_576 *Clostridium* Clostridiaceae Clostridiales yes 7 7 80 1 2.1 *Lactobacillus_rogosae* clade_522 *Lactobacillus* Lactobacillaceae Lactobacillases yes 7 7 80 0 1.1 *Clostridium*_sp_HGF2 clade_351 *Clostridium* Clostridiaceae Clostridiales yes 7 7 47 0 1.1 *Clostridium*_sp_SY8519 clade_482 *Clostridium* Clostridiaceae Clostridiales yes 7 7 47 0 1.1 *Desulfotomaculum_nigrificans* clade_560 *Desulfotomaculum* Peptococcaceae Clostridiales yes 7 7 27 0 1.1 *Eubacterium_cylindroides* clade_385 unassigned Erysipelotrichaceae Erysipelotrichales yes 7 7 7 0 1.1 *Ruminococcus*_sp_K_1 clade_309 *Ruminococcus* Ruminococcaceae Clostridiales yes 7 7 0 0 1.1 *Lachnospiraceae_bacterium*_oral_taxon_F15 clade_393 unassigned Lachnospiraceae Clostridiales yes 7 0 53 0 0.7 *Clostridium_nexile* clade_262 *Clostridium* Clostridiaceae Clostridiales yes 7 0 40 1 1.7 *Acetanaerobacterium_elongatum* clade_439 *Acetanaerobacterium* Ruminococcaceae Clostridiales yes 7 0 7 0 0.7 *Butyricicoccus_pullicaecorum* clade_572 *Butyricicoccus* Clostridiaceae Clostridiales yes 7 0 0 1 1.7 *Clostridium_butyricum* clade_252 *Clostridium* Clostridiaceae Clostridiales yes 7 0 0 0 0.7 Solobacterium_*moorei* clade_388 *Solobacterium* Erysipelotrichaceae Erysipelotrichales yes 7 0 0 0 0.7 *Bacteroides_uniformis* clade_110 *Bacteroides* Bacteroidaceae Bacteroidales no 87 13 0 4.0 *Alloscardovia*_sp_OB7196 clade_475 *Alloscardovia* Bifidobacteriaceae Bifidobacteriales no 53 33 0 2.3 *Clostridiales_bacterium*_oral_clone_P4PA clade_558 unassigned unassigned Clostridiales no 47 27 0 2.3 *Enterococcus_faecium* clade_497 *Enterococcus* Enterococcaceae Lactobacillales no 47 7 0 3.0 *Clostridiales_bacterium*_oral_taxon_F32 clade_584 unassigned unassigned Clostridiales no 40 0 0 1.7 *Bifidobacterium_breve* clade_172 *Bifidobacterium* Bifidobacteriaceae Bifidobacteriales no 33 7 0 2.4 *Bifidobacterium_longum* clade_172 *Bifidobacterium* Bifidobacteriaceae Bifidobacteriales no 33 0 1 2.7 *Bacteroides_oleiciplenus* clade_85 *Bacteroides* Bacteroidaceae Bacteroidales no 27 47 0 1.7 *Dialister_invisus* clade_506 *Dialister* Veillonellaceae Selenomonadales no 27 7 0 1.7 *Anaerobaculum_hydrogeniformans* clade_591 *Anaerobaculum* Synergistaceae Synergistales no 20 47 0 1.7 *Streptococcus*_sp_oral_taxon_G63 clade_98 *Streptococcus* Streptococcaceae Lactobacillales no 20 27 0 1.7 *Streptococcus_thermophilus* clade_98 *Streptococcus* Streptococcaceae Lactobacillales no 20 13 0 2.4 *Dialister_micraerophilus* clade_506 *Dialister* Veillonellaceae Selenomonadales no 20 13 0 1.7 *Bifidobacterium_animalis* clade_172 *Bifidobacterium* Bifidobacteriaceae Bifidobacteriales no 20 7 0 1.7 *Lactobacillus_iners* clade_398 *Lactobacillus* Lactobacillaceae Lactobacillales no 20 0 0 1.7 *Butyrivibrio_fibrisolvens* clade_444 *Butyrivibrio* Lachnospiraceae Clostridiales no 13 67 0 1.1 *Streptococcus*_sp_ACS2 clade_98 *Streptococcus* Streptococcaceae Lactobacillales no 13 33 0 1.7 *Lactococcus_lactis* clade_401 *Lactococcus* Streptococcaceae Lactobacillales no 13 13 0 1.7 *Lactobacillus_delbrueckii* clade_72 *Lactobacillus* Lactobacillaceae Lactobacillales no 13 13 0 1.7 *Cytophaga_xylanolytica* clade_561 *Cytophaga* Cytophagaceae Cytophagales no 7 53 0 1.1 *Streptococcus_gallolyticus* clade_98 *Streptococcus* Streptococcaceae Lactobacillales no 7 33 0 1.1 *Marvinbryantia_formatexigens* clade_309 unassigned Lachnospiraceae Clostridiales no 7 33 0 1.1 *Akkermansia_muciniphila* clade_583 *Akkermansia* Verrucomicrobiaceae Verrucomicrobiales no 7 20 1 2.7 *Bacteroides_dorei* clade_378 *Bacteroides* Bacteroidaceae Bacteroidales no 7 20 1 2.7 *Megasphaera*_genomosp_type_1_28L clade_506 *Megasphaera* Veillonellaceae Selenomonadales no 7 20 0 1.1 *Lactobacillus_hominis* clade_398 *Lactobacillus* Lactobacillaceae Lactobacillales no 7 13 0 1.7 *Actinomyces_oricola* clade_54 *Actinomyces* Actinomycetaceae Actinomycetales no 7 13 0 1.1 *Streptobacillus_moniliformis* clade_532 *Streptobacillus* Leptotrichiaceae Fusobacteriales no 7 13 0 1.1 *Streptococcus*_sp_oral_clone_ASCB06 clade_98 *Streptococcus* Streptococcaceae Lactobacillales no 7 7 0 1.1 *Bacteroides*_sp_D20 clade_110 *Bacteroides* Bacteroidaceae Bacteroidales no 7 7 1 2.1 *Collinsella_intestinalis* clade_553 *Collinsella* Coriobacteriaceae Coriobacteriales no 7 0 0 1.7 *Methanosphaera_stadtmanae* clade_595 *Methanosphaera* Methanobacteriaceae Methanobacteriales no 7 0 0 1.7 *Streptococcus_vestibularis* clade_98 *Streptococcus* Streptococcaceae Lactobacillales no 7 0 0 1.1 *Clostridiaceae_bacterium*_END_2 clade_368 unassigned Clostridiaceae Clostridiales no 0 53 0 0.7 *Parabacteroides_distasonis* clade_335 *Parabacteroides* Porphyromonadaceae Bacteroidales no 0 27 0 1.3 *Veillonella_dispar* clade_358 *Veillonella* Veillonellaceae Selenomonadales no 0 27 0 0.7 *Bacteroides_caccae* clade_170 *Bacteroides* Bacteroidaceae Bacteroidales no 0 20 1 1.7 *Veillonella*_sp_3_1_44 clade_358 *Veillonella* Veillonellaceae Selenomonadales no 0 20 0 0.7 *Megasphaera_micronuciformis* clade_493 *Megasphaera* Veillonellaceae Selenomonadales no 0 20 0 0.7 *Oxalobacter_formigenes* clade_357 *Oxalobacter* Oxalobacteraceae Burkholderiales no 0 20 0 0.7 *Streptococcus_parasanguinis* clade_98 *Streptococcus* Streptococcaceae Lactobacillales no 0 13 1 2.3 *Bacteroides_fragilis* clade_65 *Bacteroides* Bacteroidaceae Bacteroidales no 0 13 0 0.7 *Bacteroides*_sp_4_1_36 clade_110 *Bacteroides* Bacteroidaceae Bacteroidales no 0 13 0 0.7 *Lactobacillus*_sp_BT6 clade_373 *Lactobacillus* Lactobacillaceae Lactobacillales no 0 13 0 0.7 *Bacteroides*_sp_1_1_14 clade_65 *Bacteroides* Bacteroidaceae Bacteroidales no 0 13 0 0.7 *Escherichia_hermannii* clade_92 *Escherichia* Enterobacteriaceae Enterobacteriales no 0 13 0 0.7 *Escherichia*_sp_B4 clade_92 *Escherichia* Enterobacteriaceae Enterobacteriales no 0 13 0 0.7 *Gemella_moribillorum* clade_450 *Gemella* unassigned Bacillales no 0 13 0 0.7 *Klebsiella_variicola* clade_92 *Klebsiella* Enterobacteriaceae Enterobacteriales no 0 13 0 0.7 *Phascolarctobacterium_succinatutens* clade_556 *Phascolarctobacterium* Acidaminococcaceae Selenomonadales no 0 13 0 0.7 *Streptococcus*_sp_CM7 clade_60 *Streptococcus* Streptococcaceae Lactobacillales no 0 13 0 0.7 *Bilophila_wadsworthia* clade_521 *Bilophila* Desulfovibrionaceae Desulfovibrionales no 0 7 1 2.3 *Streptococcus*_sp_oral_clone_GM006 clade_98 *Streptococcus* Streptococcaceae Lactobacillales no 0 7 0 1.3 *Adlercreutzia_equolifaciens* clade_566 *Adlercreutzia* Coriobacteriaceae Coriobacteriales no 0 7 0 0.7 *Lactobacillus_murinus* clade_449 *Lactobacillus* Lactobacillaceae Lactobacillales no 0 7 0 0.7 *Helicobacter_pullorum* clade_489 *Helicobacter* Helicobacteraceae Campylobacterales no 0 7 0 0.7 *Alistipes_finegoldii* clade_500 *Alistipes* Rikenellaceae Bacteroidales no 0 7 0 0.7 *Averyella_dalhousiensis* clade_92 *Averyella* Enterobacteriaceae Enterobacteriales no 0 7 0 0.7 *Desulfovibrio_desulfuricans* clade_445 *Desulfovibrio* Desulfovibrionaceae Desulfovibrionales no 0 7 0 0.7 *Plesiomonas_shigelloides* clade_92 *Plesiomonas* Enterobacteriaceae Enterobacteriales no 0 7 0 0.7 *Actinomyces_israelii* clade_212 *Actinomyces* Actinomycetaceae Actinomycetales no 0 7 0 0.7 *Bacteroidales*_genomosp_P1 clade_529 unassigned unassigned Bacteroidales no 0 7 0 0.7 *Bifidobacterium_bifidum* clade_293 *Bifidobacterium* Bifidobacteriaceae Bifidobacteriales no 0 7 0 0.7 *Cedecea_davisae* clade_92 *Cedecea* Enterobacteriaceae Enterobacteriales no 0 7 0 0.7 Gardnerella_*vaginalis* clade_344 *Gardnerella* Bifidobacteriaceae Bifidobacteriales no 0 7 0 0.7 *Lactobacillus_fermentum* clade_313 *Lactobacillus* Lactobacillaceae Lactobaciliales no 0 7 0 0.7 *Lactobacillus_reuteri* clade_313 *Lactobacillus* Lactobacillaceae Lactobacillales no 0 8 0 0.7 *Lactococcus_raffinolactis* clade_524 *Lactococcus* Streptococcaceae Lactobacillales no 0 7 0 0.7 *Pediococcus_pentosaceus* clade_372 *Pediococcus* Lactobacillaceae Lactobacillales no 0 7 0 0.7 *Prevotella_denticola* clade_83 *Prevotella* Prevotellaceae Bacteroidales no 0 7 0 0.7 *Rothia_mucilaginosa* clade_271 *Rothia* Micrococcaceae Actinomycetales no 0 7 0 0.7 *Sutterella_stercoricanis* clade_432 *Sutterella* Sutterellaceae Burkholderiales no 0 7 0 0.7 *Eggerthelia*_sp_1_3_56FAA clade_566 *Eggerthella* Coriobacteriaceae Coriobacteriales no 0 0 0 1.3 *Coriobacteriaceae_bacterium*_JC110 clade_566 unassigned Coriobacteriaceae Coriobacteriales no 0 0 0 1.3 *Megamonas_funiformis* clade_542 *Megamonas* Veillonellaceae Selenomonadales no 0 0 0 1.3 *Gordonibacter_pamelaeae* clade_566 *Gordonibacter* Coriobacteriaceae Coriobacteriales no 0 0 0 1.3 *Bifidobacterium*_sp_HM2 clade_172 *Bifidobacterium* Bifidobacteriaceae Bifidobacteriales no 0 0 0 0.7 *Bacteroides_stercoris* clade_85 *Bacteroides* Bacteroidaceae Bacteroidales no 0 0 1 1.7 *Bifidobacterium_angulatum* clade_172 *Bifidobacterium* Bifidobacteriaceae Bifidobacteriales no 0 0 0 0.7 *Parasutterella_excrementihominis* clade_432 *Parasutterella* Sutterellaceae Burkholderiales no 0 0 0 0.7 *Phascolarctobacterium_faecium* clade_556 *Phascolarctobacterium* Acidaminococcaceae Selenomonadales no 0 0 0 0.7 *Cryptobacterium_curtum* clade_566 *Cryptobacterium* Coriobacteriaceae Coriobacteriales no 0 0 0 0.7 *Prevotella*_sp_BI_42 clade_168 *Prevotella* Prevotellaceae Bacteroidales no 0 0 0 0.7 *Slackia_isoflavoniconvertens* clade_566 *Slackia* Coriobacteriaceae Coriobacteriales no 0 0 0 0.7 *Acidaminococcus*_sp_D21 clade_556 *Acidaminococcus* Acidaminococcaceae Selenomonadales no 0 0 0 0.7 *Atopobium_vaginae* clade_539 *Atopobium* Coriobacteriaceae Coriobacteriales no 0 0 0 0.7 *Catabacter_hongkongensis* clade_558

*Catabacter* Catabacteriaceae Clostridiales no 0 0 0 0.7 *Lactobacillus_ruminis* clade_449 *Lactobacillus* Lactobacillaceae Lactobacillales no 0 0 0 0.7 *Lactobacillus_senioris* clade_398 *Lactobacillus* Lactobacillaceae Lactobacillales no 0 0 0 0.7 *Morganella_morganii* clade_89 *Morganella* Enterobacteriaceae Enterobacteriales no 0 0 0 0.7 *Parabacteroides_merdae* clade_286 *Parabacteroides* Porphyromonadaceae Bacteroidales no 0 0 1 1.7 *Peptoniphilus_harei* clade_389 *Peptoniphilus* Clostridiales Family XI Clostridiales no 0 0 0 0.7 *Streptococcus_downei* clade_441 *Streptococcus* Streptococcaceae Lactobacillales no 0 0 0 0.7

(321) TABLE-US-00013 TABLE 12 Percent Percent Clade of OTU3 of Dose Ecologies of Post-treatment OTC1 of Composition OTC2 of Composition OTC3 of Composition Clade of OTU1 Clade of OTU2 (if applicable) *C. diff* Inhibition Score Occurs Ecologies *Dorea longicatena Eubacterium rectale* clade_360 clade_444 ++++ 92.9 100.0 *Ruminococcus torques Ruminococcus torques* clade_262 clade_262 ++++ 92.9 93.3 *Coprococcus comes Eubacterium rectale* clade_262 clade_444 ++++ 85.7 46.7 *Coprococcus comes Ruminococcus bromii* clade_262 clade_537 ++++ 85.7 26.7 *Ruminococcus torques Coprococcus comes* clade_262 clade_262 ++++ 78.6 40.0 *Ruminococcus obeum Ruminococcus obeum* clade_309 clade_309 ++++ 64.3 33.3 *Ruminococcus obeum Coprococcus comes* clade_309 clade_262 ++++ 64.3 20.0 *Ruminococcus obeum Ruminococcus torques* clade_309 clade_262 ++++ 57.1 33.3 *Clostridium disporicum Eubacterium rectale* clade_253 clade_444 ++++ 50.0 46.7 *Ciostridium mayombei Eubacterium rectale Eubacterium rectale* clade_354 clade_444 clade_444 ++++ 50.0 20.0 *Ciostridium mayombei Faecalibacterium prausnitzii Eubacterium rectale* clade_354 clade_478 clade_444 ++++ 50.0 20.0 *Ciostridium mayombei Faecalibacterium prausnitzii Faecalibacterium prausnitzii* clade_354 clade_478 clade_478 ++++ 50.0 20.0 *Eubacterium rectale Clostridium mayombei Blautia* sp. M25 clade_444 clade_354 clade-309 ++++ 50.0 20.0 *Eubacterium rectale Clostridium mayombei Clostridium mayombei* clade_444 clade_354 clade_354 ++++ 50.0 20.0 *Eubacterium rectale Clostridium mayombei Ruminococcus bromii* clade_444 clade_354 clade_537 ++++ 50.0 20.0 *Faecalibacterium prausnitzii Clostridium mayombei Blautia* sp. 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M25 clade_444 clade_494 clade_309 ++++ 28.6 80.0 *Faecalibacterium prausnitzii Clostridium orbiscindens Blautia* sp. M25 clade_478 clade_494 clade_309 ++++ 28.6 73.3 *Clostridium disporicum Clostridium orbiscindens* clade_253 clade_494 ++++ 28.6 46.7 *Clostridium hylemonae Eubacterium rectale Eubacterium rectale* clade_260 clade_444 clade_444 ++++ 21.4 40.0 *Eubacterium rectale Clostridium hylemonae Blautia* sp. M25 clade_444 clade_260 clade_309 ++++ 21.4 40.0 *Coprococcus comes Clostridium orbiscindens Blautia* sp. 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M25 clade_408 clade_494 clade_309 ++++ 7.1 80.0 *Clostridium orbiscindens Clostridium symbiosum* Lachnospiraceae *bacterium* clade_494 clade_408 clade_260 ++++ 7.1 66.7 5_1_57FAA Lachnospiraceae *bacterium Clostridium orbiscindens Blautia* sp. 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(322) TABLE-US-00014 TABLE 13 OTU Clade clade_309 clade_309 clade_309 clade_408 clade_537 clade_444 clade_478 clade_360

clade_537 clade_478 Num. of *Blautia Blautia Blautia* sp. *Clostridium Clostridium Eubacterium Faecalibacterium* Lachnospiraceae
*Ruminococcus Subdoligranulum* OTUs *luti schinkii* M25 *citroniae leptum rectale prausnitzii bacterium* 2_1_58FAA *bromii variabile* 10 Y Y Y Y
Y Y Y Y Y Y 9 Y Y Y Y Y Y Y Y Y 9 Y Y Y Y Y Y Y Y Y 9 Y Y Y Y Y Y Y Y Y 9 Y Y Y Y Y Y Y Y Y 9 Y Y Y Y Y Y Y Y Y 9 Y Y Y Y Y Y Y Y
Y 9 Y Y Y Y Y Y Y Y Y 9 Y Y Y Y Y Y Y Y Y 9 Y Y Y Y Y Y Y Y Y 9 Y Y Y Y Y Y Y Y Y 8 Y Y Y Y Y Y Y Y 8 Y Y Y Y Y Y Y Y 8 Y Y Y Y Y Y
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Y Y Y Y Y Y Y Y 8 Y Y Y Y Y Y Y Y 8 Y Y Y Y Y Y Y Y 8 Y Y Y Y Y Y Y Y 8 Y Y Y Y Y Y Y Y 8 Y Y Y Y Y Y Y Y 8 Y Y Y Y Y Y Y Y 8 Y Y Y
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4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y
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Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y
Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y
4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y
Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y Y 4 Y Y Y
Y 4 Y Y Y Y 4 Y Y Y Y

(323) TABLE-US-00015 TABLE 14 Post-Treatment Ecologies Percent of doses Percent of post-treatment in which the patients in which the OTU 1 OTU 2 OTU 3 ternary is present ternary is present *Blautia* sp. M25 *Faecalibacterium Ruminococcus bromii* 100 75 *prausnitzii Blautia* sp. M25 *Faecalibacterium Ruminococcus obeum* 100 89 *prausnitzii Blautia* sp. M25 *Faecalibacterium Ruminococcus* sp. 100 89 *prausnitzii* 5_1_39BFAA *Blautia* sp. M25 *Faecalibacterium Ruminococcus torques* 100 93 *prausnitzii Blautia* sp. M25 *Ruminococcus bromii Ruminococcus obeum* 100 71 *Blautia* sp. M25 *Ruminococcus bromii Ruminococcus* sp. 100 71 5_1_39BFAA *Blautia* sp. M25 *Ruminococcus bromii Ruminococcus torques* 100 75 *Blautia* sp. M25 *Ruminococcus obeum Ruminococcus* sp. 100 86 5_1_39BFAA *Blautia* sp. M25 *Ruminococcus obeum Ruminococcus torques* 100 89 *Blautia* sp. M25 *Ruminococcus* sp. *Ruminococcus torques* 100 89 5_1_39BFAA *Clostridium Clostridium Escherichia coli* 0 100 *hathewayi orbiscindens Clostridium Clostridium* Lachnospiraceae bacterium 0 100 *hathewayi orbiscindens* 3_1_57FAA_CT1 *Clostridium Clostridium Ruminococcus torques* 0 100 *hathewayi orbiscindens Clostridium Escherichia coli* Lachnospiraceae bacterium 0 100 *hathewayi* 3_1_57FAA_CT1 *Clostridium Escherichia coli Ruminococcus torques* 0 100 *hathewayi Clostridium Escherichia coli* Lachnospiraceae bacterium 0 100 *orbiscindens* 3_1_57FAA_CT1 *Clostridium Escherichia coli Ruminococcus torques* 0 100 *orbiscindeas Clostridium Ruminococcus torques Streptococcus salivarius* 0 100 *orbiscindens Faecalibacterium Ruminococcus bromii Ruminococcus obeum* 100 75 *prausnitzii Faecalibacterium Ruminococcus bromii Ruminococcus* sp. 100 71 *prausnitzii* 5_1_39BFAA *Faecalibacterium Ruminococcus bromii Ruminococcus torques* 100 75 *prausnitzii Faecalibacterium Ruminococcus obeum Ruminococcus* sp. 100 86 *prausnitzii* 5_1_39BFAA Faecalibacterium *Ruminococcus obeum Ruminococcus torques* 100 93 *prausnitzii Faecalibacterium Ruminococcus* sp. *Ruminococcus torques* 100 89 *prausnitzii* 5_1_39BFAA *Ruminococcus bromii Ruminococcus obeum Ruminococcus* sp. 100 68 5_1_9BFAA *Ruminococcus bromii Ruminococcus obeum Ruminococcus torques* 100 79 *Ruminococcus bromii Ruminococcus* sp. *Ruminococcus torques* 100 71 5_1_39BFAA *Ruminococcus obeum Ruminococcus* sp. *Ruminococcus torques* 100 89 5_1_39BFAA

(324) TABLE-US-00016 TABLE 15 Engrafting OTUs OTU 1 OTU 2 OTU 3 *Blautia* sp. M25 *Clostridiales* sp. SSC/2 *Coprococcus catus Blautia* sp. M25 *Clostridiales* sp. SSC/2 *Faecalibacterium prausnitzii Blautia* sp. M25 *Clostridiales* sp. SSC/2 *Ruminococcus bromii Blautia* sp. M25 *Clostridiales* sp. SSC/2 *Ruminococcus obeum Blautia* sp. M25 *Clostridiales* sp. SSC/2 *Ruminococcus* sp. 5_1_39BFAA *Blautia* sp. M25 *Clostridiales* sp. SSC/2 *Ruminococcus torques Blautia* sp. M25 *Coproeoceus catus Faecalibacterium prausnitzii Blautia* sp. M25 *Coprococcus catus Ruminococcus bromii Blautia* sp. M25 *Coproeoceus catus Ruminococcus obeum Blautia* sp. M25 *Coprococcus catus Ruminococcus* sp. 5_1_39BFAA *Blautia* sp. M25 *Coprococcus catus Ruminococcus torques Blautia* sp. M25 *Eubacterium hallii Faecalibacterium prausnitzii Blautia* sp. M25 *Eubacterium hallii Ruminococcus bromii Blautia* sp. M25 *Eubacterium hallii Ruminococcus obeum Blautia* sp. M25 *Eubacterium hallii Ruminococcus* sp. 5_1_39BFAA *Blautia* sp. M25 *Eubacterium hallii Ruminococcus torques Blautia* sp. M25 *Faecalibacterium prausnitzii Gemmiger formicilis Blautia* sp. M25 *Gemmiger formicilis Ruminococcus bromii Blautia* sp. M25 *Gemmiger formicilis Ruminococcus obeum Blautia* sp. M25 *Gemmiger formicilis Ruminococcus* sp. 5_1_39BFAA *Blautia* sp. M25 *Gemmiger formicilis Ruminococcus torques Clostridiales* sp. SSC/2 *Coprococcus catus Faecalibacterium prausnitzii Clostridiales* sp. SSC/2 *Coprococcus catus Ruminococcus bromii Clostridiales* sp. SSC/2 *Coprococcus catus Ruminococcus obeum Clostridiales* sp. SSC/2 *Coprococcus catus Ruminococcus* sp. 5_1_39BFAA *Clostridiales* sp. SSC/2 *Coprococcus catus Ruminococcus torques Clostridiales* sp. SSC/2 *Faecalibacterium prausnitzii Ruminococcus bromii Clostridiales* sp. SSC/2 *Faecalibacterium prausnitzii Ruminococcus obeum Clostridiales* sp. SSC/2 *Faecalibacterium prausnitzii Ruminococcus* sp. 5_1_39BFAA *Clostridiales* sp. SSC/2 *Faecalibacterium prausnitzii Ruminococcus torques Clostridiales* sp. SSC/2 *Ruminococcus bromii Ruminococcus obeum Clostridiales* sp. SSC/2 *Ruminococcus bromii Ruminococcus* sp. 5_1_39BFAA *Clostridiales* sp. SSC/2 *Ruminococcus bromii Ruminococcus torques Clostridiales* sp. SSC/2 *Ruminococcus obeum Ruminococcus* sp. 5_1_39BFAA *Clostridiales* sp. SSC/2 *Ruminococcus obeum Ruminococcus torques Clostridiales* sp. SSC/2 *Ruminococcus* sp. *Ruminococcus torques* 5_1_39BFAA *Coprococcus catus Faecalibacterium prausnitzii Ruminococcus bromii Coprococcus catus Faecalibacterium prausnitzii Ruminococcus obeum Coprococcus catus Faecalibacterium prausnitzii Ruminococcus* sp. 5_1_39BFAA *Coprococcus catus Faecalibacterium prausnitzii Ruminococcus torques Coprococcus catus Ruminococcus bromii Ruminococcus obeum Coprococcus catus Ruminococcus bromii Ruminococcus* sp. 5_1_39BFAA *Coprococcus catus Ruminococcus bromii Ruminococcus torques Coprococcus catus Ruminococcus obeum Ruminococcus* sp. 5_1_39BFAA *Coprococcus catus Ruminococcus obeum Ruminococcus torques Coprococcus catus Ruminococcus* sp. *Ruminococcus torques* 5_1_39BFAA *Eubacterium hallii Faecalibacterium prausnitzii Ruminococcus bromii Eubacterium hallii Faecalibacterium prausnitzii Ruminococcus obeum Eubacterium hallii Faecalibacterium prausnitzii Ruminococcus* sp. 5_1_39BFAA *Eubacterium hallii Faecalibacterium prausnitzii Ruminococcus torques Eubacterium hallii Ruminococcus bromii Ruminococcus obeum Eubacterium hallii Ruminococcus bromii Ruminococcus* sp. 5_1_39BFAA *Eubacterium hallii Ruminococcus bromii Ruminococcus torques Eubacterium hallii Ruminococcus obeum Ruminococcus* sp. 5_1_39BFAA *Eubacterium hallii Ruminococcus obeum Ruminococcus torques Eubacterium hallii Ruminococcus* sp. *Ruminococcus torques* 5_1_39BFAA *Faecalibacterium Gemmiger formicilis Ruminococcus bromii prausnitzii Faecalibacterium Gemmiger formicilis Ruminococcus obeum* prausnitzii *Faecalibacterium Gemmiger formicilis Ruminococcus* sp. *prausnitzii* 5_1_39BFAA *Faecalibacterium Gemmiger formicilis Ruminococcus torques prausnitzii Gemmiger formicilis Ruminococcus bromii Ruminococcus obeum Gemmiger formicilis Ruminococcus bromii Ruminococcus* sp. 5_1_39BFAA *Gemmiger formicilis Ruminococcus bromii Ruminococcus torques Gemmiger formicilis Ruminococcus obeum Ruminococcus* sp. 5_1_39BFAA *Gemmiger formicilis Ruminococcus obeum Ruminococcus torques Gemmiger formicilis Ruminococcus* sp. *Ruminococcus torques* 5_1_39BFAA

(325) TABLE-US-00017 TABLE 16 Augmenting OTUs Percent of post-treatment patients in which the ternary is OTU 1 OTU 2 OTU 3 present *Anaerotruncus Clostridium Escherichia* 75 *colihominis orbiscindens coli Clostridium Clostridium Escherichia* 79 *lactatifermentans orbiscindens coli Clostridium Clostridium Streptococcus* 79 *lactatifermentans orbiscindens salivarius Clostridium Escherichia Streptococcus* 75 *lactatifermentans coli salivarius Clostridium Clostridium Escherichia* 89 *orbiscindens* sp. NML *coli* 04A032 *Clostridium Clostridium Oscillibacter* 89 *orbiscindens* sp. NML sp. G2 04A032 *Clostridium Clostridium Streptococcus* 93 *orbiscindens* sp. NML *salivarius* 04A032 *Clostridium Escherichia Klebsiella* sp. 75 *orbiscindens coli* SRC_DSD2 *Clostridium Escherichia Oscillibacter* 89 *orbiscindens coli* sp. G2 *Clostridium Escherichia Streptococcus* 96 *orbiscindens coli salivarius Clostridium Oscillibacter Streptococcus* 89 *orbiscindens* sp. G2 *salivarius Clostridium* sp. *Escherichia Oscillibacter* 82 NML 04A032 *coli* sp. G2 *Clostridium* sp. *Escherichia Streptococcus* 86 NML 04A032

*coli salivarius Clostridium* sp. *Oscillibacter Streptococcus* 86 NML 04A032 sp. G2 *salivarius Escherichia coli Oscillibacter Streptococcus* 82 sp. G2 *salivarius*

(326) TABLE-US-00018 TABLE 17 Ternary OTU combinations in administered spore ecology doses resulting in augmentation or engraftment of the OTU *Clostridiales* sp. SM4/1 Percent of doses in which the ternary is OTU 1 OTU 2 OTU 3 present *Clostridium Eubacterium rectale Faecalibacterium* 85 *saccharogumia prausnitzii Clostridium Eubacterium rectale Ruminococcus* 85 *saccharogumia torques Clostridium Faecalibacterium Ruminococcus* 85 *saccharogumia prausnitzii torques Blautia* sp. M25 *Clostridium Eubacterium rectale* 85 *saccharogumia Blautia* sp. M25 *Clostridium Faecalibacterium* 85 *saccharogumia prausnitzii Blautia* sp. M25 *Clostridium Ruminococcus* 85 *saccharogumia torques Clostridium Eubacterium rectale Ruminococcus* 85 *saccharogumia obeum Clostridium Eubacterium rectale Ruminococcus* sp. 85 *saccharogumia* 5_1_39BFAA *Clostridium Faecalibacterium Ruminococcus* 85 *saccharogumia prausnitzii obeum Clostridium Faecalibacterium Ruminococcus* sp. 85 *saccharogumia prausnitzii* 5_1_39BFAA *Clostridium Ruminococcus Ruminococcus* 85 *saccharogumia obeum torques Clostridium Ruminococcus* sp. *Ruminococcus* 85 *saccharogumia* 5_1_39BFAA *torques Blautia* sp. M25 *Clostridium Ruminococcus* sp. 85 *saccharogumia* 5_1_39BFAA *Blautia* sp. M25 *Clostridium Ruminococcus* 85 *saccharogumia obeum Clostridium Ruminococcus Ruminococcus* sp. 85 *saccharogumia obeum* 5_1_39BFAA *Clostridium Faecalibacterium Ruminococcus* 85 *saccharogumia prausnitzii bromii Blautia* sp. M25 *Clostridium Ruminococcus* 85 *saccharogumia bromii Clostridium Eubacterium rectale Ruminococcus* 85 *saccharogumia bromii Clostridium Ruminococcus Ruminococcus* 85 *saccharogumia bromii obeum Clostridium Ruminococcus Ruminococcus* sp. 85 *saccharogumia bromii* 5_1_39BFAA *Clostridium Ruminococcus Ruminococcus* 85 *saccharogumia bromii torques Clostridiales* sp. *Clostridium Eubacterium rectale* 80 SSC/2 *saccharogumia Clostridiales* sp. *Clostridium Faecalibacterium* 80 SSC/2 *saccharogumia prausnitzii Clostridiales* sp. *Clostridium Ruminococcus* 80 SSC/2 *saccharogumia torques Blautia* sp. M25 *Clostridiales* sp. *Clostridium* 80 *SSC/2 saccharogumia Blautia* sp. M25 *Clostridium Ruminococcus* 80 *saccharogumia lactaris Clostridiales* sp. *Clostridium Ruminococcus* 80 *SSC/2 saccharogumia obeum Clostridiales* sp. *Clostridium Ruminococcus* sp. 80 SSC/2 *saccharogumia* 5_1_39BFAA *Clostridium Clostridium Eubacterium rectale* 80 *lavalense saccharogumia Clostridium Clostridium Faecalibacterium* 80 *lavalense saccharogumia prausnitzii Clostridium Clostridium Ruminococcus* 80 *lavalense saccharogumia torques Clostridium Eubacterium rectale Ruminococcus* 80 *saccharogumia lactaris Clostridium Ruminococcus Ruminococcus* sp. 80 *saccharogumia lactaris* 5_1_39BFAA *Clostridium Ruminococcus Ruminococcus* 80 *saccharogumia lactaris torques Blautia* sp. M25 *Clostridium Clostridium* 80 *lavalense saccharogumia Clostridium Clostridium Eubacterium rectale* 80 *asparagiforme saccharogumia Clostridium Clostridium Faecalibacterium* 80 *asparagiforme saccharogumia prausnitzii Clostridium Clostridium Ruminococcus* 80 *asparagiforme saccharogumia torques Clostridium Clostridium Ruminococcus* 80 *lavalense saccharogumia obeum Clostridium Clostridium Ruminococcus* sp. 80 *lavalense saccharogumia* 5_1_39BFAA *Clostridium Eubacterium rectale Gemmiger* 80 *saccharogumia formicilis Clostridium Faecalibacterium Ruminococcus* 80 *saccharogumia prausnitzii lactaris Clostridium Gemmiger Ruminococcus* 80 *saccharogumia formicilis torques Clostridium Ruminococcus Ruminococcus* 80 *saccharogumia lactaris obeum Clostridium Faecalibacterium Gemmiger* 80 *saccharogumia prausnitzii formicilis Blautia* sp. M25 *Clostridium Clostridium* 80 *asparagiforme saccharogumia Blautia* sp. M25 *Clostridium Gemmiger* 80 *saccharogumia formicilis Clostridium Clostridium Ruminococcus* 80 *asparagiforme saccharogumia obeum Clostridium Clostridium Ruminococcus* sp. 80 *asparagiforme saccharogumia* 5_1_39BFAA *Clostridium Coprococcus comes Eubacterium rectale* 80 *saccharogumia Clostridium Coprococcus comes Faecalibacterium* 80 *saccharogumia prausnitzii Clostridium Coprococcus comes Ruminococcus* 80 *saccharogumia obeum Clostridium Coprococcus comes Ruminococcus* 80 *saccharogumia torques Clostridium Dorea Eubacterium rectale* 80 *saccharogumia formicigenerans Clostridium Dorea Faecalibacterium* 80 *saccharogumia formicigenerans prausnitzii Clostridium Dorea Ruminococcus* 80 *saccharogumia formicigenerans obeum Clostridium Dorea Ruminococcus* 80 *saccharogumia formicigenerans torques Clostridium Gemmiger Ruminococcus* 80 *saccharogumia formicilis obeum Clostridium Gemmiger Ruminococcus* sp. 80 *saccharogumia formicilis* 5_1_39BFAA *Clostridium Clostridium Clostridium* 80 *asparagiforme lavalense saccharogumia Blautia* sp. M25 *Clostridium Coprococcus comes* 80 *saccharogumia Blautia* sp. M25 *Clostridium Dorea* 80 *saccharogumia formicigenerans Blautia* sp. M25 *Clostridium Eubacterium hallii* 80 *saccharogumia Clostridiales* sp. *Clostridium Ruminococcus* 80 SSC/2 *saccharogumia bromii Clostridium Clostridium Ruminococcus* 80 *asparagiforme saccharogumia bromii Clostridium Clostridium Ruminococcus* 80 *asparagiforme saccharogumia lactaris Clostridium Clostridium Dorea* 80 *lavalense saccharogumia formicigenerans Clostridium Clostridium Ruminococcus* 80 *lavalense saccharogumia bromii Clostridium Clostridium Ruminococcus* 80 *lavalense saccharogumia lactaris Clostridium Coprococcus comes Ruminococcus* sp. 80 *saccharogumia* 5_1_39BFAA *Clostridium Dorea Ruminococcus* 80 *saccharogumia formicigenerans lactaris Clostridium Dorea Ruminococcus* sp. 80 *saccharogumia formicigenerans* 5_1_39BFAA *Clostridium Eubacterium hallii Eubacterium rectale* 80 *saccharogumia Clostridium Eubacterium hallii Faecalibacterium* 80 *saccharogumia prausnitzii Clostridium Eubacterium hallii Ruminococcus* 80 *saccharogumia lactaris Clostridium Eubacterium hallii Ruminococcus* 80 *saccharogumia obeum Clostridium Eubacterium hallii Ruminococcus* sp. 80 *saccharogumia* 5_1_39BFAA *Clostridium Eubacterium hallii Ruminococcus* 80 *saccharogumia torques Clostridium Ruminococcus Ruminococcus* 80 *saccharogumia bromii lactaris Clostridium Clostridium Coprococcus comes* 80 *asparagiforme saccharogumia Clostridium Clostridium Eubacterium hallii* 80 *asparagiforme saccharogumia Clostridium Clostridium Coprococcus comes* 80 *lavalense saccharogumia Clostridium Coprococcus comes Ruminococcus* 80 *saccharogumia bromii Clostridium Coprococcus comes Ruminococcus* 80 *saccharogumia lactaris Clostridium Gemmiger Ruminococcus* 80 *saccharogumia formicilis bromii Blautia* sp. M25 *Clostridium Coprococcus catus* 80 *saccharogumia Blautia* sp. M25 *Clostridium Dorea longicatena* 80 *saccharogumia Clostridium Clostridium Dorea* 80 *asparagiforme saccharogumia formicigenerans Clostridium Clostridium Dorea longicatena* 80 *asparagiforme saccharogumia Clostridium Clostridium Dorea longicatena* 80 *lavalense saccharogumia Clostridium Clostridium Eubacterium hallii* 80 *lavalense saccharogumia Clostridium Coprococcus catus Eubacterium rectale* 80 *saccharogumia Clostridium Coprococcus catus Faecalibacterium* 80 *saccharogumia prausnitzii Clostridium Coprococcus catus Ruminococcus* sp. 80 *saccharogumia* 5_1_39BFAA *Clostridium Coprococcus catus Ruminococcus* 80 *saccharogumia torques Clostridium Coprococcus comes Dorea* 80 *saccharogumia formicigenerans Clostridium Coprococcus comes Eubacterium hallii* 80 *saccharogumia Clostridium Dorea Eubacterium hallii* 80 *saccharogumia formicigenerans Clostridium Dorea Ruminococcus* 80 *saccharogumia formicigenerans bromii Clostridium Dorea longicatena Eubacterium rectale* 80 *saccharogumia Clostridium Dorea longicatena Faecalibacterium* 80 *saccharogumia prausnitzii Clostridium Dorea longicatena Ruminococcus* 80 *saccharogumia obeum Clostridium Dorea longicatena Ruminococcus* sp. 80 *saccharogumia* 5_1_39BFAA *Clostridium Dorea longicatena Ruminococcus* 80 *saccharogumia torques Clostridium Eubacterium hallii Ruminococcus* 80 *saccharogumia bromii Clostridiales* sp. *Clostridium Coprococcus catus* 80 SSC/2 *saccharogumia Clostridium Coprococcus catus Ruminococcus* 80 *saccharogumia obeum Clostridium Dorea longicatena Eubacterium hallii* 80 *saccharogumia Clostridium Dorea longicatena Ruminococcus* 80 *saccharogumia bromii Clostridium Dorea longicatena Ruminococcus* 80 *saccharogumia lactaris Clostridium Dorea Dorea longicatena* 80 *saccharogumia formicigenerans Clostridium Coprococcus catus Ruminococcus* 80 *saccharogumia bromii Clostridium Coprococcus comes Dorea longicatena* 80 *saccharogumia Blautia* sp. M25 *Clostridium Faecalibacterium* 75 *algidixylanolyticum prausnitzii Clostridium Faecalibacterium Ruminococcus* 75 *algidixylanolyticum prausnitzii obeum Clostridium Faecalibacterium Ruminococcus* 75 *algidixylanolyticum prausnitzii torques Blautia* sp. M25 *Clostridium Ruminococcus* 75 *algidixylanolyticum obeum Blautia* sp. M25 *Clostridium Ruminococcus* 75 *algidixylanolyticum torques Clostridium Faecalibacterium Ruminococcus* sp. 75 *algidixylanolyticum prausnitzii* 5_1_39BFAA *Clostridium Ruminococcus Ruminococcus* 75 *algidixylanolyticum obeum torques Blautia* sp. M25 *Clostridium Ruminococcus* sp. 75 *algidixylanolyticum* 5_1_39BFAA *Clostridium Ruminococcus* sp. *Ruminococcus* 75 *algidixylanolyticum* 5_1_39BFAA *torques Clostridium Ruminococcus Ruminococcus* sp. 75 *algidixylanolyticum obeum* 5_1_39BFAA

*Clostridiales* sp. *Clostridium Ruminococcus* 75 SSC/2 *algidixylanolyticum torques Clostridiales* sp. *Clostridium Faecalibacterium* 75 SSC/2 *algidixylanolyticum prausnitzii Clostridiales* sp. *Clostridium Ruminococcus* 75 SSC/2 *algidixylanolyticum obeum Clostridiales* sp. *Clostridium Ruminococcus* sp. 75 SSC/2 *algidixylanolyticum* 5_1_39BFAA *Clostridium Ruminococcus Ruminococcus* 75 *algidixylanolyticum bromii torques Blautia* sp. M25 *Clostridiales* sp. *Clostridium* 75 SSC/2 *algidixylanolyticum Blautia* sp. M25 *Clostridium Ruminococcus* 75 *algidixylanolyticum bromii Clostridium Faecalibacterium Ruminococcus* 75 *algidixylanolyticum prausnitzii bromii Clostridium Ruminococcus Ruminococcus* 75 *algidixylanolyticum bromii obeum Clostridium Ruminococcus Ruminococcus* sp. 75 *algidixylanolyticum bromii* 5_1_39BFAA *Clostridiales* sp. *Clostridium Ruminococcus* 75 SSC/2 *algidixylanolyticum bromii Clostridium Coprococcus catus Faecalibacterium* 75 *algidixylanolyticum prausnitzii Clostridium Coprococcus catus Ruminococcus* sp. 75 *algidixylanolyticum* 5_1_39BFAA *Blautia* sp. M25 *Clostridium Coprococcus catus* 75 *algidixylanolyticum Clostridium Coprococcus catus Ruminococcus* 75 *algidixylanolyticum obeum Clostridium Coprococcus catus Ruminococcus* 75 *algidixylanolyticum torques Clostridiales* sp. *Clostridium Ruminococcus* 75 SSC/2 *saccharogumia lactaris Clostridiales* sp. *Clostridium Coprococcus catus* 75 SSC/2 *algidixylanolyticum Clostridiales* sp. *Clostridium Gemmiger* 75 SSC/2 *saccharogumia formicilis Clostridiales* sp. *Clostridium Clostridium* 75 SSC/2 *lavalense saccharogumia Clostridiales* sp. *Clostridium Dorea* 75 SSC/2 *saccharogumia formicigenerans Clostridium Coprococcus catus Ruminococcus* 75 *algidixylanolyticum bromii Clostridium Clostridium Gemmiger* 75 *lavalense saccharogumia formicilis Clostridium Gemmiger Ruminococcus* 75 *saccharogumia formicilis lactaris Clostridiales* sp. *Clostridium Clostridium* 75 SSC/2 *asparagiforme saccharogumia Clostridiales* sp. *Clostridium Coprococcus comes* 75 SSC/2 *saccharogumia Clostridiales* sp. *Clostridium Eubacterium hallii* 75 SSC/2 *saccharogumia Clostridium Dorea Gemmiger* 75 *saccharogumia formicigenerans formicilis Clostridium Clostridium Gemmiger* 75 *asparagiforme saccharogumia formicilis Clostridium Coprococcus comes Gemmiger* 75 *saccharogumia formicilis Clostridium Clostridium Coprococcus catus* 75 *asparagiforme saccharogumia Clostridium Clostridium Coprococcus catus* 75 *lavalense saccharogumia Clostridium Eubacterium hallii Gemmiger* 75 *saccharogumia formicilis Blautia* sp. M25 *Clostridium Eubacterium* 75 *saccharogumia ramulus Clostridiales* sp. *Clostridium Dorea longicatena* 75 SSC/2 *saccharogumia Clostridiales* sp. *Clostridium Eubacterium* 75 SSC/2 *saccharogumia ramulus Clostridium Coprococcus catus Ruminococcus* 75 *saccharogumia lactaris Clostridium Eubacterium Eubacterium rectale* 75 *saccharogumia ramulus Clostridium Eubacterium Faecalibacterium* 75 *saccharogumia ramulus prausnitzii Clostridium Eubacterium Ruminococcus* 75 *saccharogumia ramulus obeum Clostridium Eubacterium Ruminococcus* sp. 75 *saccharogumia ramulus* 5_1_39BFAA *Clostridium Eubacterium Ruminococcus* 75 *saccharogumia ramulus torques Clostridium Clostridium Eubacterium* 75 *asparagiforme saccharogumia ramulus Clostridium Coprococcus catus Dorea longicatena* 75 *saccharogumia Clostridium Coprococcus catus Eubacterium hallii* 75 *saccharogumia Clostridium Coprococcus catus Gemmiger* 75 *saccharogumia formicilis Clostridium Coprococcus comes Eubacterium* 75 *saccharogumia ramulus Clostridium Dorea longicatena Gemmiger* 75 *saccharogumia formicilis Clostridium Eubacterium hallii Eubacterium* 75 *saccharogumia ramulus Clostridium Eubacterium Ruminococcus* 75 *saccharogumia ramulus lactaris Clostridium Clostridium Eubacterium* 75 *lavalense saccharogumia ramulus Clostridium Coprococcus catus Coprococcus comes* 75 *saccharogumia Clostridium Coprococcus catus Dorea* 75 *saccharogumia formicigenerans Clostridium Dorea Eubacterium* 75 *saccharogumia formicigenerans ramulus Clostridium Eubacterium Ruminococcus* 75 *saccharogumia ramulus bromii Clostridium Coprococcus catus Eubacterium* 75 *saccharogumia ramulus Clostridium Dorea longicatena Eubacterium* 75 *saccharogumia ramulus*

(327) TABLE-US-00019 TABLE 18 Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTU *Clostridiales* sp. SSC/2. Percent of doses in which the ternary is OTU 1 OTU 2 OTU 3 present *Faecalibacterium Ruminococcus Turicibacter* 85 *prausnitzii obeum sanguinis Faecalibacterium Ruminococcus Turicibacter* 85 *prausnitzii torques sanguinis Blautia* sp. M25 *Faecalibacterium Turicibacter* 85 *prausnitzii sanguinis Blautia* sp. M25 *Ruminococcus Turicibacter* 85 *torques sanguinis Faecalibacterium Ruminococcus* sp. *Turicibacter* 85 *prausnitzii* 5_1_39BFAA *sanguinis Ruminococcus Ruminococcus Turicibacter* 85 *obeum torques sanguinis Ruminococcus Ruminococcus* sp. *Turicibacter* 85 *obeum* 5_1_39BFAA *sanguinis Ruminococcus* sp. *Ruminococcus Turicibacter* 85 5_1_39BFAA *torques sanguinis Blautia* sp. M25 *Ruminococcus Turicibacter* 85 *obeum sanguinis Faecalibacterium Ruminococcus Turicibacter* 85 *prausnitzii bromii sanguinis Ruminococcus Ruminococcus Turicibacter* 85 *bromii obeum sanguinis Ruminococcus Ruminococcus Turicibacter* 85 *bromii torques sanguinis Blautia* sp. M25 *Ruminococcus* sp. *Turicibacter* 85 5_1_39BFAA *sanguinis Ruminococcus Ruminococcus* sp. *Turicibacter* 85 *bromii* 5_1_39BFAA *sanguinis Blautia* sp. M25 *Ruminococcus Turicibacter* 85 *bromii sanguinis Clostridiales* sp. *Faecalibacterium Turicibacter* 80 SSC/2 *prausnitzii sanguinis Clostridiales* sp. *Ruminococcus Turicibacter* 80 SSC/2 *obeum sanguinis Clostridiales* sp. *Ruminococcus* sp. *Turicibacter* 80 SSC/2 5_1_39BFAA *sanguinis Clostridiales* sp. *Ruminococcus Turicibacter* 80 SSC/2 *torques sanguinis Blautia* sp. M25 *Clostridiales* sp. *Turicibacter* 80 SSC/2 *sanguinis Blautia* sp. M25 *Gemmiger Turicibacter* 80 *formicilis sanguinis Gemmiger Ruminococcus Turicibacter* 80 *formicilis obeum sanguinis Gemmiger Ruminococcus* sp. *Turicibacter* 80 *formicilis* 5_1_39BFAA *sanguinis Gemmiger Ruminococcus Turicibacter* 80 *formicilis torques sanguinis Faecalibacterium Gemmiger Turicibacter* 80 *prausnitzii formicilis sanguinis Clostridiales* sp. *Ruminococcus Turicibacter* 80 SSC/2 *bromii sanguinis Eubacterium Faecalibacterium Turicibacter* 80 *hallii prausnitzii sanguinis Eubacterium Ruminococcus Turicibacter* 80 *hallii obeum sanguinis Eubacterium Ruminococcus* sp. *Turicibacter* 80 *hallii* 5_1_39BFAA *sanguinis Eubacterium Ruminococcus Turicibacter* 80 *hallii torques sanguinis Gemmiger Ruminococcus Turicibacter* 80 *formicilis bromii sanguinis Blautia* sp. M25 *Eubacterium Turicibacter* 80 *hallii sanguinis Eubacterium Ruminococcus Turicibacter* 80 *hallii bromii sanguinis Blautia* sp. M25 *Coprococcus Turicibacter* 80 *catus sanguinis Coprococcus Faecalibacterium Turicibacter* 80 *catus prausnitzii sanguinis Coprococcus Ruminococcus Turicibacter* 80 *catus obeum sanguinis Coprococcus Ruminococcus* sp. *Turicibacter* 80 *catus* 5_1_39BFAA *sanguinis Coprococcus Ruminococcus Turicibacter* 80 *catus torques sanguinis Clostridiales* sp. *Coprococcus Turicibacter* 80 SSC/2 *catus sanguinis Coprococcus Ruminococcus Turicibacter* 80 *catus bromii sanguinis Clostridium Faecalibacterium Ruminococcus* 75 *bartlettii prausnitzii obeum Clostridium Faecalibacterium Ruminococcus* 75 *bartlettii prausnitzii torques Blautia* sp. M25 *Clostridium Faecalibacterium* 75 *bartlettii prausnitzii Clostridium Ruminococcus Ruminococcus* 75 *bartlettii obeum torques Clostridium Ruminococcus* sp. *Ruminococcus* 75 *bartlettii* 5_1_39BFAA *torques Blautia* sp. M25 *Clostridium Ruminococcus* 75 *bartlettii torques Clostridium Faecalibacterium Ruminococcus* sp. 75 *bartlettii prausnitzii* 5_1_39BFAA *Blautia* sp. M25 *Clostridium Ruminococcus* 75 *bartlettii obeum Clostridium Ruminococcus Ruminococcus* sp. 75 *bartlettii obeum* 5_1_39BFAA *Blautia* sp. M25 *Clostridium Ruminococcus* sp. 75 *bartlettii* 5_1_39BFAA *Clostridium Ruminococcus Ruminococcus* 75 *bartlettii bromii torques Clostridium Faecalibacterium Ruminococcus* 75 *bartlettii prausnitzii bromii Clostridium Ruminococcus Ruminococcus* 75 *bartlettii bromii obeum Blautia* sp. M25 *Clostridium Ruminococcus* 75 *bartlettii bromii Clostridium Ruminococcus Ruminococcus* sp. 75 *bartlettii bromii* 5_1_39BFAA *Eubacterium Faecalibacterium Turicibacter* 75 *rectale prausnitzii sanguinis Eubacterium Ruminococcus Turicibacter* 75 *rectale obeum sanguinis Eubacterium Ruminococcus* sp. *Turicibacter* 75 *rectale* 5_1_39BFAA *sanguinis Eubacterium Ruminococcus Turicibacter* 75 *rectale torques sanguinis Blautia* sp. M25 *Eubacterium Turicibacter* 75 *rectale sanguinis Clostridium Faecalibacterium Turicibacter* 75 *asparagiforme prausnitzii sanguinis Clostridium Ruminococcus Turicibacter* 75 *asparagiforme obeum sanguinis Clostridium Ruminococcus* sp. *Turicibacter* 75 *asparagiforme* 5_1_39BFAA *sanguinis Eubacterium Ruminococcus Turicibacter* 75 *rectale bromii sanguinis Clostridium Ruminococcus Turicibacter* 75 *asparagiforme torques sanguinis Blautia* sp. M25 *Clostridium Turicibacter* 75 *asparagiforme sanguinis Clostridiales* sp. *Eubacterium Turicibacter* 75 SSC/2 *hallii sanguinis Clostridiales* sp. *Gemmiger Turicibacter* 75 SSC/2 *formicilis sanguinis Clostridium Ruminococcus Turicibacter* 75 *asparagiforme bromii sanguinis Clostridium* sp. *Ruminococcus Turicibacter* 75 SS2/1 *torques sanguinis Clostridium Eubacterium Turicibacter* 75 *asparagiforme hallii sanguinis Clostridium* sp. *Faecalibacterium Turicibacter* 75 SS2/1 *prausnitzii sanguinis Clostridium* sp. *Ruminococcus Turicibacter* 75 SS2/1 *obeum sanguinis Clostridium* sp. *Ruminococcus* sp. *Turicibacter* 75 SS2/1 5_1_39BFAA *sanguinis Eubacterium*

*Gemmiger Turicibacter* 75 *hallii formicilis sanguinis Clostridiales* sp. *Clostridium* sp. *Turicibacter* 75 SSC/2 SS2/1 *sanguinis Blautia* sp. M25 *Clostridium* sp. *Turicibacter* 75 SS2/1 *sanguinis Clostridium* sp. *Ruminococcus Turicibacter* 75 SS2/1 *bromii sanguinis Clostridium* sp. *Coprococcus Turicibacter* 75 SS2/1 *catus sanguinis Collinsella Faecalibacterium Turicibacter* 75 *aerofaciens prausnitzii sanguinis Collinsella Ruminococcus Turicibacter* 75 *aerofaciens bromii sanguinis Collinsella Ruminococcus Turicibacter* 75 *aerofaciens obeum sanguinis Collinsella Ruminococcus Turicibacter* 75 *aerofaciens torques sanguinis Coprococcus Eubacterium Turicibacter* 75 *catus hallii sanguinis Coprococcus Gemmiger Turicibacter* 75 *catus formicilis sanguinis Blautia* sp. M25 *Collinsella Turicibacter* 75 *aerofaciens sanguinis Collinsella Eubacterium Turicibacter* 75 *aerofaciens hallii sanguinis Collinsella Ruminococcus* sp. *Turicibacter* 75 *aerofaciens* 5_1_39BFAA *sanguinis*

(328) TABLE-US-00020 TABLE 19 Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTU *Clostridium* sp. NML 04A032. Percent of doses in which the ternary is OTU 1 OTU 2 OTU 3 present *Clostridium Faecalibacterium Ruminococcus* 80 *asparagiforme prausnitzii champanellensis Clostridium Ruminococcus Ruminococcus* 80 *asparagiforme champanellensis torques Clostridium Ruminococcus Ruminococcus* 80 *asparagiforme champanellensis obeum Blautia* sp. M25 *Clostridium Ruminococcus* 80 *asparagiforme champanellensis Clostridium Ruminococcus Ruminococcus* sp. 80 *asparagiforme champanellensis* 5_1_39BFAA *Clostridium Ruminococcus Ruminococcus* 80 *asparagiforme bromii champanellensis Clostridium Eubacterium Ruminococcus* 80 *asparagiforme hallii champanellensis Clostridiales Eubacterium Faecalibacterium* 75 *bacterium rectale prausnitzii* 1_7_47FAA *Clostridiales Eubacterium Ruminococcus* 75 *bacterium rectale obeum* 1_7_47FAA *Clostridiales Eubacterium Ruminococcus* 75 *bacterium rectale torques* 1_7_47FAA *Blautia* sp. M25 *Clostridiales Eubacterium* 75 *bacterium rectale* 1_7_47FAA *Clostridiales Eubacterium Ruminococcus* sp. 75 *bacterium rectale* 5_1_39BFAA 1_7_47FAA *Eubacterium Faecalibacterium Ruminococcus* 75 *rectale prausnitzii champanellensis Faecalibacterium Ruminococcus Ruminococcus* 75 *prausnitzii champanellensis lactaris Clostridiales Clostridium Faecalibacterium* 75 *bacterium asparagiforme prausnitzii* 1_7_47FAA *Clostridiales Clostridium Ruminococcus* 75 *bacterium asparagiforme torques* 1_7_47FAA *Blautia* sp. M25 *Ruminococcus Ruminococcus* 75 *champanellensis lactaris Eubacterium Ruminococcus Ruminococcus* 75 *rectale champanellensis obeum Eubacterium Ruminococcus Ruminococcus* 75 *rectale champanellensis torques Ruminococcus Ruminococcus Ruminococcus* 75 *champanellensis lactaris obeum Clostridiales Clostridium Ruminococcus* 75 *bacterium asparagiforme obeum* 1_7_47FAA *Ruminococcus Ruminococcus Ruminococcus* 75 *champanellensis lactaris torques Blautia* sp. M25 *Eubacterium Ruminococcus* 75 *rectale champanellensis Clostridium Faecalibacterium Ruminococcus* 75 *lavalense prausnitzii champanellensis Eubacterium Ruminococcus Ruminococcus* 75 *rectale champanellensis lactaris Clostridiales Clostridium Ruminococcus* sp. 75 *bacterium asparagiforme* 5_1_39BFAA 1_7_47FAA *Clostridiales Eubacterium Ruminococcus* 75 *bacterium rectale bromii* 1_7_47FAA *Clostridium Ruminococcus Ruminococcus* 75 *lavalense champanellensis obeum Clostridium Ruminococcus Ruminococcus* 75 *lavalense champanellensis torques Eubacterium Ruminococcus Ruminococcus* sp. 75 *rectale champanellensis* 5_1_39BFAA *Ruminococcus Ruminococcus Ruminococcus* 75 *bromii champanellensis lactaris Ruminococcus Ruminococcus Ruminococcus* sp. 75 *champanellensis lactaris* 5_1_39BFAA *Blautia* sp. M25 *Clostridiales Clostridium* 75 *bacterium asparagiforme* 1_7_47FAA *Clostridium Eubacterium Ruminococcus* 75 *lavalense rectale champanellensis Clostridium Ruminococcus Ruminococcus* 75 *lavalense champanellensis lactaris Clostridium Ruminococcus Ruminococcus* sp. 75 *lavalense champanellensis* 5_1_39BFAA *Blautia* sp. M25 *Clostridium Ruminococcus* 75 *lavalense champanellensis Clostridium Eubacterium Ruminococcus* 75 *asparagiforme rectale champanellensis Clostridium Ruminococcus Ruminococcus* 75 *lavalense bromii champanellensis Eubacterium Ruminococcus Ruminococcus* 75 *rectale bromii champanellensis Clostridiales Clostridium Ruminococcus* 75 *bacterium asparagiforme bromii* 1_7_47FAA *Clostridium Ruminococcus Ruminococcus* 75 *asparagiforme champanellensis lactaris Clostridiales Clostridium Eubacterium* 75 *bacterium asparagiforme hallii* 1_7_47FAA *Clostridium Clostridium Ruminococcus* 75 *asparagiforme lavalense champanellensis Coprococcus Eubacterium Ruminococcus* 75 *comes rectale champanellensis Coprococcus Faecalibacterium Ruminococcus* 75 *comes prausnitzii champanellensis Coprococcus Ruminococcus Ruminococcus* 75 *comes champanellensis obeum Coprococcus Ruminococcus Ruminococcus* 75 *comes champanellensis torques Dorea Eubacterium Ruminococcus* 75 *formicigenerans rectale champanellensis Dorea Faecalibacterium Ruminococcus* 75 *formicigenerans prausnitzii champanellensis Dorea Ruminococcus Ruminococcus* 75 *formicigenerans champanellensis obeum Dorea Ruminococcus Ruminococcus* 75 *formicigenerans champanellensis torques Dorea Ruminococcus Ruminococcus* 75 *longicatena champanellensis obeum Blautia* sp. M25 *Coprococcus Ruminococcus* 75 *comes champanellensis Blautia* sp. M25 *Dorea Ruminococcus* 75 *longicatena champanellensis Dorea Ruminococcus Ruminococcus* 75 *formicigenerans champanellensis lactaris Dorea Faecalibacterium Ruminococcus* 75 *longicatena prausnitzii champanellensis Dorea Ruminococcus Ruminococcus* 75 *longicatena champanellensis torques Eubacterium Eubacterium Ruminococcus* 75 *hallii rectale champanellensis Eubacterium Ruminococcus Ruminococcus* 75 *hallii champanellensis lactaris Blautia* sp. M25 *Dorea Ruminococcus* 75 *formicigenerans champanellensis Clostridium Dorea Ruminococcus* 75 *asparagiforme longicatena champanellensis Clostridium Gemmiger Ruminococcus* 75 *asparagiforme formicilis champanellensis Clostridium Dorea Ruminococcus* 75 *lavalense formicigenerans champanellensis Dorea Ruminococcus Ruminococcus* sp. 75 *formicigenerans champanellensis* 5_1_39BFAA *Dorea Eubacterium Ruminococcus* 75 *longicatena rectale champanellensis Dorea Ruminococcus Ruminococcus* 75 *longicatena champanellensis lactaris Clostridiales* sp. *Clostridium Ruminococcus* 75 SSC/2 *asparagiforme champanellensis Clostridium Coprococcus Ruminococcus* 75 *asparagiforme comes champanellensis Clostridium Dorea Ruminococcus* 75 *asparagiforme formicigenerans champanellensis Clostridium Dorea Ruminococcus* 75 *lavalense longicatena champanellensis Coprococcus Ruminococcus Ruminococcus* 75 *comes bromii champanellensis Coprococcus Ruminococcus Ruminococcus* 75 *comes champanellensis lactaris Coprococcus Ruminococcus Ruminococcus* sp. 75 *comes champanellensis* 5_1_39BFAA *Dorea Ruminococcus Ruminococcus* 75 *longicatena bromii champanellensis Dorea Ruminococcus Ruminococcus* sp. 75 *longicatena champanellensis* 5_1_39BFAA *Clostridium Coprococcus Ruminococcus* 75 *asparagiforme catus champanellensis Clostridium Eubacterium Ruminococcus* 75 *lavalense hallii champanellensis Dorea Eubacterium Ruminococcus* 75 *formicigenerans hallii champanellensis Dorea Dorea Ruminococcus* 75 *formicigenerans longicatena champanellensis Clostridium Coprococcus Ruminococcus* 75 *lavalense comes champanellensis Coprococcus Dorea Ruminococcus* 75 *comes formicigenerans champanellensis Coprococcus Eubacterium Ruminococcus* 75 *comes hallii champanellensis Dorea Ruminococcus Ruminococcus* 75 *formicigenerans bromii champanellensis Coprococcus Dorea Ruminococcus* 75 *comes longicatena champanellensis Dorea Eubacterium Ruminococcus* 75 *longicatena hallii champanellensis Clostridium Collinsella Ruminococcus* 75 *asparagiforme aerofaciens champanellensis*

(329) TABLE-US-00021 TABLE 20 Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTUs *Clostridium* sp. NML 04A032, *Ruminococcus lactaris*, and *Ruminococcus torques*. Percent of doses in which the ternary is OTU 1 OTU 2 OTU 3 present *Clostridiales* sp. *Faecalibacterium Ruminococcus* 75 SM4/1 *prausnitzii obeum Clostridiales* sp. *Faecalibacterium Ruminococcus* 75 SM4/1 *prausnitzii torques Clostridiales* sp. *Ruminococcus Ruminococcus* 75 SM4/1 *obeum torques Blautia* sp. *Clostridiales* sp. *Faecalibacterium* 75 M25 SM4/1 *prausnitzii Clostridiales* sp. *Ruminococcus* sp. *Ruminococcus* 75 SM4/1 5_1_39BFAA *torques Blautia* sp. *Clostridiales* sp. *Ruminococcus* 75 M25 SM4/1 *torques Clostridiales* sp. *Faecalibacterium Ruminococcus* sp. 75 SM4/1 *prausnitzii* 5_1_39BFAA *Blautia* sp. *Clostridiales* sp. *Ruminococcus* 75 M25 SM4/1 *obeum Blautia* sp. *Clostridiales* sp. *Ruminococcus* sp. 75 M25 SM4/1 5_1_39BFAA *Clostridiales* sp. *Ruminococcus Ruminococcus* sp. 75 SM4/1 *obeum* 5_1_39BFAA *Clostridiales* sp. *Clostridium Faecalibacterium* 75 SM4/1 *asparagiforme prausnitzii Clostridiales* sp. *Clostridium Ruminococcus* 75 SM4/1 *asparagiforme torques Clostridiales* sp. *Clostridium Ruminococcus* 75 SM4/1 *asparagiforme obeum Blautia* sp. *Clostridiales* sp. *Ruminococcus* 75 M25 SM4/1 *bromii Clostridiales* sp. *Faecalibacterium Ruminococcus* 75 SM4/1 *prausnitzii bromii Clostridiales* sp. *Ruminococcus Ruminococcus* 75 SM4/1 *bromii obeum Clostridiales* sp. *Ruminococcus Ruminococcus* sp. 75 SM4/1 *bromii* 5_1_39BFAA *Clostridiales* sp. *Ruminococcus Ruminococcus* 75 SM4/1

*bromii torques Blautia* sp. *Clostridiales* sp. *Clostridium* 75 M25 SM4/1 *asparagiforme Clostridiales* sp. *Clostridium Ruminococcus* sp. 75 SM4/1 *asparagiforme* 5_1_39BFAA *Clostridiales* sp. *Eubacterium Faecalibacterium* 75 SM4/1 *hallii prausnitzii Clostridiales* sp. *Eubacterium Ruminococcus* 75 SM4/1 *hallii obeum Clostridiales* sp. *Eubacterium Ruminococcus* 75 SM4/1 *hallii torques Blautia* sp. *Clostridiales* sp. *Eubacterium* 75 M25 SM4/1 *hallii Clostridiales* sp. *Eubacterium Ruminococcus* sp. 75 SM4/1 *hallii* 5_1_39BFAA *Clostridiales* sp. *Clostridium Ruminococcus* 75 SM4/1 *asparagiforme bromii Clostridiales* sp. *Clostridium Eubacterium* 75 SM4/1 *asparagiforme hallii Clostridiales* sp. *Eubacterium Ruminococcus* 75 SM4/1 *hallii bromii*
(330) TABLE-US-00022 TABLE 21 Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTUs *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Oscillibacter* sp. G2, *Ruminococcus lactaris*, and *Ruminococcus torques*. Percent of doses in which the ternary is OTU 1 OTU 2 OTU 3 present *Clostridium Faecalibacterium Ruminococcus* 75 *bifermentans prausnitzii obeum Clostridium Ruminococcus Ruminococcus* 75 *bifermentans obeum torques Clostridium Ruminococcus Ruminococcus* sp. 75 *bifermentans obeum* 5_1_39BFAA *Blautia* sp. *Clostridium Faecalibacterium* 75 M25 *bifermentans prausnitzii Blautia* sp. *Clostridium Ruminococcus* 75 M25 *bifermentans torques Clostridium Faecalibacterium Ruminococcus* sp. 75 *bifermentans prausnitzii* 5_1_39BFAA *Clostridium Faecalibacterium Ruminococcus* 75 *bifermentans prausnitzii torques Clostridium Ruminococcus* sp. *Ruminococcus* 75 *bifermentans* 5_1_39BFAA *torques Blautia* sp. *Clostridium Ruminococcus* 75 M25 *bifermentans obeum Blautia* sp. *Clostridium Ruminococcus* sp. 75 M25 *bifermentans* 5_1_39BFAA *Clostridium Ruminococcus Ruminococcus* 75 *bifermentans bromii obeum Blautia* sp. *Clostridium Gemmiger* 75 M25 *bifermentans formicilis Clostridium Faecalibacterium Ruminococcus* 75 *bifermentans prausnitzii bromii Clostridium Gemmiger Ruminococcus* 75 *bifermentans formicilis obeum Clostridium Gemmiger Ruminococcus* sp. 75 *bifermentans formicilis* 5_1_39BFAA *Clostridium Ruminococcus Ruminococcus* 75 *bifermentans bromii torques Clostridium Faecalibacterium Gemmiger* 75 *bifermentans prausnitzii formicilis Blautia* sp. *Clostridium Ruminococcus* 75 M25 *bifermentans bromii Clostridium Gemmiger Ruminococcus* 75 *bifermentans formicilis torques Clostridium Ruminococcus Ruminococcus* sp. 75 *bifermentans bromii* 5_1_39BFAA *Clostridium Gemmiger Ruminococcus* 75 *bifermentans formicilis bromii*
(331) TABLE-US-00023 TABLE 22 Selected OTUs that may be present in the tables, specification, or in the art with their alternate names, e.g., the current name used per NCBI. Reference 1 is Kaur et al., "*Hungatella effluvii* gen. nov., sp. nov., an obligately anaerobic bacterium isolated from an effluent treatment plant, and reclassification of *Clostridium hathewayi* as *Hungatella hathewayi* gen. nov., comb. nov.", Int J Sys Evol Microbiol, March 2014, vol. 64, pp. 710-718. Reference 2 is Gerritsen et al., "Characterization of *Romboutsia ilealis* gen. nov., sp. nov., isolated from the gastro-intestinal tract of a rat, and proposal for the reclassification of five closely related members of the genus *Clostridium* into the genera *Romboutsia* gen. nov., *Intestinibacter* gen, nov., *Terrisporobacter* gen. nov. and *Asaccharospora* gen. nov.", Int J Sys Evol Microbiol, May 2014, vol. 64, pp. 1600-1616. Alternate OTU OTU name name Reference *Clostridium Hungatella* 1 *hathewayi hathewayi Clostridium Romboutsia* 2 *lituseburense lituseburense Clostridium Intestinibacter* 2 *bartlettii bartlettii Clostridium Terrisporobacter* 2 *glycolicum glycolicus Clostridium Terrisporobacter* 2 *mayombei mayombei Clostridium Asaccharospora* 2 *irregulare irregularis*

## Claims

1. A method of functionally populating the gastrointestinal tract of a human subject in need thereof, comprising administering to the subject a composition comprising: (a) a first species of an isolated bacterium capable of forming a spore, (b) a second species of isolated bacterium capable of forming a spore, and (c) a third species of isolated bacterium capable of forming a spore, and (d) a capsule; wherein the capsule encapsulates the first species, the second species, and third species; wherein the first species, the second species, and the third species are not identical; wherein the first species is *Clostridium orbiscindens*, the second species is *Clostridium bolteae*, and the third species is *Lachnospiraceae bacterium* 5_1_57FAA, wherein the *Clostridium orbiscindens* comprises a 16S rDNA sequence that is at least 97% identical to a 16S rDNA the sequence set forth in SEQ ID NO: 609; and wherein a combination of the first species, the second species, and the third species is capable of decreasing and/or inhibiting the growth and/or colonization of at least one type of pathogenic bacteria.
2. The method of claim 1, wherein the combination of the first species, the second species, and the third species is capable of synergistically inhibiting the growth and/or colonization of the at least one type of pathogenic bacteria in a CivSim assay with a confidence interval of >99% (++++).
3. The method of claim 1, wherein one or more of the first species, the second species, and the third species are in the form of spores.
4. The method of claim 3, wherein the first species, the second species, and the third species are in the form of spores.
5. The method of claim 1, wherein the composition further comprises a capsule, which encapsulates one or more of the first species, the second species, and the third species.
6. The method of claim 5, wherein the capsule is enterically coated.
7. The method of claim 5, wherein the capsule encapsulates the first species, the second species, and the third species.
8. The method of claim 1, wherein the composition further comprises glycerol.
9. The method of claim 7, wherein the composition further comprises glycerol.
10. The method of claim 1, wherein the *Clostridium orbiscindens* comprises a 16S rDNA sequence that is at least 98% identical to the sequence set forth in SEQ ID NO: 609.
11. The method of claim 1, wherein the *Clostridium bolteae* comprises a 16S rDNA sequence that is at least 97% identical to the sequence set forth in SEQ ID NO: 559.
12. The method of claim 1, wherein the *Lachnospiraceae bacterium* 5_1_57FAA comprises a 16S rDNA sequence that is at least 97% identical to the sequence set forth in SEQ ID NO: 1054.
13. The method of claim 1, wherein the *Clostridium orbiscindens* comprises a 16S rDNA sequence that is at least 99% identical to the sequence set forth in SEQ ID NO: 609.
14. The method of claim 1, wherein the *Clostridium orbiscindens* comprises the 16S rDNA sequence set forth in SEQ ID NO: 609.
15. The method of claim 1, wherein the *Clostridium bolteae* comprises a 16S rDNA sequence that is at least 98% identical to the sequence set forth in SEQ ID NO: 559.
16. The method of claim 1, wherein the *Clostridium bolteae* comprises a 16S rDNA sequence that is at least 99% identical to the sequence set forth in SEQ ID NO: 559.
17. The method of claim 1, wherein the *Clostridium bolteae* comprises the 16S rDNA sequence set forth in SEQ ID NO: 559.
18. The method of claim 1, wherein the *Lachnospiraceae bacterium* 5_1_57FAA comprises a 16S rDNA sequence that is at least 98% identical to the sequence set forth in SEQ ID NO: 1054.
19. The method of claim 1, wherein the *Lachnospiraceae bacterium* 5_1_57FAA comprises a 16S rDNA sequence that is at least 99% identical to the sequence set forth in SEQ ID NO: 1054.
20. The method of claim 1, wherein the *Lachnospiraceae bacterium* 5_1_57FAA comprises the 16S rDNA sequence set forth in SEQ ID NO: 1054.